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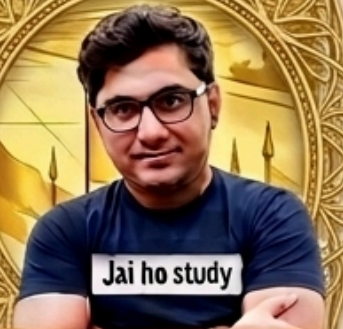


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अंगद सामान्य अध्ययन

प्रतियोगी
परीक्षाओं हेतु

सफलता
की सही दिशा

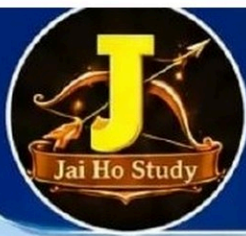


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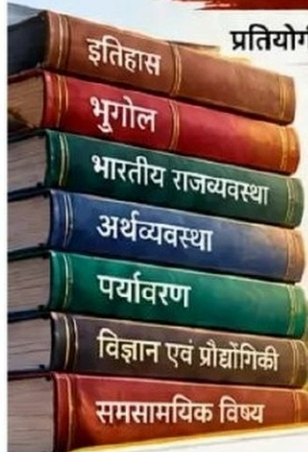


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अंगद

सामान्य अध्ययन

प्रतियोगी परीक्षाओं के लिए समग्र एवं अद्यतन अध्ययन सामग्री



Utkarsh Dwivedi सर
Founder & Author

आपका लक्ष्य हमारी जिम्मेदारी



परीक्षा केंद्रित
सामग्री



सरल एवं
सटीक भाषा



अद्यतन
तथ्यों का समावेश



चार्ट, तालिका
एवं चित्रों सहित



सफलता की
दिशा में एक मजबूत कदम

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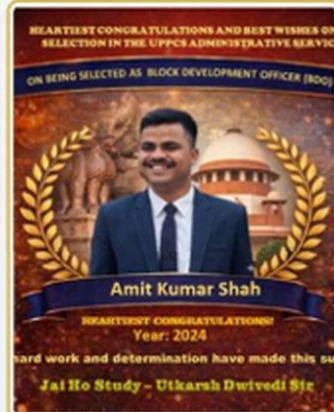
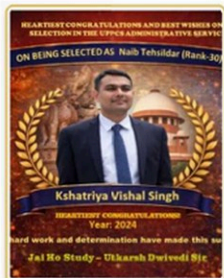
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सफलता इच्छाफा नहीं, सही चुनाव, सही मार्गदर्शन और निरंतर मेहनत का परिणाम है।

हजारों घंटों की रिसर्च, सटीक नोट्स और परीक्षा के पैटर्न पर पैनी नजर — यही जय हो स्टडी को अलग बनाती है। हमारे सफल छात्र मेहनत, सही रणनीति और भरोसेमंद मार्गदर्शन का प्रमाण हैं।



हम वादों में नहीं, परिणामों में विश्वास रखते हैं।
अगली बारी आपकी है — बस सही दिशा में पहला कदम बढ़ाएं।

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01

History

UPPCS PRELIMINARY EXAMINATION (PRELIMS): HISTORY

1 Periods of History

Period	Main Characteristic	Principal Example / Era
Prehistoric Period	No written evidence is available.	Stone Age
Proto-historic Period	Evidence of a script exists, but it remains undeciphered.	Indus Valley Civilisation
Historical Period	Written evidence is available and has been deciphered.	Begins from the 6th century BCE (Mahajanapada period)

2 Stone Age: Tools, Characteristics and Major Sites

Period	Tools and Main Characteristics	Major Sites (India/Pakistan)
Lower Palaeolithic	Heavy, crude stone tools; handaxe , cleaver	Pallavaram and Attirampakkam (Tamil Nadu), Hunsgi-Isampur (Karnataka), Soan Valley (Pakistan), Narmada Valley (MP)
Middle Palaeolithic	Flake-based tools ; scrapers and other small tools	Nevasa (Maharashtra), Bhimbetka (MP), Belan Valley (UP) , Didwana (Rajasthan)
Upper Palaeolithic	Blades and blade-based tools ; use of bone implements	Bhimbetka (MP), Patne (Maharashtra), Kurnool Caves (Andhra Pradesh)
Mesolithic	Microliths (tiny stone tools); main occupation: hunting and food-gathering	Bagor (Rajasthan), Langhnaj (Gujarat), Sarai Nahar Rai, Mahadaha, Damdama (UP) , Adamgarh (MP)
Neolithic	Ground and polished stone tools; development of agriculture, animal husbandry and permanent settlement	Mehrgarh, Burzahom, Gufkral, Koldihwa, Mahagara, Lahuradewa (UP) , Chirand, Sanganakallu-Piklihal

3 Major Neolithic and Mesolithic Sites (UPPCS Special)

Site	State/Region	Evidence Found and Exam Facts
Lahuradewa	Uttar Pradesh	Earliest evidence of rice (paddy) and agriculture (as per the latest findings).
Koldihwa	Uttar Pradesh	On the bank of the Belan river (south of Allahabad/Prayagraj); early evidence of paddy cultivation .
Mahadaha	Uttar Pradesh	Circular huts, cultivation of paddy and barley, animal husbandry; also important for Mesolithic burials .
Sarai Nahar Rai, Damdama	Uttar Pradesh	Important Mesolithic sites of Uttar Pradesh.
Chopani Mando	Uttar Pradesh	Located in the Belan Valley; evidence of microliths and handmade pottery (a Mesolithic–Neolithic transition site).
Burzahom	Kashmir	Famous for pit dwellings ; evidence of dogs buried with their masters in graves.
Gufkral	Kashmir	Bone tools, grey pottery and grinding stones (silbatta).
Mehrgarh	Baluchistan (Pakistan)	Earliest evidence of wheat , cultivation of two varieties of barley; the earliest site of settled life , agriculture and animal husbandry.
Chirand	Bihar	Especially famous for Neolithic bone implements/tools .
Sanganakallu and Piklihal	Karnataka	Evidence of ash mounds , ash heaps and habitation sites (Neolithic sites of South India).
Sarutaru and Daojali	Assam	Primitive axes, painted pottery (Sarutaru) and evidence of agriculture and animal husbandry (Daojali).

4 Indus Valley Civilisation (IVC): Geographical Extent and Coastal Cities

Boundary/Direction	Principal Site	River Bank	Geographical Position
North	Manda (Jammu)	Chenab river	Northernmost point of the civilisation
South	Daimabad (Ahmednagar, Maharashtra)	Pravara river	Southernmost point of the civilisation
East	Alamgirpur (Meerut, Uttar Pradesh)	Hindon river	Easternmost point of the civilisation
West	Sutkagendor (Baluchistan, Pakistan)	Dasht river	Westernmost point of the civilisation

Geographical Extent Facts: The Indus Valley Civilisation was **triangular** in shape and covered a total area of about **1.3 million sq km (13 lakh sq km)**. Its extent included Punjab, Sindh, Baluchistan, Rajasthan, Gujarat and western Uttar Pradesh.

> Major Ports / Coastal Towns of the Indus Civilisation

- **Lothal (Gujarat):** World-famous trading centre known for its **dockyard**.
- **Sutkagendor (Baluchistan):** Principal trading centre of the western frontier of the Indus Civilisation.
- **Nageshwar (Gujarat):** Especially famous for its **shell (conch) industry**.
- **Balakot (Baluchistan):** Coastal Harappan site; centre of **marine resources and the shell industry**.

5 Major Harappan Sites, Excavations and Important Finds

Site (River Bank / Location)	Excavator (Year)	Main Evidence and Remains Found (Exam Points)
Harappa (left bank of the Ravi, Montgomery, Punjab–Pakistan)	Daya Ram Sahni (1921)	• Great Granary • Cemetery R-37 • Broken torso in a dancing posture • Rectangular seal showing a plant growing out of a woman's womb • Bronze human figure and seals with the one-horned animal (unicorn)
Mohenjo-daro (right bank of the Indus, Larkana, Sindh–Pakistan)	Rakhal Das Banerji (1922)	• Great Bath • Great Granary • Bronze statue of the Dancing Girl • The famous Pashupati Seal (a three-faced yogi seated in padmasana)
Lothal (Bhogava river, Ahmedabad, Gujarat)	S.R. Rao (1955)	• Dockyard • Three paired (joint) burials • Bead-making factories • Ivory scale / measuring rod and fire altars
Dholavira (Rann of Kutch, Gujarat)	J.P. Joshi (1967–68) and R.S. Bisht (1990–91)	• Advanced water-management system (rock-cut tanks / huge reservoirs) • Earliest inscription (signboard) with 10 signs
Kalibangan (Hanumangarh, Rajasthan)	B.B. Lal and B.K. Thapar (1961)	• Evidence of a ploughed field • Buildings of mud bricks • Earliest evidence of an earthquake • Wells and fire altars in almost every house

> Other Harappan Sites and Important Objects:

- **Banawali (Haryana): Toy model of a plough (terracotta plough)**, beads of lapis lazuli and carnelian, seals and weights.
- **Daimabad (Maharashtra):** Bronze figures of **a man driving a chariot**, a bull and a rhinoceros; a seal bearing the Indus script and pottery.
- **Nausharo (near Mehrgarh):** Evidence of **vermilion (sindoor) in the hair-parting** of women.
- **Balakot (Pakistan):** Fragments of **bangles made of shell** (shell industry).
- **Manda (Jammu):** Bronze **spiral-headed pin** (northernmost site).
- **Hulas (UP):** A **stamp/sealing** bearing the Indus script.
- **Sutkagendor (Pakistan):** Remains of clay bangles.

6 Vedic Literature: Vedas, Priests and Upavedas

Veda	Priest (Ritvik)	Upaveda	Nature of Compilation / Main Subject
Rigveda	Hota (Hotri)	Ayurveda	Collection of richas (hymns/mantras).
Yajurveda (two parts: <i>Shukla</i> and <i>Krishna</i>)	Adhvaryu	Dhanurveda	Hymns and rules related to sacrificial rites (in both prose and verse).
Samaveda	Udgata	Gandharvaveda	Collection of songs (chants); regarded as the father of Indian music.
Atharvaveda	Brahma (the senior priest who supervises the sacrifice)	Shilpaveda	Contains charms and spells, magic, remedies against disease and medicines.

1 Evolution of Vedic Literature

Stage (Sequence)	Main Characteristic and Definition
1. Samhita	Comprises the four Vedas: Rigveda, Yajurveda, Samaveda and Atharvaveda.
2. Brahmana	Prose texts written as commentaries (exegesis) on the Samhitas.
3. Aranyaka	Appendices to the Brahmana texts, composed in the forests (aranya).
4. Upanishad	The special and concluding part of the Aranyakas, regarded as ' Vedanta ' and the source of Indian philosophy.

2 Vedas, Shakhas, Brahmanas, Aranyakas and Upanishads (Complete Compendium)

Veda	Principal Shakhas (Recensions)	Related Brahmanas	Related Aranyakas	Related Upanishads	Special Historical Facts (Exam Facts)
Rigveda	<i>Not mentioned</i>	Aitareya, Kaushitaki (Shankhayana)	Aitareya, Kaushitaki	Aitareya, Kaushitaki	• Aitareya Brahmana: first description of the Rajasuya sacrifice and of the four yugas (Satyuga, Treta, Dvapara, Kaliyuga).
Yajurveda	• Krishna Yajurveda: Kathaka, Kapishtala, Maitrayani, Taittiriya • Shukla Yajurveda: Kanva (Vajasaneyi Samhita), Madhyandina	Taittiriya, Shatapatha	Taittiriya, Maitrayani, Brihadaranyaka	Taittiriya, Kathopanishad, Maitrayani, Shvetashvatara, Ishopanishad, Brihadaranyaka	• Shatapatha Brahmana: Yajnavalkya–Maitreyi dialogue, the story of the Great Flood, the story of Videgha Madhava, the Pururavas–Urvashi legend. • Brihadaranyaka Upanishad: the saying "Tamaso ma jyotirgamaya" and the Yajnavalkya–Gargi dialogue. • Kathopanishad: the famous Yama–Nachiketa dialogue.
Samaveda	Kauthuma, Ranayaniya, Jaiminiya	Panchavimsha (Tandya), Chhandogya, Shadvimsha, Jaiminiya	Chhandogya, Jaiminiya	Chhandogya, Kena	• Chhandogya Upanishad: the oldest Upanishad; first mention of 'Krishna'; the story of Satyakama Jabala, the dialogue of Uddalaka Aruni and Shvetaketu. • Three Ashramas: Brahmacharya, Grihastha and Vanaprastha are mentioned in the Chhandogya. • Taittiriya Upanishad: gives women the right to wear the sacred thread (yajnopavita).

Veda	Principal Shakhas (Recensions)	Related Brahmanas	Related Aranyakas	Related Upanishads	Special Historical Facts (Exam Facts)
Atharvaveda	Shaunakiya, Pippalada	Gopatha	<i>No Aranyaka</i>	Mundakopanishad, Mandukya, Prashna	• Mundakopanishad: India's national motto 'Satyameva Jayate' is taken from here. • Mandukya Upanishad: the shortest Upanishad.

3 Mandalas of the Rigveda and their Composers/Features

Mandala	Composer (Rishi)	Main Feature / Evidence
First	Madhuchhanda, Medhatithi, Gotama, etc.	Early mandala
Second	Gritsamada	-
Third	Vishvamitra	Contains the famous Gayatri Mantra (dedicated to the deity Savitri/Sun).
Fourth	Vamadeva	Mentions mantras related to agriculture .
Fifth	Atri	-
Sixth	Bharadvaja	-
Seventh	Vasishtha	Describes the famous Dasharajna War (Battle of Ten Kings) .
Eighth	Kanva and Angiras	-
Ninth	Various rishis	Called the ' Soma Mandala ' (dedicated to the deity Soma).
Tenth	Vimada, Indra, Shachi and others	Contains the Purusha Sukta , which first gives the theory of the origin of the four varnas (Brahmana, Kshatriya, Vaishya, Shudra).

4 Principal Deities of the Rigvedic Period

Deity	Importance / Feature in the Rigvedic Period
Indra (Purandara)	The most prominent deity of the Rigveda; leader in war and god of rain .
Agni	Intermediary between gods and men .

Deity	Importance / Feature in the Rigvedic Period
Varuna	Creator of the sky, earth and sun; god of the ocean; controller of the world; sustainer of the gods; upholder of cosmic order (Ritasya Gopa) , symbol of truth; agent of the change of seasons and of day and night.
Dyaus	God of the sky (the oldest deity).
Soma	God of plants (vegetation).
Ashvins	Deities who remove calamities (gods of medicine).
Pushan	God of cattle (became the god of the Shudras in the Later Vedic period).
Usha	Goddess of progress and rising (dawn).

5 Major Vedic Sacrifices

Name of Sacrifice	Main Purpose and Description
Ashvamedha	The most important sacrifice ; performed by the king to extend the boundaries of his kingdom (sovereignty).
Rajasuya	Sacred sacrifice related to the king's coronation .
Vajapeya	Sacrifice related to the king's display of valour and a chariot race (entertainment).
Agnishtoma	Sacrifice in which animals were offered to Agni and Soma juice was drunk to please the gods.

6 Regional Systems of Rule Described in the Aitareya Brahmana

Direction/Region	Type of Rule	Title of the Ruler
East	Samrajya	Samrat
West	Svarajya	Svarat
North	Vairajya	Virat
South	Bhojya	Bhoja
Madhya Desha (Centre)	Rajya	Raja

7 Ancient and Modern Names of Vedic Rivers (Match the Following Special)

Ancient Name	Modern Name	Ancient Name	Modern Name
Vitasta	Jhelum	Kubha	Kabul
Parushni	Ravi	Krumu	Kurram
Asikni	Chenab	Suvastu	Swat
Vipasha	Beas	Gomati	Gomal
Shutudri	Sutlej	Sarayu	Ghaghara
Sadanira	Gandak	Yamuna	Yamuna
Drishadvati	Associated with the Ghaggar region	Marudvridha	Associated with the Ravi/Chenab region

8 Vedic Administration and Principal Officials (Ratnins/Officials)

Administrative Post (Officer)	Department / Main Function
Kulapati	Head of the family (kula); officer of family administration.
Gramani	Head of the village; officer concerned with village administration and military leadership.
Spasha	Spy / secret agent ; messenger who gathered information for the king.
Vrajapati	Chief officer in charge of pastures and livestock .
Bhagadugha	Revenue collector (tax collector) ; collected taxes for the king.
Jivagribha	Police officer ; caught criminals and maintained law and order.
Akshavapa	Accountant ; the king's partner in the game of dice and officer of income and expenditure.
Adhipati	Chief Justice ; highest officer of the judicial system.
Sangrihitri	Treasurer ; officer in charge of managing the royal treasury.
Takshan	Carpenter; skilled artisan who made objects from wood.
Palagala	Messenger; carried royal messages from one place to another.
Govikartana	Keeper of forests and of the hunt (game) .

9 Important Facts Related to the Puranas

- **Composers:** Sage Lomaharsha and his son Ugrashrava.
- **Total number:** 18 Puranas.
- **Period of redaction:** The Puranas were recast into their present form during the **Gupta period**.
- **Social rule:** Women and Shudras were not permitted to read the Vedas, but they could listen to the Puranas.

> Principal Puranas and the Dynasties/Details Associated with Them:

Name of Purana	Historical Importance / Associated Dynasty / Details
Matsya Purana	• The oldest and most authentic of the Puranas. • Historical account of the Satavahana dynasty.
Vishnu Purana	• Source of information on the history of the Maurya dynasty.
Vayu Purana	• Mentions the history and genealogy of the Gupta dynasty.
Agni Purana	• Important Purana that gives details of agriculture along with polity (monarchy).

Note

Note: Of the 18 Puranas, the genealogies of kings are found in only **five Puranas** (Matsya, Vayu, Vishnu, Brahmanda and Bhagavata Purana). The best systematic account of Indian historical traditions is found in the Puranas.

1 Principal Mottos / Sayings and their Authentic Sources

Motto / Saying	Authentic Source	Special Exam Note (Exam Warning/Fact)
Satyameva Jayate	Mundakopanishad	• India's national motto .
Gayatri Mantra	Rigveda (Third Mandala)	• Composer: Sage Vishvamitra. • Dedicated to: the deity Savitri (Sun).
Tamaso ma jyotirgamaya	Brihadaranyaka Upanishad (1.3.28) / Rigveda	• A principal great saying (mahavakya) of the Upanishad.
Aham Brahmasmi	Brihadaranyaka Upanishad	• A principal Vedic mahavakya.
Sarve bhavantu sukhinah	Brihadaranyaka	• A popular peace prayer; do not write it as the original mantra of the Brihadaranyaka Upanishad.
Yama–Nachiketa dialogue	Kathopanishad	• Nachiketa receives from Yama the knowledge of the soul and of death .

Motto / Saying	Authentic Source	Special Exam Note (Exam Warning/Fact)
Vasudhaiva Kutumbakam	Maha Upanishad (6.71–75)	• The whole earth is one family.
Janani janmabhumishcha svargadapi gariyasi	Ramayana (Yuddha Kanda)	• Mother and motherland are greater even than heaven.
Karmanyevadhikaraste...	Shrimad Bhagavad Gita	• Message of doing one's duty without worrying about the fruits.
Vidya dadati vinayam	Hitopadesha	• Note: do not write it as a line from the Gita; it is from the Hitopadesha .
Atithi Devo Bhava	Taittiriya Upanishad (1.11.2) / Taittiriya Brahmana	• Instruction to treat the guest as equal to a god.
Tat Tvam Asi	Chhandogya Upanishad	• The famous dialogue of Sage Uddalaka and Shvetaketu .
Satyam jnanam anantam Brahma	Taittiriya Upanishad	• Note: do not write 'Satyam Shivam Sundaram' as a line from the Kathopanishad.

2

Indian Philosophies and their Founders (Shad-darshana and Philosophical Schools)

Philosophy / Philosophical School	Founder
Sankhya	Sage Kapila
Yoga	Maharishi Patanjali
Nyaya	Sage Gautama
Vaisheshika	Maharishi Kanada (or Uluka)
Purva Mimamsa	Maharishi Jaimini
Uttara Mimamsa	Maharishi Badarayana
Charvaka	Maharishi Charvaka
Advaita (Non-dualism)	Adi Shankaracharya
Dvaita (Dualism)	Madhvacharya
Dvaitadvaita (Dualistic non-dualism)	Nimbarkacharya
Vishishtadvaita (Qualified non-dualism)	Ramanujacharya
Shuddhadvaita (Pure non-dualism)	Vallabhacharya

3 The 16 Mahajanapadas: Capitals and Present Geographical Location

Mahajanapada	Capital	Present Geographical Region
Kashi	Varanasi	Eastern Uttar Pradesh (Varanasi region)
Koshala	• North Koshala: Shravasti • South Koshala: Saketa / Ayodhya	Awadh and eastern Uttar Pradesh
Malla	Kushinara (Kushinagar) and Pava	Eastern Uttar Pradesh (Kushinagar–Gorakhpur region)
Vatsa	Kaushambi	Region around Prayagraj
Kuru	Indraprastha (<i>other city: Hastinapur</i>)	Delhi, Haryana and western Uttar Pradesh
Panchala	• North Panchala: Ahichchhatra • South Panchala: Kampilya	Western Uttar Pradesh (Bareilly–Farrukhabad region)
Shurasena	Mathura	Braj region (western Uttar Pradesh)
Chedi	Shuktimati / Sotthivati	Bundelkhand (Uttar Pradesh and Madhya Pradesh)
Anga	Champa	Eastern Bihar (Bhagalpur–Munger region)
Magadha	Girivraja / Rajagriha (<i>later Pataliputra</i>)	Southern Bihar (Patna–Gaya–Nalanda region)
Vajji	Vaishali	Northern Bihar (Vaishali–Muzaffarpur region)
Matsya	Viratnagar	Eastern Rajasthan (Jaipur–Alwar region)
Avanti	• North Avanti: Ujjayini • South Avanti: Mahishmati	Malwa and western Madhya Pradesh
Gandhara	Taxila (or Pushkalavati)	North-western Pakistan (Peshawar–Rawalpindi)
Kamboja	Rajapura (or Hataka)	North-western Pakistan and north-eastern Afghanistan
Ashmaka	Potana / Potali (or Paithan)	Godavari valley (Maharashtra–Telangana region)

4 Jainism: Life of Mahavira Swami (Fact Sheet)

Event of Life	Details and Exam Facts
Birth	Born in 599 BCE (traditional) at Kundagrama / Kundalpur (Vaishali) .

Event of Life	Details and Exam Facts
Childhood name	Vardhamana.
Clan/Family	Father Siddhartha (head of the Jnatrika Kshatriya clan).
Mother	Trishala / Videhadatta (sister of Chetaka, chief of the Lichchhavi republic of Vaishali).
Wife and daughter	Wife: Yashoda (a girl of the Kaundinya gotra). Daughter: Anojja / Priyadarshana.
Attainment of Kaivalya (knowledge)	Under a sal tree on the bank of the Rijupalika river , near Jrimbhikagrama.
New names after enlightenment	Came to be called Kevalin, Arhat (the worthy one), Jina (the conqueror) and Nirgrantha (free from bonds).
First disciple	His son-in-law Jamali (husband of Priyadarshana).
Philosophical doctrine propounded	Anekantavada / Syadvada (Saptabhangi Naya).
Mahaparinirvana / death	At Pavapuri near Rajgir, according to the traditional date of 527 BCE.
Meaning of the word Tirthankara	One who shows the way to cross the ocean of worldly existence (related to Jainism).
Place in the order of Tirthankaras	Mahavira Swami was the 24th and last Tirthankara of Jainism.

5 Important Facts Related to the Puranas (Quick Revision Link)

- **Total number:** 18
- **Composers:** Sage Lomaharsha and his son Ugrashrava.
- **Redaction:** Took place in the Gupta period. Women and Shudras had the right to listen to them, but not to read them.
- **Principal dynastic links:**
 - **Vishnu Purana** → Maurya dynasty
 - **Vayu Purana** → Gupta dynasty
 - **Matsya Purana** → Satavahana dynasty (the oldest and most authentic Purana)
 - **Agni Purana** → Account of agriculture along with polity (monarchy).

1 Jainism: Chaturyama Teaching, Pancha Mahavrata and the Tri-ratna

Vow / Teaching	Meaning / Description	Special Exam Facts
1. Satya	Always speaking the truth.	• Chaturyama teaching: these first four vows were propounded by the 23rd Tirthankara, Parshvanatha .
2. Ahimsa	Complete abstention from violence.	
3. Asteya	Not stealing (theft forbidden).	
4. Aparigraha	Not accumulating wealth/property.	
5. Brahmacharya	Control of the senses (the fifth great vow).	• This fifth vow was added by the 24th Tirthankara, Mahavira Swami , after which they came to be called the 'Pancha Mahavrata' (Five Great Vows).

> Tri-ratna (Three Jewels) of Jainism

- **Samyak Darshana (Right Faith):** Belief in truth (faith).
- **Samyak Jnana (Right Knowledge):** Real knowledge free from doubt.
- **Samyak Acharana (Right Conduct):** Equanimity (detachment) towards the pleasures and pains arising from worldly objects.

> Jainism: Ananta Chatushtaya (the four infinite powers attained after moksha)

1. Ananta Jnana (Infinite Knowledge) | 2. Ananta Darshana (Infinite Perception) | 3. Ananta Virya (Infinite Energy/Power) | 4. Ananta Sukha (Infinite Bliss)

2 The 24 Tirthankaras of Jainism and their Emblems

(Most important for 'Match the Following' questions in UPPCS Prelims)

No.	Name of Tirthankara	Emblem (Lanchhana)	No.	Name of Tirthankara	Emblem (Lanchhana)
1	Rishabhadeva (Adinatha)	Bull	13	Vimalanatha	Boar
2	Ajitanatha	Elephant	14	Anantanatha	Falcon (hawk)
3	Sambhavanatha	Horse	15	Dharmanatha	Thunderbolt (vajra)
4	Abhinandana	Monkey	16	Shantinatha	Deer
5	Sumatinatha	Chakravaka (ruddy shelduck)	17	Kunthunatha	Goat

No.	Name of Tirthankara	Emblem (Lanchhana)	No.	Name of Tirthankara	Emblem (Lanchhana)
6	Padmaprabha	Lotus	18	Aranatha	Fish
7	Suparshvanatha	Swastika	19	Mallinatha	Pitcher (kalasha)
8	Chandraprabhu	Crescent moon	20	Munisuvrata	Tortoise
9	Pushpadanta (Suvidhinatha)	Crocodile	21	Naminatha	Blue lotus
10	Shitalanatha	Shrivatsa	22	Neminatha (Arishtanemi)	Conch
11	Shreyamsanatha	Rhinoceros	23	Parshvanatha	Serpent (snake)
12	Vasupujya	Buffalo	24	Mahavira Swami	Lion

3 Principal Jain Tirthankaras and the Places of their Nirvana (Moksha)

Name of Tirthankara	Place of Nirvana	Present Geographical Location / Special Fact
Rishabhanatha (Adinatha)	Ashtapada	Also associated with Mount Kailash.
Vasupujya	Champapuri	Present-day Bhagalpur region of Bihar.
Neminatha (Arishtanemi)	Urjayanta	Present-day Girnar hills, Gujarat.
Mahavira Swami	Pavapuri	Present-day Bihar (near Rajgir).

4 Jain Councils

Council	Time (Year)	Place	President	Main Outcome / Work
First Council	322 BCE / 300 BCE	Pataliputra (Bihar)	Sthulabhadra	- The 12 Angas of Jainism were compiled. - The Jain order split into two sects—Shvetambara and Digambara.
Second Council	513 / 526 CE	Vallabhi (Gujarat)	Devardhi Kshamashramana	Scattered Jain texts were collected and the 11 Angas were finally committed to writing.

5 Fundamental Doctrines of Jain Philosophy

Philosophical Doctrine	Main Meaning and Feature
Anekantavada	Doctrine of many-sidedness/plurality of reality (an object has infinite attributes).

Philosophical Doctrine	Main Meaning and Feature
Syadvada (Saptabhengi Naya)	Doctrine of relativity (the law of the relativity of knowledge).
Anishvaravada (Non-theism)	Denial of any role of God in the creation of the world (creation is beginningless and eternal).
Atmavada / Rebirth	Acceptance of the existence of the soul and belief in rebirth according to the fruits of karma.

6 Principal Jain Texts and their Authors

Jain Text	Author	Main Theme of the Text
Kalpasutra	Bhadrabahu	Related to the life histories of the Tirthankaras (especially Mahavira and Parshvanatha) (in Sanskrit).
Parishishta Parvan	Hemachandra	Important epic related to Jain history, kings and Jain tradition.
Tattvartha Sutra	Umasvami	A highly important, systematic and authoritative sutra text of Jain philosophy.
Syadvadamanjari	Mallisena	Detailed exposition of the Syadvada and Anekantavada of Jain philosophy.
Nyayavatara	Siddhasena Divakara	Early and foundational text of Jain Nyaya (logic).
Pravachanasara	Kundakunda	A great text of Digambara Jain philosophy (original language: Prakrit).
Dravya Sangraha	Nemichandra	Scientific treatment of substance (dravya), qualities (guna) and modes (paryaya) in Jain philosophy.
Nyaya Dipika	Dharmabhushana	Related to the Jain Nyaya tradition and logic.
Panchavimshatika	Padmanandi	Related to Jain philosophical literature and spirituality.
Shloka Vartika	Vidyananda Swami	Related to Jain logic and the refutation of Mimamsa.

7 Buddhism: Life of Mahatma Buddha (The Light of Asia)

Event / Fact of Life	Details and Exam Evidence
Birth	Born in 563 BCE at Lumbini (near Kapilavastu, present-day Nepal).
Original/childhood name	Siddhartha (gotra name - Gautama).

Event / Fact of Life	Details and Exam Evidence
Parents and clan	Father: Shuddhodana (king of the Shakya republic of Kapilavastu). Mother: Mahamaya (of the Koliya republic).
Upbringing	As his mother died after his birth, he was brought up by his maternal aunt Mahaprajapati Gautami .
Family details	Wife: Yashodhara (<i>other names: Bimba, Gopa, Bhaddakachchha</i>). Son: Rahula .
Mahabhinishkramana (Great Renunciation)	Left home at the age of 29 . Horse: Kanthaka ; charioteer: Channa .
First teacher (Vaishali)	Alara Kalama (from whom Siddhartha received initiation in Sankhya philosophy).
Second teacher (Rajgir)	Rudraka Ramaputta (from whom he gained knowledge of yoga).
Enlightenment (Bodh Gaya)	At the age of 35 , on the bank of the Niranjana river (Punpun/Phalgu) , under a peepal tree (Bodhi tree) .
Dharmachakra Pravartana (First Sermon)	Gave his first sermon to five Brahmin ascetics at Rishipatana (Sarnath) .
Most sermons	Delivered at Shravasti , capital of Kosala.
Last sermon	Given at Kushinagar to his disciple Subhadda .
Mahaparinirvana (death)	In 483 BCE at Kushinagar , capital of the Malla republic, at the age of 80 .

8 Buddhist Symbols Related to the Life of the Buddha

Event / Place of Life	Buddhist Symbol	Specific Name of the Event / Place
Birth	Lotus (and bull)	Place: Lumbini
Renunciation of home	Horse	Event: Mahabhinishkramana (name of the horse: Kanthaka)
Enlightenment	Peepal (Bodhi tree)	Place: Bodh Gaya
First sermon	Wheel	Event: Dharmachakra Pravartana (place: Sarnath)
Nirvana (footprints)	Footprints	Symbol of liberation from the bonds of life.
Death (Mahaparinirvana)	Stupa	Place: Kushinagar

9 The 'Ashtangika Marga' (Eightfold Path) of Buddhism

The Eightfold Path is related to the fourth Noble Truth of Buddhism, '**Dukkha Nirodha Gamini Patipada**' (the path leading to the cessation of suffering). It is also called the '**Madhyama Pratipada**' (Middle Path) and comprises the following 8 steps (stages):

Step of the Eightfold Path	Real Meaning and Practical Form
1. Samyak Drishti (Right View)	Understanding an object in its real (true) form.
2. Samyak Sankalpa (Right Resolve)	Holding pure thoughts free from greed, hatred and violence.
3. Samyak Vak (Right Speech)	Complete renunciation of unpleasant, harsh and false words.
4. Samyak Karmanta (Right Action)	Always following good deeds in life.
5. Samyak Ajiva (Right Livelihood)	Earning a just livelihood marked by good conduct and honesty.
6. Samyak Vyayama (Right Effort)	Continuous wholesome effort for mental and physical well-being.
7. Samyak Smriti (Right Mindfulness)	Always maintaining a pure and alert state of mind (keeping away from false notions).
8. Samyak Samadhi (Right Concentration)	Concentration of the mind (attaining the higher state of meditation).

1 Fundamental Doctrines of Buddhist Philosophy

Philosophical Doctrine	Main Meaning and Practical Form
Anishvaravada (Non-theism)	No belief in a divine being (God as creator).
Kshanikavada (Momentariness)	The world is changing; nothing here is permanent or stable.
Anatmavada (No-soul)	Denial of the eternal existence of the soul, while laying the greatest emphasis on self-awareness.
Shunyavada (Emptiness)	All things in the world are relative and without inherent existence (empty of own-nature).
Rebirth	Note: Although Buddhism does not accept the soul, it still believes in rebirth (rebirth of consciousness/karma).

2 Buddhist Councils: Time, Place, Ruler, President and Main Work

(This table falls in the 'most important' category for UPPCS Prelims)

Council (Time)	Place / Cave	Ruler (Dynasty)	President / Vice-President	Main Work and Historical Outcome (Exam Facts)
First (483 BCE)	Rajagriha (Saptaparni Cave)	Ajatashatru (Haryanka dynasty)	• President: Mahakashyapa (Mahakassapa) • Principal associate: Ananda	• The tradition of compiling the Buddha's teachings began. • Ananda compiled/recited the Sutta Pitaka (teachings) and Upali the Vinaya Pitaka (rules).
Second (383 BCE)	Vaishali	Kalashoka (Shishunaga dynasty)	• President: Sabakami (Sabbakami / Sabakami)	• Main dispute among monks over 10 rules of the Vinaya. • First split of the Buddhist Sangha into the Sthaviras (Theravada) and the Mahasanghikas.
Third (250 BCE)	Pataliputra	Ashoka (Maurya dynasty)	• President: Moggaliputta Tissa	• The Abhidhamma Pitaka was compiled and the 'Kathavatthu' was composed. • Attempt to remove unorthodox views from the Sangha. • Missionaries were sent abroad (Mahendra and Sanghamitra were sent to Sri Lanka).
Fourth (72 CE / 1st century)	Kundalavana (Kashmir)	Kanishka (Kushana dynasty)	• President: Vasumitra • Vice-President: Ashvaghosha	• Final, complete division of Buddhism into Hinayana (basically Sthaviravadin) and Mahayana (basically Mahasanghika). • Use of the Sanskrit language increased in Buddhist literature.

3 The 'Tripitaka' and Other Important Texts of Buddhism

Tripitaka / Text	Original Language	Subject Matter / Internal Division (Core Content)
Sutta Pitaka	Pali	• Contains the principal teachings of the Buddha and the Dhamma. • Divided into 5 Nikayas: 1. Digha Nikaya, 2. Majjhima Nikaya, 3. Samyutta Nikaya, 4. Anguttara Nikaya, 5. Khuddaka Nikaya. • <i>(Note: the principal texts belonging to the Khuddaka Nikaya are the Dhammapada, Jatakas, Suttanipata, Theragatha and Therigatha).</i>
Vinaya Pitaka	Pali	• Rules of conduct, discipline and of the Sangha applicable to the monastic life of monks and nuns.
Abhidhamma Pitaka	Pali	• Philosophical analysis and systematisation of the doctrines taught to monks (contains a total of 7 texts).

Tripitaka / Text	Original Language	Subject Matter / Internal Division (Core Content)
Milindapanho (<i>Questions of Milinda</i>)	Pali	• Famous philosophical dialogue between the Indo-Greek ruler Milinda (Menander) and the Kashmiri Sarvastivadin Buddhist monk Nagasena .
Other texts	Pali/Sanskrit	• The Divyavadana , Dipavamsa , Mahavamsa , etc. are other important Buddhist texts.

4 Principal Buddhist Texts and their Authors

Buddhist Text	Author
Milindapanho	Nagasena
Buddhacharita, Saundarananda, Shariputra Prakarana	Ashvaghosha
Madhyamakakarika	Nagarjuna
Visuddhimagga	Buddhaghosha
Abhidharmakosha	Vasubandhu

5 Principal Dynasties of Ancient India: Founders and Last Rulers

Dynasty	Founder	Last Ruler	Special Exam Note (UPPCS Facts)
Nanda dynasty	Mahapadma Nanda (or Ugrasena)	Dhanananda	Chandragupta Maurya defeated Dhanananda and laid the foundation of the Maurya Empire.
Maurya dynasty	Chandragupta Maurya	Brihadratha	The commander Pushyamitra Shunga killed Brihadratha and founded the Shunga dynasty.
Shunga dynasty	Pushyamitra Shunga	Devabhuti	Brahmin dynasty established over Magadha immediately after the end of Mauryan power.
Satavahana dynasty	Simuka	Yajna Shri Satakarni	Also called the Andhra dynasty; Yajnashri was among the prominent last rulers.
Gupta dynasty	Shri Gupta	Kumaragupta III / Vishnugupta	After Shrigupta came Ghatotkacha, then Chandragupta I ascended the throne. (Last known ruler: Vishnugupta)
Chalukya dynasty (<i>Vatapi / Badami</i>)	Pulakeshin I	Kirtivarman II	Capital: Vatapi (Badami) .

6 Ancient Rulers / Personalities and their Other Names (Titles)

Real Name	Other Names in Greek Sources / Inscriptions	Specific Meaning / Description
Chandragupta Maurya	Sandrokottos / Androkottos	The Greek writer Justin called Chandragupta Maurya 'Androkottos'.
Bindusara	Amitraghata / Amitrochates	Literally means ' slayer of enemies '.
Ashoka	Piyadassi (Priyadarshi) / Devanampriya	The title " Devanampiya " (beloved of the gods) is found in the inscriptions.
Chanakya	Vishnugupta / Kautilya	Prime Minister of Chandragupta Maurya and author of the famous text 'Arthashastra'.

1 Maurya Dynasty: The Founder, Chandragupta Maurya and Bindusara

Ruler / Personality	Important Events / Historical Evidence (Exam Facts)
Chandragupta Maurya (founder)	• Teacher and Prime Minister: Kautilya (Chanakya). • Religious end: In the last days of his life he became a disciple of the Jain monk Bhadrabahu. In 298 BCE he gave up his body by the Sallekhana method at Shravanabelagola (Karnataka). • War (305 BCE): War and treaty with Seleucus Nicator. Seleucus ceded the regions of Kabul, Kandahar, Herat and Makran; in return Chandragupta gave 500 war elephants.
Megasthenes (Greek ambassador)	• Sent by Seleucus to the court of Chandragupta Maurya. Wrote the famous book 'Indica' (lost in its original form; fragments survive in the works of later Greco-Roman writers). • 7 Indian castes: according to him society had 7 castes—1. philosophers, 2. cultivators, 3. herdsmen, 4. artisans, 5. soldiers, 6. overseers (inspectors), 7. councillors.
Bindusara (297–273 BCE)	• Name in Greek sources: Amitrochates / Amitraghata (slayer of enemies). • Asked the Syrian ruler Antiochus for 3 things: 1. sweet wine, 2. dried figs, 3. a philosopher. (Note: Antiochus refused to send a philosopher because Greek law did not permit philosophers to be sold).

2 Kautilya's 'Arthashastra' and the Saptanga Theory

- **Subject matter:** Detailed and practical description of the state, administration, economy, justice and the espionage system.

- **Saptanga theory:** For the efficient running of the state, Kautilya described 7 limbs (constituents) of the state:

Limb of the State	Symbol in the Human Body (Analogy)	Limb of the State	Symbol in the Human Body (Analogy)
1. Raja (Svami – King)	Like the head	5. Janapada (territory and people)	The thighs (legs) of the state
2. Amatya (Minister)	The eye of the state	6. Danda (Army)	The brain (mind) of the state
3. Kosha (Treasury)	The mouth of the state	7. Mitra (Ally)	The ears of the state
4. Durga (Fort)	The arms (hands) of the state		

3 Emperor Ashoka: Life and Basic Facts about his Inscriptions

Fact / Position	Details and Exam Evidence (Core Facts)
Parents	Father: Bindusara . Mother: Subhadrangi (chief queen of Bindusara, also called 'Dharma').
Coronation	In 269 BCE (4 years after his accession to the throne).
Kalinga War	261 BCE (in the 8th/9th year after his coronation). After this Ashoka adopted the policy of Dhamma Vijaya (conquest by righteousness).
Language of the inscriptions	Basically the Prakrit language. (Exception: the Kandahar inscription is bilingual—Greek and Aramaic).
Script of the inscriptions	Brahmi (in most rock edicts), Kharoshthi (in the north-western region), Aramaic and Greek .

4 Names and Titles of Ashoka in the Sources

Title / Name	Source / Inscription	Historical Importance / Special Note
Ashoka	• Maski, Gujarra, Nittur, Udegolam minor rock edicts.	• The name first occurs in the Maski edict . " Devanampiya Asoka " is clearly engraved at Maski.
Priyadarshi	• Bhabru (Bairat) edict, Kandahar inscription and the Dipavamsa .	• Generally used in the inscriptions as " Devanampriya Priyadarshi ".
Devanampriya	• In the 12th and 13th major rock edicts.	• It means 'beloved of the gods'.

Title / Name	Source / Inscription	Historical Importance / Special Note
Devanampriya Priyadarshi	• Girnar, Kalsi, Erragudi and the Nigliva inscription.	• Note: the Kharoshthi script is used in the Shahbazgarhi and Mansehra inscriptions.
Ashokavardhana	• In the Vishnu Purana .	• This name occurs in the Buddhist and Puranic traditions.
Magadhadhiraja	• In the Bhabru (Bairat) edict .	• This inscription is the greatest proof of Ashoka's firm faith in Buddhism .
<i>Buddha Shakya (Note)</i>	<i>Mentioned in the Maski inscription.</i>	Exam warning: "Buddha Shakya" is not a title of Ashoka; it is a reference to Mahatma Buddha (Shakyamuni).

5 Discoverers of Ashoka's Important Inscriptions / Pillar Edicts

Rock Edict / Pillar Edict	Discoverer (Year)	Present Geographical Location / Special Fact
Girnar Rock Edict	Colonel Tod (1822)	Junagadh, Gujarat .
Shahbazgarhi Rock Edict	General Court (1836)	Peshawar, Pakistan (in the Kharoshthi script).
Dhauli Rock Edict	Kittoe (1837)	Puri, Odisha (principal rock edict of the Kalinga region).
Jaugada Rock Edict	Walter Elliot (1850)	Ganjam, Odisha (site of the Separate Kalinga Edicts).
Kalsi Rock Edict	Forrest (1860)	Dehradun, Uttarakhand (includes the text of the 13th rock edict).
Mansehra Rock Edict	<i>Year of discovery: 1889</i>	Hazara, Pakistan (Note: naming "General Cunningham" as its discoverer is incorrect; do not write it).

6 Subject Matter of Ashoka's 14 Major Rock Edicts (Core Themes)

Rock Edict	Main Theme Propounded (Core Content)
1 (First)	Prohibition of the killing of living beings ; ban on animal sacrifice and festive gatherings, and strict control over animal slaughter.
2 (Second)	Mentions medical arrangements for men and animals in the kingdoms of the Cholas, Pandyas, Satiyaputras, Keralaputras, Tamraparni (Sri Lanka) and the Yavana king Antiochus.

Rock Edict	Main Theme Propounded (Core Content)
3 (Third)	Order to royal officials (Yuktas, Rajjukas and Pradeshikas) to tour the country every 5th year.
4 (Fourth)	The sound of the war-drum (bherighosha) replaced by the sound of Dhamma (dhammaghosha) (moral/religious conquest in place of war).
5 (Fifth)	Mentions the appointment, in the 13th year of the coronation, of new officers called ' Dhamma Mahamattas '.
6 (Sixth)	Teaching of self-control and order to the Prativedakas (reporters) to report the condition of the people at all times.
7 (Seventh)	Religious tolerance ; people of all sects may reside peacefully everywhere.
8 (Eighth)	Beginning of Ashoka's Dhamma Yatras (e.g., the journey to Bodh Gaya).
9 (Ninth)	True courtesy and declaring 'Dhamma Mangala' (ceremony of Dhamma) the best ; avoiding extravagance and useless ceremonies.
10 (Tenth)	Condemnation of fame and glory; emphasis on the superiority of the policy of Dhamma .
11 (Eleventh)	Explanation of Dhamma Dana (the gift of Dhamma is the best gift).
12 (Twelfth)	Religious tolerance and appointment of Stri Mahamattas (women's officers) ; emphasis on the growth of the essence (core elements) of all sects.
13 (Thirteenth)	• Moving description of the terrible war of the Kalinga conquest (261 BCE) and of Ashoka's change of heart. • Mentions the 5 Yavana kings and the southern kingdoms (Cholas, Pandyas, Satiyaputras, Keralaputras, Tamraparni). • It is the largest and conclusive in character among all the major rock edicts.
14 (Fourteenth)	Inspiration to lead a religious life and the main purpose of having the edicts engraved.

7 Find-spots of the 14 Major Rock Edicts (Geographical Locations)

Find-spot	District / Present State	Country
Kalsi	Dehradun, Uttarakhand	India
Girnar	Junagadh, Gujarat	India
Sopara	Thane, Maharashtra	India
Dhauili	Puri, Odisha	India
Jaugada	Ganjam, Odisha	India

Find-spot	District / Present State	Country
Erragudi	Kurnool, Andhra Pradesh	India
Sannati	Gulbarga, Karnataka	India
Shahbazgarhi	Peshawar, Khyber Pakhtunkhwa	Pakistan (Kharoshthi script)
Mansehra	Hazara, Khyber Pakhtunkhwa	Pakistan (Kharoshthi script)

8

The Five Yavana Kings Mentioned in the 13th Rock Edict (Contemporary Hellenic Kings)

Name in the Inscription (Pali/Prakrit)	Greek / Modern Name	Related Region / Country (Kingdom)
Antiyoka	Antiochus II Theos	Syria
Turamaya	Ptolemy II Philadelphus	Egypt
Antikina	Antigonus Gonatas	Macedonia
Maga	Magas	Cyrene
Alikasudara	Alexander	Epirus

1

Hierarchical Units of Mauryan Administration (Administrative Units)

Administrative Unit	Head / Officer	Administrative Level / Feature (Core Facts)
Empire (Samrajya)	King	Supreme central authority.
Province	Viceroy (Kumara / Aryaputra)	• Principal provinces: Taxila, Ujjayini, Tosali, Suvarnagiri.
Mandala	Pradeshta	Officer concerned with the judicial and penal (criminal) system.
Ahara or Vishaya	Vishayapati	The principal district-level unit of administration.
Sthaniya	-	A group of 800 villages .
Dronamukha	-	A group of 400 villages .
Kharvatika	-	A group of 200 villages .
Sangrahana	-	A group of 10 villages .

Administrative Unit	Head / Officer	Administrative Level / Feature (Core Facts)
Grama (Village)	Gramika	The smallest unit of Mauryan administration; head of the village.

2 Central Administration: The 18 Tirthas (Highest Administrative Officers)

(According to Kautilya's Arthashastra, the top-ranking officials were called 'Tirthas' or 'Mahamatras')

No.	Tirtha (Officer)	Department / Main Function (Core Responsibility)
1	Purohita / Prime Minister	Chief adviser and administrative counsellor of the king.
2	Yuvaraja	Heir to the king (crown prince).
3	Samaharta	Revenue minister / collector of income (supreme head of the revenue department).
4	Sannidhata	Treasurer (supreme head of the royal treasury, treasury-house and storehouses).
5	Senapati	Minister of war / supreme military officer of the army.
6	Pradeshta	Judge / commissioner of the criminal courts (Kantakashodhana).
7	Vyavaharika	Chief judge of the civil courts (Dharmasthiya).
8	Dandapala	Officer who collected the army's provisions, materials and military resources.
9	Antapala	Head in charge of the frontier (border) regions and forts.
10	Karmantika	Chief minister of the country's industries and factories.
11	Durgapala	Head in charge of the forts inside the country (internal forts).
12	Nagaraka	Supreme head of city administration and civic arrangements.
13	Prashasta	Head who preserved royal documents/edicts (registrar-general).
14	Nayaka	Chief of city defence / officer leading the army in the battlefield.
15	Atavika	Supreme head of the forest department.
16	Mantriparishad Adhyaksha	President of the council of ministers.
17	Sena	Direct commander of the army.
18	Antarvamshika	In charge of the king's bodyguard and the security of the royal household (harem).

3 Principal 'Adhyakshas' (Heads of Departments) of the Mauryan Period

Adhyaksha (Superintendent)	Department / Portfolio
Akaradhyaksha	Supreme superintendent of mines and minerals (mining).
Lakshanadhyaksha	Superintendent of the mint ; supervised the minting of coins and the royal currency.
Sitadhyaksha	Supervised the royal agricultural land (sita land) and headed the agriculture department.
Vivitadhyaksha	Head in charge of pastures and the protection of cattle and of uninhabited tracts.
Pautavadhyaksha	Superintendent of weights and measures ; officer for the standardisation of weights and measures.
Manadhyaksha	Officer who fixed the standards of time, distance, direction and measurement.
Suradhyaksha	Superintendent of liquor (sale of liquor and excise control).
Panyadhyaksha	Superintendent of commerce and trade ; controlled the market system.
Shulkadhyaksha	Officer who collected tolls and customs duties.
Navadhyaksha	Supreme officer of shipping and waterways.
Mudradhyaksha	Officer for royal passports/seals and permits (passes) for movement.
Koshthagaradhyaksha	Chief superintendent of the royal granary and storehouse.
Ashvadhyaksha / Hastyadhyaksha / Rathadhyaksha	Chief officers of horses (cavalry), elephants (elephant corps) and chariots (chariot corps) respectively.
Ganikadhyaksha	Officer who supervised the department concerning courtesans (ganikas).

4 Kautilya's 'Saptanga Theory' (Organic Theory of State)

Kautilya regarded the state as **equivalent to a human body**. He called these 7 limbs "**Prakritis**", of which the **king (Svami)** is regarded as the supreme and principal limb of the state:

Limb (Saptanga)	Symbol in the Human Body (Analogy)	Special UPPCS Exam Note (Core Conceptual Facts)
1. Svami (King)	Head	• The supreme, principal and most important limb of the state; supports the monarchical system.
2. Amatya (Minister)	Eyes	• A body of qualified officials; the king's principal associates in administration.
3. Janapada	Thighs (legs)	• Kautilya used the word " Janapada " for it (in notes 'Rashtra' is given as the symbol of this limb).
4. Durga (Fort)	Arms	• The main basis of the security, defence and military fortification of the state.
5. Kosha (Treasury)	Mouth	• The basis of the economic strength of the state (royal treasury).
6. Danda (Army)	Brain (mind)	• Force/army relates to the punitive and defensive power of government.
7. Mitra (Ally)	Ears	• External associates or friendly states who help in times of calamity.

5 The Most Important Inscriptions of Ancient India (Rock Edicts and Prashastis)

Name of Inscription	Related Ruler (Dynasty)	Composer / Specific Evidence and Historical Importance (Exam Points)
Hathigumpha inscription	Kharavela, king of Kalinga (Chedi dynasty)	• Located at Udayagiri in Odisha; the principal and systematic source of information on the events of Kharavela's reign.
Junagadh (Girnar) inscription	Rudradaman (Shaka Kshatrapa)	• Located in Gujarat ; one of the first long and authentic inscriptions engraved in the Sanskrit language .
Nasik inscription	Gautami Balashri (Satavahana)	• Detailed description of the military victories and achievements of the Satavahana emperor Gautamiputra Satakarni .
Prayag (Allahabad) Pillar inscription	Samudragupta (Gupta dynasty)	• Composer: Samudragupta's court poet Harishena (a famous prashasti composed in the champu style).
Aihole inscription	Pulakeshin II (Chalukya dynasty)	• Composer: Ravikirti ; it gives evidence of Pulakeshin II defeating Harshavardhana.
Mandsaur inscription	Yashovarman, king of Malwa	• Historical mention of Yashodharman defeating the famous Huna ruler Mihirakula .

Name of Inscription	Related Ruler (Dynasty)	Composer / Specific Evidence and Historical Importance (Exam Points)
Gwalior inscription	Bhoja, the Pratihara king (Gurjara-Pratihara)	• Important evidence relating to the history of the Gurjara-Pratihara dynasty and the achievements of King Bhoja.
Bhitari and Junagadh inscriptions	Skandagupta (Gupta dynasty)	• Authentic mention of the barbaric invasion of the Hunas and of Skandagupta's great successes/victories against them.
Deopara inscription	Vijayasena (Sena dynasty)	• Historical account of the achievements of Vijayasena, the powerful ruler of the Sena dynasty of Bengal.

6 Other Important Inscriptions of the Post-Mauryan Period

Name of Inscription	Related Ruler / Personality	Historical Evidence Useful for the Exam (Core Facts)
Ayodhya inscription	Dhanadeva	• The first written evidence of the history of the Shunga dynasty and of Pushyamitra Shunga and the Ashvamedha sacrifices performed by him.
Besnagar inscription	Heliodorus (Yavana ambassador)	• Garuda pillar located at Vidisha (MP); the oldest archaeological evidence of Heliodorus adopting the Bhagavata (Vaishnava) religion .
Shinkot inscription	Menander (Milinda)	• Epigraphic evidence relating to the famous Indo-Greek ruler Menander.
Naneghat inscription	Naganika (Satavahana queen)	• Important information on the history of the Satavahana dynasty and on Queen Naganika's royal acts and sacrifices (land grants).
Panjtar inscription	Kushanas	• Important epigraphic evidence relating to the expansion of the Kushana empire and its administration in India.

1 Foreign Invaders of the Post-Mauryan Period: Sequence, Rulers and Original Homeland

No.	Invading Dynasty	Most Famous Ruler	Capital / Important Facts	Original Homeland
1	Indo-Greeks	Menander / Milinda	• Capital: Shakala (present-day Sialkot). • Philosophical dialogue with the Kashmiri Buddhist monk Nagasena (text: Milindapanho).	Bactria, Afghanistan
2	Shakas (Scythians)	Nahapana (in western India) <i>(Note: Rudradaman was also a Shaka ruler, whose capital was Ujjayini)</i>	• A historical mention of Nahapana is found in Satavahana inscriptions. • Rudradaman had the famous Sanskrit inscription of Junagadh engraved.	Scythia, Iran
3	Pahlavas (Parthians)	Gondophernes	• Evidence of their coins has been found at Begram (Afghanistan).	Parthia, Iran
4	Kushanas (Yuezhi)	Kanishka <i>(real founder: Kujula Kadphises)</i>	• Capitals: Purushapura (present-day Peshawar) and Mathura (two capitals of the empire). • Started the Shaka Samvat (Shaka era) in 78 CE (used by the Government of India).	Chinese Turkestan

2 Important Rulers of the Kushana Empire

Kushana Ruler	Reign (approx.)	Important Historical Facts and Evidence (Exam Facts)
Kujula Kadphises	30 CE - 80 CE	• Regarded as the real founder of Kushana power in India; he was the father of Vima Kadphises.
Vima Kadphises	95 CE - 127 CE	• The first Kushana ruler to issue pure gold coins. His coins depict Shiva and Nandi. • Rabatak inscription evidence: according to it he was the son of Sadashkana and the father of Kanishka.
Kanishka	127 CE - 150 CE	• The most famous Kushana ruler and a great patron of Buddhism (convened the Fourth Buddhist Council). • Started the Shaka Samvat calendar in 78 CE. • Built the Kanishka stupa at Purushapura. Gandhara and Mathura art developed unprecedentedly in his time.

3 Distinctive Titles of Post-Mauryan Kings

Ruler (Dynasty)	Titles Held	Historical Sources and Evidence Useful for the Exam
Pushyamitra Shunga (Shunga dynasty)	'Senani' (commander) 'Audbhijja' (one who rises suddenly)	- The Harivamsha Purana calls him 'Ashvamedha Yajin' (performer of two Ashvamedha sacrifices). - An authentic mention of his sacrifices is found in Dhanadeva's Ayodhya inscription.
Kharavela (Chedi dynasty, Kalinga)	'Rajarshi', 'Lemaraja', 'Vridddharaja' 'Bhikshuraja', 'Dharmaraja', 'Mahavijayaraja'	The only source of information on all these titles and on his life is the Hathigumpha inscription at Udayagiri.
Satakarni I (Satavahana dynasty)	'Lord of Dakshinapatha' 'Wielder of the unchecked wheel (of sovereignty)' 'Pashchimapati', 'Pratishthanapati'	An important ruler who established Satavahana power.
Gautamiputra Satakarni (Satavahana dynasty)	'Advitiya Brahmana' (the unique Brahmana) 'Refuge of the Vedas' (Dvijakulavardhana)	These titles are clearly mentioned in the Nasik prashasti (composed by Gautami Balashri).

4 History of Coins (Currency) in Ancient India

Type of Coins / Metal	Issuing Dynasty / Ruler	Special Exam Note (Exam Warning)
First gold coins (gold)	By the Indo-Greek rulers	The earliest evidence of gold coinage in India comes from here.
First pure gold coins	By the Kushana rulers (by Vima Kadphises)	Kushana gold coins bear images of Greek, Iranian and Indian deities.
First inscribed coins	By the Greek (Indo-Greek) rulers	Started the tradition of writing the names, portraits and titles of kings on coins.
Largest number of gold coins	By the Gupta rulers	Note: the Guptas issued them in the largest number, but Chandragupta I was the first Gupta ruler to issue a gold coin.
Largest number of copper coins	By the Kushana rulers	Kushana copper coins have been found in large numbers, especially in rural areas.
Lead and tin (potin) coins	By the Satavahana rulers	Lead was the most prominent metal in the trade currency of the Satavahanas.

Type of Coins / Metal	Issuing Dynasty / Ruler	Special Exam Note (Exam Warning)
Largest number of coins (on a wide scale)	By the Mauryan rulers	• Punch-marked (ahata) coins were widely in circulation in the Mauryan period.

5 Gupta Dynasty: Sequence, Historical Facts and Queens

- **Gupta rule:** from about 275 CE to 550 CE.
- **Successive family tree of the rulers:** Shrigupta (founder) → Ghatotkacha → Chandragupta I → Samudragupta → Ramagupta → Chandragupta II → Kumaragupta I → Skandagupta → Purugupta → Kumaragupta II → Budhagupta → ... → Vishnugupta (last known Gupta ruler).

> Gupta Rulers, their Queens and Principal Facts

Gupta Ruler	Queen / Marriage Alliance	Titles and Principal Exam Facts (Core Facts)
Chandragupta I	Kumaradevi (<i>Lichchhavi princess</i>)	• The real founder of the Gupta dynasty. • The first Gupta ruler to issue a gold coin. Started the 'Gupta Samvat' (Gupta era) in 319 CE.
Samudragupta	Dattadevi	• Title: 'Napoleon of India' (given by the historian Vincent Smith). • His conquests are mentioned in Harishena's 'Prayag Prashasti'. Issued gold coins of the Ashvamedha type.
Chandragupta II	Kuberanaga (<i>Naga dynasty</i>) and Dhruvasvamini (Dhruvadevi)	• Title: 'Vikramaditya'. The contemporary Chinese traveller Fa-Hien came to India. • The first Gupta ruler to issue silver coins. The Mehrauli iron pillar is associated with him.
Kumaragupta I	Anantadevi	• Title: 'Mahendraditya'. Founder of the famous Nalanda University in Bihar.
Rudrasena II (<i>Vakataka king</i>)	Prabhavati Gupta (<i>daughter of Chandragupta II</i>)	• Note: he was not a Gupta ruler but a Vakataka king contemporary with the Gupta empire, with whom Chandragupta II formed a matrimonial alliance.

> The 'Navaratnas' (Nine Gems) of the Court of Chandragupta II (Vikramaditya)

1. **Kalidasa** (great poet and writer) | 2. **Dhanvantari** (famous physician) | 3. **Varahamihira** (astronomer) | 4. **Amarasimha** (lexicographer - author of the Amarakosha) | 5. **Kshapanaka** (astrologer) | 6. **Shanku** (architect) | 7. **Vetala Bhatta** (magician) | 8. **Ghatakarpara** (skilled diplomat) | 9. **Vararuchi** (great grammarian)

> Principal Kings of the Gupta Period and Related Inscriptions

King / Ruler	Related Inscription	Excavator / Composer / Historical Evidence Found
Samudragupta	Eran inscription (and the Prayag Prashasti)	• The Eran inscription describes Samudragupta as benevolent and just, like Kubera , the god of wealth, and Yama , the god of death.
Chandragupta II	Udayagiri Cave inscription	• Engraved by his minister Virasena (Shaba) . (The Mehrauli iron pillar of Delhi is also associated with him).
Kumaragupta I	Mandsaur inscription	• Composer: Vatsabhatti . This inscription gives important information on the activities of the guild of silk weavers in the Gupta period.
Patika	Taxila inscription	• Note: he was not a Gupta ruler; this is an authentic copper-plate inscription of the Kushana/Shaka period.

7 Principal Administrative Officers of the Gupta Period and their Offices

(The most important table for the administrative terminology of UPPCS Prelims)

Administrative Post (Officer)	Department / Main Area of Work (Core Responsibility)
Kumaramatyā	The highest administrative officer of the Gupta empire (equivalent to the IAS).
Mahadandanayaka	Chief Justice ; a post carrying high responsibilities, both military and judicial.
Maha-sandhivigrahika	Minister of war, peace and foreign affairs (the court poet Harishena held this post).
Maha-baladhikrita	Commander-in-chief / supreme head of the military department.
Maha-akshapatalika	Supreme head of the accounts department / superintendent of records and accounts.
Pratihara	Guard of the security of only the inner part (harem/antahpura) of the royal palace.
Mahapratihara	High officer in charge of the security of the whole palace and of the king (head of the bodyguards).
Ashvapati / Mahashvapati	Officer in charge/head of the cavalry (horses).
Pilupati	Supreme officer in charge of the elephant corps (army of elephants).
Narapati	Chief military officer concerned with the infantry .

Administrative Post (Officer)	Department / Main Area of Work (Core Responsibility)
Ranabhandagarika	Officer in charge of military supplies (provisions) and stores of military material.
Nagara Shreshthi	Leading merchant who supervised city administration; representative of the mercantile community.
Gramadhyaksha / Gramika	Village headman ; supreme head of rural administration.
Dutakarma / Dutaka	Officer who conveyed the king's royal orders, gifts and land-grant charters to the persons concerned.

1 Administrative Units of the Gupta Period

Arranged in *ascending order*

> **Grama (smallest) → Petha → Vithi → Vishaya → Bhukti → Desha or Rashtra (largest)**

Administrative Unit	Head / Administrator	Special Exam Facts
Grama (<i>smallest unit</i>)	Gramika	The highest officer of village-level administration.
Petha	—	A group of several villages (a unit larger than the village).
Vithi	—	An administrative unit larger than the Petha and smaller than the Vishaya (district).
Vishaya (<i>district</i>)	Vishayapati	The principal district-level unit of Gupta administration.
Bhukti (<i>province</i>)	Uparika	The largest administrative division (province) of the empire.
Desha or Rashtra	King / Emperor	The supreme administrative and central unit of the Gupta empire.

2 Administrative Terminology Related to Land in the Gupta Period

Type of Land	Real Meaning and Use (Core Meaning)
Kshetra	Land fit for cultivation (suitable for farming).
Vastu	Land fit for habitation (for building houses).
Charagaha land	Grasslands suitable for the grazing of cattle.

Type of Land	Real Meaning and Use (Core Meaning)
Khila	Land that was not fit for ploughing (barren or infertile).
Aprahata	Land that was forest or uncultivated (unploughed) .

3 Important Literary Works of the Gupta Period and of Ancient India

Literary Work (Book)	Author	Specific Details / Exam Note
• Abhijnanashakuntalam, - Malavikagnimitram, - Vikramorvashiyam • Raghuvamsham, - Kumarasambhavam • Meghaduta, Ritusamhara	Kalidasa	• The Meghaduta has two parts: Purvamegha and Uttaramegha. • Malavikagnimitram, Abhijnanashakuntalam and Vikramorvashiyam are his famous plays.
• Mudrarakshasa • Devichandraguptam	Vishakhadatta	• Mudrarakshasa throws light on Mauryan history and Devichandraguptam on Gupta history.
• Brihatsamhita • Panchasiddhantika	Varahamihira	• Famous Gupta-period texts on astronomy and astrology.
Brahmasiddhanta	Brahmagupta	• He is also called the Newton of India.
• Aryabhatiyam • Suryasiddhanta	Aryabhata	• The great mathematician and astronomer of ancient India.
Nyayavatara	Siddhasena (Divakara)	• The early systematic text of Jain Nyaya philosophy.
Panchatantra	Vishnu Sharma	• A world-famous collection of moral fables.
• Nitisara • Nitishastra	Kamandaka	• Text based on Gupta-period government and politics.
Mrichchhakatikam	Shudraka	• Literally means 'The Little Clay Cart' (depicts the social system of the time).
Kavyadarsha Dashakumaracharita	Dandin	• Court poet of Narasimhavarman II (Pallava dynasty).
Svapnavasavadatta Charudatta	Bhasa	• A great dramatist of ancient India.
Kiratarjuniyam	Bharavi	• Court poet of Simhavishnu (Pallava dynasty).
Ravanavadha	Bhatti	• Also known as the 'Bhattikavya'.

Literary Work (Book)	Author	Specific Details / Exam Note
Amarakosha	Amarasimha	• One of the Navaratnas of Chandragupta II (famous Sanskrit lexicographer).
Visuddhimagga	Buddhaghosha	• A highly important text of Buddhism.
Kamasutra	Vatsyayana	• The authoritative text of ancient Indian erotics (Kamashastra).
Charaka Samhita	Charaka	• Famous Ayurvedic text related to the Kushana period (Kanishka).
Ramayana	Maharishi Valmiki	• Sanskrit epic.
Mahabharata	Maharishi Vedavyasa	• The largest epic of the world.
Nitishataka	Bhartrihari	• Famous Sanskrit shataka (century of verses) poem.
Yogachara	Asanga	• Related to the Vijnanavada/Yogachara school of Buddhist philosophy.

4 Gupta Architecture: Principal Temples and their Location

Name of Temple	Location	District and State
Dashavatara Temple of Deogarh	Deogarh	Lalitpur district, Uttar Pradesh (UP Special)
Krishna/Brick Temple of Bhitargaon	Bhitargaon	Kanpur district, Uttar Pradesh (UP Special)
Vishnu Temple of Eran	Eran	Sagar district, Madhya Pradesh
Shiva Temple of Bhumara	Bhumara	Satna district, Madhya Pradesh
Vishnu Temple of Pipariya	Pipariya	Satna district, Madhya Pradesh
Vishnu Temple of Tigawa	Tigawa	Jabalpur, Madhya Pradesh
Parvati Temple of Nachna-Kuthara	Nachna-Kuthara	Panna, Madhya Pradesh
Lakshmana Temple of Sirpur	Sirpur	Raipur district, Chhattisgarh

5 Vardhana Dynasty (Pushyabhuti Dynasty)

- **Founder:** Pushyabhuti.
- **Sequence of rulers:** Naravardhana → Rajyavardhana I → Adityavardhana → Prabhakaravardhana → Rajyavardhana II → Harshavardhana.

> Emperor Harshavardhana (606–647 CE)

- **Family background:** Father—Prabhakaravardhana, Mother—Yashomati.
- **Other name and title:** Other name—**Shiladitya**. Title—**Sakalottarapathanatha** (Lord of the whole of North India).
- **Change of capital:** His capital was first **Thaneshwar**, which he later shifted to **Kannauj (Mahodayashri, Uttar Pradesh)**.
- **Literary works:** He himself was the author of three famous Sanskrit plays—1. **Priyadarshika**, 2. **Ratnavali**, 3. **Nagananda**.
- **Aihole inscription evidence (historic battle):** According to the Aihole inscription composed by Ravikirti, court poet of the Chalukya king **Pulakeshin II**, a fierce battle took place between Harsha and Pulakeshin II on the bank of the Narmada, in which **Pulakeshin II was victorious** and Harsha's southern conquest was halted.

6 Pallava Dynasty: Founder, Rulers and Cultural Contribution

- **Founder:** **Simhavishnu** (some historians also regard Bappadeva as the founder).
- **Capital:** **Kanchi (Kanchipuram)**.
- **Last ruler:** **Aparajitavarman**.

> Principal Pallava Rulers and their Works

Pallava Ruler (Period)	Titles / Court Scholars	Architecture and Historical Achievements (Exam Facts)
Simhavishnu (575-600 CE)	● Bharavi, author of the 'Kirtarjuniyam', lived at his court.	● Pioneer of the revival of Pallava power; extended the kingdom up to the Chola country (Kaveri river).
Mahendravarman I (600-630 CE)	● He himself wrote the famous Sanskrit play 'Mattavilasa Prahasana'.	● Began rock-cut (cut out of hills) cave-temple architecture (monolithic temples) in South India.
Narasimhavarman I (630-668 CE)	● Titles: 'Vatapikonda' and 'Mahamalla'.	● In 642 CE he captured Vatapi (Badami), the capital of the Chalukyas. ● Began the construction of the famous 'Ratha temples' (Seven Pagodas) of Mahabalipuram (Mamallapuram).
Mahendravarman II and Parameshvaravarman I	—	● Period of continuous conflict with the Chalukya ruler Vikramaditya I and of Pallava–Chalukya wars.

Pallava Ruler (Period)	Titles / Court Scholars	Architecture and Historical Achievements (Exam Facts)
Narasimhavarman II (<i>Rajasimha</i>) (695-722 CE)	• Dandin , the famous author of the ' Dashakumaracharita ', lived at his court.	- Used the new 'Rajasimha style' of temple building. - Built the famous 'Kailashanatha Temple' of Kanchi and the 'Shore Temple' of Mahabalipuram.
Parameshvaravarman II (722-731 CE)	—	The final period of the decline of the Pallava dynasty and of intense succession struggles.

7 Administrative and Local Self-Government System of the Chola Period

> Administrative Divisions of the Chola Empire (in descending order of seniority)

Mandalam (province) → Valanadu (commissionery) → Nadu (district) → Kurram / Grama (smallest unit)

> Principal Chola Officials and Bodies

Administrative Post / Body	Department / Actual Responsibility (Core Responsibility)
Olai	Principal duty of implementing the king's oral and written orders on the ground.
Tirumandira Olai Nayagam	The chief high officer who recorded on palm leaf the oral orders given by the king in the public assembly.
Perundaram	Administrative officers of the upper grade (first class) of the Chola central administration.
Sirudaram	Administrative officers of the lower grade (second class) of the Chola administration.
Viceroy	Governors/leaders of the provinces, who were usually members of the royal family or close to the king.
Nattar	Leading persons who headed the local administrative assembly of the Nadu (district).
Nattuvagai Nadu Council	Local council supervising revenue, justice and administration at the level of the Nadu (district).
Nattuvais	Officials connected with local agriculture, land survey and rural agricultural administration.

> Local Self-Government of the Chola Period (Village Assembly and Committees)

(According to the famous **Uttaramerur inscription**, the working of Chola village government was as follows)

Village Assembly / Committee	Practical Form and Method of Selection (Functional Domain)
Ur	The basic assembly of the common residents of ordinary (non-Brahmin) villages.
Sabha or Mahasabha	The central assembly of learned Brahmins of Brahmadeya villages (villages granted to Brahmins) .
Nagaram	Mainly the administrative assembly of merchants, artisans and trading towns .
Kudavolai	The ' lottery method ' (system of slips) used to select the members of the committees (variya).
Variyam	Special working committees of men formed to run the various administrative functions of village government.
↳ <i>Eri Variyam</i>	Committee for the maintenance of irrigation and tanks .
↳ <i>Totta Variyam</i>	Committee for the management of gardens (orchards) .
↳ <i>Pon Variyam</i>	Committee that examined gold, currency and finance (revenue) .

> Chola Dynasty: Rulers, Titles and Historical Facts

- **Founder:** Vijayalaya.
- **Last ruler:** Rajendra III.
- **Capital:** Thanjavur (Tanjore) → later **Gangaikonda Cholapuram**.

Chola Ruler	Titles	Important Facts and Evidence Useful for the Exam (Exam Facts)
Parantaka I	• 'Maduraikonda'	• Completely eliminated the Pallavas. • Battle of Takkolam (949 CE): defeated by the Rashtrakuta ruler Krishna III.
Rajaraja I	—	• Built the famous Brihadishvara Temple (Rajarajeshvara Temple) at Thanjavur.
Rajendra I	• 'Gangaikonda'	• Defeated the Pala ruler Mahipala and the Eastern Ganga ruler in order to bring Ganga water. • Founded the new capital Gangaikonda Cholapuram and built a Brihadishvara temple there.
Kulottunga II	—	• Had the image of Govindaraja (Vishnu) removed from the Chidambaram Nataraja temple and thrown into the sea.

> The Principal Feature of Chola Administration: Autonomy of Village Administration

Committees (**variya**) were set up to supervise the activities of the villages:

- **Eri Variyam:** Tank committee (the notes say 'Uri Variyam'; the standard form is 'Eri')
- **Pon Variyam:** Gold committee (financial system)
- **Totta Variyam:** Garden/orchard committee (the notes say 'Toh Variyam'; the standard form is 'Totta')
- **Udasina Variyam:** Committee to supervise royal directions/rules

2 Principal Sun Temples of India and their Builders

Name of Sun Temple	Location	Builder / Ruler (Dynasty)	Special Historical Fact
Martand Sun Temple	Kashmir	Lalitaditya Muktapida (Karkota dynasty)	Famous ancient temple of Kashmir.
Konark Sun Temple	Odisha	Narasimhadeva I (Eastern Ganga dynasty)	Also called the 'Black Pagoda'.
Modhera Sun Temple	Gujarat	Bhimadeva I (Solanki/Chalukya dynasty)	India's largest Surya Kund (sun tank) is located here.
Tanjore Sun Temple	Tamil Nadu	Related to the Chola period	The only sun temple of South India.

3 Outline of the Three Sangams (Tamil Assemblies)

 **Note**

Note: All three Sangams were held under the royal patronage of the **Pandya kings**.

Sangam	Venue	President	Members / Literature Compiled
First	Madurai	Sage Agastya	• 549 members. • Texts compiled: Akattiyam, Mudunarai, Paripadal, Mudukurukku.
Second	Kapatapuram (Alaivai)	Sage Agastya <i>(later his disciple Tolkappiyar)</i>	• 59 members. • Literature compiled: Tolkappiyam (a text on Tamil grammar).
Third	Madurai	Nakkirar	• 49 members. • Works compiled: Nedunthogai, Kurunthogai, Ainkurunuru, Paripadal.

4 Important Sangam Literature and their Authors

Sangam Text	Author	Subject Matter / Internal Feature (Core Themes)
Silappadikaram (<i>The Tale of the Anklet</i>)	Ilango Adigal	- One of the 5 great epics of Tamil literature. - The love story of the hero Kovalan, the heroine Kannagi and the dancer Madhavi. Later the 'Kannagi (Pattini) cult' developed in South India.
Manimekalai	Sittalai Sattanar	- A Tamil epic related to Buddhism. - It is the sequel to the Silappadikaram; its heroine is Manimekalai (daughter of Kovalan and Madhavi).
Tolkappiyam	Tolkappiyar	- The only surviving work of the Second Sangam; the oldest text on Tamil grammar and poetics. - It has three parts—letters (ezhuttu), words (sol) and meaning (porul).
Tirukkural (or Kural)	Tiruvalluvar	- The 'Fifth Veda' of Tamil literature; related to philosophy (1330 couplets, 133 chapters). - It has three parts—Aram (dharma/virtue), Porul (artha/wealth) and Inbam (kama/love).
Jivaka Chintamani	Tiruttakkadevar	Counted among the great epics of Tamil literature; related to Jainism (the author was a Jain ascetic of Madurai).
Kundalakesi	Nathakuthanar (Nagakuthanar)	A Buddhist Tamil epic; the tragic love story of Kundalakesi, a girl of the merchant caste, and the Buddhist criminal Kalan.
Ahananuru (Ahanar)	Rudrasharma	A collection of famous love poems of the Sangam age.
Murugarruppadai	Nakkirar	Religious text in praise of Lord Murugan (Kartikeya).

5 Principal Coastal and Administrative Towns of the Sangam Age

Name of Town	Dynasty	Historical Position and Geographical Importance (Exam Points)
Uraiyur	Chola	Early inland capital of the Cholas; in ancient times a very large centre of cotton cloth.
Puhar (Kaveripattinam)	Chola	Second and coastal capital of the Cholas; the famous international port of the Sangam age.
Madurai	Pandya	Main capital of the Pandya rulers and the central venue of the Sangam assemblies.
Korkai	Pandya	Second and coastal capital of the Pandya dynasty (famous for pearls).

Name of Town	Dynasty	Historical Position and Geographical Importance (Exam Points)
Vanji (or Karur)	Chera	• Main administrative capital of the Chera dynasty.
Mushiri (Muziris)	Chera	• A highly famous and busy sea port of the Chera period.

6 Principal Texts and Musicians of Medieval Indian Music

Musician	Music Text (Book)	Philosophical Feature of the Text / System Propounded (Core Concept)
Pundarika Vitthala	• Shadragachandrodaya • Ragamala • Ragamanjari	• He composed the 'Ragamanjari' at the instance of Raja Man Singh. • The Ragamanjari contains the first classification in Indian music of 15 ragas of Persian origin. The 'Sadragachandrodaya' is also his work.
Shrikantha	• Rasa Kaumudi	• He was a worthy pupil of Pundarika Vitthala and court poet of Satrusalya. • This text deals with vocal music, instrumental music and dance.
Somanatha	• Raga-vibodha	• He was a Telugu Brahmin of Rajahmundry. • This text contains a scientific treatment of the 'Janya-Janaka' or 'Thaat-Raga system' for classifying ragas. It has a scheme of 23 melas (thaats).
Venkatamakhin	• Chaturdandiprakashika (composed c. 1660 CE)	• He was a great musicologist of the Carnatic music tradition. • He introduced the 'scientific system of 72 melas (thaats)' to organise the ragas of Carnatic music, which is still regarded as authoritative.

1 Important Foreign Travellers and Ambassadors

Foreign Traveller	Period	Country of Origin	Contemporary Indian Ruler	Travel Account / Book and Main Facts
Megasthenes	302-298 BCE	Greece	Chandragupta Maurya	• Was the ambassador of Seleucus Nicator. • Famous book: 'Indica'.
Fa-Hien	399-414 CE	China	Chandragupta II (Vikramaditya)	• The first Chinese Buddhist monk to come to India. • Famous book: 'Fo-Guo-Ji' (Record of the Buddhist Kingdoms).

Foreign Traveller	Period	Country of Origin	Contemporary Indian Ruler	Travel Account / Book and Main Facts
Hiuen Tsang (<i>Xuanzang</i>)	630-645 CE (or 636-645 CE)	China	Harshavardhana (also at the court of <i>Bhaskaravarman</i>)	- Famous book: 'Si-Yu-Ki' (Record of the Western Regions). - Studied at Nalanda University.
I-Tsing	671-695 CE	China	Post-Gupta period	- Came to India by the sea route. - Studied at Nalanda; book: '<i>Nanhai Jigui Neifa Zhuan</i>'.
Matwajin	—	China	Chalukya ruler	Mentioned the cultural relations between the Chalukya rulers and China.
Al-Masudi	915 CE	Arabia	Rashtrakuta/Gurjara-Pratihara period	- Famous book: 'Muruj-ul-Zahab'. - He is also called the 'Herodotus of the Arabs'.
Al-Biruni (<i>Abu Raihan</i>)	1024-1030 CE	Khwarizm (Arabia)	Mahmud of Ghazni (came with him)	- Famous book: 'Kitab-ul-Hind' (Tahqiq-i-Hind). - The first Muslim scholar to study Sanskrit.
Ma Huan	1451 CE	China	Medieval period (Sayyid/Lodi period)	A Chinese traveller who visited India in the medieval period.

2 Principal Historical Battles of Ancient India

(*Most important for matching dates and victorious sides in UPPCS Prelims*)

Name of Battle	Time (Period)	Fought Between (Contenders)	Victorious Side	Special UPPCS Note (Core Facts)
Dasharajna War (Battle of Ten Kings)	Rigvedic period	King Sudas vs. a confederacy of 10 kings	King Sudas	Fought on the bank of the Parushni (Ravi) river.
Battle of the Hydaspes	326 BCE	Alexander vs. King Porus	Alexander	Fought on the plain of the Jhelum river (Vitasta).
Chandragupta–Seleucus War	303 BCE (or 305 BCE)	Chandragupta Maurya vs. Seleucus Nicator	Chandragupta Maurya	As a result of the treaty, four provinces were added to the Maurya Empire.
Kalinga War	261 BCE	Emperor Ashoka vs. Kalinga (Padmanabha)	Emperor Ashoka	In the 8th year after Ashoka's coronation (described in the 13th rock edict).
Kanishka–China War	1st century	Kanishka vs. the Chinese emperor (Pan Chao)	Kanishka	After defeat in the first phase, Kanishka was ultimately victorious.

Name of Battle	Time (Period)	Fought Between (Contenders)	Victorious Side	Special UPPCS Note (Core Facts)
Skandagupta–Huna War	466 CE	Skandagupta vs. the barbarian Huna invaders	Skandagupta	This victory is described in detail in the Bhitari pillar inscription .
Battle of the Narmada	630 CE	Harshavardhana vs. Pulakeshin II	Pulakeshin II	The Chalukya king Pulakeshin II halted Harsha's southern conquest (Aihole inscription).
Battles of Suramara, Parisala and Manimangalai	641-642 CE	Narasimhavarman I vs. Pulakeshin II	Narasimhavarman I	The Pallava ruler Narasimhavarman I killed Pulakeshin II and took the title ' Vatapikonda '.
Battle of Sindh	712 CE	Raja Dahir Sen vs. Muhammad bin Qasim	Muhammad bin Qasim	The first successful Arab/Muslim invasion of India.
Battle of Venni (Venuka)	Chola period	Chola king vs. 11 kings including the Cheras and Pandyas	Chola ruler	The early supremacy of the Cholas was established.
Battle of Takkolam	949 CE	Parantaka I (Chola) vs. the Rashtrakutas (Krishna III) and the Western Gangas	Parantaka Chola	The Chola crown prince Rajaditya was killed in this battle against the Rashtrakuta Krishna III.
Battle of Vellore	955 CE	Parantaka I (Chola) vs. the Pandyas and Sri Lanka	Parantaka Chola	Parantaka I defeated the combined army of the Pandyas and Simhala (Sri Lanka).
Battle of Tilakaid	—	Ganga dynasty vs. the Chalukyas (Vishnuvardhana)	Vishnuvardhana	The Chalukya ruler of Vengi, Vishnuvardhana, was victorious.
Battle of Koppam	1052 CE	Rajadhiraja (Chola) vs. Someshvara (Chalukya)	Chola ruler	When Rajadhiraja was killed on the battlefield, Virarajendra led the Cholas to victory.

3 Principal Historical Eras (Important Eras/Calendars)

*(The favourite UPPCS Prelims section for arranging in 'chronology', i.e. **chronological order**)*

Name of Era	Starting Year	Founder / Ruler (Dynasty)	Special Historical Exam Facts
1. Vikrama Samvat	57 BCE	The traditional King Vikramaditya of Ujjain	• Also called the Krita Samvat or Malava Samvat ; its association with Vikramaditya is traditional.
2. Shaka Samvat	78 CE	The Kushana king Kanishka	• Associated with the victory over the Shakas or with Kanishka's accession. The official national calendar of the Government of India.
3. Kalachuri Samvat	248 CE	The Abhira king Ishvarasena	• Historically also known as the ' Chedi Samvat '.
4. Gupta Samvat	319–320 CE	Chandragupta I (Gupta dynasty)	• Started to commemorate the accession of Chandragupta I.
5. Harsha Samvat	606 CE	Harshavardhana (Vardhana/Pushyabhuti dynasty)	• Associated with the accession of Harshavardhana, ruler of Kannauj.
6. Hijri Samvat	622 CE	Prophet Muhammad	• In memory of the Prophet Muhammad's migration from Mecca to Medina (Hijrat) (622 CE = 1 Hijri).
7. Kollam Samvat	825 CE	Adi Shankaracharya (Malabar)	• In the regional language it is also called the ' Malayalam Samvat '.
8. Simha / Siddha Samvat	1113 CE	Jayasimha Siddharaja (Chalukya/Solanki)	• Issued by Jayasimha Siddharaja, the powerful Chalukya ruler of Gujarat.
9. Lakshmana Samvat	1190 CE	Lakshmana Sena (Sena dynasty, Bengal)	• Issued in the Mithila/Bengal region by the ruler of the Sena dynasty of Bengal.
10. Ilahi Samvat	1583 CE	Mughal emperor Akbar	• A solar calendar issued after the 'Din-i-Ilahi' propounded by Akbar.
11. Rajashaka Samvat	1673 CE	Chhatrapati Shivaji Maharaj	• Began on the occasion of the coronation of Chhatrapati Shivaji Maharaj of the Maratha empire.

1 Delhi Sultanate: Dynasties and Rulers (1206 - 1526)

Dynasty (Tenure)	Rulers (in sequence)
Slave (Mamluk) dynasty (1206-1290)	Qutbuddin Aibak → Aram Shah → Iltutmish → Razia Sultan → Bahram Shah → Balban → Kayumars
Khilji dynasty (1290-1320)	Jalaluddin Khilji → Alauddin Khilji → Shihabuddin Umar → Qutbuddin Mubarak → Nasiruddin Khusrau

Dynasty (Tenure)	Rulers (in sequence)
Tughlaq dynasty (1320-1412)	Ghiyasuddin Tughlaq → Muhammad bin Tughlaq → Firoz Shah Tughlaq → Ghiyasuddin Tughlaq II → Muhammad Shah III <i>Note: at the end there were two independent rulers — Nasiruddin Mahmud (Delhi) and Nusrat Shah (Ferozabad)</i>
Sayyid dynasty (1414-1451)	Khizr Khan → Mubarak Shah → Muhammad Shah → Alauddin Shah
Lodi dynasty (1451-1526)	Bahlol Lodi → Sikandar Lodi → Ibrahim Lodi

- **Additional fact (Tughlaq dynasty):** It was a Muslim dynasty of Turkic origin. It was founded by Ghazi Malik (Ghiyath al-Din Tughluq) in 1320.

2 Buildings of the Delhi Sultanate Period and their Builders

Ruler / Builder	Principal Buildings / Architecture
Qutbuddin Aibak	Adhai Din ka Jhonpra (Ajmer), Quwwat-ul-Islam Mosque (Delhi)
Iltutmish	Tomb of Iltutmish, Jama Masjid, Atarkin ka Darwaza, Sultan Ghari
Balban	Lal Mahal (Red Palace)
Alauddin Khilji	Jamat Khana Masjid, Alai Darwaza, Hazar Sutun (palace of a thousand pillars)
Ghiyasuddin Tughlaq	Tughlaqabad city / fort
Muhammad bin Tughlaq	Jahanpanah city, tomb of Sheikh Nizamuddin Auliya
Firoz Shah Tughlaq	Tomb of Firoz Shah Tughlaq (<i>at Hauz Khas, New Delhi, near Alauddin's tank</i>)
Junan Shah Khan-i-Jahan	Tomb of Firoz Shah, Khirki Masjid, Kali Masjid
Miyan Bhuwa	Moth ki Masjid

- **Special note:** Firoz Shah Tughlaq (reign: 1351-1388) is also called the **Shah Jahan of the Sultanate period** because of his vast contribution to architecture. He wrote his autobiography, the '**Futuhat-i-Firozshahi**'.

3 Principal Books of Medieval India and their Authors

Name of Book	Author / Composer
Akbarnama	Abul Fazl

Name of Book	Author / Composer
Shahnama	Firdausi
Baburnama	Babur
Humayunnama	Gulbadan Begum
Buddhacharitam	Ashvaghosha
Kavyamimamsa	Rajashekhara
Harshacharita	Banabhatta
Meghaduta	Kalidasa
Mudrarakshasa	Vishakhadatta
Rajatarangini	Kalhana
Indica	Megasthenes
Kamasutra	Vatsyayana
Vinaya Patrika	Tulsidas
Gita Govinda	Jayadeva
Hitopadesha	Narayana Pandit
Malgudi Days	R.K. Narayan
My Truth	Indira Gandhi

4 Books Written by Emperors and their Language

Ruler / Emperor	Name of Book	Language
Harshavardhana	Ratnavali, Priyadarshika, Nagananda	Sanskrit
Shri Krishnadevaraya	Amuktamalyada	Telugu
Jahangir	Tuzuk-i-Jahangiri	Persian
Babur	Baburnama (Tuzuk-i-Baburi)	Chaghatai (Turkish)

5 Principal Rulers and their Wives / Begums

Name of Begum / Wife	Ruler / Husband
Shah Turkan	Itutmish
Malika-i-Jahan	Alauddin Khilji
Hamida Banu Begum	Humayun
Mariam-uz-Zamani (Hirabai / Jodha Bai)	Akbar
Nur Jahan (Mehrunnisa)	Jahangir
Mumtaz Mahal (Arjumand Banu Begum)	Shah Jahan

6 Principal Battles of the Medieval Period (Chronology)

Year	Name of Battle	Between Whom and Result
1556	Second Battle of Panipat	The Mughal emperor Akbar defeated the Hindu king Hemu .
1567	Battle of Thanesar	Akbar defeated two groups of sanyasis (ascetics).
1575	Battle of Tukaroi	Akbar defeated the Sultanate of Bihar and Bengal.
1576	Battle of Haldighati	Akbar (Man Singh) defeated the valiant Maharana Pratap .
1671	Battle of Saraighat	The Ahom commander Lachit Borphukan defeated the Mughal army (Ram Singh).
1739	Battle of Karnal	The Iranian ruler Nadir Shah defeated the Mughal emperor Muhammad Shah (looted the Kohinoor diamond and the Peacock Throne).

7 Principal Battles Fought by Humayun (Special Table)

Year	Name of Battle	Place / River Bank	Result / Special Fact
1532	Battle of Dauhariya (Dadrah)	On the bank of the Gomti river	Humayun defeated the combined army of Mahmud Lodi and Sher Shah.
1539	Battle of Chausa (26 June)	Buxar, Bihar	Sher Shah Suri defeated Humayun. A water-carrier (bhishti) named Nizam saved Humayun's life. Sher Khan took the title ' Sher Shah '.

Year	Name of Battle	Place / River Bank	Result / Special Fact
1540	Battle of Kannauj (Bilgram)	Kannauj, U.P.	Humayun suffered a decisive defeat and Sher Shah captured Delhi.
1555	Battle of Machhiwara (15 May)	Punjab	Humayun defeated the army of Sikandar Sur.
1555	Battle of Sirhind (22 June)	Punjab	Humayun again defeated Sikandar Sur and recaptured Delhi in July 1555 .

8 Principal Religious Schools (Philosophies) and their Founders

(Asked repeatedly in UPPCS Prelims for matching)

School / Philosophy	Founder
Advaitavada	Shankaracharya
Vishishtadvaitavada	Ramanujacharya
Dvaitadvaitavada	Nimbarkacharya
Shuddhadvaitavada	Vallabhacharya
Dvaitavada	Madhvacharya
Bhedabhedavada	Bhaskaracharya
Shaiva Vishishtadvaita	Shrikantha
Achintya Bhedabhedavada	Baladeva
Vira Shaiva Vishishtadvaita	Shripati

9 Mughal Architecture and Buildings

Building / Monument	Place	Builder / Ruler
Kabuli Bagh Mosque	Panipat	Babur
Jami Masjid	Sambhal (Rohilkhand)	Babur
Mosque of Agra	Agra (inside the Lodi fort)	Babur
Mosque of Fatehabad	Punjab (Hisar)	Humayun

Building / Monument	Place	Builder / Ruler
Dinpanah city	Delhi	Humayun
Qila-i-Kuhna Mosque	Delhi (inside the Old Fort/Purana Qila)	Sher Shah Suri
Shersur city	Kannauj	Sher Shah Suri
Rohtasgarh Fort	North-West Frontier Province	Sher Shah Suri
Tomb of Sher Shah	Sasaram (Bihar)	Sher Shah Suri
Humayun's Tomb	Delhi	Haji Begum (Akbar's stepmother)
Agra Fort	Agra	Akbar
Jahangiri Mahal and Akbari Mahal	Agra Fort	Akbar
Buildings of Fatehpur Sikri (Diwan-i-Am, Diwan-i-Khas, Panch Mahal, Palace of the Turkish Sultana, Mariam Mahal, Buland Darwaza, Tomb of Sheikh Salim Chishti, Tomb of Islam Shah)	Fatehpur Sikri	Akbar
Lahore Fort	Lahore	Akbar
Fort with 40 pillars (the last building of Akbar's period)	Allahabad (Prayagraj)	Akbar
Akbar's Tomb	Sikandra (Agra)	Jahangir
Tomb of Itimad-ud-Daulah	Agra	Nur Jahan
Jahangir's Tomb	Shahdara (Lahore)	Nur Jahan
Tomb of Rabia-ud-Daurani (Bibi ka Maqbara)	Aurangabad	Aurangzeb
Badshahi Mosque	Lahore	Aurangzeb
Moti Masjid (Pearl Mosque) (inside the Red Fort)	Delhi	Aurangzeb

1 The Ashtadiggajas of Krishnadevaraya (Vijayanagara Empire)

(UPPCS asks directly, giving names, which of these were or were not among the Ashtadiggajas)

- **Allasani Peddana** (also called 'Andhra Kavita Pitamaha' – grandfather of Telugu poetry)
- **Nandi Timmana**
- **Bhattumurti**

- Pingali Surana
- Dhurjati
- Tenali Ramakrishna

2 The Ten Gurus of the Sikhs and their Important Works (Chronology)

No.	Name of the Guru (Tenure)	Important Works / Exam Facts
1.	Guru Nanak Dev (1469-1539)	Founder of Sikhism, the first Guru.
2.	Guru Angad (Lehna) (1504-1552)	Invented / introduced the Gurmukhi script .
3.	Guru Amar Das (1479-1574)	Made the langar system permanent; established 22 gaddis (seats).
4.	Guru Ram Das (1534-1581)	Akbar granted 500 bighas of land, on which the city of Amritsar was founded. Made the office of Guru hereditary .
5.	Guru Arjan Dev (1563-1606)	Founded the town of Tarn Taran. Compiled the Adi Granth (Guru Granth Sahib) . Sentenced to death by Jahangir .
6.	Guru Hargobind (1594-1644)	Transformed the Sikhs into a martial community. Built the Akal Takht .
7.	Guru Har Rai (1630-1661)	Had blessed Dara Shikoh.
8.	Guru Har Krishan (1656-1664)	Died young of smallpox.
9.	Guru Tegh Bahadur (1621-1675)	Executed in Delhi by Aurangzeb (Sis Ganj Gurudwara).
10.	Guru Gobind Singh (1666-1708)	Founded the Khalsa Panth (1699) at Anandpur. The last Guru.

3 Delhi Sultanate: Descending Order of Length of Rule (Chronology)

(A pet question of UPPCS Prelims - from longest to shortest rule)

Dynasty	Duration of Rule (Years)	Period of Rule (CE)
1. Tughlaq dynasty	94 years	1320 - 1414 CE
2. Slave dynasty	84 years	1206 - 1290 CE

Dynasty	Duration of Rule (Years)	Period of Rule (CE)
3. Lodi dynasty	76 years	1451 - 1526 CE
4. Sayyid dynasty	36 years	1414 - 1451 CE
5. Khilji dynasty	30 years	1290 - 1320 CE

- The last three rulers of the Sayyid dynasty (in sequence): Mubarak Shah (1421-34) → Muhammad Shah (1434-45) → Alauddin Alam Shah (1445-51).

4 Administrative Units and Officers of the Mughal Period

Administrative Unit	Name of Officer	Main Function / Department
Province (Suba)	Sipahsalar / Subedar / Nazim	Chief executive (Sipahsalar under Akbar).
	Diwan	In charge of the revenue department (finance head of the suba).
	Bakhshi	In charge of the military department.
District (Sarkar)	Faujdar	Administrative head (maintenance of peace).
	Amal / Amalguzar	Revenue collection (tax collection) officer.
	Kotwal	Law and order, disposal of criminal cases, price control.
Pargana	Shiqdar	Administrative head (his duties also included those of the kotwal).
	Amin / Qanungo	Revenue officer (land records and tax assessment).
Village	Muqaddam	Village headman.
	Patwari	Accountant (maintained the village land/revenue accounts).
	Chaukidar	Security (watchman).

5 Principal Books of Medieval History and their Authors

Name of Book	Author	Special Details
Taj-ul-Maasir	Hasan Nizami	The first official history of the Delhi Sultanate.

Name of Book	Author	Special Details
Tabaqat-i-Nasiri	Minhaj-us-Siraj	Account of the period of Iltutmish and Nasiruddin Mahmud.
Fawa'id-ul-Fuad	Amir Hasan Dehlavi	Collection of the sayings of the Sufi saint Nizamuddin Auliya.
Tarikh-i-Firozshahi	Ziauddin Barani	An important source for the period of Alauddin and the Tughlaqs.
Futuhat-i-Firozshahi	Firoz Shah Tughlaq	Autobiography of Sultan Firoz Shah Tughlaq.
Futuh-us-Salatin	Abdul Malik Isami	History of the founding of the Bahmani kingdom and of the Sultanate period.

6 Principal Works of Amir Khusrau and their Subject Matter

(Highly important for Match the Following)

Name of Work	Main Subject Matter (what it describes)
Qiran-us-Sadain	Description of the historic meeting of Bughra Khan and his son Kaiqubad.
Miftah-ul-Futuh	Description of the military campaigns and victories of Jalaluddin Khilji.
Khazain-ul-Futuh	Description of the reign of Alauddin Khilji and his conquest of the South.
Nuh Sipihr	Description of the period of Mubarak Khilji, and of the climate of India and its praise.
Ashiqi	The love story of Deval Rani, daughter of King Karan of Gujarat, and Khizr Khan, son of Alauddin.
Tughlaqnama	Depiction of the victory of Ghiyasuddin Tughlaq and his struggle with Khusrau Shah.

7 Principal Sufi Silsilas (Orders) of India and their Founders

Sufi Silsila	Founder (in India)
Chishti Silsila	Khwaja Muinuddin Chishti (Dargah: Ajmer)
Suhrawardi Silsila	Bahauddin Zakariya
Qadiri Silsila	Sayyid Makhdum Muhammad Jilani
Shattari Silsila	Shah Abdullah Shattari
Firdausi Silsila	Sheikh Badruddin Samarqandi (active in Bihar)

Sufi Silsila	Founder (in India)
Naqshbandi Silsila	Baqi Billah (Aurangzeb was a follower of this silsila)

8 Administrative System of the Delhi Sultanate (Departments and Sultans)

Name of Department	Related Sultan / Creator	Type / Function of the Department
Diwan-i-Arz	Balban	Military department
Diwan-i-Mustakhraj	Alauddin Khilji	Finance department (for realising arrears of revenue)
Diwan-i-Riyasat	Alauddin Khilji	Market control department / regulation of trade
Diwan-i-Amir Kohi	Muhammad bin Tughlaq	Agriculture department (for improvement in agriculture)
Diwan-i-Bandagan	Firoz Shah Tughlaq	Department of slaves
Diwan-i-Khairat	Firoz Shah Tughlaq	Charity department (for helping orphans and widows)
Diwan-i-Istihqaq	Firoz Shah Tughlaq	Pension department

9 Principal Ministers / Officers of the Sultanate Period

Name of Officer	Related Department / Function
Ariz-i-Mumalik	Head of the military department (head of the Diwan-i-Arz).
Barid-i-Mumalik	Chief officer of the intelligence department.
Qazi-ul-Quzat	Highest officer of the judicial department (after the Sultan).
Mushrif-i-Mumalik	Accountant-General (kept accounts of income and expenditure received from the provinces).
Wakil-i-Dar	Looked after the Sultan's personal services and the royal karkhanas (workshops).
Khazin	Treasurer.

10 Tax (Revenue) System of the Delhi Sultanate

Name of Tax	Type of Tax / From Whom it was Collected
Ushr	Land tax collected from Muslims .
Kharaj	Land tax collected from non-Muslims (Hindus).

Name of Tax	Type of Tax / From Whom it was Collected
Zakat	Religious tax on Muslims (one-fortieth of property, or 2.5%).
Jizya	Religious / protection tax collected from non-Muslims .

11 Foreign Travellers who Came in the Time of Shah Jahan

Name of Traveller	Country of Origin	Period (CE)	Special Details (UPPCS Factory)
Peter Mundy	Italy (Italian)	1630 - 1634 CE	Described the severe famine that occurred in the time of Shah Jahan.
Niccolao Manucci	Italy (Italian)	1653 - 1708 CE	Was first an artilleryman in the army of Dara Shikoh , later became a physician. (Book: Storia do Mogor).
Jean-Baptiste Tavernier	France (French)	1638 - 1663 CE	A jeweller by profession. Visited India 6 times .
François Bernier	France (French)	1656 - 1717 CE	A physician by profession. Eyewitness of the war of succession.

12 Projects of Muhammad bin Tughlaq (MBT) (Chronology)

(UPPCS asks for their correct chronological order every second year)

- 1. Increase of taxation in the Doab (1325 CE)** - The very first scheme, which failed because of famine.
- 2. Transfer of capital (1326-27 CE)** - Shifted the capital from Delhi to Devagiri (Daulatabad).
- 3. Introduction of token currency (1330 CE)** - Issued copper/bronze coins.
- 4. Khurasan expedition (1332 CE)** - Gave the soldiers 1 year's salary in advance, but the expedition was called off.
- 5. Qarachil expedition (1337 CE)** - Sent an army into the Kumaon hills (Uttarakhand), which failed with heavy destruction.

13 Principal Events of Babur's Time (Chronology)

Date / Year	Name of Battle / Event	Against Whom / Result
21 April 1526	First Battle of Panipat	Defeated Ibrahim Lodi (foundation of the Mughal Empire in India).
16 March 1527	Battle of Khanwa	Defeated Rana Sanga of Mewar (gave the call of jihad).
29 January 1528	Battle of Chanderi	Defeated Medini Rai.

Date / Year	Name of Battle / Event	Against Whom / Result
05 May 1529	Battle of Ghaghra	Defeated the Afghans (Mahmud Lodi) (fought both on water and on land).
26 December 1530	Death of Babur	Died at Agra (later buried at Kabul).

14 Principal Military Invasions of Alauddin Khilji

(UPPCS mainly asks for matching the king and the year)

Year (CE)	Place of Invasion	Contemporary Ruler / King
1296	Devagiri (first invasion)	Ramachandra Deva
1299	Gujarat	Rai Karan (Karnadeva Baghela)
1301	Ranthambhore	Hammir Deva
1303	Chittor	Raja Ratan Singh
1305	Malwa	Mahlak Deva
1308	Devagiri (second invasion)	Ramachandra Deva
1310	Warangal	Rudradeva (Prataparudra Deva - the Kohinoor was obtained from here)
1311	Dwarasamudra	Vira Ballala III (Hoysala dynasty)

15 Administrative Units of the Vijayanagara Empire (in descending order)

- Province (Mandala)** - The largest unit (under a governor)
- Kottam or Valanadu** - District-level unit
- Nadu** - Pargana/tehsil-level unit
- Melagrama** - A group of 50 villages
- Uragrama** - The smallest unit (village)

16

Independent States that Broke away from the Bahmani Kingdom (Deccan Sultanates) and their Dynasties

Name of State	Name of Dynasty	Founder (additional information)
Bijapur	Adil Shahi	Yusuf Adil Shah
Ahmednagar	Nizam Shahi	Malik Ahmad
Berar	Imad Shahi	Fathullah Imad Shah
Golconda	Qutb Shahi	Quli Qutb Shah
Bidar	Barid Shahi	Amir Ali Barid

1 Wars of Succession (Aurangzeb's Period) – Chronology

No.	Name of Battle	Time / Date	Main Result
1.	Battle near Banaras	February 1658	Defeat of Shah Shuja.
2.	Battle of Dharmat	April 1658	Aurangzeb and Murad defeated (the army of) Dara Shikoh (Jaswant Singh).
3.	Battle of Samugarh	May 1658	Aurangzeb decisively defeated Dara Shikoh.
4.	Battle of Khajwa	January 1659	Aurangzeb finally defeated Shah Shuja.
5.	Battle of Deorai	April 1659	Aurangzeb's final decisive victory over Dara Shikoh.

2 Later Mughal Rulers (Tenure and Principal Facts)

Name of Ruler	Reign	Important Exam Facts / Nicknames
Bahadur Shah I	1707 – 1712 CE	Son of Aurangzeb (Shah Alam I); met the Sikh Guru Gobind Singh. Nickname: ' Shah-i-Bekhabar ' (the heedless king).
Jahandar Shah	1712 – 1713 CE	Married Lal Kunwar; Zulfiqar Khan was influential. Nickname: ' Lampat Murkh ' (the profligate fool).
Farrukhsiyar	1713 – 1719 CE	Got the throne with the help of the Sayyid brothers (king-makers). Nickname: ' Ghrinit Kayar ' (the contemptible coward).
Rafi-ud-Darajat	1719 CE	The Mughal emperor with the shortest reign; influence of the Sayyid brothers.
Rafi-ud-Daula	1719 CE	Took the title 'Shah Jahan II'; brother of Rafi-ud-Darajat.

Name of Ruler	Reign	Important Exam Facts / Nicknames
Muhammad Shah 'Rangeela'	1719 – 1748 CE	In his time Nadir Shah invaded in 1739 (the Kohinoor diamond was looted).
Ahmad Shah Bahadur	1748 – 1754 CE	Son of Muhammad Shah.
Alamgir II	1754 – 1759 CE	Ascended the throne after Ahmad Shah Bahadur (ruler at the time of the Battle of Plassey).
Shah Alam II	1759 – 1806 CE	In his time the Battle of Buxar (1764) and the Treaty of Allahabad (1765) took place.
Akbar II	1806 – 1837 CE	Sent Rammohan Roy to England after giving him the title of 'Raja' .

3 Early Medieval Period: The Tripartite (Triangular) Struggle

Main Point	Description
Dynasties involved	The Gurjara-Pratiharas , the Rashtrakutas and the Pala dynasty
Cause of the struggle	To establish control over Kannauj , the principal centre of North India
Duration of the struggle	Lasted about 100 years
Final victory	Of the Gurjara-Pratiharas

4 Principal Dynasties of the Tripartite Struggle

Dynasty	Founder	Capital	Principal Exam Facts
Gurjara-Pratihara dynasty	Nagabhata I (730 CE)	Bhinmal (Jalore, Rajasthan)	* The triangular struggle began in the time of Vatsaraja . * At the time of Mahmud of Ghazni's invasion the ruler was Rajyapala .
Rashtrakuta dynasty	Dantidurga (752 CE)	Manyakheta	* Krishna I built the famous Kailasha Temple of Ellora. * The last ruler was Karka II .
Pala dynasty	Gopala (750 CE)	Munger	* Dharmapala founded Vikramashila University . * The last ruler was Govindapala .

5 Early Muslim Invaders of India

Invader	Time / Period	Principal Historical Facts
Muhammad bin Qasim	712 CE	The first Muslim ruler to invade India (Sindh).
Mahmud of Ghazni	998–1030 CE	* Son of Subuktigin; invaded India 17 times (1000–1027 CE). * Made his famous attack on the Somnath Temple in 1026 CE .
Muhammad Ghori	1175–1206 CE	* Ruler of the province of Ghor; first invasion on Multan in 1175 CE. * In 1178 CE he was defeated by Bhima II (Mularaja II) , the ruler of Gujarat.

> Court Scholars of Mahmud of Ghazni (asked directly):

- **Al-Biruni:** Kitab-ul-Hind
- **Utbi:** Kitab-ul-Yamini
- **Firdausi:** Shahnama
- **Baihaqi:** Tarikh-i-Subuktigin

6 Principal Battles of Muhammad Ghori

Year	Name of Battle	Against Whom	Result / Special Fact
1191 CE	First Battle of Tarain	Prithviraj Chauhan	Prithviraj Chauhan was victorious (Ghori lost).
1192 CE	Second Battle of Tarain	Prithviraj Chauhan	Muhammad Ghori victorious (foundation of Muslim power in India).
1194 CE	Battle of Chandawar	Raja Jaichand	Muhammad Ghori victorious (Jaichand was killed).

7 Delhi Sultanate: Principal Dynasties and the Works of the Sultans

> (a) Slave Dynasty (1206 - 1290 CE)

Sultan	Reign	Principal Works and Important Facts
Qutbuddin Aibak	1206–1210 CE	* Founder of the Slave dynasty; nicknamed ' Lakh Bakhsh ' (giver of lakhs) and ' Quran Khan '. * Constructions: Quwwat-ul-Islam Mosque (Delhi), Adhai Din ka Jhonpra (Ajmer); began the Qutub Minar.

Sultan	Reign	Principal Works and Important Facts
Iltutmish	1211–1236 CE	* Moved the capital from Lahore to Delhi; completed the Qutub Minar. * Started the iqta system; formed the 'Turkan-i-Chahalgani' of 40 nobles. * Issued the silver 'tanka' and the copper 'jital' coins.
Razia Sultan	1236–1240 CE	* The only woman ruler to become Sultan of Delhi. * Altunia rebelled against her; Razia later married him.
Balban	1266–1286 CE	* Real name: Bahauddin; ended the 'Turkan-i-Chahalgani'. * Took the title 'Zill-i-Ilahi' (shadow of God); called himself 'Niyabat-i-Khudai'. * Introduced the Iranian customs of 'sijda' and 'paibos' and the Nauroz festival. Adopted the policy of 'blood and iron'.

> (b) Khilji Dynasty (1290 - 1320 CE)

Sultan	Reign	Principal Works and Important Facts
Jalaluddin Firoz Khilji	1290–1296 CE	* Founded the Khilji dynasty after killing Kayumars. The revolt of Malik Chhajju took place in his time.
Alauddin Khilji	1296–1316 CE	* Took the title 'Sikandar II (Sikandar-i-Sani)'; obtained a slave named Malik Kafur from Gujarat. * Works: created the 'Diwan-i-Mustakhraj' for land revenue based on measurement of land; implemented the policy of market control (price control); started the practice of branding horses (dagh) and recording the descriptive roll (huliya) of soldiers.

- **Last ruler:** Nasiruddin Khusrau Shah.

All the Military Campaigns of Alauddin Khilji (North and South India) Correct Chronological Order

Campaign / Region	Year	Ruler	Location of the Region
Gujarat	1299 CE	Karnadeva Vaghela (Rai Karan)	North India
Ranthambhore	1301 CE	Hamir Deva Chauhan	North India (Rajputana)
Mewar (Chittor)	1303 CE	Rana Ratan Singh	North India (Rajputana)
Malwa	1305 CE	Mahlak Deva Parmar	North India
Devagiri	1307 - 1308 CE	Ramachandra Deva (Yadava dynasty)	South India
Warangal	1309 - 1310 CE	Prataparudradeva II (Kakatiya dynasty)	South India
Dwarasamudra	1310 - 1311 CE	Vira Ballala III (Hoysala dynasty)	South India
Jalore	1311 CE	Kanhadadeva Chauhan	North India (Rajputana)

(Note: before becoming Sultan, Alauddin Khilji had also made a successful raid on Devagiri in **1296 CE**, but his formal southern campaign as Sultan began in **1307 CE** under the leadership of Malik Kafur.)

> (c) Tughlaq Dynasty (1320 - 1412 CE)

Sultan	Reign	Principal Works and Important Facts
Ghiyasuddin Tughlaq	1320–1325 CE	* Founder of the Tughlaq dynasty; founded the city of Tughlaqabad. * He was the first Sultan of Delhi to have canals constructed.
Muhammad bin Tughlaq	1325–1351 CE	* Real name: Ulugh Khan (Juna Khan). Created the agriculture department 'Diwan-i-Amir Kohi'. * 5 schemes (in order): increase of taxation in the Doab (1325) → transfer of capital (1326) → token currency (1330) → Khurasan and Qarachil expeditions. * In 1333 CE the Moroccan traveller Ibn Battuta came to India (book: 'Rehla'); he was appointed Qazi of Delhi.
Firoz Shah Tughlaq	1351–1388 CE	* Imposed only 4 Sharia taxes—Kharaj, Khums, Jizya, Zakat. Imposed jizya on Brahmins. * Issued the 'Shashgani' coin; the first Sultan to levy an irrigation tax (Haq-i-Sharb). * First created a 'Public Works Department (PWD)'. Founded towns: Fatehabad, Hisar, Firozpur, Jaunpur, Firozabad. * New departments: Diwan-i-Khairat (charity), Dar-ul-Shafa (hospital), Diwan-i-Bandagan (slaves).

- **Last ruler of the Tughlaq dynasty**: was **Nasiruddin Mahmud (1394-1412 CE)**. In his time Malik Sarwar made **Jaunpur** independent and **Timur (Tamerlane)** invaded **Delhi in 1398 CE**.

> (d) Sayyid and Lodi Dynasties

Sultan / Dynasty	Reign	Principal Works and Important Facts
Khizr Khan (Sayyid dynasty)	1414–1421 CE	* Founder of the Sayyid dynasty; he was a commander of Timur. (Last ruler: Alauddin Alam Shah).
Bahlol Lodi (Lodi dynasty)	1451–1489 CE	* Founder of the Lodi dynasty; issued a coin called the 'Bahloli'. Annexed Jaunpur to the Delhi Sultanate.
Sikandar Lodi	1489–1517 CE	* Founded the city of 'Agra' in 1504 CE and made it the capital. Introduced the 'Gaz-i-Sikandari' for land measurement. * Wrote Persian poetry under the pen-name 'Gulrukhi'. Persian translations: 'Farhang-i-Sikandari' (Ayurveda) and 'Lazzat-i-Sikandar Shahi' (music).

Sultan / Dynasty	Reign	Principal Works and Important Facts
Ibrahim Lodi	1517– 1526 CE	* The last ruler of the Delhi Sultanate. Lost the Battle of Khatoli in 1518 CE to Rana Sanga. * Was defeated and killed by Babur in the First Battle of Panipat on 21 April 1526.

8 Principal Administrative Officers of the Sultanate Period and their Functions

(Extremely important for UPPCS Prelims 'Match the Following')

Designation (Officer)	Related Department / Administrative Function
Wazir	Head of the finance department (prime minister).
Ariz-i-Mumalik	Head of the military department.
Diwan-i-Insha / Dabir-i-Risalat	Chief officer of the department of royal correspondence.
Diwan-i-Risalat	Head of the Foreign Department.
Sadr-us-Sudur	Chief officer of the department of religion and charity.
Qazi-ul-Quzat	Chief officer of the judicial department (chief justice).
Barid-i-Mumalik	Chief officer of the intelligence department (intelligence chief).
Wakil-i-Hajib	Chief officer in charge of court etiquette.
Sar-i-Jandar	Head of the Sultan's personal bodyguards.
Mustaufi-i-Mumalik	Auditor General (who examined expenditure).
Mushrif-i-Mumalik	Accountant General (who kept account of income).

1 Principal Administrative Departments of the Delhi Sultanate and their Founders

Name of Department	Type / Function of the Department	Founder (Sultan)
Naib-i-Mamlikat	Running of the entire administration	Bahram Shah
Diwan-i-Arz	Military department	Balban
Diwan-i-Riyasat	Market control / regulation of commerce	Alauddin Khilji
Diwan-i-Mustakhraj	Realisation of arrears of revenue (tax)	Alauddin Khilji

Name of Department	Type / Function of the Department	Founder (Sultan)
Diwan-i-Amir Kohi	Agriculture department (for agricultural improvement)	Muhammad bin Tughlaq
Diwan-i-Bandagan	Care of slaves	Firoz Shah Tughlaq
Diwan-i-Khairat	Charity department (for orphans and widows)	Firoz Shah Tughlaq
Dar-ul-Shafa	Department of free medical treatment (charitable hospital)	Firoz Shah Tughlaq

2 Dynasties of the Vijayanagara Empire (Chronology)

Dynasty	Time / Period	Principal Rulers (in sequence)
1. Sangama dynasty	1336–1485 CE	Harihara I → Bukka I → Harihara II → Virupaksha I → Bukka II → Devaraya I → Devaraya II → Mallikarjuna → Virupaksha II
2. Saluva dynasty	1485–1505 CE	Narasimha Saluva → Narasa Nayaka
3. Tuluva dynasty	1505–1565 CE	Vira Narasimha → Krishnadevaraya → Achyuta Deva Raya → Venkata I → Sadashiva Raya
4. Aravidu dynasty	1570–1647 CE	Tirumala → Sriranga I → Venkata II → Sriranga III → Ramadeva → Venkata III → Sriranga III

3 Extremely Important Exam Facts Related to the Vijayanagara Empire

Subject / Event	Ruler / Character	Details and Exam Facts
First ruler	Harihara I	The first ruler of the Vijayanagara Empire (founded in 1336 CE by Harihara and Bukka).
Title: Veda Marga Pratishthapaka	Bukka I	Assumed this title for promoting the Vedic religion.
The war of the 'goldsmith's daughter'	Devaraya I	Fought between Devaraya I and the Bahmani ruler Firoz Shah.
Title: Gajabetekara and Immadi Devaraya	Devaraya II	Gajabetekara means 'hunter of elephants'. He is also called Immadi Devaraya.

Subject / Event	Ruler / Character	Details and Exam Facts
Title: Andhra Bhoja and Yavanarajya Sthapanacharya	Krishnadevaraya	The most powerful ruler of the Tuluva dynasty.
Literary works: Amuktamalyada and Jambavati Kalyanam	Krishnadevaraya	Wrote the Amuktamalyada (in Telugu) and the Jambavati Kalyanam (in Sanskrit).
Construction of temples and towns	Krishnadevaraya	Built the Hazara Ramaswamy and Vitthala Swamy temples; founded the town of ' Nagalapura ' in the name of his mother.
Text: Madhura Vijayam	Gangadevi	Composed by Gangadevi, wife of Kumara Kampana, to celebrate his conquest of Madurai.
Battle of Talikota <i>(Battle of Rakshasi-Tangadi / Banni Hatti)</i>	Rama Raya <i>(leading Vijayanagara)</i>	Took place in January 1565. The Muslim confederacy of the Deccan (Bijapur, Ahmednagar, Golconda, Bidar) defeated and destroyed Vijayanagara. <i>Note: Berar was not part of this confederacy.</i>

- **Administrative units of the Vijayanagara Empire (in descending order):** Kottam / Valanadu (district) → Nadu (pargana) → Melagrama (a group of 50 villages) → Ur (village).

4 Foreign Travellers who Came to Vijayanagara (a pet question for Match the Following)

Foreign Traveller	Country of Origin	Contemporary Vijayanagara Ruler
Nicolo Conti	Italy	Devaraya I
Nikitin	Russia	Devaraya I
Abdur Razzaq	Persia	Devaraya II
Duarte Barbosa	Portugal	Krishnadevaraya
Domingo Paes	Portugal	Krishnadevaraya
Fernao Nuniz	Portugal	Achyuta Devaraya
Caesar Frederick	Italy	Sadashiva Raya

5 History of Medieval India: The Jizya Tax (Chronology)

Year / Period	Name of Ruler	Historical Details Regarding the Jizya Tax
712 CE	Muhammad bin Qasim	The first ruler to impose the jizya tax in India (Sindh).

Year / Period	Name of Ruler	Historical Details Regarding the Jizya Tax
After 1351 CE	Firoz Shah Tughlaq	The first Sultan of Delhi to impose jizya even on Brahmins .
1564 CE	Akbar	The ruler who completely abolished jizya in the Mughal Empire.
1674 CE	Aurangzeb	The ruler who reimposed jizya in the Mughal period.
1713 CE	Farrukhsiyar	Abolished jizya once again.
1717 CE	Farrukhsiyar	Reimposed jizya.
1719 CE	Muhammad Shah 'Rangeela'	Finally (completely) abolished jizya in India.

6 Principal Sufi Silsilas and their Founders (with reference to India)

Sufi Silsila	Founder (in India / original)
Chishti Silsila	Khwaja Muinuddin Chishti (Dargah: Ajmer)
Suhrawardi Silsila	Shihabuddin Suhrawardi / Bahauddin Zakariya (made it famous in India)
Qadiri Silsila	Sheikh Abdul Qadir Jilani
Shattari Silsila	Shah Abdullah Shattari
Firdausi Silsila	Sheikh Badruddin (most active in the Bihar region)
Naqshbandi Silsila	Baqi Billah

7 The Four Principal Peethas (Mathas) Established by Adi Shankaracharya

Note: according to the source, the Sharada Peetha has been given here in place of the fourth peetha; remember it in the order of the states.

Name of Peetha	Site / Place	State	Associated Goddess/God
Shringeri Peetha	Mysore	Karnataka	Shiva
Govardhana Peetha	Puri	Odisha	Balabhadra and Subhadra
Sharada Peetha	Dwarka	Gujarat	Krishna

8 Principal Battles Fought by the Mughals – Chronology

Name of Battle	Exact Date	Between Whom and Result
First Battle of Panipat	21 April 1526 CE	Babur defeated Ibrahim Lodi.
Battle of Khanwa	16 March 1527 CE	Babur defeated Rana Sanga.
Battle of Chanderi	29 January 1528 CE	Babur defeated Medini Rai.
Battle of Ghaghra	5 May 1529 CE	Babur defeated the Afghans (Mahmud Lodi).
Battle of Chausa	26 June 1539 CE	Sher Shah Suri defeated Humayun.
Battle of Kannauj (Bilgram)	17 May 1540 CE	Sher Shah Suri defeated Humayun again and took possession of Delhi.

9 The Five Independent States Formed by the Disintegration of the Bahmani Kingdom

Name of State	Year of Foundation	Founder	Name of Dynasty
Bijapur	1489 CE	Yusuf Adil Shah	Adil Shahi
Ahmednagar	1490 CE	Malik Ahmad	Nizam Shahi
Berar	1490 CE	Fathullah	Imad Shahi
Golconda	1518 CE	Qutb Shah	Qutb Shahi
Bidar	1527 CE	Amir Ali Barid	Barid Shahi

> Special Exam Facts of the Adil Shahi Dynasty (Bijapur):

- **Ibrahim Adil Shah II:** He was called '**Abla Baba**' for helping the poor and '**Jagat Guru**' for patronising scholars. He composed the '**Kitab-i-Nauras**', a collection of Hindi songs.
- **Tarikh-i-Firishta:** The historian Firishta composed this work under the patronage of Ibrahim Adil Shah II.
- **Muhammad Adil Shah:** He built the world-famous '**Gol Gumbaz**' at Bijapur.

10 Medieval Historical Literature (Books and Authors)

(This table has been prepared by combining both the literary sections given by you, so that revision is easy)

Name of Book / Text	Author / Composer	Main Subject / Special Fact
Kitab-ul-Hind / Tahqiq-i-Hind	Al-Biruni	A view of India in the period of Mahmud of Ghazni.

Name of Book / Text	Author / Composer	Main Subject / Special Fact
Tarikh-i-Yamini	Utbi	History of Mahmud of Ghazni.
Shahnama	Firdausi	He is also called 'the Homer of the East' (Ghaznavid period).
Taj-ul-Maasir	Hasan Nizami	Early administrative information on the Delhi Sultanate.
Tabaqat-i-Nasiri	Minhaj-us-Siraj	The period of Iltutmish and Nasiruddin Mahmud.
Siyasatnama	Nizam-ul-Mulk Tusi	A book of administrative rules.
Tarikh-i-Firozshahi	Ziauddin Barani	Detailed account of the period of Alauddin Khilji and the Tughlaqs.
Fatawa-i-Jahandari	Ziauddin Barani	Political philosophy of the Sultanate period.
Futuh-i-Firozshahi	Firoz Shah Tughlaq	Autobiography of Sultan Firoz Shah Tughlaq.
Kitab-ul-Rehla (Rehla)	Ibn Battuta	Travel account of the Moroccan traveller (MBT period).
Tarikh-i-Mubarakshahi	Yahya bin Ahmad Sirhindi	The only authentic source for the history of the Sayyid dynasty.
Works of Amir Khusrau (<i>Khazain-ul-Futuh, Nuh Sipahr, Ashiqa, Tughlaqnama</i>)	Amir Khusrau	History of various rulers of the Khilji and Tughlaq dynasties.
Futuh-us-Salatin	Isami / Malik Isami	Important history related to the Bahmani kingdom.
Tarikh-i-Shershahi	Abbas Khan Sherwani	History of the administration and life of Sher Shah Suri.
Akbarnama	Abul Fazl	Official history of the Mughal emperor Akbar.
Tabaqat-i-Akbari	Nizamuddin Ahmad	History of Akbar's time.
Muntakhab-ut-Tawarikh	Abdul Qadir Badauni	A critical account of Akbar's policies.
Tuzuk-i-Baburi (Baburnama)	Babur	Autobiography of Babur (in the Chaghatai Turkish language).
Humayunnama	Gulbadan Begum	Biography written by Humayun's sister.

Name of Book / Text	Author / Composer	Main Subject / Special Fact
Tuzuk-i-Jahangiri	Jahangir, Mutamid Khan and Hadi Khan	Autobiography of Jahangir (later completed by Mutamid Khan).
Alamgirnama	Mirza Muhammad Kazim	History of Aurangzeb's reign.
Fatawa-i-Alamgiri	Sheikh Nizamuddin Ahmad	A famous compilation of Islamic law of Aurangzeb's period.
Nuskha-i-Dilkusha	Bhimsen	History of the Deccan (South India) in Aurangzeb's time.
Tarikh-i-Rashidi	Mirza Haidar Dughlat	History of Central Asia and Kashmir/the Mughals.
Rajatarangini	Kalhana	A systematic history of Kashmir.
Gaudavaho	Vakpati	Description of the conquests of King Yashovarman of Kannauj.
Ramacharita	Sandhyakara Nandi	History of Ramapala, ruler of the Pala dynasty.
Ballalacharita	Ananda Bhatta	History of Ballalasena, ruler of the Sena dynasty of Bengal.

1 Bhakti Movement: Saints, Schools and Sects

Saint / Philosopher	Philosophical School / Doctrine
Shankaracharya	Advaitavada
Ramanujacharya	Vishishta Advaitavada
Madhvacharya	Dvaitavada
Nimbarkacharya (<i>Nimbakacharya</i>)	Dvaitadvaitavada

> The Bhakti Tradition in South India

- **Alvar saints:** They were devotees of **Vishnu** (Vaishnava).
- **Nayanar saints:** They were devotees of **Shiva** (Shaiva).

> The Vaishnava Bhakti Tradition in Maharashtra

- **Varkari sect:** Worship of **Vitthala or Vithoba** (a form of Shri Krishna).
- **Dharkari sect:** Worship of **Shri Rama**.

2 Sufi Movement: Categories and Silsilas

> Main Categories of Sufism

- **Ba-shara (those who followed the Islamic Sharia law):** Chishti, Suhrawardi, Qadiri, Naqshbandi, Firdausi.
- **Be-shara (free from the bonds of Islamic law):** Qalandar, Madari.

> Principal Sufi Silsilas and their Founders in India

Sufi Silsila	Founder in India / Principal Saint
Chishti	Sheikh Muinuddin Chishti (Dargah: Ajmer)
Suhrawardi	Sheikh Bahauddin Zakariya (<i>originally: Shihabuddin Suhrawardi</i>)
Qadiri	Abdul Qadir Jilani
Naqshbandi	Khwaja Baqi Billah (<i>originally: Khwaja Bahauddin Naqshband</i>)
Firdausi	Sheikh Jalaluddin Surkh
Shattari	Abdullah Shattari

3 Maratha Empire: Chhatrapati Shivaji Maharaj and the Ashtapradhan

Event of Life	Year / Date	Historical Details and Exam Facts
Birth	April 1627	At Shivneri Fort (<i>Mother: Jijabai, Father: Shahji Bhonsle</i>)
Guru / Guardian	—	Guardian-teacher: Dadaji Kondadev ; spiritual guru: Samarth Ramdas
Killing of Afzal Khan	1659 CE	Killed Afzal Khan (Abdullah Bhatari), the commander of Bijapur.
Defeat of Shaista Khan	1663 CE	Defeated the Mughal subedar Shaista Khan.
First sack of Surat	1664 CE	First plunder of Surat, the trading city of the Mughals.
Treaty of Purandar	1665 CE	Between the Mughal commander Mirza Raja Jai Singh and Shivaji .
Escape from Agra	1666 CE	Succeeded in escaping from Aurangzeb's captivity (Jaipur Bhawan, Agra).
Second sack of Surat	1670 CE	Plundered the city of Surat a second time.

Event of Life	Year / Date	Historical Details and Exam Facts
Sons and coronation	June and September 1674	Crowned at Raigad Fort. (<i>Sons: Sambhaji and Rajaram</i>)
Death	12 April 1680	Death of Chhatrapati Shivaji Maharaj.

➤ Shivaji's Ashtapradhan (Council of Ministers):

(Matching of the departments of administrative officers comes in the exam every time)

Designation	Administrative Function / Department
1. Peshwa	Prime Minister (general administration and acting in the king's absence)
2. Amatya (Majumdar)	Finance Minister (revenue and accounts of income and expenditure)
3. Mantri (Waqianavis)	Kept a record of the king's daily activities and court proceedings
4. Sachiv	Supervised the department of correspondence
5. Sumant (Dabir)	Foreign Minister (supervised foreign affairs and ambassadors)
6. Senapati (Sar-i-Naubat)	Military Minister (recruitment, organisation and discipline of the army)
7. Panditrao	Dharmadhikari (head of religious affairs, charity and grants)
8. Nyayadhis	Head of Justice (highest officer for civil and criminal cases)

4 Successors of Shivaji Maharaj (Chronology)

No.	Ruler (Chhatrapati)	Reign (CE)
1.	Sambhaji	1680 – 1689 CE
2.	Rajaram	1689 – 1700 CE
3.	Tarabai	1700 – 1707 CE
4.	Shahuji	1707 – 1749 CE
5.	Rajaram II	1749 – 1777 CE

5 Areas of Influence / Control of the Maratha Chiefs

Maratha Dynasty / Chief	Area of Influence (Headquarters)
Peshwa	Poona (Pune)

Maratha Dynasty / Chief	Area of Influence (Headquarters)
Scindia	Gwalior
Holkar	Indore
Bhonsle	Nagpur
Gaekwad	Baroda (Vadodara)

6 Anglo-Maratha Wars and Important Treaties (Chronology)

> Anglo-Maratha Wars:

- **First Anglo-Maratha War:** 1775 – 1782 CE
- **Second Anglo-Maratha War:** 1803 – 1805 CE
- **Third Anglo-Maratha War:** 1817 – 1818 CE

> Historic Treaties and their Signatories (Match the Following):

Name of Treaty	Year (CE)	Between Whom (Signatories)
Treaty of Purandar	1665 CE	Mughal commander Jai Singh and Shivaji
Treaty of Sangola	1750 CE	Peshwa Balaji Bajirao and Rajaram II
Treaty of Surat	1775 CE	Raghunath Rao (Raghoba) and the East India Company
Second Treaty of Purandar	1776 CE	Peshwa Madhav Rao Narayan and the English
Convention of Wadgaon	1779 CE	Peshwa Madhav Rao Narayan and the English
Treaty of Salbai	1782 CE	Peshwa Madhav Rao Narayan and the English
Treaty of Bassein	1802 CE	Bajirao II and the English
Treaty of Deogaon	17 December 1803	Raghuji Bhonsle II and the English
Treaty of Surji-Anjangaon	30 December 1803	Scindia and the English
Treaty of Rajpurghat	1805 CE	Holkar and the English
Treaty of Poona	June 1817	Bajirao II and the English
Treaty of Gwalior	November 1817	Daulatrao Scindia and the English
Treaty of Mandsaur	January 1818	Malharrao Holkar and the English

7 Mughal Empire: Babur, Humayun, Sher Shah and Akbar (Battles and Policies)

> (a) Babur (1526 - 1530 CE) – Founder of the Mughal Dynasty

- **Family details:** Father: Umar Sheikh Mirza; Mother: Qutlugh Nigar Khanum. Sons: Humayun, Kamran, Askari and Hindal.
- **Autobiography:** 'Tuzuk-i-Baburi' or 'Baburnama'.

Name of Battle	Exact Date / Year	Between Whom	Victor
First Battle of Panipat	21 April 1526	Ibrahim Lodi and Babur	Babur
Battle of Khanwa	16 March 1527	Babur and Rana Sanga	Babur
Battle of Chanderi	29 January 1528	Babur and Medini Rai	Babur
Battle of Ghaghra	5 May 1529	Babur and Mahmud Lodi	Babur

> (b) Humayun (1530 - 1556 CE)

- **Life facts:** Born: 1508 CE at Kabul. Crowned at Agra in 1530 CE. Died: in 1556 CE.
- **Literature and architecture:** 'Humayun's Tomb' in Delhi was built by his wife **Haji Begum**. The biography 'Humayunnama' was written by his step-sister **Gulbadan Begum**.

Name of Battle	Year (CE)	Between Whom	Victor
Battle of Devra (Dauhariya)	1532	Humayun and the Afghan Mahmud Lodi	Humayun
Battle of Chausa	1539	Humayun and Sher Shah Suri	Sher Shah
Battle of Kannauj (Bilgram)	1540	Humayun and Sher Shah Suri	Sher Shah
Battle of Sirhind	1555	Humayun and Sikandar Sur	Humayun

> (c) Sher Shah Suri (1540 - 1545 CE)

- **Childhood name:** Farid Khan. The subedar of Bihar, '**Muhammad Bahar Khan Lohani**', gave him the title '**Sher Khan**'.
- **Throne and death:** Ascended the throne of Delhi in 1540. Died in 1545 during the **conquest of Kalinjar**.
- **Administrative work:** Introduced the '**Qabuliyat and Patta system**' to help the peasants.

> (d) Mughal Emperor Akbar (1556 - 1605 CE)

- **Life facts:** Born: 1542 CE in the palace of Raja Virsal of Amarkot. Parents: Hamida Banu and Humayun. Childhood name: **Jalal**.
- **Coronation and death:** Crowned in 1556 CE at **Kalanaur** in Punjab. Guardian: **Bairam Khan**. Died: in 1605. Tomb: located at **Sikandra** (near Agra).

> Principal Battles of Akbar's Period:

Name of Battle	Exact Date	Between Whom	Result / Victor
Second Battle of Panipat	5 November 1556	Akbar and Hemu	Akbar was victorious
Battle of Haldighati	18 June 1576	Akbar and Maharana Pratap	Akbar was victorious

> Principal Policy / Administrative Measures of Akbar (Chronology - extremely important for UPPCS):

Administrative Measure / Reform	Year (CE)
Abolition of slavery	1562 CE
Abolition of the pilgrimage tax	1563 CE
Abolition of jizya	1564 CE
Capital shifted from Agra to Fatehpur Sikri	1571 CE
Construction of the Ibadatkhana	1575 CE
Doors of the Ibadatkhana opened to all religions	1578 CE
Proclamation of the Mahzar	1579 CE
Ibadatkhana closed	1582 CE
Founding of the Din-i-Ilahi	1582 CE

1 Akbar's Navaratnas (Nine Gems)

- Abul Fazl
- Faizi
- Tansen
- Birbal
- Todar Mal
- Raja Man Singh
- Abdul Rahim Khan-i-Khanan
- Faqir Azizuddin
- Mulla Do Piyaza

2 Mughal Emperors: Jahangir, Shah Jahan and Aurangzeb

Name of Ruler (Tenure)	Real Name and Parents	Wives and Sons	Important Exam Facts / Architecture
Jahangir (1605–1627 CE)	* Real name: Salim * Parents: Harkha Bai - Akbar	* Wives: Man Bai, Jagat Gosain, Mehrunnisa (Nur Jahan) * Sons: Khusrau, Parvez, Shahryar, Jahandar	* His reign is called the ' golden age of painting '. * Introduced the do-aspa and sih-aspa system. * Foreign travellers: Hawkins, Sir Thomas Roe and Francisco Pelsaert came to his court. * Tomb: at Shahdara (Lahore). * Architecture: Akbar's Tomb (Sikandra).
Shah Jahan (1628–1657 CE)	* Real name: Khurram * Parents: Jagat Gosain - Jahangir	* Wife: Arjumand Banu Begum (Mumtaz Mahal) * Sons: Dara Shikoh, Aurangzeb, Murad, Shah Shuja	* His reign is called the ' golden age of architecture '. * In 1638 CE shifted the capital from Agra to Delhi . * Gave Dara Shikoh the title ' Shah Buland Iqbal '. * Dara Shikoh translated the Upanishads into Persian under the name ' Sirr-i-Akbar '. * Tomb: Agra (Taj Mahal). * Architecture: Jama Masjid and Red Fort of Delhi; Moti Masjid and Taj Mahal of Agra.
Aurangzeb (1658–1707 CE)	* Parents: Mumtaz Mahal - Shah Jahan	* Wife: Dilras Banu Begum (Rabia-ud-Daurani) * Sons: Muhammad Azam, Muhammad Akbar, Kam Bakhsh	* Crowned twice: first on 31 July 1658 (Agra), second on 15 June 1659 (Delhi). * Death: February 1707. * Tomb: Khuldabad / Aurangabad (Maharashtra). * Architecture: Moti Masjid inside the Red Fort (Delhi).

3 Wars of Succession in the Time of Shah Jahan (Exact Chronology)

(The favourite question of UPPCS Prelims - for arranging in correct order by date)

No.	Name of Battle	Exact Date	Between Whom	Victor
1.	Battle of Bahadurpur	24 February 1658	Between Shah Shuja and the imperial army	Imperial army
2.	Battle of Dharmat	15 April 1658	Combined army of Aurangzeb + Murad Bakhsh vs. the imperial army of Jaswant Singh and Qasim Khan	Aurangzeb
3.	Battle of Samugarh	29 May 1658	Combined army of Aurangzeb + Murad Bakhsh vs. Dara Shikoh	Aurangzeb

No.	Name of Battle	Exact Date	Between Whom	Victor
4.	Battle of Khajwa	5 January 1659	Between Aurangzeb and Shah Shuja	Aurangzeb
5.	Battle of Deorai	12-14 April 1659	Between Dara Shikoh and Aurangzeb	Aurangzeb (finally)

4 Administrative Divisions of the Mughal Empire and Principal Officers

Administrative Unit	Principal Officer / Person Responsible
Suba (province)	Subedar or Nazim
Sarkar (district)	Faujdar
Pargana or Mahal	Shiqdar
Mauza / Diha / Grama (village)	Khut, Muqaddam or Chaudhari

5 Principal Literature / Books of the Mughal Period and their Authors

Name of Book / Text	Author / Composer
Baburnama (Tuzuk-i-Baburi)	Babur
Humayunnama	Gulbadan Begum
Tabaqat-i-Akbari	Nizamuddin Ahmad
Tarikh-i-Shershahi	Abbas Khan Sherwani
Alamgirnama	Muhammad Kazim
Qanun-i-Humayuni	Khwandamir
Nuskha-i-Dilkusha	Bhimsen
Siyar-ul-Mutakhkherin	Ghulam Husain
Iqbalnama-i-Jahangiri	Mutamid Khan
Futuhāt-i-Alamgiri	Ishwar Das Nagar
Khulasat-ut-Tawarikh	Sujan Rai Bhandari

6 Important Policy Measures of Akbar (Chronology)

Year (CE)	Important Measure Taken by Akbar
1562	Abolition of slavery
1562	Akbar's complete freedom from the 'harem party' (petticoat government)
1563	Abolition of the pilgrimage tax
1564	Abolition of jizya
1571	Founding of Fatehpur Sikri and transfer of the capital from Agra to Fatehpur
1575	Establishment of the Ibadatkhana (Fatehpur Sikri)
1579	Proclamation of the Mahzar (supremacy of the emperor in religious matters)
1580	' Dahsala system ' (settlement system by Todar Mal) implemented
1582	Proclamation of the Din-i-Ilahi (Tauhid-i-Ilahi)

7 Important Administrative and Economic Terminology of Medieval History

(A question asking the meaning of a single term, or a matching question, comes directly from here in UPPCS Prelims)

> (a) Land Revenue, Tax and Economic Terminology

Term / Terminology	Administrative or Economic Meaning
Iqta	In the Delhi Sultanate, a special right to collect revenue (tax) from land.
Iqtadar	The officer or provincial noble who received an iqta.
Jagir	A tract of land assigned for revenue collection in the Mughal period.
Jagirdar	The Mughal mansabdar/officer who received a jagir.
Khalisa	Land whose revenue went directly to the royal treasury rather than to any intermediary.
Madad-i-Maash	Revenue-free land or grant given to scholars or religious persons.
Kharaj	Land revenue tax levied on the land of non-Muslims.
Ushr	Land tax collected from Muslim cultivators.
Jizya	Protection/religious tax levied on non-Muslim subjects (zimmis).

Term / Terminology	Administrative or Economic Meaning
Zakat	Religious tax compulsorily levied on Muslims.
Khums	Share of the booty obtained in war or of buried treasure (of which 1/5 went to the royal treasury).
Hundi	A written credit instrument used in the medieval period for the transfer of money or payment.
Banjara	A nomadic community that transported and traded grain and other goods over long distances.
Waqf	Immovable property dedicated to religious, educational or charitable purposes.

> (b) Military and Royal Control Terminology

Term / Terminology	Administrative or Military Meaning
Mansab	The rank of an officer in the Mughal administration, denoting his status, pay and military obligation (zat and sawar).
Dagh	Official mark branded on horses in the army to prevent their substitution.
Chehra	System of recording the descriptive roll of soldiers to prevent corruption.
Sulh-i-Kul	Akbar's policy of peace and tolerance towards all religions (universal peace).
Din-i-Ilahi	A moral-religious ideology/code of conduct introduced by Akbar.
Farman	The highest royal order issued by the Mughal emperor or ruler.
Sanad	A royal document (certificate) confirming a right, post or land grant.
Muhar (Seal)	Official mark (seal) used on royal orders or important documents for authentication.
Tughra	A special artistic/calligraphic style of writing the Sultan's name or royal titles.
Fatwa	A religious ruling given by a mufti or qazi on the basis of the Islamic Sharia law.
Ulema	The learned scholars of theology and Islamic law in Muslim society.

> (c) Classes of Peasants and Methods of Land Revenue Assessment

Term / Terminology	Meaning of the Type of Peasant or Land Revenue Method
Raiyat	The ordinary peasant or the subjects who cultivated the land.

Term / Terminology	Meaning of the Type of Peasant or Land Revenue Method
Khudkasht	Peasants who cultivated their own land in their own village.
Pahikasht	Peasants who went temporarily to other villages and cultivated land on a sharecropping basis.
Muzara	A landless peasant or sharecropping labourer who cultivated another's land.
Batai	System of dividing the harvested crop/produce between the cultivator and the landowner in a fixed ratio.
Dahsala	System of fixing land revenue on the basis of the average produce and average prices of the previous ten years.
Zabti	System of collecting tax on the basis of actual measurement of land and a fixed cash rate.
Kankut	Method of fixing rent by merely estimating (notional assessment of) the standing crop in the fields.
Nasaq	Method of fixing land revenue on the basis of a previous assessment or old records, without fresh measurement.

1 Literature of the Sultanate Period, Regional Dynasties and the Vijayanagara Empire

No.	Name of Text / Book	Author / Composer	Special Details Useful for the Exam (where available)
1.	Tabaqat-i-Nasiri	Minhaj-us-Siraj	Period of Iltutmish and Nasiruddin Mahmud
2.	Taj-ul-Maasir	Hasan Nizami	The first official history of the Delhi Sultanate
3.	Khazain-ul-Futuh	Amir Khusrau	Description of Alauddin Khilji's conquest of the South
4.	Tarikh-i-Alai	Amir Khusrau	History of the reign of Alauddin Khilji
5.	Ashiqi	Amir Khusrau	Love story of Khizr Khan and Deval Rani
6.	Nuh Sipihr	Amir Khusrau	Period of Mubarak Khilji and praise of India
7.	Tughlaqnama	Amir Khusrau	Depiction of the victory of Ghiyasuddin Tughlaq
8.	Futuh-us-Salatin	Isami (Abdul Malik Isami)	History of the Bahmani kingdom
9.	Tarikh-i-Firozshahi	Ziauddin Barani	Period of Alauddin Khilji and the Tughlaqs
10.	Fatawa-i-Jahandari	Ziauddin Barani	Political philosophy of the Sultanate period

No.	Name of Text / Book	Author / Composer	Special Details Useful for the Exam (where available)
11.	Futuhāt-i-Fīrozshahī	Fīroz Shah Tughlaq	Autobiography of Sultan Fīroz Shah Tughlaq
12.	Tārīkh-i-Fīrozshahī	Shams-i-Siraj Afīf	Continued Barani's history
13.	Tārīkh-i-Mubarakshahī	Yahya bin Ahmad Sirhindi	The only authentic source on the Sayyid dynasty
14.	Rihla (Kitāb-ul-Rehla)	Ibn Battuta	Travel account of the Moroccan traveller (MBT period)
15.	Chachnama	Ali bin Hamid Kufi	History of the Arab conquest of Sindh
16.	Kitāb-ul-Hind	Al-Biruni	A view of India in the period of Mahmud of Ghazni
17.	Madhuravijayam	Gangadevi	Description of Kumara Kampana's conquest of Madurai
18.	Amuktamalyada	Krishnadevaraya	An administrative text written in the Telugu language
19.	Jambavati Kalyanam	Krishnadevaraya	A drama written in the Sanskrit language
20.	Manucharitam	Allasani Peddana	Principal among the Ashtadiggajas of Krishnadevaraya
21.	Panduranga Mahatmyam	Tenali Ramakrishna	Famous poet included among the Ashtadiggajas
22.	Varadambika Parinayam	Tirumalamba	Woman writer of the Vijayanagara period
23.	Achyutarayabhyudayam	Rajanatha Dindima	Biography of King Achyutadeva Raya
24.	Saluvabhyudayam	Rajanatha Dindima	History of the Saluva dynasty
25.	Krishnaraya Vijayam	Kumara Dhurjati	Description of the victories of Krishnadevaraya
26.	Rayavachakamu	Unknown author	A Telugu prose source on the history of Vijayanagara
27.	Tārīkh-i-Fīrishta	Muhammad Qasim Hindu Shah (Firishta)	Dedicated to Ibrahim Adil Shah of Bijapur
28.	Kitāb-i-Nauras	Ibrahim Adil Shah II	Collection of Hindi songs (ruler of Bijapur)
29.	Burhan-i-Maasir	Sayyid Ali bin Azizullah Tabatabai	History of Ahmednagar (Nizam Shahi)
30.	Adab-ul-Harb	Fakhr-i-Mudabbir	Book on the art of war and military organisation
31.	Dalail-i-Fīrozshahī	Azizuddin Khan	Persian translation of Sanskrit philosophical texts

No.	Name of Text / Book	Author / Composer	Special Details Useful for the Exam (where available)
32.	Mirat-i-Sikandari	Sikandar bin Muhammad Arif	History of the Gujarat Sultanate
33.	Riyaz-us-Salatin	Ghulam Husain Salim	History of Bengal
34.	Riyaz-ul-Insha	Mahmud Gawan	Collection of the letters of the Bahmani wazir Mahmud Gawan

2 History of the Mughal Period and the Later Mughal Period

No.	Name of Text / Book	Author / Composer	Special Details Useful for the Exam (where available)
35.	Baburnama	Babur	Autobiography of Babur (in the Chaghatai Turkish language)
36.	Humayunnama	Gulbadan Begum	Biography written by Humayun's sister
37.	Tarikh-i-Shershahi	Abbas Khan Sarwani	Principal source for the history of Sher Shah Suri
38.	Akbarnama	Abul Fazl	Official history of Akbar
39.	Ain-i-Akbari	Abul Fazl	The third and statistical part of the Akbarnama itself
40.	Muntakhab-ut-Tawarikh	Abdul Qadir Badauni	A critical work on Akbar's religious policies
41.	Tabaqat-i-Akbari	Nizamuddin Ahmad	General history of India in the Mughal period
42.	Tarikh-i-Akbari	Arif Qandhari	Account of Akbar's early years
43.	Tuzuk-i-Jahangiri	Jahangir	Autobiography of Jahangir (Persian language)
44.	Iqbalnama-i-Jahangiri	Mutamid Khan	History of Jahangir (completed the autobiography)
45.	Padshahnama (I)	Abdul Hamid Lahori	Official history of Shah Jahan's reign
46.	Padshahnama (II)	Muhammad Amin Qazvini	Another authentic text of Shah Jahan's period
47.	Amal-i-Salih	Muhammad Salih Kamboh	Account of the whole life of Shah Jahan
48.	Alamgirnama	Mirza Muhammad Kazim	History of the first 10 years of Aurangzeb's reign
49.	Maasir-i-Alamgiri	Saqi Mustaid Khan	Also called the ' Gazetteer of the Mughal State '
50.	Futuh-i-Alamgiri	Ishwar Das Nagar	An important source for the history of Aurangzeb's period
51.	Nuskha-i-Dilkusha	Bhimsen	History of the Deccan (South) in Aurangzeb's time
52.	Muntakhab-ul-Lubab	Khafi Khan	History of Aurangzeb's period (written secretly)

No.	Name of Text / Book	Author / Composer	Special Details Useful for the Exam (where available)
53.	Mirat-i-Alam	Bakhtawar Khan	Account of Aurangzeb's early reign
54.	Ruqaat-i-Alamgiri	Aurangzeb	Collection of Aurangzeb's letters
55.	Qanun-i-Humayuni	Khwandamir	Description of Humayun's administrative rules and inventions
56.	Siyar-ul-Mutakhkherin	Ghulam Husain Tabatabai	History of the later Mughal period and of Bengal
57.	Shahjahannama	Inayat Khan	History of Shah Jahan's reign
58.	Chahar Chaman	Chandrabhan Brahman	Description of Mughal court life and of Shah Jahan's period
59.	Khulasat-ut-Tawarikh	Sujan Rai Bhandari	Geographical and economic facts of India in the Mughal period

3 Maratha History, the Rajput Period and Other Regional Literature

No.	Name of Text / Book	Author / Composer	Special Details Useful for the Exam (where available)
60.	Shivabharata	Paramananda	Sanskrit epic on the life of Shivaji Maharaj
61.	Sabhasad Bakhar	Krishnaji Anant Sabhasad	The oldest and most authentic bakhar of Shivaji's period
62.	Chitnis Bakhar	Malhar Ramrao Chitnis	A source for later Maratha history
63.	Shivadigvijay	Unknown author	Bakhar based on the victories of Shivaji Maharaj
64.	Ajnapatra	Ramchandrapant Amatya	A text on Maratha administrative and political principles
65.	Prithvirajavijaya	Jayanaka	Sanskrit poem on the victories of Prithviraj Chauhan
66.	Prithviraj Raso	Chand Bardai	The first epic of Hindi (of Prithviraj Chauhan's period)
67.	Hammira Mahakavya	Nayachandra Suri	History of Hammir Deva, the Chauhan ruler of Ranthambhore
68.	Hammir Raso	Jodhraj	Description of the heroic battles of Hammir Deva
69.	Rajatarangini	Kalhana	The first systematic historical text of Kashmir
70.	Prabandhachintamani	Merutunga	History of the Solanki/Chalukya kings of Gujarat
71.	Prabandhakosha	Rajashekhara Suri	Biographies of medieval kings and Jain acharyas
72.	Tutinama	Khwaja Ziauddin Nakhshabi	Persian translation of Sanskrit tales (Shukasaptati)

4 Bhakti Movement, Sufi Saints and Cultural Literature

No.	Name of Text / Book	Author / Composer	Nature of the Bhakti/Sufi Tradition
73.	Ramcharitmanas	Tulsidas	Written in the Awadhi language (contemporary of Akbar)
74.	Sursagar	Surdas	The principal text of Krishna bhakti in Braj Bhasha
75.	Bijak	Kabir	Collection of the teachings/sakhis of Kabirdas
76.	Guru Granth Sahib (Adi Granth)	Guru Arjan Dev	The holy book of Sikhism (compiled in 1604 CE)
77.	Chaurasi Vaishnavan ki Varta	Gokulnath	Account of the Vaishnava saints of the Vallabha sect
78.	Bhaktamal	Nabhadas	Biographies of medieval Vaishnava devotees and saints
79.	Premvatika	Raskhan	Poetry based on Krishna bhakti
80.	Fawaid-ul-Fuad	Amir Hasan Sijzi	Collection of the sayings of the Sufi saint Nizamuddin Auliya
81.	Padmavat	Malik Muhammad Jayasi	Written in the Awadhi language (contemporary of Sher Shah Suri)

1 Arrival of the European Companies in India, their Establishment and Trading Centres

European Company	Year of Arrival/Establishment	First Factory in India (Year)	Principal Trading Centres / Headquarters
1. Portuguese	1498 (Vasco da Gama came to Calicut) / company established: 1505	Cochin (1503)	Goa, Diu, Daman
2. Dutch	1596 (arrival) / company established: 1602 (VOC)	Masulipatnam (1605 CE)	Pulicat, Nagapattinam, Chinsura
3. English	1608 (arrival) / company established: 1600	Surat (1613 CE)	Surat, Madras, Bombay, Calcutta
4. Danes (Denmark)	1616	Tranquebar (1620, Tamil Nadu)	Tranquebar (Tamil Nadu), Serampore (Bengal)
5. French	1664	Surat (1668 CE)	Pondicherry, Chandernagore, Mahe, Karaikal, Yanam

- Correct order of arrival of the European powers: Portuguese → Dutch → English → Danes → French.

2 Two Historic Battles that Decided the Destiny of India

Name of Battle	Exact Date	Between Whom	Result and Historical Importance
Battle of Plassey	23 June 1757	The Nawab of Bengal, Siraj-ud-Daulah , vs. the English commander Robert Clive	The English were victorious. This battle laid the foundation of the British Empire in India.
Battle of Buxar	22 October 1764	The combined front of Mir Qasim (Bengal), Shuja-ud-Daulah (Awadh) and Shah Alam II (Mughal emperor) vs. the English commander Hector Munro	The English were victorious. This battle established the real and sovereign power of the English in India.

3 Principal Social Reform Acts of the British Period

Act / Law	Year (CE)	Governor-General Responsible	Main Subject / Legal Provision
Prohibition of infanticide	1795 and 1802	Sir John Shore and Lord Wellesley	Complete ban on the killing of newborn infants (especially girls).
Prohibition of Sati	1829 CE	Lord William Bentinck	Regulation XVII completely banned the practice of Sati and declared it a punishable offence.
Prohibition of slavery	1843 CE	Lord Ellenborough	Under the provisions of the Charter Act of 1833, slavery was declared illegal in India in 1843.
Hindu Widows' Remarriage Act	1856 CE	Lord Canning (<i>draft prepared in Dalhousie's time</i>)	Gave legal recognition to widow remarriage and treated the children born of it as legitimate (effort of Ishwar Chandra Vidyasagar).
Native Marriage Act	1872 CE	Lord Northbrook	Gave legal recognition to inter-caste marriages.
Age of Consent Act	1891 CE	Lord Lansdowne	The minimum age of marriage for girls was fixed at 12 years .
Sarda Act	1929 CE	Lord Irwin	Minimum age of marriage fixed: 14 years for girls and 18 years for boys.

4 Principal Historic Treaties of Modern India (1639 - 1846)

Name of Treaty	Year (CE)	Between Whom (Parties)	Main Terms / Result of the Treaty
Treaty of Asurar Ali	1639 CE	Mughal Empire and Ahom kingdom	The boundary between the Mughal and Ahom kingdoms was fixed; Mughal attempts to conquer the Ahoms ended.

Name of Treaty	Year (CE)	Between Whom (Parties)	Main Terms / Result of the Treaty
Treaty of Purandar (First)	1665 CE	Mughal commander Jai Singh I and Chhatrapati Shivaji Maharaj	After the siege of Purandar fort, Shivaji Maharaj had to hand over 23 forts to the Mughals.
Treaty of Alinagar	1757 (February)	Nawab Siraj-ud-Daulah and Robert Clive	The English were permitted to fortify Calcutta and to trade in Bengal without taxes.
First Treaty of Allahabad	12 August 1765	Robert Clive and the Nawab of Awadh, Shuja-ud-Daulah	* The Nawab gave the districts of Allahabad and Kara to the Mughal emperor Shah Alam II. * Paid the Company an indemnity of ₹50 lakh. * The jagir of Balwant Singh, zamindar of Banaras, was restored.
Second Treaty of Allahabad	16 August 1765	Robert Clive and the Mughal emperor Shah Alam II	* The Company obtained the permanent Diwani of Bengal, Bihar and Orissa. * An annual pension of ₹26 lakh was fixed for the emperor. * The emperor was to live at Allahabad under the Company's protection.
Treaty of Madras	1769 CE	Hyder Ali (Mysore) and the English	The First Anglo-Mysore War ended; conquered territories were returned and a defensive agreement was made.
Treaty of Purandar (Second)	1776 CE	The Marathas (Nana Phadnavis) and the English (Calcutta Council)	Made in the first phase of the First Anglo-Maratha War to annul the Treaty of Surat.
Convention of Wadgaon	1779 CE	The Marathas and the English	A humiliating treaty after the defeat of the English during the First Anglo-Maratha War.
Treaty of Salbai	1782 CE	The Marathas and the English (mediation of Mahadji Scindia)	The First Anglo-Maratha War ended; peace was established for the next 20 years.
Treaty of Seringapatam	1792 CE	Tipu Sultan and the English (Lord Cornwallis + Marathas + Nizam)	The Third Anglo-Mysore War ended; Tipu had to give up half his kingdom and pay an indemnity of ₹3 crore.
Treaty of Lahore	1846 (March)	Governor-General Henry Hardinge and Maharaja Dalip Singh (Lahore Darbar)	The First Anglo-Sikh War ended; a heavy indemnity was imposed on the Sikhs and a British Resident was appointed.
Treaty of Amritsar	1846 (March)	The British East India Company and Maharaja Gulab Singh	The English sold Kashmir to the Dogra Raja Gulab Singh for ₹75 lakh (ruled until 1947).

5 Chronology of the Peshwas (Rise of the Maratha Empire)

No.	Name of Peshwa	Tenure	Special Exam Points
1.	Balaji Vishwanath	1713 – 1720 CE	He is also called 'the second founder of the Maratha Empire'.
2.	Bajirao I	1720 – 1740 CE	'The fighting Peshwa'; gave the principle of Hindu Pad-Padshahi; seized Salsette and Bassein from the Portuguese.
3.	Balaji Bajirao	1740 – 1761 CE	Famous as ' Nana Saheb '; the Third Battle of Panipat took place on 14 January 1761 in his time.
4.	Madhavrao I	1761 – 1772 CE	Re-established Maratha power after the defeat at Panipat.
5.	Narayan Rao	1772 – 1773 CE	His uncle Raghunath Rao (Raghoba) had him murdered.
6.	Madhav Narayan Rao	1774 – 1795 CE	Minor Peshwa; the 'Barbhai Council' (Nana Phadnavis) ran the government.
7.	Bajirao II	1795 – 1818 CE	The last Peshwa; signed the Treaty of Bassein (1802) with the English. The office was abolished after the Third Anglo-Maratha War.

6 Anglo-Mysore Wars (Chronology and Treaties)

War	Duration (Years)	Ruler of Mysore	English Governor / Leadership	Ended by which Treaty / Result
First Anglo-Mysore War	1767 – 1769 CE	Hyder Ali	Lord Verelst	Treaty of Madras (1769) : the English were defeated.
Second Anglo-Mysore War	1780 – 1784 CE	Hyder Ali (<i>after his death in 1782 Tipu took over</i>)	Warren Hastings	Treaty of Mangalore (1784) : both sides returned each other's conquered territories.
Third Anglo-Mysore War	1790 – 1792 CE	Tipu Sultan	Lord Cornwallis	Treaty of Seringapatam (1792) : heavy defeat of Tipu; half his kingdom went to the Company and its allies.
Fourth Anglo-Mysore War	1799 CE	Tipu Sultan	Lord Wellesley	No treaty : Tipu Sultan died fighting. The independent power of Mysore fell and the throne was handed to the Wodeyar dynasty.

7 Principal Civil, Peasant and Tribal Revolts of Modern India (Chronology)

(UPPCS Prelims asks direct matching of years or the correct chronological order of 4 revolts)

No.	Name of Revolt / Movement	Exact Year	Geographical Area / Main Feature
1.	Sanyasi Revolt	1763 CE	Bengal; the plot of Bankim Chandra Chatterjee's novel 'Anandamath' is based on it.
2.	Paika Rebellion	1817 CE	Odisha; a strong revolt against the English under the leadership of Bakshi Jagabandhu.
3.	Bhil Revolt	1818 CE	Western Ghats (Khandesh/Maharashtra); under the leadership of Sewaram, against agrarian hardships.
4.	Ramosi Revolt	1822 CE	Western Ghats (Maharashtra); under the leadership of Chittur Singh.
5.	Ahom Revolt	1828 CE	Assam; under the leadership of Gomdhar Konwar, because the English did not vacate the region.
6.	Wahabi Movement	1831 CE	In India Sayyid Ahmad of Rae Bareilly was its chief propounder (main centre: Patna).
7.	Kol Revolt	1831 CE	Chotanagpur region (Jharkhand); under the leadership of Buddho Bhagat.
8.	Chotanagpur Revolt	1831-32 CE	Collective land discontent of the tribals in the Chotanagpur plateau region.
9.	Kuka Movement	1840 CE	Punjab; a religio-political movement started by Bhagat Jawahar Mal (Sian Sahib) and Baba Ram Singh Kuka.
10.	Santhal Rebellion	1855-56 CE	Daman-i-Koh (Bhagalpur to the Rajmahal hills); a major revolt under the leadership of Sidhu and Kanhu .
11.	Munda Rebellion (<i>Ulgulan</i>)	1899-1900 CE	Ranchi and Khunti (Jharkhand); under the leadership of Birsa Munda ('Dharti Aba') to save collective farming.
12.	Tana Bhagat Movement	1914 CE	Chotanagpur (Jharkhand); a cultural reform movement under the leadership of Jatra Bhagat and Balram Bhagat.
13.	Eka Movement	1921 CE	Awadh (Hardoi, Bahraich, Sitapur); under the leadership of Madari Pasi , against the increase in rent.
14.	Moplah Rebellion	1921 CE	Malabar coast (Kerala); revolt of Muslim tenants against Hindu landlords and the English under the leadership of Ali Musliyar.
15.	Rampa Rebellion	1922 CE	Andhra Pradesh (Godavari region); under the leadership of Alluri Sitarama Raju , against the forest laws.

8 Important Political Organisations Established before the Founding of the Congress (1885)

Name of the Organisation	Year of Establishment	Place of Establishment	Principal Founders
East India Association	1866 CE	London	Dadabhai Naoroji (to acquaint the British public with Indian issues)
Poona Sarvajanik Sabha	1870 CE	Poona	S.H. Chiplunkar, G.V. Joshi and Mahadev Govind (M.G.) Ranade
Indian Society	1872 CE	London	Anand Mohan Bose
Indian Association (<i>Bharat Sangh</i>)	1876 CE	Kolkata	Surendranath (S.N.) Banerjee and Anand Mohan Bose (an important precursor organisation of the Congress)
Madras Mahajan Sabha	1884 CE	Chennai (Madras)	G.S. Iyer, M. Viraraghavachari and P. Ananda Charlu
Bombay Presidency Association	1885 CE	Mumbai (Bombay)	Pherozechah Mehta, K.T. Telang and Badruddin Tyabji

1 The Revolt of 1857: Principal Centres, Leadership and Date of Outbreak

(UPPCS Prelims asks for their chronological order or direct matching)

Centre of the Revolt	Date of Outbreak of the Revolt	Principal Indian Leader
Barrackpore	29 March 1857	Mangal Pandey (the first spark of the revolt)
Meerut	10 May 1857	Kadam Singh (open revolt by the soldiers and march to Delhi)
Lucknow	4 June 1857	Begum Hazrat Mahal
Jhansi	June 1857	Rani Lakshmi Bai
Kanpur	5 June 1857	Nana Saheb

2 Raja Rammohan Roy: Chronology of Life and Reform Work

Year (CE)	Historical Event / Social Reform Work
1809	Wrote the book ' A Gift to Monotheists ' (Tuhfat-ul-Muwahhidin) in Persian.
1815	Founded the Atmiya Sabha for religious and social discussion.

Year (CE)	Historical Event / Social Reform Work
1817	Founded the Hindu College in Calcutta with the help of David Hare.
1820	Wrote the book ' The Precepts of Jesus '.
1825	Founded the Vedanta College to synthesise Western and Indian learning.
1828	Founded the Brahma Sabha , which later became the Brahmo Samaj .
1829	Brought about the complete abolition of Sati (Regulation XVII) with the help of Governor-General Lord William Bentinck.

3 The Kisan Sabha Movement (especially with reference to Uttar Pradesh)

(Extremely important for both the 'UP Special' and 'Modern History' sections of UPPCS)

Kisan Sabha / Movement	Year of Establishment	Place of Establishment	Principal Founders and Associated Leaders
United Provinces Kisan Sabha	1918 CE	Lucknow	Indra Narayan Dwivedi, Madan Mohan Malaviya and Gauri Shankar Mishra.
Awadh Kisan Sabha	1920 CE	Pratapgarh	Baba Ramchandra and Jawaharlal Nehru.
All India Kisan Sabha	1936 CE	Lucknow	* President: Swami Sahajanand Saraswati. * Secretary: N.G. Ranga. * Other leaders: E.M.S. Namboodiripad, Pandit Narendra Dev.

4 Important Sessions of the Indian National Congress (INC) and their Features

(A coded matching question from here is asked in UPPCS Prelims almost every year)

Year (CE)	Venue of the Session	Name of the President	Historical Feature / Event Occurring for the First Time
1887	Madras	Badruddin Tyabji	Became the first Muslim President of the Congress.
1888	Allahabad	George Yule	Became the first English/European President of the Congress.
1896	Kolkata	Rahimtulla M. Sayani	The national song ' Vande Mataram ' was sung for the first time.
1906	Kolkata	Dadabhai Naoroji	The word ' Swaraj ' was used from the platform for the first time.
1911	Kolkata	Pandit Bishan Narayan Dar	The national anthem ' Jana Gana Mana ' was sung for the first time.

Year (CE)	Venue of the Session	Name of the President	Historical Feature / Event Occurring for the First Time
1917	Kolkata	Annie Besant	Became the first woman President of the Congress.
1923	Delhi (<i>special</i>)	Abul Kalam Azad	Became the youngest President in the history of the Congress.
1925	Kanpur	Sarojini Naidu	Became the first Indian woman President of the Congress.
1929	Lahore	Jawaharlal Nehru	The historic resolution of ' Purna Swaraj ' was passed.
1931	Karachi	Vallabhbhai Patel	Resolution demanding Fundamental Rights was passed.
1937	Faizpur	Jawaharlal Nehru	The first session of the Congress held in a village.

5 Principal 'First' Newspapers and Journals of India

Newspaper / Journal	Year of Founding	Place of Publication	Founder / Editor	Language and Special Exam Facts
Bengal Gazette	1780 CE	Kolkata	James Augustus Hicky	India's first newspaper (English language).
Madras Courier	1784 CE	Madras	—	The first newspaper published from Madras (Chennai).
Samachar Darpan	1818 CE	Kolkata	William Carey	The first Bengali weekly newspaper.
Mirat-ul-Akhbar	1822 CE	Kolkata	Raja Rammohan Roy	India's first Persian journal .
Jam-i-Jahan Numa	1822 CE	Kolkata	—	The first newspaper in the Urdu language .
Udant Martand	1826 CE	Kolkata	Jugal Kishore Shukla	The first newspaper in the Hindi language .
Bombay Samachar	1830 CE	Mumbai	—	The first newspaper in the Gujarati language .

6 Principal Social Reformers and their Organisational Work / Posts

Name of Social Reformer	Related Important Work / Administrative Post
D.K. Karve (<i>Dhondo Keshav Karve</i>)	Secretary, Widow Remarriage Association.

Name of Social Reformer	Related Important Work / Administrative Post
J.E.D. Bethune (<i>John Elliot Drinkwater Bethune</i>)	Founded the famous ' Bethune Girls' School' (1849) in Calcutta.
Ishwar Chandra Vidyasagar	Was Principal of the famous Sanskrit College of Calcutta.
B.M. Malabari (<i>Behramji M. Malabari</i>)	Parsi social reformer; began an intensive struggle against child marriage (the Age of Consent Act of 1891 came about through his efforts).

7 Principal Education Commissions and Committees of the British Period (Chronology)

(The favourite section of UPPCS Prelims for 'Match the Following' and chronology)

Education Commission / Policy	Year	Chairman / Governor	Main Recommendations and Exam Facts
Macaulay's Minute	1835	Lord Macaulay (Bentinck's period)	The foundation of English education was laid in India and a fixed sum was allotted for education. (Downward Filtration Theory).
Wood's Despatch	1854	Charles Wood (Dalhousie's period)	* It is called the ' Magna Carta ' of English education in India. * Recommended English as the medium for primary, secondary and graduate-level education, and the opening of education departments in the provinces and of universities on the model of London University.
Hunter Commission	1882	William Hunter (Ripon's period)	To study the progress of education since 1854. It laid special emphasis on primary education .
Universities Commission	1902	Thomas Raleigh (Curzon's period)	To implement improvements in the condition of Indian universities. The Indian Universities Act, 1904 was passed on its recommendation.
Sadler Commission (<i>Calcutta University Commission</i>)	1917	Michael Sadler (Chelmsford's period)	Study of the problems of Calcutta University. It emphasised separating secondary education from university education .
Hartog Committee	1929	Philip Hartog (Irwin's period)	Study of the democratisation of primary education and of the problem of wastage and stagnation in education.
Sargent Plan	1944	John Sargent (Wavell's period)	A long-term (40-year) target was set for implementing universal, free and compulsory primary education in India.

8 Principal Trade Unions (Central Labour Organisations) and their Related Political Parties

(Highly important for current political coded-matching questions)

Name of the Trade Union (Labour Organisation)	Related Parent Political Party
Bharatiya Mazdoor Sangh (BMS)	Bharatiya Janata Party (BJP)
Indian National Trade Union Congress (INTUC)	Indian National Congress (INC)
All India Trade Union Congress (AITUC)	Communist Party of India (CPI)
United Trade Union Congress (UTUC)	Communist Party of India - Marxist (CPI-M)

9 Principal National Newspapers / Books of Modern India and their Authors

Name of Book / Newspaper	Author / Editor / Composer
Commonweal	Annie Besant
New India	Annie Besant
Comrade	Mohammad Ali
Hamdard	Mohammad Ali
Som Prakash	Ishwar Chandra Vidyasagar (<i>the first newspaper against which the Vernacular Press Act was applied</i>)
Independent	Motilal Nehru (newspaper)
Sambad Kaumudi	Raja Rammohan Roy
Free Hindustan	Tarakanath Das
Krantikari	Sachindranath Sanyal
Amrita Bazar Patrika	Sisir Kumar Ghosh (<i>converted into English overnight to escape the Vernacular Press Act</i>)
Satyarth Prakash	Dayanand Saraswati
Gulamgiri	Jyotiba Phule
The Indian Struggle	Subhas Chandra Bose
'India is for the Indians' (slogan/statement)	Chittaranjan Das (C.R. Das)

1 Important Events of Indian History (1911 – 2013)

(UPPCS Prelims asks every year from this section to arrange 4 events in correct chronological order)

> (a) Pre-Independence Period (1911 – 1947)

Year	Important Historical Event / Exam Facts
1911	* Arrival in India of the British Emperor King George V. * Announcement of the transfer of the capital from Calcutta to Delhi. * 'Jana Gana Mana' was sung for the first time at the Calcutta session of the INC. * India's (and the world's) first airmail service began between Bamrauli and Allahabad.
1919	* Dyarchy was introduced in the provinces by the Government of India Act, 1919. * The repressive Rowlatt Act 1919 was enforced. * The Jallianwala Bagh tragedy took place on 13 April.
1920	* Beginning of the Khilafat Movement and of the Non-Cooperation Movement by Mahatma Gandhi.
1922	* The Chauri Chaura incident (outburst) took place at Gorakhpur in Uttar Pradesh. * Gandhiji withdrew (suspended) the Non-Cooperation Movement.
1928	* Visit of the Simon Commission to India (nationwide protest). * Death of Lala Lajpat Rai from a lathi-charge during a protest demonstration.
1929	* Historic resolution of complete independence (Swaraj) at the Lahore session of the INC (President: Nehru).
1930	* Beginning of the Dandi March (Salt Satyagraha) and the Civil Disobedience Movement by Gandhiji.
1931	* The Gandhi-Irwin Pact was concluded. * Bhagat Singh, Sukhdev and Rajguru were hanged on 23 March.
1935	* The Government of India Act, 1935 was passed (dyarchy at the centre, provincial autonomy).
1942	* Visit of the Cripps Mission to India. * The Quit India Movement began in August. * Formation of the Azad Hind Fauj by Captain Mohan Singh and Rash Behari Bose.
1946	* Visit of the three-member Cabinet Mission to India (formation of the Constituent Assembly).
1947	* 15 August: Partition of India and attainment of independence.

> (b) Post-Independence Period (1948 – 2013)

Year	Important National and International Events
1948	* Assassination of Mahatma Gandhi on 30 January. * Pakistan's first attack on India over the Kashmir issue.
1950	* With the Constitution coming into force on 26 January, India became a sovereign republic.

Year	Important National and International Events
1951	* Beginning of the First Five-Year Plan (1951-56). * The first Asian Games were held in New Delhi.
1952	* The first general elections in independent India were held.
1954	* India and China signed the historic Panchsheel Agreement.
1956	* Reorganisation of Indian states on a linguistic basis by the States Reorganisation Act.
1957	* Introduction of the decimal system in the Indian currency system.
1959	* India's first television service began experimentally in New Delhi.
1961	* Liberation of Goa from the Portuguese through Operation Vijay.
1962	* Sino-Indian War (China's sudden attack on India).
1964	* Death of Jawaharlal Nehru, India's first Prime Minister.
1965	* Second Indo-Pak War (over the Kashmir border dispute).
1966	* Death of Lal Bahadur Shastri, the then Prime Minister, during the Tashkent Agreement.
1969	* Split in the Indian National Congress due to the Syndicate dispute. * Nationalisation of 14 major banks by the Indira Gandhi government. * Commercial operation began at India's first nuclear power station at Tarapur (Maharashtra).
1971	* Third Indo-Pak War, which resulted in the birth of Bangladesh.
1972	* Signing of the historic Shimla Agreement between India and Pakistan.
1974	* India's first successful nuclear test at Pokhran (Rajasthan) on 18 May (codename: Smiling Buddha).
1975	* India's very first artificial satellite 'Aryabhata' was launched into space. * Proclamation of Emergency on the ground of internal disturbance in the country.
1977	* After defeat in the general election, the Congress lost power at the Centre for the first time (Morarji Desai became Prime Minister).
1980	* Return of the Congress to power with a huge majority in the general election. * Six more commercial banks were nationalised.
1982	* Beginning of colour television and national broadcasting in India. * The ninth Asian Games were held in New Delhi.
1984	* Assassination of Smt. Indira Gandhi, the then Prime Minister, on 31 October. * The Indian Army established full control over the Siachen Glacier under Operation Meghdoot.
1991	* Assassination of former Prime Minister Rajiv Gandhi at Sriperumbudur in Tamil Nadu. * Beginning of economic liberalisation (LPG reforms) in India under the Rao-Manmohan model.

Year	Important National and International Events
1992	* Demolition of the disputed Babri Masjid structure at Ayodhya. * Constitutional beginning of Panchayati Raj and urban local bodies through the 73rd and 74th Constitutional Amendments.
1995	* Commercial launch of internet service in India by VSNL.
1998	* India's second successful nuclear test at Pokhran (under the codename Operation Shakti).
1999	* Kargil War against Pakistani infiltrators (India's victory through Operation Vijay).
2000	* India's population crossed 1 billion (100 crore) officially.
2001	* January: devastating earthquake at Bhuj in Gujarat. * December: terrorist attack by Lashkar/Jaish on the Indian Parliament in New Delhi.
2002	* The Godhra incident in Gujarat and the riots that broke out afterwards.
2004	* India's coastal areas were affected by the devastating tsunami in the Indian Ocean caused by the Sumatra earthquake.
2008	* India's first spacecraft to the Moon, Chandrayaan-1 , was successfully sent.
2013	* ISRO sent India's first Mars Orbiter Mission (MOM) 'Mangalyaan' towards Mars.

2 States that Accepted Lord Wellesley's Subsidiary Alliance

(UPPCS Prelims repeatedly asks for the correct chronological order in which the states accepted the alliance)

No.	Name of State	Year / Time of Acceptance of the Alliance
1.	Hyderabad	1798 CE <i>(revised treaty: 1800 CE)</i>
2.	Mysore	1799 CE <i>(after the Fourth Anglo-Mysore War)</i>
3.	Tanjore	1799 CE <i>(helpful for chronology in addition to the states mentioned in the source)</i>
4.	Awadh	November 1801 CE
5.	Peshwa (Maratha)	December 1801 CE <i>(Treaty of Bassein)</i>
6.	Bhonsle of Berar	December 1803 CE <i>(Treaty of Deogaon)</i>
7.	Scindia	February 1804 CE <i>(Treaty of Surji-Anjangaon)</i>
8.	Rajput states <i>(Jodhpur, Jaipur, Macheri, Bundi, Bharatpur)</i>	1818 CE

3 Principal Administrative Committees and Commissions of the British Period

(Extremely important for the matching questions of the UPPCS examination)

Name of Committee / Commission	Year of Formation	Governor-General / Viceroy of the Time	Main Subject / Area of Inquiry of the Committee
Scott-Moncrieff Commission	1901 CE	Lord Curzon	For improvement of the irrigation system
Hilton Young Commission	1926 CE	Lord Reading	To advise on the Indian currency and financial system (basis of the establishment of the RBI)
Babington Smith Commission	1919 CE	Lord Chelmsford	Reform of currency and the exchange rate
Hunter Committee Report	1919 CE	Lord Chelmsford	Punjab disturbances (to inquire into the Jallianwala Bagh massacre)
Muddiman Committee	1924 CE	Lord Reading	To examine the working of dyarchy introduced under the Montagu-Chelmsford reforms
Butler Commission	1927 CE	Lord Irwin	To examine the relations between the British Crown and the Indian princely states
Simon Commission	1928 CE	Lord Irwin	To review the Act of 1919 and suggest new steps of administrative reform
Whitley Commission	1929 CE	Lord Irwin	To study labour conditions in industrial establishments
Sapru Commission	1935 CE	Lord Linlithgow	To inquire into the causes of unemployment in the country
Chatfield Commission	1939 CE	Lord Linlithgow	For the modernisation of the Indian Army and assessment of military expenditure
Floud Commission	1940 CE	Lord Linlithgow	Tenancy in Bengal and the land revenue system (basis of the Tebhaga movement)

4 Important Schemes Related to Education in Modern India (Post-Independence)

(UPPCS Prelims asks for the correct matching of the years of these national schemes)

No.	Name of the Education Scheme / Mission	Year / Session of Commencement
1.	Non-formal Education Scheme	1975 CE
2.	Operation Blackboard	1987 CE
3.	Total Literacy Campaign	1991 CE

No.	Name of the Education Scheme / Mission	Year / Session of Commencement
4.	Mid-Day Meal Scheme	1995 CE
5.	Kalp Shiksha Yojana	1998 CE
6.	Shiksha Mitra Yojana	2000 - 01 CE
7.	Sarva Shiksha Abhiyan (SSA)	2001 CE
8.	Vidya Vahini Project	2003 CE
9.	Kasturba Gandhi Balika Vidyalaya Scheme (KGBV)	2004 CE
10.	Saakshar Bharat Mission	2009 CE

5 Principal Peasant Movements and Satyagrahas of Modern India

Name of the Movement (Year)	Principal Leader	Place / Region of the Movement	Main Cause and Exam-Useful Result
Champaran Satyagraha (1917 CE)	Mahatma Gandhi	Champaran, Bihar	* Cause: against the 'Tinkathia' system imposed by the English (compulsory cultivation of indigo on 3/20 of the land). * Result: Gandhiji's first successful satyagraha in India; the British government had to abolish the Tinkathia system.
Kheda Satyagraha (1918 CE)	Mahatma Gandhi and Sardar Vallabhbhai Patel	Kheda district, Gujarat	* Cause: forcible collection of land revenue (rent) by the British government despite a severe famine. * Result: a successful satyagraha; the government had to suspend tax collection and the peasants got great relief.
Moplah Rebellion (1921 CE)	Ali Musliyar and local leaders	Malabar region, Kerala	* Cause: discontent of the Muslim Moplah tenants against Hindu landlords and the exploitative taxes of the British government. * Result: the movement later turned into a communal/religious conflict and was harshly suppressed by the British army.
Bardoli Satyagraha (1928 CE)	Sardar Vallabhbhai Patel	Bardoli, Gujarat	* Cause: a sudden 22% increase in rent (land tax) on the peasants by the provincial government. * Result: the peasants, united, refused to pay the tax; the government had to withdraw the increase. On the success of this movement, Gandhiji, on behalf of the women of Bardoli, gave Vallabhbhai Patel the title of "Sardar".

Name of the Movement (Year)	Principal Leader	Place / Region of the Movement	Main Cause and Exam-Useful Result
Tebhaga Movement (1946-47 CE)	Sharecropping peasants and communist leaders	Bengal province	* Cause: the sharecroppers (tenants) demanded to give the landlords only one-third of the crop instead of half. * Result: largely successful; under its pressure major reforms were made in the landlord and tenancy laws of Bengal.

1 Principal Tribal Revolts: Causes, Leadership and Results

Name of the Revolt (Year)	Geographical Region (Place)	Principal Leader	Main Cause of the Revolt and Exam-Useful Result
Ho Revolt (1820 - 1827 CE)	Chotanagpur (Jharkhand)	Local Ho tribals	* Cause: economic exploitation of the tribals by British colonialism and local landlords. * Result: the British army suppressed it harshly, but it inspired later movements.
Kol Revolt (1831 - 1832 CE)	Chotanagpur, Jharkhand	Budhu Bhagat and associates	* Cause: seizure by the English of the traditional land of the Kol tribals and the oppression of outside landlords (dikus). * Result: the revolt was suppressed, but the English had to think of a policy of reducing outside interference in the tribal areas.
Santhal Rebellion (1855 - 1856 CE)	Santhal Parganas (Bihar and W. Bengal)	Sidho and Kanhu Murmu	* Cause: excessive tax policies, the interest trap of moneylenders and intense economic exploitation by landlords. * Result: suppressed after widespread violent resistance. As a reform, the 'Santhal Pargana' was constituted for administrative convenience.
Bhil Revolt (1818 - 1831 CE)	Rajasthan, Gujarat and Madhya Pradesh	Govind Guru	* Cause: crisis in the traditional forest rights and economic life of the Bhils due to the new forest and tax policies of the British government. * Result: suppressed by the British army after a long struggle, but the movement became a symbol of the defence of tribal rights.

2 Evolution of the Highest Office of British India (Governor-General and Viceroy)

Act / Law	Changed Designation (New Name)	First Holder	Special Exam Details
Regulating Act of 1773	Governor of Bengal → Governor-General of Bengal	Warren Hastings	The Bombay and Madras Presidencies were placed under Bengal (beginning of centralisation).

Act / Law	Changed Designation (New Name)	First Holder	Special Exam Details
Charter Act of 1833	Governor-General of Bengal → Governor-General of India	Lord William Bentinck	A single Governor-General was fixed for the whole of British India. He became the first Governor-General of India .
Government of India Act of 1858	Governor-General of India → Viceroy of India	Lord Canning	Company rule ended and power passed into the hands of the British Crown. Canning became the first Viceroy of India.

- **Note:** In **1948** the office of Governor-General/Viceroy of India was completely abolished. **Lord Mountbatten** was the **last Viceroy** of India.

3 The First Anglo-Maratha War (1775-82) and Maratha Treaties

> Main Details of the War:

- **Main cause:** An internal war of succession for the office of Peshwa. **Raghunath Rao (Raghoba)**, eager to become Peshwa, sought help from the English (Bombay Council), which sparked the war.
- **Principal parties involved:** Raghunath Rao, Nana Phadnavis (Maratha confederacy) and Warren Hastings (British Governor-General).
- **Treaty and conclusion:** The **Treaty of Purandar in 1776** (between Nana Phadnavis and the English) was signed. Finally the war ended with the **Treaty of Salbai in 1782**, under which the English abandoned Raghoba and peace prevailed between the two sides for the next two decades (20 years).

> Principal Treaties of the Anglo-Maratha Conflict (exact chronology - a UPPCS pet question):

(Their year-matching or chronology is certainly asked in the exam)

No.	Name of Treaty	Exact Year (CE)	No.	Name of Treaty	Exact Year (CE)
1.	Treaty of Surat	1775 CE	7.	Treaty of Surji-Anjangaon	1803 CE
2.	Treaty of Purandar	1776 CE	8.	Treaty of Rajpurghat	1804 CE
3.	Convention of Wadgaon	1779 CE	9.	Treaty of Nagpur	1816 CE
4.	Treaty of Salbai	1782 CE	10.	Treaty of Gwalior	1817 CE <i>(November)</i>
5.	Treaty of Bassein	1802 CE	11.	Treaty of Poona	1817 CE <i>(June)</i>
6.	Treaty of Deogaon	1803 CE	12.	Treaty of Mandsaur	1818 CE

4 Principal National Writers, Books and Journals of Modern India

(A complete integrated compilation of both the literary sources given - the biggest scoring section of UPPCS Prelims)

Writer / National Leader	Their Principal Books / Newspapers / Journals
Mahatma Gandhi	The Story of My Experiments with Truth (autobiography), Young India, Harijan, Navajivan, Hind Swaraj
B.R. Ambedkar	Mook Nayak, Bahishkrit Bharat, Annihilation of Caste, Pakistan or the Partition of India, Who were the Shudras, The Untouchables, Buddha or Karl Marx, The Buddha and his Dhamma
Jawaharlal Nehru	The Discovery of India, Glimpses of World History
Lala Lajpat Rai	The Story of My Deportation, Unhappy India
Bal Gangadhar Tilak	Gita Rahasya, Kesari (in Marathi), Mahratta (in English)
Surendranath Banerjee	A Nation in Making, Bengalee (newspaper)
Maulana Abul Kalam Azad	India Wins Freedom, Al-Hilal (journal)
Motilal Nehru	Independent (newspaper)
Dadabhai Naoroji	Poverty and Un-British Rule in India, Rast Goftar (journal)
Aurobindo Ghosh	Karmayogin, Yugantar, Savitri, The Life Divine, Essays on the Gita, Bande Mataram
Bankim Chandra Chattopadhyay	Anandamath (Vande Mataram is in it), Devi Chaudhurani
Madan Mohan Malaviya	Abhyudaya, Hindustan, Leader
Bhagat Singh	Why I am an Atheist, An Introduction to the Dreamland
Rajendra Prasad	India Divided
Rabindranath Tagore	Gitanjali, The Home and the World
Ram Manohar Lohia	Guilty Men of India's Partition
Sachindranath Sanyal	Bandi Jeevan (<i>regarded as a textbook of the revolutionaries</i>)
Bhagwati Charan Vohra	The Philosophy of the Bomb
Valentine Chirol	Indian Unrest (<i>it was he who called Tilak 'the father of Indian unrest'</i>)
Lady Katherine Mayo	Mother India
Larry Collins and Dominique Lapierre	Freedom at Midnight
Gopal Krishna Gokhale	Nation

Writer / National Leader	Their Principal Books / Newspapers / Journals
Keshab Chandra Sen	Indian Mirror
Sir Syed Ahmad Khan	Tahzib-ul-Akhlaq
Dinabandhu Mitra	Nil Darpan (a play based on the Indigo Revolt)
Bharatendu Harishchandra	Bharat Durdasha
R.C. Dutt	The Economic History of India
Bibhutibhushan Banerjee	Pather Panchali
Lord Curzon	Problems of the Far East

➤ Special Books Related to the Revolt of 1857:

Name of Book	Author / Composer
The Indian War of Independence, 1857	V.D. Savarkar
The Sepoy Mutiny and the Revolt of 1857	R.C. Majumdar
Civil Rebellion in the Indian Mutinies (1857-1859)	S.B. Chaudhuri
1857, The Great Rebellion	Asoka Mehta

5 The Nawabs of Bengal and the Establishment of British Rule (Important Facts and Chronology)

(Political changes of the Nawabs, especially useful for UPPCS)

Year / Date	Historical Event and Exam Facts
1717 CE	The Mughal emperor Farrukhsiyar appointed Murshid Quli Khan as Subedar (Nawab) of Bengal .
Change of capital	Murshid Quli Khan shifted the capital from Dhaka to Murshidabad , while later Mir Qasim moved the capital from Murshidabad to Munger .
1740 CE	Alivardi Khan became Nawab of Bengal after killing Sarfaraz Khan in the Battle of Giria (he stopped paying revenue to the Mughals).
1756 CE	After the death of Alivardi, Siraj-ud-Daulah became Nawab of Bengal.
20 June 1756	The famous ' Black Hole Tragedy ' took place in Calcutta, described by J.Z. Holwell (according to the source, the reference is to an account mentioned by George Orwell).

Year / Date	Historical Event and Exam Facts
23 June 1757	The famous Battle of Plassey was fought between Nawab Siraj-ud-Daulah and Robert Clive (the Nawab was defeated through Mir Jafar's treachery).
Puppet rulers	After Plassey the English made first Mir Jafar and then, in 1760, Mir Qasim the puppet Nawab of Bengal.
22 October 1764	The Battle of Buxar was fought between the combined front of Mir Qasim, Shah Alam II (Mughal emperor) and Shuja-ud-Daulah (Nawab of Awadh) and the English commander Hector Munro . Complete victory of the English.
12 August 1765	First Treaty of Allahabad — between Robert Clive and the Mughal emperor Shah Alam II (the English got the Diwani of Bengal, Bihar and Orissa).
16 August 1765	Second Treaty of Allahabad — concluded between Robert Clive and the Nawab of Awadh, Shuja-ud-Daulah.
1765 – 1772 CE	The period of Dual Government in Bengal, which was ended by Warren Hastings.

6 The Three Historic Round Table Conferences of the British Period

Conference	Date of Commencement	Viceroy of the Time	Main Purpose and Reason for the Failure of Talks (Exam Facts)
First Round Table Conference	12 November 1930	Lord Irwin	* Purpose: held in London to consider the recommendations of the Simon Commission. * Outcome: the Congress did not take part in it, as a result of which the talks failed completely.
Second Round Table Conference	7 September 1931	Lord Willingdon	* Purpose: under the Gandhi-Irwin Pact, Mahatma Gandhi took part in it as the sole representative of the Congress. * Outcome: the talks failed because of Gandhiji's disagreement with Dr. B.R. Ambedkar's demand for a separate electorate for the Dalits.
Third Round Table Conference	17 November 1932	Lord Willingdon	* Purpose: discussion of the final draft of the constitutional reforms. * Outcome: the conference was completely boycotted by the Congress. After the White Paper of this conference, the Government of India Bill was introduced in the British Parliament, which later became the Government of India Act, 1935.

1 Principal Women Related to the History of Modern India and their Work

Woman Revolutionary / Leader	Main Exam Facts Related to the National Movement, Organisations and History
Madame Bhikaji Cama	* Parsi revolutionary; founded the 'Paris Indian Society'. * In 1907 at Stuttgart (Germany) she hoisted the first version of the national flag (she is also called 'the mother of the Indian revolution').
Annie Besant	* First president of the 'Women's Indian Association'. * Founder of the 'Indian Home Rule League' (1916). * Made a central contribution to the development of the Theosophical Society.
Captain Lakshmi Sahgal	* Was a leading officer of the 'Rani of Jhansi Regiment' of the Azad Hind Fauj. * Founder member of the 'All India Democratic Women's Association' (AIDWA).
Kamaladevi Chattopadhyay	* Inspired by Gandhian ideas; founder of the 'Indian Cooperative Union'. * Was actively associated with the All India Women's Conference (AIWC).
Kasturba Gandhi	* Wife of Mahatma Gandhi; struggled for the rights of Indian immigrants in South Africa. * Took an active part in the Borsad Civil Disobedience Movement in Gujarat.
Aruna Asaf Ali	* Hoisted the Indian national flag at the Gowalia Tank Maidan in Mumbai during the Quit India Movement (1942). * After independence she became the first Mayor of Delhi.
Sarojini Naidu	* Became the first Indian woman President of the Indian National Congress (Kanpur session, 1925). * Went to London with Mahatma Gandhi to take part in the Second Round Table Conference (1931).
Usha Mehta	* During the Quit India Movement she began operating an underground (secret) radio programme to spread national consciousness.
Shanti Ghosh	* A great young revolutionary heroine of the freedom struggle. * Together with her classmate Suniti Chaudhuri, in December 1931 she shot dead C.G.B. Stevens, the oppressive British Collector of Tripura (Comilla). The youngest woman revolutionary pair in the world.

- *Note: According to the original source, there is a partial mention of the Chittagong Armoury Raid (1930) in the accounts of Aruna Asaf Ali and Usha Mehta, but historically those mainly associated with the Chittagong Armoury Raid, along with Surya Sen, were Pritilata Waddedar and Kalpana Dutt.*

2 Principal Revolutionary and Political Organisations of the Freedom Movement (Chronology)

(UPPCS Prelims asks every year for their year-wise order or direct coded matching)

Name of the Organisation / Samiti	Year of Founding	Place of Founding	Principal Founders / Leaders
Mitra Mela	1899 CE	Poona	The Savarkar brothers (Ganesh Savarkar and V.D. Savarkar)
Anushilan Samiti (Calcutta and Dhaka)	1902 CE	Bengal	Barindra Kumar Ghosh, Jatindranath Banerjee, Pramatha Nath Mitra (P. Mitra) and Pulin Das
Anushilan Samiti (Midnapore)	1902 CE	Midnapore (Bengal)	Jnanendra Nath Bose
Abhinav Bharat Society	1904 CE	Poona	Vinayak Damodar (V.D.) Savarkar (<i>a transformed form of the Mitra Mela</i>)
Bharat Mata Samiti	1904 CE	South India	J.M. Chatterjee, Nilakantha Brahmachari and Vanchi Iyer
Swadesh Bandhab Samiti	1905 CE	Barisal (Bengal)	Ashwini Kumar Dutt (formed during the Swadeshi Movement)
Bharat Mata Society	1907 CE	Punjab	Sufi Amba Prasad and Sardar Ajit Singh (<i>Bhagat Singh's uncle</i>)
Ghadar Dal (Ghadar Party)	1913 CE	San Francisco (USA)	Lala Har Dayal and Sohan Singh Bhakna
Hindi Association of America	1913 CE	America	Sohan Singh Bhakna (<i>the Ghadar movement developed out of it</i>)
Hindustan Republican Association (HRA)	1924 CE	Kanpur	Sachindranath Sanyal, Ramprasad Bismil and Jogesh Chandra Chatterjee (<i>supported by Bhagat Singh and Udham Singh</i>)
Hindustan Socialist Republican Association (HSRA)	1928 CE	Delhi (Feroz Shah Kotla)	Chandrashekhar Azad and Bhagat Singh
Hindu Dharma Sangh	—	Maharashtra	The Chapekar brothers (Damodar and Balkrishna Chapekar)
Arya Bandhav Samaj	—	Maharashtra	Bal Gangadhar Tilak
Atmonnati Samiti	—	Bengal	Bipin Behari Ganguly
Bengal Volunteers	—	Bengal	Hemchandra Ghosh and Leela Nag (Leela Roy)
Indian Republican Army (IRA)	1930 CE	Chittagong (Bengal)	Surya Sen ('Master Da')

3 Principal Ashrams of Mahatma Gandhi and their Years of Founding (Chronology)

(UPPCS Prelims repeatedly asks for their correct chronological order)

No.	Name of the Ashram	Place of Founding	Year of Founding
1.	Phoenix Farm (Phoenix Settlement)	Durban (South Africa)	1904 CE
2.	Tolstoy Farm	Johannesburg (South Africa)	1910 CE
3.	Satyagraha Ashram	Kochrab, Ahmedabad (Gujarat)	1915 CE <i>(May)</i>
4.	Sabarmati Ashram	On the bank of the Sabarmati river, Ahmedabad	1915 CE <i>(later shifted in 1917)</i>
5.	Anasakti Ashram	Kausani (Uttarakhand)	1929 CE
6.	Sevagram Ashram	Wardha (Maharashtra)	1936 CE

4 Principal Historic Titles Conferred on Mahatma Gandhi

Title / Name	Person / Group who Gave the Title	On what Occasion / Reason
Kaiser-i-Hind	British government	For cooperation in the British war effort at the time of the First World War.
Recruiting Sergeant	Indian public/critics	For encouraging Indians to join the British army at the time of the First World War.
Coolie Barrister	English magistrate of South Africa	As a racial and racist remark in a South African court.
Half-naked fakir / seditious fakir	Winston Churchill	Said on seeing him climbing the steps of St James's Palace during the Second Round Table Conference (1931).
Malang Baba	The tribesmen of the North-West Frontier Province	To express their deep reverence for him during the Civil Disobedience Movement.
Bapu	C.F. Andrews and Jawaharlal Nehru	Because of his simple nature and fatherly protection.
King of Beggars	Pandit Madan Mohan Malaviya	For his life of sacrifice and his mass awakening for the nation.
Father of the Nation	Subhas Chandra Bose	First said on 6 July 1944 from Singapore Radio while seeking his blessings for the soldiers of the Azad Hind Fauj.
One Man Boundary Force	Lord Mountbatten	Said on seeing his courage in single-handedly calming the riots at Noakhali (Bengal) at the time of the partition of the country.

Title / Name	Person / Group who Gave the Title	On what Occasion / Reason
Mahatma	Rabindranath Tagore	Gave this title after the historic success of the Champaran Satyagraha (1917) .
Ajatashatru of the Modern Age	Dr. Rajendra Prasad	To show his highest respect for Gandhiji.

5 Principal Educational Institutions Established in the British Period

(Extremely important for both the 'UP Special' and general history sections of UPPCS Prelims)

Name of the Educational Institution	Year of Founding	Place of Founding	Founder / Governor	Main Purpose / Feature of Founding
First Madrasa in Calcutta	1781 CE	Kolkata	Warren Hastings	To promote the study of Muslim laws and the Arabic-Persian languages.
Asiatic Society of Bengal	1784 CE	Kolkata	Sir William Jones	To encourage oriental studies, the understanding of Indian history and culture, and translation work.
Banaras Sanskrit College	1791 CE	Varanasi (Banaras)	Jonathan Duncan	For the systematic study of Hindu religion, philosophy and laws (U.P. special fact).
Fort William College	1801 CE	Kolkata	Lord Wellesley	To train the new civil servants (ICS) of the East India Company in Indian languages, culture and local customs.
Mayo College	1875 CE	Ajmer (Rajasthan)	Lord Mayo	To give higher education on Western lines to the children of Indian princes and nobles.
Muhammadan Anglo-Oriental College	1875 CE	Aligarh	Sir Syed Ahmad Khan	To spread modern Western education among Muslims (it later became Aligarh Muslim University / AMU in 1920).

6 Principal Wars and Historic Treaties of Modern India (1746 – 1849 CE)

(The biggest core area of the history section of UPPCS Prelims - the best reference for chronology and coded matching)

Name of War	Period / Time	Historical Background, Important Facts, Treaty and Result
First Carnatic War	1746 - 1748 CE	* The famous Battle of St. Thome was fought between the French and the English. * The war ended with the Treaty of Aix-la-Chapelle (1748); the English got back Madras.
Second Carnatic War	1749 - 1754 CE	* Internal dispute over succession in the Deccan and the Carnatic. * The war ended with the Treaty of Pondicherry (1754); Dupleix was recalled to France.
Third Carnatic War	1758 - 1763 CE	* An extension of the Seven Years' War of Europe. Under it was fought the famous Battle of Wandiwash (22 January 1760), in which Sir Eyre Coote decisively defeated the French. * The war ended with the Treaty of Paris (1763).
Battle of Plassey	23 June 1757	* Between the Nawab of Bengal, Siraj-ud-Daulah, and Robert Clive. * On the Nawab's side, Mir Jafar and Diwan Rai Durlabh committed treachery. * The Nawab's loyal commanders Mir Madan and Mohanlal died fighting bravely.
Battle of Buxar	22 October 1764	* On one side was the combined front of Nawab Mir Qasim (Bengal), Shuja-ud-Daulah (Awadh) and Shah Alam II (Mughal emperor), and on the other the English commander Hector Munro. * Mir Qasim's main army was led by Gurgin Khan. Complete victory of the English.
First Anglo-Mysore War	1767 - 1769 CE	* The English began the war against Hyder Ali after making a triple alliance with the Marathas and the Nizam of Hyderabad, but the clever Hyder Ali won the Nizam over to his side. * The war ended with the Treaty of Madras (1769) (the English were defeated).
Second Anglo-Mysore War	1780 - 1784 CE	* Battle of Porto Novo (1781): the British commander Sir Eyre Coote defeated Hyder Ali. * In 1782 Hyder Ali died of wounds received during the war. * His son Tipu Sultan continued the war and finally it ended with the Treaty of Mangalore (1784). Warren Hastings was the Governor-General at this time.
Third Anglo-Mysore War	1790 - 1792 CE	* Governor-General Lord Cornwallis, with the cooperation of the Nizam of Hyderabad and the Marathas, laid siege to Tipu's capital Seringapatam. * The war ended with the Treaty of Seringapatam (1792) (Tipu had to lose half his kingdom). * Famous statement of Cornwallis: <i>"We have effectually crippled our enemy without making our friends too formidable."</i>

Name of War	Period / Time	Historical Background, Important Facts, Treaty and Result
Fourth Anglo-Mysore War	1799 CE	* On 4 May 1799 Tipu Sultan died a hero's death at the gate of the fort of Seringapatam, fighting bravely against the combined English army of Lord Wellesley. * The English placed the minor boy Krishnaraja III of the old Wodeyar dynasty on the throne of Mysore and imposed the Subsidiary Alliance.
First Anglo-Maratha War	1775 - 1782 CE	* The Treaty of Purandar in 1776 was made between the English government of Calcutta and the Poona Darbar (Nana Phadnavis), but the Bombay Council opposed it. * The Convention of Wadgaon (1779) was made during the war. * Finally the war ended with the Treaty of Salbai (1782) and there was peace for 20 years between the Marathas and the English.
Second Anglo-Maratha War	1803 - 1805 CE	* Background: Peshwa Bajirao II accepted the subsidiary alliance by signing the Treaty of Bassein (1802) with the English. * Lord Wellesley (Arthur Wellesley in the Deccan and General Lake in North India) conducted successful campaigns against the Maratha chiefs. * Important treaties: the Treaty of Deogaon (1803) with the Bhonsle, the Treaty of Surji-Anjangaon (1803) with Scindia and the Treaty of Rajpurghat (1805) with Holkar.
Third Anglo-Maratha War	1816 - 1819 CE	* The repressive military action against the Pindaris by Governor-General Lord Hastings hurt the self-respect of the Marathas and the war broke out. * Treaties: the Treaty of Nagpur (1816) with the Bhonsle, the Treaty of Gwalior (1817) with Scindia, the Treaty of Poona (1817) with Peshwa Bajirao II and the Treaty of Mandsaur (1818) with Holkar. The office of Peshwa was completely abolished and he was sent to Bithoor.
First Anglo-Sikh War	1845 - 1846 CE	* Governor-General of the time: Lord Hardinge; English army chief: Lord Gough. * Decisive battle: Battle of Sobraon (February 1846). * The war ended with the Treaty of Lahore (March 1846) (the territories south of the Sutlej went to the English). This was followed by the Treaty of Bhyrowal (December 1846), under which a British army was stationed at Lahore until Raja Dalip Singh came of age.
Second Anglo-Sikh War	1848 - 1849 CE	* Principal battles: Battle of Ramnagar (November 1848) → Battle of Chillianwala (January 1849) → Battle of Gujrat (February 1849 - also called 'the battle of the guns'). * The Sikh army was completely defeated under the leadership of General Charles Napier. * In March 1849 Lord Dalhousie completely annexed Punjab to the British Empire. The Kohinoor diamond was sent to the Queen.
Anglo-Nepal War	1814 - 1816 CE	* Fought in the time of Lord Hastings. The Treaty of Sugauli (1816) ended the border dispute with Nepal and the Gurkhas joined the British army.

Name of War	Period / Time	Historical Background, Important Facts, Treaty and Result
Anglo-Burmese War	1824 - 1826 CE	* The First Anglo-Burmese War took place in the time of Lord Amherst and ended with the Treaty of Yandabo (1826) .
Anglo-Afghan Wars	1839 - 1880 CE	* First war (1839-42) : fought in the time of Lord Auckland; it was disastrous for the English. * Second war (1878-80) : fought in the time of Lord Lytton (under the Forward Policy).
Complete annexation of Sindh	1843 CE	* In 1843, during the tenure of Governor-General Lord Ellenborough , Sindh was completely annexed to the British Empire under the military leadership of Charles Napier .

7 Principal Historical Events of the 18th Century (Chronology)

(UPPCS Prelims often asks for the direct chronological order of these historical dates)

1. **Black Hole incident (Black Hole Tragedy)**: 20 June 1756 CE
2. **Treaty of Alinagar**: 9 February 1757 CE
3. **Battle of Plassey**: 23 June 1757 CE
4. **Battle of Buxar**: 22 October 1764 CE
5. **First Treaty of Allahabad**: 12 August 1765 CE
6. **Second Treaty of Allahabad**: 16 August 1765 CE
7. **Treaty of Banaras**: the year 1773 CE

1 Governors-General and Viceroys of British India (Chronology and Important Events)

Governor-General / Viceroy (Tenure)	Status of the Office	Historical Events and Important Exam Facts
Warren Hastings (1773-1785 CE)	First Governor-General of Bengal	* The Regulating Act of 1773 was passed. * Associated with the 'Ring Fence Policy'. * Establishment of the Supreme Court at Calcutta. * The First Anglo-Maratha War (1775-82), the Second Anglo-Mysore War (1780-84) and the Treaty of Salbai (1782) took place. * Regarded as the father of the judicial service in India.

Governor-General / Viceroy (Tenure)	Status of the Office	Historical Events and Important Exam Facts
Lord Cornwallis (1786–1793 CE)	Governor-General of Bengal	* The Third Anglo-Mysore War (1790-92) and the Treaty of Seringapatam (1792) took place. * Introduced the Permanent Settlement (1793 CE). * Father of the civil service and the police service in India. * Special fact: he died on 5 October 1805 at Ghazipur (Uttar Pradesh), where his tomb is located.
Lord Wellesley (1798–1805 CE)	Governor-General of Bengal	* Introduced the Subsidiary Alliance (1798). * The Fourth Anglo-Mysore War (1799) and the Treaty of Bassein (1802) took place. * The Second Anglo-Maratha War (1803-05) was fought. * Founded Fort William College in Calcutta (1801).
Lord Hastings (1813–1823 CE)	Governor-General of Bengal	* After the Third Anglo-Maratha War (1817-19) he adopted the policy of non-intervention and abolished the office of Peshwa in 1818. * Anglo-Nepal War (1814-16) and the Treaty of Sugauli. * The Ryotwari system was introduced in Bombay and Madras by Thomas Munro (1820). * Approval of the Mahalwari system in Awadh and the North-Western Provinces.
Lord William Bentinck (1828–1835 CE)	Last Governor-General of Bengal and first Governor-General of India	* By the Charter Act of 1833 the Governor-General of Bengal was made the Governor-General of India (thus he became the first Governor-General of India). * Abolished Sati in 1829 by Regulation XVII. * Suppressed thuggee in 1830 with the help of Colonel Sleeman.
Lord Ellenborough (1842–1844 CE)	Governor-General of India	* The First Anglo-Afghan War ended. * In 1843 Sindh was completely annexed to the British Empire. * Complete abolition of slavery by Act V of 1843.

Governor-General / Viceroy (Tenure)	Status of the Office	Historical Events and Important Exam Facts
Lord Dalhousie (1848–1856 CE)	Governor-General of India	* The Second Anglo-Sikh War (1848-49) and the annexation of Punjab on 29 March 1849 (the Kohinoor diamond was sent to the Queen). * Annexation of states under the Doctrine of Lapse (see the order below). * Father of the railways in India; the first train ran in 1853 (Bombay to Thane, 34 km) and the second train in 1854 (Calcutta to Raniganj). * Established the P.W.D. (Public Works Department) for the first time in India.
Lord Canning (1856–1862 CE)	Last Governor-General of India and first Viceroy	* The First War of Independence (Revolt) of 1857 took place. * By the Act of 1858, Company rule came to an end and the office of the Mughal emperor was abolished. * In 1857 three new universities were founded at Calcutta, Bombay and Madras.
Lord Mayo (1869–1872 CE)	Viceroy of India	* The Archaeological Survey of India (ASI) was set up in 1871 CE. * The first (non-synchronous) census of India in 1872 CE was held in his tenure.
Lord Lytton (1876–1880 CE)	Viceroy of India	* Appointment of the first Famine Commission in 1878 under the chairmanship of Richard Strachey. * Vernacular Press Act (1878) passed; Surendranath Banerjee called it '<i>a thunderbolt from the blue</i>'. * The Second Anglo-Afghan War (1878-80) took place and the Statutory Civil Service was constituted in 1878. * Reduced the maximum age for entry into the civil service examination from 21 to 19 years.
Lord Ripon (1880–1884 CE)	Viceroy of India	* Introduced the regular (decennial) census in India in 1881 CE. * The resolution on local self-government (1882) was passed; hence he is called 'the father of local self-government'. * Constitution of the William Hunter Education Commission (1882) related to primary education. * Repealed the Vernacular Press Act (1882). * The Ilbert Bill controversy (1883) took place, also known as the 'White Mutiny'.

Governor-General / Viceroy (Tenure)	Status of the Office	Historical Events and Important Exam Facts
Lord Dufferin (1884–1888 CE)	Viceroy of India	* Founding of the Indian National Congress (INC) in 1885 CE by A.O. Hume. * Allahabad University was founded in 1887 CE (U.P. special fact).
Lord Minto II (1905–1910 CE)	Viceroy of India	* Split in the Congress at the Surat session of 1907 (Moderates and Extremists). * Under the Morley-Minto Reforms (1909), separate electorates were established for Muslims for the first time.
Lord Hardinge II (1910–1916 CE)	Viceroy of India	* Delhi Durbar (1911): announcement of the annulment of the Partition of Bengal and of the transfer of the capital from Calcutta to Delhi. * In 1912 Delhi became the official capital of India. * A bomb was thrown at him in Delhi on 23 December 1912 (Delhi Conspiracy Case), in which Bhai Bal Mukund was hanged.
Lord Chelmsford (1916–1921 CE)	Viceroy of India	* The August Declaration of 20 August 1917, famous as the 'Montagu-Chelmsford Declaration'.
Lord Reading (1921–1926 CE)	Viceroy of India	* The only Jewish Viceroy of India. * In 1922 CE the I.C.S. examination was held for the first time simultaneously in London and Allahabad. * The 'Public Service Commission (PSC)' was established on 1 October 1926.
Lord Irwin (1926–1931 CE)	Viceroy of India	* Appointment of the Harcourt Butler Committee (1927) regarding the princely states. * Arrival of the Simon Commission in India on 3 February 1928. * The Gandhi-Irwin Pact was concluded on 5 March 1931.
Lord Linlithgow (1936–1944 CE)	Viceroy of India	* The first provincial general elections (1936–37) were held; the Congress formed governments in 8 of the 11 provinces (full majority in 6, coalition in 2). * The Cripps Mission came to India in 1942; Gandhiji called it "<i>a post-dated cheque</i>". * The Quit India Movement (August Revolution) began on 8 August 1942 (slogan 'Do or Die').

Governor-General / Viceroy (Tenure)	Status of the Office	Historical Events and Important Exam Facts
Lord Wavell (1944–1947 CE)	Viceroy of India	* The Sargent Plan (1944 CE) for educational reform was announced. * The Wavell Plan / Simla Conference (1945 CE) was presented. * On 20 February 1947 the British PM Attlee announced the transfer of power to India by June 1948.
Lord Mountbatten (1947–1948 CE)	Last Viceroy of British India and first Governor-General of independent India	* The Mountbatten Plan (Dickie Bird Plan), containing the outline of the partition of India, was announced on 3 June 1947. * After independence on 15 August 1947 he became the first Governor-General of independent India.
C. Rajagopalachari (1948–1950 CE)	Governor-General of independent India	* He was the last Governor-General of India and the first and only Indian Governor-General of independent India.

2 States Annexed under Dalhousie's 'Doctrine of Lapse' – Chronology

(UPPCS Prelims asks for its correct chronological order every second year)

1. **Satara** — 1848 CE
2. **Jaitpur** — 1849 CE
3. **Sambalpur** — 1849 CE
4. **Baghat** — 1850 CE
5. **Udaipur** — 1852 CE
6. **Jhansi** — 1853 CE
7. **Nagpur** — 1854 CE
8. **Awadh** — 1856 CE (*Special: it was annexed not under the Doctrine of Lapse but on the charge of misgovernment (Outram Report)*)

3 Principal Commissions Constituted during Lord Curzon's Tenure (1899 - 1905 CE)

(The favourite coded matching question of UPPCS)

Year (CE)	Name of the Commission	Name of the Chairman	Main Purpose of Formation / Area of Inquiry
1900 CE	Famine Commission	Sir Antony MacDonnell	To review famine relief works and frame policy.
1901 CE	Irrigation Commission	Sir Colin Scott-Moncrieff	To increase agricultural productivity and for irrigation reforms.

Year (CE)	Name of the Commission	Name of the Chairman	Main Purpose of Formation / Area of Inquiry
1901 CE	Railway Commission	Thomas Robertson	To reform the administrative and financial structure of the railways.
1902 CE	Police Commission	Sir Andrew Fraser	To recommend efficiency and reforms in police administration.
1902 CE	Universities Commission	Sir Thomas Raleigh	To improve the condition of higher education/universities (on its basis came the Universities Act of 1904).

- **Special note:** On **19 July 1905 CE** Curzon made the '**announcement of the Partition of Bengal**' at Shimla; it was approved by the British sovereign in September 1905 and came into effect on 16 October 1905.

4 Principal Land Revenue Systems of the British Period

(The most important comparative table for the UPPCS Prelims examination)

Feature / Point	1. Permanent Settlement (Zamindari)	2. Ryotwari System	3. Mahalwari System
Year of introduction	1793 CE	1820 CE	1822 CE
Principal leader / father	Lord Cornwallis <i>(and John Shore)</i>	Thomas Munro <i>(and Captain Read)</i>	Holt Mackenzie <i>(and Martin Bird)</i>
Geographical area	Bengal, Bihar, Orissa, Uttar Pradesh (Banaras division), northern Karnataka.	Bombay, Madras, Assam.	Uttar Pradesh (United Provinces), Central Provinces, Punjab.
Total area (percentage)	19% of the area of British India	51% of the area of British India <i>(the largest)</i>	30% of the area of British India
Real owner of the land	The zamindar was regarded as owner.	The peasant (raiyyat) was regarded as owner.	The village (mahal / collective community) was regarded as owner.
Responsibility for revenue	The responsibility for paying tax to the government lay with the zamindar .	The responsibility for paying tax to the government lay directly with the peasant .	The responsibility for paying tax to the government lay with the village headman (lambardar) .

1 Principal Commissions and Committees of the British Period

Name of Commission / Committee	Year of Establishment	Viceroy of the Time	Main Subject / Purpose of Inquiry of the Committee
Inam Commission	1852 CE	Lord Dalhousie	To examine the jagirs and titles of landholders.

Name of Commission / Committee	Year of Establishment	Viceroy of the Time	Main Subject / Purpose of Inquiry of the Committee
Strachey Commission	1880 CE	Lord Lytton	The first Famine Commission; for a policy of providing relief to famine victims.
Hunter Commission	1882 CE	Lord Ripon	For a review of the progress of education after Wood's Despatch of 1854.
Herschell Committee	1893 CE	Lord Lansdowne	To give suggestions on the mint/currency and finance.
Lyll Commission	1897 CE	Lord Elgin II	To study the report of the Famine Commission of 1880.
MacDonnell Commission	1900 CE	Lord Curzon	The second Famine Commission; to give its suggestions on the report of the Strachey Commission.
Moncrieff Commission	1901 CE	Lord Curzon	To plan irrigation expenditure and outline the development of canals.
Raleigh Commission	1902 CE	Lord Curzon	For reforms and work relating to university education .
Fraser Commission	1902 CE	Lord Curzon	To examine the working and efficiency of police administration .
Sadler Commission	1917 CE	Lord Chelmsford	To examine the working of Calcutta University .
Butler Committee	1927 CE	Lord Irwin	Suggestions for improving the relations between the British Crown and the princely states .
Hartog Committee	1929 CE	Lord Irwin	For the development and improvement of primary and missionary education.
Whitley Commission	1929 CE	Lord Irwin	For labour reforms in industrial workshops, mills and plantations.
Sapru Committee	1934 CE	Lord Willingdon	Study of the causes of unemployment in the United Provinces (Uttar Pradesh).
Woodhead Commission	1943-44 CE	Lord Wavell	To inquire into the causes of the severe Bengal Famine of 1943 .

2 The Great Revolt of 1857: Principal Exam Facts and the Cycle of Suppression

> Important Historical Points:

- The first spark of the revolt:** On **29 March 1857 Mangal Pandey** of the 34th Native Infantry at Barrackpore (West Bengal, Hooghly/Parganas district) rebelled. He fired at his Sergeant-Major Hewson and the Adjutant. Mangal Pandey was hanged on **8 April 1857**.

- **Collective beginning of the revolt:** Began on **10 May 1857 at Meerut** with the soldiers' open revolt and march to Delhi.
- **Top office-holders of the time:** At the time of the revolt the Governor-General of India was **Lord Canning**, and the symbolic Mughal ruler of Delhi was **Bahadur Shah II 'Zafar'**.

➤ **Principal Centres of the Revolt of 1857 and the British Suppressors (Match the Following):**

Centre of the Revolt	Date of Outbreak	Principal Indian Leader (Hero)	Principal British Suppressor	Date of Suppression / Control
Delhi	11 May 1857	Bahadur Shah II (real command: Bakht Khan)	John Nicholson, Hodson	20 September 1857
Lucknow	30 May 1857	Begum Hazrat Mahal , Birjis Qadr	Colin Campbell	March 1858
Jhansi	4 June 1857	Rani Lakshmibai	General Hugh Rose	17 June 1858
Kanpur	5 June 1857	Nana Saheb (Tatya Tope)	Colin Campbell	6 December 1857
Jagdishpur (Bihar)	12 June 1857	Kunwar Singh (and Amar Singh)	Major William Taylor and Vincent Eyre	December 1858
Faizabad	June 1857	Maulvi Ahmadullah (<i>gave the call of jihad</i>)	Colin Campbell	5 June 1858
Bareilly	June 1857	Khan Bahadur Khan	Colin Campbell	1858
Allahabad (Prayagraj)	June 1857	Liaquat Ali	Colonel Neill	1858
Fatehpur	1857	Azimullah	General Renaud	1858

3 Pioneers of the Socio-Religious Reform Movements

➤ **(a) Raja Rammohan Roy and the Brahmo Samaj**

Subject Section	Historical Details and Exam Facts
Life	Born on 22 May 1772 at Radhanagar, Hooghly (Bengal). Died on 27 September 1833 at Bristol (England).
Titles and honours	* Raja: given by the Mughal ruler Akbar II. * Yugadoot (Messenger of the Age): given by Subhas Chandra Bose. * Father of the constitutional agitation: so called by Surendranath Banerjee. * Other names: herald/father of the Indian Renaissance, father of modern India, father of Indian nationalism, morning star of the new dawn, bridge between the past and the future, father of the Indian awakening, messiah of new India, the first modern man and man of the age.

Subject Section	Historical Details and Exam Facts
Newspapers published	* Sambad Kaumudi (Bengali language) — 1821 CE * Mirat-ul-Akhbar (Persian language) — 1822 CE
Principal books / works	* Tuhfat-ul-Muwahhidin (A Gift to Monotheists) — 1809 CE * The Precepts of Jesus — 1820 CE
Principal institutions founded	* Atmiya Sabha (1814-15 CE) — Calcutta * Brahmo Samaj (1828 CE) — Calcutta * Hindu College and David Hare College (1817 CE) — Calcutta * Vedanta College (1825 CE) — Calcutta

> (b) Swami Dayanand Saraswati and the Arya Samaj

Subject Section	Historical Details and Exam Facts
Life	Born on 12 February 1824 at Tankara (Gujarat). His childhood name was Mool Shankar . His guru Purnananda named him 'Swami Dayanand Saraswati'.
Title	He is called ' the Martin Luther of India '.
Institution founded and slogans	* Founded the Arya Samaj in 1875 CE at Bombay (headquarters later shifted to Lahore). * Slogans: "Go back to the Vedas", "India is for the Indians".
Principal movements	Started the Shuddhi movement (return of those who had become non-Hindus) and the cow-protection movement .
Principal books / works	* Satyarth Prakash (Hindi/Sanskrit medium) — 1874 CE (<i>it is called the Bible of the Arya Samaj</i>). * Pakhand Khandan (1866 CE) and Advaitamat ka Khandan (1873 CE).
Special exam facts	* He was the first to use the word 'Swaraj' and the first person to accept Hindi as the national language. * His famous statement is: <i>"The worst indigenous rule is better than the best foreign rule."</i>

4 Hindu Socio-Religious Movements and Organisations (Chronology)

Name of the Organisation / Movement	Year of Founding	Principal Founder / Propounder	Place of Founding
Asiatic Society of Bengal	1784 CE	Sir William Jones	Calcutta (Bengal)
Young Bengal Movement	1826 CE	Henry Louis Vivian Derozio	Calcutta (Bengal)
Brahmo Samaj	1828 CE	Raja Rammohan Roy	Calcutta (Bengal)
Tattvabodhini Sabha	1839 CE	Debendranath Tagore	Calcutta (Bengal)

Name of the Organisation / Movement	Year of Founding	Principal Founder / Propounder	Place of Founding
Paramahansa Mandali	1840 CE	Gopal Hari Deshmukh (<i>Lokahitavadi</i>)	Bombay
Rahnumai Mazdayasnan Sabha	1851 CE	Dadabhai Naoroji, S.S. Bengalee	Bombay (Parsi reform)
Prarthana Samaj	1867 CE	Dr. Atmaram Pandurang, M.G. Ranade	Bombay
Arya Samaj	1875 CE	Swami Dayanand Saraswati	Bombay
Theosophical Society	1875 CE	Madame H.P. Blavatsky, Colonel Olcott	New York (centre in India: Adyar)
Deccan Education Society	1884 CE	G.G. Agarkar, Bal Gangadhar Tilak	Poona
Indian National Social Conference	1887 CE	Mahadev Govind (M.G.) Ranade	Bombay
Dev Samaj	1887 CE	Shivnarayan Agnihotri	Lahore
Ramakrishna Mission	1897 CE	Swami Vivekananda	Belur (Bengal)
Servants of India Society	1905 CE	Gopal Krishna Gokhale	Bombay
Poona Seva Sadan	1909 CE	Smt. Ramabai Ranade	Poona
Social Service League	1911 CE	N.M. Joshi	Bombay
Seva Samiti	1914 CE	Hriday Nath (H.N.) Kunzru	Allahabad (Prayagraj)
Hindu Mahasabha	1915 CE	Pandit Madan Mohan Malaviya	Haridwar
Women's Indian Association	1917 CE	Lady Sadasiva Iyer, Annie Besant	Madras
Vishva Bharati	1921 CE	Rabindranath Tagore	Calcutta

5 Dalit, Backward and Lower-Caste Movements / Organisations

Name of the Movement / Organisation	Year of Founding	Principal Founder / Leader	Place of Founding
Satyashodhak Samaj	1873 CE	Jyotiba Phule (<i>book: Gulamgiri</i>)	Maharashtra
Aruvippuram Movement	1888 CE	Sree Narayana Guru (<i>slogan: one caste, one religion, one God</i>)	Aruvippuram (Kerala)
Sree Narayana Dharma Paripalana Yogam	1903 CE	Sree Narayana Guru, Dr. Palpu and Kumaran Asan	Kerala

Name of the Movement / Organisation	Year of Founding	Principal Founder / Leader	Place of Founding
Depressed Classes Mission Society	1906 CE	V.R. Shinde	Bombay
Bahujan Samaj	1910 CE	Mukundrao Patil	Satara (Maharashtra)
Justice Party Movement	1915-16 CE	C.N. Mudaliar, T.M. Nair and P. Theagaraya	Madras
Bahishkrit Hitakarini Sabha	1924 CE	Dr. Bhimrao Ambedkar	Bombay

6 Muslim Socio-Religious Movements and Organisations

Name of the Movement / Organisation	Year of Founding	Principal Founder / Leader	Main Centre / Place
Darul-Uloom (Deoband Movement)	1866 CE	Muhammad Qasim Nanautavi and Rashid Ahmad Gangohi	Deoband (Saharanpur, U.P.)
Aligarh Movement	1875 CE	Sir Syed Ahmad Khan	Aligarh (U.P.)
Ahmadiya Movement	1889-90 CE	Mirza Ghulam Ahmad (<i>called himself Masih-ul-Mauood, the promised Messiah</i>)	Qadian / Faridkot (Punjab)
Nadwatul Ulama Movement	1894-95 CE	Maulana Shibli Numani	Lucknow (U.P.)

7 Political Organisations before the Founding of the Indian National Congress (1885)

(The favourite section of UPPCS Prelims for coded matching and chronology)

Name of the Organisation	Principal Founder	Year of Founding	Headquarters / Place
Landholders' Society (<i>Zamindari Association</i>)	Dwarkanath Tagore	1838 CE	Calcutta
British India Society	William Adams	1839 CE	London / Calcutta
Bengal British India Society	George Thompson	1843 CE	Calcutta
British Indian Association (<i>merger of the Landholders' + Bengal British India Society</i>)	Debendranath Tagore	1851 CE	Calcutta
Bombay Association	Dadabhai Naoroji	1852 CE	Bombay
Madras Native Association	Gazulu Lakshminarasu Chetty	1852 CE	Madras

Name of the Organisation	Principal Founder	Year of Founding	Headquarters / Place
East India Association (merger of the London India Committee + Society)	Dadabhai Naoroji	1866 CE	London
Poona Sarvajanik Sabha	Ganesh Vasudev Joshi, M.G. Ranade	1870 CE	Poona
India Society	Anand Mohan Bose	1872 CE	London
Indian Association (Bharat Sangh)	Anand Mohan Bose, Surendranath Banerjee	1876 CE	Calcutta
Native Press Association	Surendranath Banerjee	1877 CE	Delhi
Madras Mahajan Sabha	P. Ananda Charlu, M. Viraraghavachari	1884 CE	Madras
Bombay Presidency Association	Badruddin Tyabji, Pherozeshah Mehta, K.T. Telang	1885 CE	Bombay

1 Principal Sessions of the Indian National Congress (INC) and Historical Facts

(UPPCS Prelims often asks for matching the 'President' and 'special fact' from this table)

Venue	Year (CE)	President of the Session	Historical Feature / Exam Fact
Bombay	1885 CE	Womesh Chandra Bonnerjee	The first session of the Congress; 72 delegates from all over the country took part.
Calcutta	1886 CE	Dadabhai Naoroji	The Indian National Association (Conference) merged with the Congress.
Madras	1887 CE	Badruddin Tyabji	Became the first Muslim President of the Congress.
Allahabad	1888 CE	George Yule	Became the first European (English) elected President of the Congress.
Calcutta	1896 CE	Rahimtulla M. Sayani	The national song ' Vande Mataram ' was sung for the first time.
Banaras	1905 CE	Gopal Krishna Gokhale	Support for the Swadeshi and Boycott Movement against the Partition of Bengal.
Surat	1907 CE	Rash Behari Ghosh	The historic split of the Congress into Moderates and Extremists .
Calcutta	1911 CE	Bishan Narayan Dar	The national anthem ' Jana Gana Mana ' was sung for the first time.

Venue	Year (CE)	President of the Session	Historical Feature / Exam Fact
Lucknow	1916 CE	Ambika Charan Majumdar	' Lucknow Pact ' — the Moderates and Extremists reunited and an agreement was made between the Congress and the Muslim League.
Calcutta	1917 CE	Annie Besant	Became the first woman President in the history of the Congress.
Delhi	1923 CE	Maulana Abul Kalam Azad	Special session; became the youngest President at the age of 35.
Belgaum	1924 CE	Mahatma Gandhi	The only session held under the presidency of Mahatma Gandhi.
Kanpur	1925 CE	Sarojini Naidu	Became the first Indian woman President of the Congress (U.P. special fact).
Lahore	1929 CE	Pandit Jawaharlal Nehru	On the bank of the Ravi the resolution demanding 'Purna Swaraj' was passed for the first time.
Karachi	1931 CE	Sardar Vallabhbhai Patel	Fundamental Rights were demanded from the Congress platform for the first time.
Faizpur	1937 CE	Jawaharlal Nehru	The first session held in a village (rural area).

2 Exact Chronology of the Principal Events of the Freedom Movement

(The most core area of the UPPCS Prelims examination; a question arranging events by date certainly comes from here)

> (a) First Phase: From the Partition of Bengal to the Non-Cooperation Movement (1905 – 1922 CE)

No.	Name of the Historical Event	Exact Date / Period
1.	Formal proclamation of the Swadeshi Movement	7 August 1905 CE
2.	Partition of Bengal comes into effect (day of mourning throughout Bengal)	16 October 1905 CE
3.	Founding of the Muslim League at Dhaka	30 December 1906 CE
4.	Madame Bhikaji Cama hoists the first Indian flag at Stuttgart (Germany)	1907 CE
5.	Split of the Congress (Surat session)	1907 CE
6.	Founding of a branch of the Muslim League in London (by Syed Ameer Ali)	1908 CE
7.	Holding of the Delhi Durbar (announcement of the annulment of the Partition of Bengal and change of capital)	12 December 1911 CE

No.	Name of the Historical Event	Exact Date / Period
8.	Founding of the Ghadar Party at San Francisco	1 November 1913 CE
9.	Komagata Maru ship incident	1914 CE
10.	Mahatma Gandhi's permanent return to India from South Africa (Pravasi Bharatiya Divas)	9 January 1915 CE
11.	British government confers the title 'Kaiser-i-Hind' on Gandhiji	1915 CE
12.	Bal Gangadhar Tilak founds the Home Rule League at Poona	April 1916 CE
13.	Annie Besant starts the Home Rule League at Madras	September 1916 CE
14.	Lucknow Pact (Congress-League Pact)	December 1916 CE
15.	Champaran Satyagraha (Gandhiji's first satyagraha in India)	10 April 1917 CE
16.	'Montagu Declaration' by the Secretary of State Edwin Montagu	20 August 1917 CE
17.	Ahmedabad mill workers' movement (Gandhiji's first hunger strike)	February - March 1918 CE
18.	Kheda Satyagraha (movement in favour of the peasants of Gujarat)	March 1918 CE
19.	Enforcement of the repressive Rowlatt Act (black law)	March 1919 CE
20.	Jallianwala Bagh massacre (Amritsar tragedy)	13 April 1919 CE
21.	Formation of the Khilafat Committee at Bombay	September 1919 CE
22.	All India Khilafat Conference at Delhi (Gandhiji became its president)	November 1919 CE
23.	Beginning of the Non-Cooperation Movement and death of Bal Gangadhar Tilak	1 August 1920 CE
24.	Special Congress session at Calcutta (resolution on non-cooperation moved for the first time)	September 1920 CE
25.	Resolution on the Non-Cooperation Movement finally passed at the Nagpur annual session	December 1920 CE
26.	Chauri Chaura incident (Gorakhpur, Uttar Pradesh)	4 - 5 February 1922 CE
27.	Suspension (withdrawal) of the Non-Cooperation Movement by the Bardoli resolution	12 February 1922 CE
28.	Gandhiji's arrest by the English after the withdrawal of the movement	10 March 1922 CE

> (b) Second Phase: From the Swaraj Party to the Attainment of Independence (1922 – 1946 CE)

No.	Name of the Historical Event	Exact Date / Period
29.	Peshawar Communist (leftist) Conspiracy Case	1922 - 1923 CE
30.	Founding of the Swaraj Party at Allahabad by C.R. Das and Motilal Nehru	January 1923 CE
31.	Kanpur Bolshevik (Communist) Conspiracy Case	1924 CE
32.	Formation of the Hindustan Republican Association (HRA) at Kanpur	1924 CE
33.	Agreement between Gandhi and Chittaranjan Das (Gandhi-Das Pact)	November 1924 CE
34.	Kakori Train Action (the changed name of the Kakori train incident, Lucknow)	9 August 1925 CE
35.	Vithalbhai Patel becomes President of the Central Legislative Assembly	1925 CE
36.	Arrival of the Simon Commission in India (Bombay)	3 February 1928 CE
37.	Reorganisation of the Hindustan Socialist Republican Association (HSRA) at Delhi	1928 CE
38.	Presentation of Motilal Nehru's 'Nehru Report' by the All-Parties Committee	August 1928 CE
39.	Bardoli Satyagraha in the Surat district of Gujarat	October 1928 CE
40.	Killing of the British officer Saunders by revolutionaries at Lahore	17 December 1928 CE
41.	Central Assembly bomb case by Bhagat Singh and Batukeshwar Dutt	8 April 1929 CE
42.	Lahore Conspiracy Case conducted against the revolutionaries	1929 CE
43.	Gandhiji begins the historic Dandi March from Sabarmati	12 March 1930 CE
44.	Beginning of the Civil Disobedience Movement by breaking the salt law at Dandi	6 April 1930 CE
45.	First Round Table Conference in London (without the participation of the Congress)	1930 - 1931 CE
46.	The historic Gandhi-Irwin Pact at Delhi	5 March 1931 CE
47.	Bhagat Singh, Rajguru and Sukhdev hanged in Lahore jail	23 March 1931 CE
48.	Second Round Table Conference (in which Gandhiji represented the Congress)	September - December 1931 CE
49.	Announcement of separate electorates for the Dalits by the British PM (Communal Award)	16 August 1932 CE

No.	Name of the Historical Event	Exact Date / Period
50.	The historic Poona Pact between Gandhiji and Dr. Ambedkar	24 September 1932 CE
51.	Third Round Table Conference (complete boycott by the Congress)	November - December 1932 CE
52.	Formation of the Congress Socialist Party by Jayaprakash Narayan and Acharya Narendra Dev	1934 CE
53.	Formation of the Forward Bloc by Subhas Chandra Bose	May 1939 CE
54.	The Muslim League celebrates ' Deliverance Day ' after the resignation of the Congress ministries	22 December 1939 CE
55.	Official 'demand for Pakistan' at the Lahore session of the Muslim League	March 1940 CE
56.	'August Offer' by Viceroy Linlithgow for cooperation in the war	8 August 1940 CE
57.	Arrival in India of the Cripps Mission for wartime constitutional reforms	March 1942 CE
58.	Formal beginning of the Quit India Movement from the Gowalia Tank Maidan, Bombay	9 August 1942 CE
59.	Holding of the Simla Conference to remove the deadlock over the Wavell Plan	1945 CE
60.	The historic Royal Indian Navy Mutiny (RIN Mutiny) by the sailors of the warship Talwar at Bombay	1946 CE

3 Important Social Reform Acts of British India

Year	Social Reform Act / Law	Governor-General Responsible	Legal Subject / Main Social Reform
1795 and 1804	Prohibition of Infanticide Act	Lord Wellesley <i>(and Sir John Shore)</i>	Complete legal ban on killing/murdering newborn infants (especially girls).
1829	Prohibition of Sati Act	Lord William Bentinck	Regulation XVII declared Sati illegal and punishable (through the intensive efforts of Raja Rammohan Roy).
1829	Prohibition of Child Marriage Act	Lord William Bentinck	An initial attempt to prohibit the marriage of boys below 18 years of age.
1837	Hindu Women's / Property Act	Lord Auckland	Hindu women were legally given a limited right to property.
1843	Prohibition of Slavery Act	Lord Ellenborough	In keeping with the spirit of the Charter Act of 1833, slavery was completely prohibited in India.

Year	Social Reform Act / Law	Governor-General Responsible	Legal Subject / Main Social Reform
1856	Hindu Widows' Remarriage Act	Lord Canning	Widow remarriage was declared legally valid and the children received inheritance (through the efforts of Ishwar Chandra Vidyasagar).
1872	Native Marriage Act (<i>Brahmo Marriage Act</i>)	Lord Northbrook	Legality of inter-caste marriages and an attempt to stop polygamy (with the cooperation of Keshab Chandra Sen).
1891	Age of Consent Act	Lord Lansdowne	The minimum age of consent/marriage for girls was raised from 10 to 12 years (through the efforts of the Parsi reformer Behramji Malabari).
1930	Sarda Act	Lord Irwin	To stop child marriage the minimum age of marriage was fixed at 18 years for boys and 14 years for girls.
1931	Child Marriage Restraint Act	Lord Irwin	Child marriage was legally prohibited in order to enforce the provisions of the Sarda Act strictly.

4 Principal Peasant and Civil Movements of Modern India

(The most important and highest-scoring table for the 'Match the Following' section of UPPCS Prelims)

Name of the Revolt / Movement	Affected Geographical Area	Principal Leaders	Main Cause and Result of the Revolt
Moplah Rebellion (1836-1854) and (1921-22 CE)	Malabar coast (Kerala)	K.N. Das, Sayyid Qazi, Ali Musliyar	* Cause: oppression by local Hindu landlords (jenmis), the repressive rent policies of the English government and the influence of the Khilafat movement. * Result: turned into violent communal riots; brutally suppressed by the British army.
Pagalpanthi Revolt	North Bengal	Karam Shah and Tipu Shah	* Cause: revolt of the peasants of the semi-religious Pagalpanthi sect against the excessive rent and oppression of the local landlords. * Result: Tipu Shah tried to set up an independent administrative system; later brought under control by the army.
Faraizi Movement (1838 CE)	Faridpur (Bengal)	Haji Shariatullah and Dudu Miyan	* Cause: protection of the religio-social order of the Faraizi sect and strong opposition to the illegal taxes imposed by the landlords.

Name of the Revolt / Movement	Affected Geographical Area	Principal Leaders	Main Cause and Result of the Revolt
Indigo Revolt (1859 - 1860 CE)	Bengal (Nadia district)	Bishnu Biswas and Digambar Biswas	* Cause: forcible, loss-making cultivation of indigo imposed on the peasants by European planters (dadni system). * Result: a highly organised and successful movement; the government had to set up the Indigo Commission. <i>(It is accurately depicted in Dinabandhu Mitra's play 'Nil Darpan').</i>
Pabna Revolt (1873 - 1876 CE)	Bengal	Ishan Chandra Roy, Keshab Chandra Roy, Shambhunath Pal	* Cause: attempts by landlords to illegally seize the land of tenants/artisans, eviction and excessive increases in rent. * Result: a peaceful and legal struggle; it was under its pressure that the Bengal Tenancy Act of 1885 later came.
Deccan Riots (1874 - 1875 CE)	Poona, Ahmednagar, Sholapur and Satara (Maharashtra)	Baba Saheb Deshmukh and local leaders	* Cause: heavy increase in rent even after the slump in cotton, and the eviction of peasants from their land by outside Marwari/Gujarati moneylenders after trapping them in debt. * Result: the moneylenders' account books were burnt; the government passed the Deccan Agriculturists' Relief Act (1879) to protect the peasants.
Bijolia Movement (1905, 1913, 1916, 1927)	Mewar / Rajasthan	Sadhu Sitaram Das, Vijay Singh Pathik, Manikya Lal Verma	* Cause: 84 different kinds of illegal cesses (lag-bag) imposed on the peasants of the Bijolia jagir by the feudal administration. * Result: the longest-running non-violent peasant movement of India; finally there was a large reduction in taxes.
Champaran Satyagraha (1917 CE)	Champaran, Motihari (Bihar)	Mahatma Gandhi , Rajendra Prasad, Rajkumar Shukla	* Cause: against the exploitative 'Tinkathia system' imposed by European indigo planters (compulsion to sow indigo on 3/20 of the land). * Result: Gandhiji's first successful movement in India; the Tinkathia system was abolished and 25% of the illegal collection was refunded.
Kheda Satyagraha (1918 CE)	Kheda (Gujarat)	Mahatma Gandhi and Vallabhbhai Patel	* Cause: forcible collection of tax, contrary to the revenue rules, despite the whole crop being destroyed by severe famine. * Result: the government bowed down and secretly ordered that tax be taken only from those peasants who were able to pay.

Name of the Revolt / Movement	Affected Geographical Area	Principal Leaders	Main Cause and Result of the Revolt
Awadh Kisan Movement (1920 CE)	Pratapgarh, Rae Bareli, Sultanpur (Uttar Pradesh)	Baba Ramchandra	* Cause: illegal rent collection (nazrana) by taluqdars and landlords and enforcement of eviction rules (U.P. special fact). * Result: unprecedented unity was established among the peasants; the Awadh Kisan Sabha was formed.
Eka Movement (1921-22 CE)	Barabanki, Sitapur, Hardoi, Bahraich (Uttar Pradesh)	Madari Pasi and Sahdev	* Cause: against a rent increase of 50% above the recorded rate and forcible tax collection by thekedars (contractors) (U.P. special fact). * Result: a militant movement of small peasants and sharecroppers; later quelled by repression.
Bardoli Satyagraha (1928 CE)	Bardoli taluka of Surat (Gujarat)	Sardar Vallabhbhai Patel	* Cause: a sudden 22% increase in rent imposed on the peasants by the Bombay government. * Result: the movement was completely successful; the rent increase was reduced to 6.03%. Vallabhbhai received the title "Sardar" on behalf of the women of Bardoli.
Tebhaga Movement (1946 CE)	Bengal province	Kamparam, Bhawan Singh	* Cause: on the basis of the recommendation of the provincial Fلود Commission, the sharecroppers decided that instead of giving 50% of the crop to the landlords they would give only one-third (1/3 or 33%). * Result: major social and legal improvements in the condition of the sharecropping peasants.

1 First and Second Phases of the Revolutionary Movement (Important Events and Coded Facts)

Revolutionary Event / Organisation	Exact Year / Date	Principal Mastermind / Leader	Special Details Useful for the Exam (UPPCS Coded Facts)
Mitra Mela and Abhinav Bharat	1899 CE / 1904 CE	Vinayak Damodar (V.D.) Savarkar	The 'Mitra Mela' founded in 1899 itself became the secret society ' Abhinav Bharat ' in 1904.
Anushilan Samiti (Bengal)	1902 CE	Pramatha Nath (P.) Mitra	Formed to spread revolutionary nationalism in Bengal.

Revolutionary Event / Organisation	Exact Year / Date	Principal Mastermind / Leader	Special Details Useful for the Exam (UPPCS Coded Facts)
Muzaffarpur bomb case	30 April 1908	Khudiram Bose and Prafulla Chaki	* Threw a bomb at the carriage of Judge Kingsford (he escaped). * Prafulla Chaki committed suicide to avoid arrest. * Khudiram Bose was hanged in 1908 at the age of only 15.
Delhi Hardinge bomb case	23 December 1912	Rash Behari Bose and Sachin Sanyal	A bomb was thrown at the procession of Viceroy Lord Hardinge II at Chandni Chowk.
Hindustan Republican Association (HRA)	October 1924	Sachindra Sanyal, Ramprasad Bismil, Jogesh Chatterjee	Place of founding: Kanpur (Uttar Pradesh).
Kakori Train Action (<i>Kakori incident</i>)	9 August 1925	Revolutionaries of the HRA	* The 8 Down train was looted near Lucknow for the government treasury. * 29 revolutionaries were tried; Chandrashekhar Azad remained absconding.
Hindustan Socialist Republican Association (HSRA)	1928 CE	Chandrashekhar Azad and Bhagat Singh	Place of founding: Feroz Shah Kotla ground, Delhi (new form of the HRA).
Saunders murder (Lahore)	17 December 1928	Bhagat Singh, Chandrashekhar Azad, Rajguru	To avenge the death of Lala Lajpat Rai caused by a lathi-charge.
Central Assembly bomb case	8 April 1929	Bhagat Singh and Batukeshwar Dutt	Threw bombs on the empty benches of the Central Legislature in Delhi "to make the deaf hear".
Alfred Park encounter	27 February 1931	Chandrashekhar Azad	While fighting the police in Alfred Park, Allahabad (Prayagraj), he shot himself with the last bullet.
Lahore jail hanging	23 March 1931	Bhagat Singh, Rajguru and Sukhdev	Hanged after being found guilty in the Saunders murder (Lahore Conspiracy Case).

➤ Places of Hanging of the Martyrs of the Kakori Train Action (1925) (Match the Following):

- Ram Prasad Bismil → Gorakhpur Jail
- Rajendra Lahiri → Gonda Jail
- Roshan Lal → Naini Jail (Allahabad)
- Ashfaqulla Khan → Faizabad Jail (Ayodhya)

> The Jallianwala Bagh Massacre (13 April 1919) and the Cycle of Repression

Stage of Events	Date / Time	Principal Character / Officer	Historical Details and Exam Facts
Passing of the Rowlatt Act	18 March 1919	—	A repressive 'black law' of no vakil (lawyer), no appeal, no daleel (argument).
Arrest of popular leaders	9 April 1919	Dr. Satyapal and Dr. Saifuddin Kitchlew	Arrested on the orders of the Governor of Punjab, Michael O'Dwyer ; great public anger.
Handing over of military command	10 April 1919	Brigadier-General R. Dyer	The administration and security of the city of Amritsar were handed over to the military officer R. Dyer.
Jallianwala Bagh massacre	13 April 1919	General R. Dyer	On the day of Baisakhi the British army fired indiscriminately on a huge peaceful gathering.
Renunciation of title / post	After the massacre	Rabindranath Tagore and Sankaran Nair	* Rabindranath Tagore renounced the title of ' Knighthood ' (Sir). * Sankaran Nair resigned from the Viceroy's Executive Council .
Formation of an inquiry committee	After the massacre	Pandit Madan Mohan Malaviya	The Congress formed a ' Tahqiqat Committee ' (inquiry committee) to investigate this heinous massacre.

3 The Non-Cooperation Movement (1920-22) and the Chauri Chaura Incident

Stage of the Event	Date / Period	Principal Mastermind / Character	Important Exam Facts Related to the Movement
Special session at Calcutta	September 1920	Lala Lajpat Rai (president)	Initial approval was given for launching the Non-Cooperation Movement for the attainment of Swaraj.
Beginning of the movement	1 August 1920	Mahatma Gandhi	The movement began on this day; unfortunately, Bal Gangadhar Tilak died on the same day.
Nagpur annual session	December 1920	C.R. Das (mover)	The resolution on the Non-Cooperation Movement was finally passed; C.R. Das was its mover.
Formation of a financial fund	1921 CE	—	The ' Tilak Swaraj Fund ' was created for the financial support of the movement.
Renunciation of titles	During the movement	Mahatma Gandhi and Jamnalal Bajaj	* Gandhiji renounced the title ' Kaiser-i-Hind '. * Jamnalal Bajaj renounced the title ' Rai Bahadur '. * Note: Rabindranath Tagore called the burning of foreign cloth 'heartless waste'.

Stage of the Event	Date / Period	Principal Mastermind / Character	Important Exam Facts Related to the Movement
Chauri Chaura incident (Gorakhpur)	4 - 5 February 1922	Militant peasant crowd and police	Because the police fired on the demonstrators, the crowd set fire to the police station; 22 policemen were killed .
Suspension of the movement	12 February 1922	Mahatma Gandhi	Saddened by the violent incident of Chauri Chaura, Gandhiji withdrew the Non-Cooperation Movement.

4 Formation of the Swaraj Party (March 1923 CE)

Main Point	Details and Exam Facts
Founders and posts	C.R. Das (President) and Motilal Nehru (General Secretary).
Place of founding	Allahabad (Prayagraj), Uttar Pradesh.
Main objective	To oppose the policies of the British government from within by contesting the elections to the councils / legislatures.
Historic success	Through the party's efforts, Vithalbhai Patel was elected in 1925 CE as the first Indian President (Speaker) of the Central Legislative Assembly.

5 Mahatma Gandhi: The Historic Dandi March (Civil Disobedience Movement)

Main Point / Feature	Statistical and Historical Details (UPPCS Facts)
Date of commencement	12 March 1930
Place of commencement	Sabarmati Ashram , Ahmedabad (Gujarat)
Number of followers	A total of 78 chosen followers accompanied Gandhiji.
Date and place of arrival	Reached Dandi (Navsari district, Gujarat) on 5 April 1930 .
Date of breaking the salt law	On 6 April 1930 he began the Civil Disobedience Movement by making salt on the seashore.
Total distance and time	Covered a total distance of 390 kilometres (about 241 miles) in a march of 24 days .
Accompanying foreign journalist	The American journalist Webb Miller was with Gandhiji on this historic march.

6 The Three Historic Round Table Conferences

> Exact Period of the Conferences:

- **First Round Table Conference:** from November 1930 to January 1931. *(Congress absent)*
- **Second Round Table Conference:** from 7 September 1931 to 1 December 1931. *(Congress present)*
- **Third Round Table Conference:** from 17 November 1932 to 24 December 1932. *(boycotted by the Congress)*

> Special Coded Facts Related to the Second Round Table Conference:

- **British PM and Viceroy:** At this time the Prime Minister of Britain was **Ramsay MacDonald** and the Viceroy of India was **Lord Willingdon**.
- **Gandhiji's journey:** As the sole representative of the Congress, Gandhiji went to London on 29 August 1931 by the ship '**S.S. Rajputana**'.
- **Other prominent Indian personalities who attended:** Mahadev Desai, Madan Mohan Malaviya, Devdas Gandhi, Ghanshyam Das Birla, Mirabehn, Annie Besant and Sarojini Naidu.
- **Participant in all three conferences:** **Dr. Bhimrao Ambedkar** was the only prominent leader who took part in all three Round Table Conferences and raised the demand for a separate electorate for the Dalits.

7 The Communal Award and the Historic Poona Pact (1932 CE)

Event / Stage	Exact Date	Principal Mastermind / Signatory	Historical Details and Exam Facts
Communal Award	16 August 1932	Ramsay MacDonald (British PM)	Treating the depressed classes as separate from the Hindus, a separate electorate was announced for them.
Gandhiji's fast	20 September 1932	Mahatma Gandhi	In protest against this divisive decision Gandhiji began a fast unto death in Yerwada Jail (Poona) .
Mediation	September 1932	Madan Mohan Malaviya, Dr. Rajendra Prasad, C. Rajagopalachari	These leaders played the main role in bringing about an agreement between Gandhiji and Dr. Ambedkar.
Poona Pact <i>(Poona Agreement)</i>	24 September 1932	Dr. B.R. Ambedkar and Madan Mohan Malaviya	* Signatories: Dr. Ambedkar signed on behalf of the depressed classes and Madan Mohan Malaviya on behalf of the Hindus. * Special: the final draft of the agreement was prepared by C. Rajagopalachari. The separate electorate for the Dalits was abolished and the seats in the provincial assemblies were increased.

8 Formation and Development of the Azad Hind Fauj (Indian National Army)

Stage of Development	Exact Date / Year	Principal Mastermind and Commander	Organisational Facts and Features
Original idea	—	Giani Pritam Singh and Fujiwara	The original idea of forming the Azad Hind Fauj first came to their minds.
Initial formation	March 1942	Captain Mohan Singh	Formation of the ' Indian National Army ' began in Malaya (Singapore island) under Mohan Singh's military leadership.
Formation of the first division	1 September 1942	Captain Mohan Singh	The first division of the army was formally formed; Mohan Singh became its first commander .
Transfer of command	4 July 1943	Rash Behari Bose and Subhas Chandra Bose	Rash Behari Bose officially handed over the supreme command of the Azad Hind Fauj in Tokyo/Singapore to Netaji Subhas Chandra Bose .

- *Note: Among post-independence developments, the non-violent "**Bhoodan Movement**" for land reform in the country was successfully led by **Acharya Vinoba Bhave** in 1951 (Pochampalli).*

1 Important Events and Coded Facts of the Freedom Movement

Historical Event / Mission	Exact Date / Period	Principal Leader / President	Important UPPCS Exam Facts (Statement Facts)
Chittagong Armoury Raid	18 April 1930	Surya Sen ('Master Da')	The Chittagong branch of the Indian Republican Army stormed and captured the armoury. Pritilata Waddedar and Kalpana Dutt were also involved. Master Da was hanged on 12 January 1934.
Arrival of the Simon Commission	3 February 1928	Sir John Simon	Reached Bombay; there was intense protest throughout the country with slogans of "Simon Go Back".
Pakistan Resolution	22-24 March 1940	Mohammad Ali Jinnah	The 'Pakistan Resolution' was officially passed at the Lahore annual session of the Muslim League.
August Offer	8 August 1940	Lord Linlithgow	A declaration made by the Viceroy to obtain the cooperation of Indians in the Second World War.
Individual Satyagraha (<i>Delhi Chalo Satyagraha</i>)	Began on 17 October 1940	Mahatma Gandhi (initiator)	* Started by the Congress after the complete rejection of the August Offer. * First satyagrahi: Acharya Vinoba Bhave * Second satyagrahi: Pandit Jawaharlal Nehru * Third satyagrahi: Sardar Vallabhbhai Patel

Historical Event / Mission	Exact Date / Period	Principal Leader / President	Important UPPCS Exam Facts (Statement Facts)
Cripps Mission	March 1942	Sir Stafford Cripps	A one-member mission led by a Labour leader/Member of Parliament, sent by the British PM Winston Churchill to break the political deadlock.

2 The Quit India Movement (1942) and Parallel Governments

> Sequence of Events of the Movement (Chronology):

- **7 August 1942:** The annual meeting of the Congress was held at the historic **Gowalia Tank Maidan** in Bombay under the presidency of **Abul Kalam Azad**.
- **Moving of the resolution:** At this meeting **Jawaharlal Nehru** moved the historic resolution of the "Quit India Movement".
- **Drafters:** The draft of the movement was prepared jointly by **Mahatma Gandhi, Jawaharlal Nehru and Abul Kalam Azad**.
- **8 August 1942:** The Quit India resolution was passed by a full majority; at this time Gandhiji gave the historic slogan "**Do or Die**".
- **9 August 1942 (morning):** Under "**Operation Zero Hour**" the British government arrested Gandhiji and all the other top leaders.
- **Accompanying journalist:** During the movement the American journalist **Louis Fischer** (biographer of Mahatma Gandhi) was with him.

> Principal Parallel Governments Established during the Movement (Match the Following):

(UPPCS Prelims asks a direct matching question from here)

Place / Region	Leader	Duration and Special Facts
Ballia (Uttar Pradesh)	Chittu Pande	The very first parallel government (U.P. special fact).
Tamluk (Midnapore, Bengal)	Local leader Sati Samanta and party	It was known as the " Jatiya Sarkar ". It remained active from December 1942 to September 1944.
Satara (Maharashtra)	Y.V. Chavan and Nana Patil	It was the longest-running government , functioning until 1945.

3 First Phase of the Revolutionary Movement in India (1897 – 1915 CE)

(An infallible table for UPPCS Prelims 'Match the Following' and for fixing the correct year of events)

Year (CE)	Revolutionary Event / Organisation	Principal Revolutionaries	Special Details Useful for the Exam (Coded Facts)
1897	Murder of Plague Commissioner Rand	Chapekar brothers (<i>Damodar and Balkrishna</i>)	Plague Commissioner W.C. Rand was shot dead at Poona (Maharashtra). Beginning of the first phase in India.
1899	Mitra Mela	Vinayak Damodar (V.D.) Savarkar	Founded in Maharashtra as a secret revolutionary organisation.
1902	Anushilan Samiti	Pramathanath (P.) Mitra	The first secret organisation for conducting revolutionary activities in Bengal.
1904	Abhinav Bharat	V.D. Savarkar	The reorganised name of the 'Mitra Mela' of 1899 (Poona).
1906	Yugantar organisation and paper	Barindra Kumar Ghosh and associates	Developed to spread revolutionary ideas after the Partition of Bengal (1905).
1908	Muzaffarpur bomb case	Khudiram Bose and Prafulla Chaki	* Attempt to kill Judge Kingsford. * Prafulla Chaki committed suicide to avoid arrest. * Khudiram Bose was hanged at the youngest age on 11 August 1908 .
1908	Alipore bomb case	Barindra Ghosh and Aurobindo Ghosh	Many revolutionaries were tried after a bomb factory was caught at Maniktola in Kolkata.
1909	Murder of Curzon Wylie	Madan Lal Dhingra	Sir Curzon Wylie, political adviser to the Secretary of State for India, was shot dead in London.
1909	Jackson murder case (Nasik)	Anant Laxman Kanhere	A.M.T. Jackson, Collector of Nasik, was shot dead.
1912	Delhi Hardinge bomb case	Rash Behari Bose and Sachin Sanyal	On 23 December 1912 a bomb was thrown at Chandni Chowk on the arrival of Viceroy Lord Hardinge in Delhi.
1913	Ghadar Movement (Ghadar Party)	Lala Har Dayal and Sohan Singh Bhakna	Organisation of the Ghadar movement in America (San Francisco) and Canada.
1914	Komagata Maru incident	Baba Gurdit Singh	Revolutionary nationalism flared up abroad because the Canadian government stopped the ship of Indian emigrants.
1915	Failure of the armed revolt	Rash Behari Bose and the Ghadar revolutionaries	The first phase ended with the failure of the Ghadar armed revolt planned during the First World War.

4 Second Phase of the Revolutionary Movement (1920 – 1934 CE)

(Complete chronological summary of the Lahore Conspiracy, the Kakori Action and the Assembly bomb case)

Revolutionary Organisation / Event	Exact Date / Year	Principal Masterminds / Martyred Revolutionaries	Socialist / Revolutionary Philosophy and UPPCS Exam Facts
Hindustan Republican Association (HRA)	October 1924	Sachindranath Sanyal, Ramprasad Bismil, Jogesh Chandra Chatterjee	* Place: Kanpur (U.P.). * Objective: to end British rule through armed revolution and establish a federal republic in India.
Kakori Train Action	9 August 1925	Revolutionaries of the HRA (Bismil, Ashfaq, Roshan, Lahiri, Azad)	* Place: Kakori near Lucknow. The 8 Down train was stopped and the government treasury was looted. * Dates of hanging (1927): 1. 17 December: Rajendranath Lahiri (Gonda Jail) 2. 19 December: Ramprasad Bismil (Gorakhpur), Ashfaqulla Khan (Faizabad), Roshan Singh (Naini/Prayagraj). * Note: Chandrashekhar Azad remained absconding. Heavy loss to the HRA.
Hindustan Socialist Republican Association (HSRA)	1928 CE	Chandrashekhar Azad, Bhagat Singh, Sukhdev, Rajguru, Bhagwati Charan Vohra	* Place: Feroz Shah Kotla ground, Delhi (new form of the HRA). * Philosophy: included socialist ideas and the establishment of an exploitation-free society among its goals.
Lahore Saunders murder	17 December 1928	Bhagat Singh, Rajguru and Chandrashekhar Azad	Killed Assistant Superintendent of Police J.P. Saunders at Lahore to avenge the death of Lala Lajpat Rai (17 November 1928) , who had been injured in a lathi-charge during the anti-Simon protest on 30 October 1928.
Central Legislative Assembly bomb case	8 April 1929	Bhagat Singh and Batukeshwar Dutt	* Place: Central Assembly, Delhi. * Objective: not to kill anyone, but <i>'to make the deaf government hear'</i> against the repressive Trade Disputes Bill and Public Safety Bill. Both deliberately courted arrest.
Ultimate martyrdom (final phase)	1931 CE	Chandrashekhar Azad, Bhagat Singh, Sukhdev, Rajguru	* 27 February 1931: during a police encounter in Alfred Park, Allahabad, Chandrashekhar Azad shot himself. * 23 March 1931: Bhagat Singh, Sukhdev and Rajguru were hanged under the Lahore Conspiracy Case (murder of Saunders).

1 Sequence of Events of the Transfer of Power and Independence (1945 – 1947 CE)

(UPPCS Prelims asks every year from this historical chronology for arranging events directly by date)

Historical Event / Mission	Exact Date / Period	Principal Mastermind / Viceroy	Important Exam Facts (Code-based Facts)
Wavell Plan	14 June 1945	Lord Wavell	A plan presented by the Conservative government of British PM Winston Churchill to resolve the constitutional deadlock.
Arrival of the Cabinet Mission	24 March 1946	* Chairman: Pethick-Lawrence * Members: A.V. Alexander, Stafford Cripps	A three-member mission constituted by British PM Clement Attlee. Under it the elections to the Constituent Assembly were held in July 1946 . (<i>Note: the idea of a Constituent Assembly first came up at the Lucknow session of the Congress in 1936</i>).
Invitation to form the Interim Government	12 August 1946	Viceroy and Jawaharlal Nehru	The Viceroy sent an invitation to Nehru to form the government, which he accepted. The Muslim League strongly opposed it.
Direct Action Day	16 August 1946	Muslim League	At the League's call terrible communal bloodshed took place in the country (especially in Bengal). Mahatma Gandhi undertook a padayatra at Noakhali to restore peace.
Formation of the Interim Government	2 September 1946	Pt. Jawaharlal Nehru and other members	Under Nehru's leadership the cabinet ministers took the oath of office and secrecy (the list of portfolios is given below).
Attlee's historic declaration	20 February 1947	Clement Attlee (British PM)	Declared in the House of Commons that the English would transfer power to Indians by June 1948 . Mountbatten was appointed the new Viceroy in place of Wavell.
Lord Mountbatten Plan	3 June 1947	Lord Mountbatten (<i>34th Governor-General</i>)	After mutual agreement among Indian leaders, the British government presented the final plan for the partition of India .
Introduction of the Independence Bill	4 July 1947	British Parliament	The Indian Independence Bill was introduced in the British Parliament.
Royal assent to the Independence Bill	18 July 1947	British sovereign	The Bill received royal assent. According to it, two dominions named India and Pakistan were to be established on 15 August.
Attainment of independence	14 - 15 August 1947	—	Pakistan on 14 August and India on 15 August became independent as sovereign nations.

2 The Interim Government (1946 CE): At a Glance

(The most important and indispensable table for UPPCS Prelims 'Match the Following')

Name of the Cabinet Minister	Administrative Department / Ministry Allotted
Pt. Jawaharlal Nehru	Vice-President (Executive Council), External Affairs and Commonwealth Relations
Sardar Vallabhbhai Patel	Home, Information and Broadcasting, and matters relating to the princely states
Dr. Rajendra Prasad	Minister of Food and Agriculture
Baldev Singh	Minister of Defence
Liaquat Ali Khan	Minister of Finance (<i>from the Muslim League's quota</i>)
Jogendra Nath Mandal	Minister of Law (<i>from the Muslim League's quota</i>)
Aruna Asaf Ali	Minister of Railways
C. Rajagopalachari	Minister of Education
Jagjivan Ram	Minister of Labour
John Mathai	Minister of Industries and Supplies
I.I. Chundrigar	Minister of Commerce
Abdur Rab Nishtar	Minister of Communications
Ghazanfar Ali Khan	Minister of Health
C.H. Bhabha	Minister of Works, Mines and Ports

3 Netaji Subhas Chandra Bose: Brief Introduction and Titles

Subject Section	Historical Details and Coded Exam Facts
Birth and early life	Born on 23 January 1897 at Cuttack (Odisha). In 1920 he secured fourth rank in the Indian Civil Service (ICS), but resigned to serve the country.
Political guru and beginning	Joined the Congress in 1921 inspired by Gandhiji. His political guru was Deshbandhu Chittaranjan Das (C.R. Das) . In 1927 he formed the 'Independence League' together with Nehru.
Resignation from the Congress and organisation	In 1939 (after the Tripuri crisis) he resigned from the presidency of the Congress and formed a new organisation named the Forward Bloc .
Title: 'Netaji'	He was first addressed by the name 'Netaji' by the people of Germany/Hitler .
Title: 'Desh Nayak'	Rabindranath Tagore honoured him with the title 'Desh Nayak'.

Subject Section	Historical Details and Coded Exam Facts
Title: 'Patriot of patriots'	Mahatma Gandhi called Subhas Chandra Bose 'the patriot of patriots'.

4 Principal National Newspapers, Books of Modern India and their Authors

(An infallible integrated compilation of all 41 literary sources given by you - the best reference for one-hundred-per-cent matching in the exam)

Writer / National Leader / Historian	Their Principal Books / Newspapers / Journals
Mahatma Gandhi	Young India, Hind Swaraj, Gokhale: My Political Guru, My Experiments with Truth (autobiography)
Dr. Rajendra Prasad	India Divided <i>(Note: with partial correction of the source list, India Divided is his principal work)</i>
Jawaharlal Nehru	Glimpses of World History, The Discovery of India
Lala Lajpat Rai	The Story of My Deportation, Unhappy India
Bal Gangadhar Tilak	Gita Rahasya, Kesari (Marathi), Mahratta (English)
Abul Kalam Azad	India Wins Freedom, Al-Hilal
Annie Besant	Commonweal, New India
Aurobindo Ghosh	The Life Divine, Karmayogin, Essays on the Gita
Bankim Chandra Chatterjee	Anandamath (based on the Sanyasi Revolt), Durgeshnandini
Raja Rammohan Roy	The Precepts of Jesus, Sambad Kaumudi, Mirat-ul-Akhbar
B.R. Ambedkar	Annihilation of Caste, Mook Nayak, Bahishkrit Bharat
Bhagat Singh	Why I am an Atheist, An Introduction to the Dreamland
Sachindra Nath Sanyal	Bandi Jeevan <i>(the textbook of the revolutionaries)</i>
Bhagwati Charan Vohra	The Philosophy of the Bomb
Dr. Ram Manohar Lohia	Guilty Men of India's Partition
Surendra Nath Banerjee	Bengalee (newspaper), A Nation in Making
Subhas Chandra Bose	The Indian Struggle
Sarojini Naidu	The Broken Wing
Jayaprakash Narayan	Towards Struggle

Writer / National Leader / Historian	Their Principal Books / Newspapers / Journals
Michael Madhusudan Dutt	The Captive Ladie
Manubehn Gandhi	Bapu: My Mother
Dr. Rafiq Zakaria	The Man Who Divided India
Durga Das	India from Curzon to Nehru and After
Valentine Chirol	Indian Unrest (<i>called Tilak the father of Indian unrest</i>)
W.W. Hunter	Our Indian Musalmans
Katherine Mayo	Mother India
Louis Fischer (<i>American journalist</i>)	The Life of Mahatma Gandhi
Bipan Chandra (<i>historian</i>)	India's Struggle for Independence
Vinayak Damodar (V.D.) Savarkar	The Indian War of Independence, 1857
R.C. Majumdar (<i>historian</i>)	The Sepoy Mutiny and the Revolt of 1857
Asoka Mehta	The Great Rebellion: 1857
S.B. Chaudhuri	Civil Rebellion in the Indian Mutinies (1857-1859)
Rudrangshu Mukherjee	Awadh in Revolt: 1857-1858
Syed Ahmad Khan	Tahzib-ul-Akhlaq (journal), Asbab-i-Baghawat-i-Hind
Hugh Toye	The Springing Tiger (biography of Subhas Chandra Bose)
Dilip Mukherjee	Terrorist

- *Note: The 'Amrita Bazar Patrika' was originally edited by Sisir Kumar Ghosh and Motilal Ghosh, and according to the source list the name 'Moti Lal' is associated with it.*

02

World Geography

WORLD GEOGRAPHY

> Principal Geographers and their Theories/Concepts

Name of the Theory / Concept	Name of the Propounder / Geographer (Direct Match Standard)
Continental Drift Theory	Alfred Wegener
Theory of Plate Tectonics	Parker and Morgan (and Harry Hess)
Concept of Isostasy	Pratt and Airy
Convection Current Theory	Arthur Holmes

> Principal Theories Related to the Origin of the Earth (Cosmogony)

Name of the Theory / Hypothesis	Name of the Propounder / Scientist (Direct Match Standard)
Gaseous Mass Hypothesis	Immanuel Kant
Nebular Hypothesis	Pierre Laplace
Planetesimal Hypothesis	Chamberlin and Moulton
Tidal Hypothesis	James Jeans and Jeffreys
Binary Star Hypothesis	Henry Russell
Big Bang Theory	Georges Lemaître

> Principal Theories Related to Mountain Building and their Propounders

Name of the Theory / Hypothesis	Name of the Propounder / Scholar (Direct Match)
Geosynclinal (landslide) theory of mountain building	Kober
Thermal Contraction Theory	Jeffreys
Continental Displacement Theory	Alfred Wegener
Convection Current Theory	Arthur Holmes
Plate Tectonic Theory	Morgan, Harry Hess, McKenzie, Parker
Radioactivity Theory	Joly

Name of the Theory / Hypothesis	Name of the Propounder / Scholar (Direct Match)
Continental Sliding Theory	Daly

> Major Tectonic Plates of the Earth

Name of the Major Plate	Geographical Parts / Regions Included
Eurasian Plate	Europe + Asia + part of the eastern Atlantic Ocean.
Indo-Australian Plate	India + Australia + Indian Ocean.
African Plate	African continent + eastern Atlantic Ocean + western Indian Ocean.
Pacific Plate	The entire Pacific Ocean floor (it is mainly an oceanic plate).
North American Plate	North-western Atlantic Ocean + North American continent.
South American Plate	South American continent + south-western Atlantic Ocean.
Antarctic Plate	Antarctic continent + the oceanic plate surrounding Antarctica on all sides.

> Internal Structure of the Earth: Discontinuities

Name of the Discontinuity	Between which layers is it located? (correct order from top to bottom)
1. Conrad	• Between the upper crust and the lower crust .
2. Mohorovičić / Moho	• Between the lower crust and the upper mantle .
3. Repetti	• Between the upper mantle and the lower mantle .
4. Gutenberg	• Between the lower mantle and the outer core .
5. Lehmann	• Between the outer core and the inner core (<i>the deepest</i>).

> Geological Time Scale: Details of the Principal Eras

Era	Principal Geological Events and Development of Life (Core Pointers)
1. Precambrian	• After the formation of the oceans, the first life was formed. • In India the Aravalli mountains and the rocks of the Dharwar system were formed at this time.
2. Palaeozoic	• The period of primary life; marine life developed abundantly and some reptiles were born on land.
3. Mesozoic	• It is called the period of 'middle life'. It was mainly the age of the dinosaurs. • It includes the Triassic, Jurassic and Cretaceous periods.

Era	Principal Geological Events and Development of Life (Core Pointers)
4. Cenozoic	- This is the present era, called the 'Age of Mammals'. - During it new fold mountain ranges such as the Rockies of North America and the Andes of South America were formed.

> The Four Main Positions of the Earth's Revolution

Position / Event	Exact Date / Time	Position of the Sun (Vertical Latitude)	Condition of the Hemispheres and Features (UPPCS Pointers)
1. Summer Solstice	20-21 June	Tropic of Cancer (23½° N latitude)	Longest day in the Northern Hemisphere and the shortest night (it is summer here). - Winter in the Southern Hemisphere; continuous day for 6 months at the North Pole.
2. Autumnal Equinox	22-23 September	Equator (0° latitude)	- The sun shines equally on both hemispheres. Day and night are equal (12 hours each) all over the world. - Autumn north of the Equator and spring to the south.
3. Winter Solstice	21-22 December	Tropic of Capricorn (23½° S latitude)	- The sun's rays fall vertically on the Tropic of Capricorn. Longest day in the Southern Hemisphere and the shortest night (it is summer here). - Winter in the Northern Hemisphere, where days are short and nights long.
4. Vernal / Spring Equinox	20-21 March	Equator (0° latitude)	- The sun shines equally on both hemispheres. Day and night are equal. - Spring north of the Equator and autumn to the south.

> Important Facts Related to the Solar System

Component of the Solar System / Question	Exact Answer and Special Explanation (Core Facts)
Brightest planet	Venus: regarded as the brightest planet seen from the Earth.
Earth's twin	Venus: because it is similar to the Earth in size and mass.
Rotation in the opposite direction	Venus: unlike the other planets, it rotates from east to west (clockwise).
Morning / evening star	Venus: it is also called the 'morning star' or the 'evening star'.
Hottest planet	Venus: because of its dense CO ₂ atmosphere (maximum temperature: 462°C).
Planet with the longest day	Venus: one of its days equals 243 Earth days .
Planets with no satellites	Mercury and Venus: they have no natural moon.

Component of the Solar System / Question	Exact Answer and Special Explanation (Core Facts)
Planet with prominent rings	Saturn: distinct ring systems are found around it.
Planet with the shortest day	Jupiter: it rotates fastest on its axis (one rotation in 9 hours 55 minutes).
Coldest planet	Uranus: its effective temperature stays at about -216°C .
Nearest star to the Earth	Proxima Centauri: the nearest star to the Earth after the Sun.

> Layers of the Atmosphere and their Main Features (Atmospheric Layers)

Name of the Layer	Main Features and Atmospheric Phenomena (Core Pointers)
1. Troposphere	• All kinds of climatic and weather phenomena (clouds, rain, storms) occur in this layer.
2. Stratosphere	• The ozone layer is found in it. Being calm in weather, it is ideal for flying aircraft .
3. Mesosphere	• Greenhouse gases are absent. Meteors / shooting stars burn up in this layer as soon as they enter from space.
4. Ionosphere	• Electrically charged particles are found. Radio waves transmitted from the Earth are reflected back from here.
5. Exosphere	• The outermost layer of the atmosphere. Owing to the weak force of gravity, gases are present in extremely rarefied/small quantities here.

> El Niño versus La Niña: Climatic Phenomena (Comparative Table)

Criterion / Effect	El Niño	La Niña
Basic concept	• Sea surface in the central and eastern tropical Pacific Ocean becoming warmer than average .	• Sea surface in the central and eastern Pacific Ocean becoming colder than average .
Effect on the Indian monsoon	• The Indian monsoon becomes weak (drought conditions arise in India).	• The Indian monsoon becomes very strong (good rainfall/floods in India).
Temperature and rainfall	• Heavy rain/floods in the Horn of Africa and in southern America. • Drought conditions in northern South America.	• Heavy rain and floods in the north-western part of the Pacific Ocean and in Canada. • Drought in southern America; colder than normal in North America and Canada.

Criterion / Effect	El Niño	La Niña
Hurricanes	• More hurricanes are encouraged in the eastern and central Pacific region.	• More severe than average hurricanes in the Atlantic Ocean, while fewer than average in the eastern Pacific.
Agriculture and marine life	• A heavy decline/reduction in the number of marine organisms such as fish and plankton.	• Cold-water species (salmon, squid) come to the California coast (increase in marine life). • Yields of wheat and soybean may fall in America.

› Correct Order of the Continents of the World (by Area and Population)

Rank	Descending Order of Continents by Area	Descending Order of Continents by Population
1	Asia (<i>largest</i>)	Asia (<i>most populous</i>)
2	Africa	Africa
3	North America	Europe
4	South America	North America
5	Antarctica	South America
6	Europe	Australia
7	Australia (<i>smallest</i>)	Antarctica (<i>uninhabited/least</i>)

› Countries of the World with the Longest Coastlines (in Descending Order)

Rank	Name of the Country	UPPCS Special Note (Coastline)
1	Canada	<i>The country with the longest coastline in the world.</i>
2	Norway	—
3	Indonesia	—
4	Russia	—
5	Philippines	—

Position of India: With a total coastline of **11,098 kilometres** (on the basis of compiled estimates including the island territories), India ranks **13th** in the world.

> Highest Mountain Peaks by Continent (Quick Pointers)

Name of the Continent	Name of the Highest Peak	Country / Specific Geographical Location
Asia	Mount Everest	Nepal (the highest mountain of the world and of Asia)
Europe	Mount Elbrus	Russia (the highest mountain of Europe)
Africa	Mount Kilimanjaro	Tanzania (the highest mountain of the African continent)
North America	Mount McKinley / Denali	United States of America (USA)
South America	Ojos del Salado / Aconcagua	Andes range <i>(Note: in the data both names are described as principal peaks)</i>
Antarctica	Vinson Massif	Antarctic continent
Australia	Mount Kosciuszko	Australian mainland
Other famous peaks	• Jura Mountains: France • Mount Titlis: Switzerland	—

> Principal Mountain Ranges of the World and their Highest Points

Name of the Mountain Range	Geographical Location (Place/Country)	Highest Point / Peak
Great Dividing Range	Eastern Australia	Kosciuszko
Transantarctic	Antarctica	Mount Vinson Massif
Caucasus Mountains	Russia / Georgia	Mount Elbrus <i>(the highest peak of Europe)</i>
Alaska Range	USA (Alaska)	Mount McKinley (Denali) <i>(the highest peak of North America)</i>
Sierra Madre Oriental	Mexico	Orizaba
Atlas Range	North-west Africa	Mount Toubkal
Tien Shan Range	South-central Asia	Pik Pobeda
Altai Mountains	Central Asia (Russia/Mongolia)	Gora Belukha
Zagros Mountains	Iran	Zard Kuh
Elburz Mountains	Iran	Damavand

Name of the Mountain Range	Geographical Location (Place/Country)	Highest Point / Peak
Appalachian Mountains	Eastern USA and Canada	Mount Mitchell
Alps	Central Europe	Mont Blanc
Cascade Range	USA and Canada	Mount Rainier
Apennine Mountains	Italy	Corno Grande
Drakensberg	South-east Africa	The highest range of South Africa
Pyrenees	Border of France and Spain	<i>A fold mountain of the Alpine system</i>
Ural / Verkhoyansk	Central Russia / Eastern Russia	—
Pennines / Cambrian	England	—

➤ Major Plateaus of the World

Name of the Plateau	Location / Related Country	Name of the Plateau	Location / Related Country
Pamir or Tibetan Plateau	Tibet (China) <i>(the largest plateau in the world)</i>	Antarctic Plateau	Antarctica <i>(the second largest plateau in the world)</i>
Djado Plateau	Niger	Libyan Plateau	Libya and Egypt
Laurentian Plateau	Canada	Mongolian Plateau	Mongolia and China
Taklamakan / Loess	China	Anatolia / Asia Minor	Turkey
Malwa / Deccan	India	Abyssinian Plateau	Ethiopia
Chhota Nagpur / Meghalaya	India	Columbia / Colorado	United States of America (USA)
Iranian Plateau	Iran	Ozark / Piedmont	United States of America (USA)
Ahaggar / Tassili	Algeria / Libya	Great Basin Plateau	United States of America (USA)
Kimberley Plateau	Australia	Patagonian Plateau	Argentina
Ubangi Plateau	Central Africa	Guiana Highlands	Venezuela, Guyana, Suriname, French Guiana

Name of the Plateau	Location / Related Country	Name of the Plateau	Location / Related Country
Bié Plateau	Angola	Mato Grosso / Brazilian	Brazil
Adamawa Plateau	Nigeria	Bolivian Plateau	Bolivia (Andes range)
Shan Plateau	Myanmar	Potwar / Khorat	Pakistan (Potwar), Thailand (Khorat)
Meseta Plateau	Iberian Peninsula (Spain)	Chiapas Plateau	Southern Mexico
Alaska / Yukon	Canada		

> Major Deserts of the World

Name of the Desert	Geographical Location / Country (UPPCS Matching Special)
Sahara Desert	Algeria, Chad, Egypt, Libya, Mali, Mauritania, Morocco, Niger, Sudan, Tunisia and Western Sahara.
Nubian / Bayuda Desert	Sudan (northern/north-eastern region)
Namib Desert	Angola, Namibia and South Africa
Kalahari Desert	Namibia, Botswana and South Africa
Gobi Desert	China and Mongolia (<i>temperate desert</i>)
Atacama Desert	Northern Chile and southern Peru
Al Nafud / Rub al Khali / Hamad	Saudi Arabia (part of the Arabian Desert)
Dasht-e-Lut / Dasht-e-Kavir	Iran
Lop / Taklamakan	China
Kyzyl Kum	Kazakhstan, Uzbekistan, Turkmenistan
Karakum	Turkmenistan, Kazakhstan
Mojave / Red Desert	United States of America (USA)
Sonoran	USA and Mexico
Thar / Kharan	India-Pakistan (Thar), Pakistan (Kharan)
Dasht-e-Margo / Libyan	Afghanistan (Margo), Libya (Libyan)

> Various Seas, Oceans and the Salinity of Water

- **Measurement of salinity:** It is expressed as the amount of salt dissolved per thousand grams of water (‰).

Sea / Ocean / Lake	Salinity (‰)	UPPCS Core Exam Fact
Lake Van of Türkiye	330‰	The lake with the highest salinity in the world .
Dead Sea	238‰	Because of its very high density a person does not sink in it.
Great Salt Lake (USA)	220‰	The most saline lake of North America.
Atlantic Ocean	36‰	Its average salinity is slightly higher among the oceans.
Arabian Sea	36‰	More saline than the Bay of Bengal because of higher evaporation.
Indian Ocean	35‰	Equal to the global average.
Average salinity of the oceans	35‰	The standard point of the global oceanic average .
Pacific Ocean	31 to 35‰	Regional variation in salinity because of its vast extent.
Bay of Bengal	30‰	Low salinity because of the large supply of fresh water by rivers (Ganga, Brahmaputra).

> Principal Salts Dissolved in Ocean Water

Name of the Salt (Chemical Compound)	Percentage of Total Amount in Ocean Water
1. Sodium chloride (NaCl)	77.80% (<i>largest amount</i>)
2. Magnesium chloride (MgCl ₂)	10.90%
3. Magnesium sulphate	4.70%
4. Calcium sulphate	3.60%
5. Potassium sulphate	2.50%
6. Calcium carbonate	0.30%
7. Magnesium bromide	0.20%
Total	100%

> Principal Lakes of the World by Continent (Regional Lakes Mapping)

Continent	Names of the Principal Lakes	Littoral Countries / Specific Geographical Location (UPPCS Matching Special)
Asia	• Caspian Sea • Lake Baikal • Aral Sea • Lake Van • Lake Balkhash • Toba / Pangong Lake • Tonle Sap / Lop Nur • Issyk Kul / Urmia • Onega and Ladoga	• Russia, Kazakhstan, Turkmenistan, Iran, Azerbaijan (<i>the largest lake in the world</i>) • Russia (Siberia) (<i>the deepest and oldest lake in the world</i>) • Kazakhstan and Uzbekistan (<i>now drying up rapidly</i>) • Türkiye (<i>most saline</i>) • Kazakhstan (Balkhash) • Indonesia (Toba), India-China (Pangong) • Cambodia (Tonle Sap), in the Tarim Basin of China (Lop Nur) • Kyrgyzstan (Issyk Kul), Iran (Urmia) • European part of Russia (<i>also included in the Europe mapping</i>)
Africa	• Lake Victoria • Lake Tanganyika • Lake Chad • Lake Malawi (Nyasa) • Lake Nasser / Tana • Volta / Kariba / Albert	• Kenya, Uganda, Tanzania (<i>the Equator passes through it</i>) • Tanzania, Zaire (DRC), Burundi, Zambia • Cameroon, Nigeria, Chad, Niger • Malawi, Mozambique and Tanzania • Egypt and Sudan (Nasser - artificial), Ethiopia (Tana) • Ghana (Volta), Zambia-Zimbabwe (Kariba - artificial), Uganda-Congo (Albert)
North America	• Great Lakes • Great Bear / Great Slave • Athabasca / Winnipeg / Reindeer • Great Salt Lake / Lake of the Woods	• A group of 5 lakes (H.O.M.E.S): Superior, Huron, Ontario, Michigan (<i>only in the USA</i>), Erie. (<i>Canada + USA border</i>) • Canada (<i>the Arctic Circle passes through Great Bear</i>) • Other principal freshwater sources of Canada • United States of America (USA); Canada-USA border (Lake of the Woods)
South America	• Lake Titicaca • Lake Maracaibo • Lake Poopó / Red Lagoon	• Located between Bolivia and Peru (<i>the highest navigable lake in the world</i>) • Located in the north of Venezuela (famous for petroleum reserves) • Principal lakes located in Bolivia
Europe	• Lake Constance • Ladoga / Onega • Vättern and Vänern	• It is the meeting point of Germany, Austria and Switzerland. • The largest lakes of Europe, located in Russia. • Principal lakes located in Sweden.
Australia	• Lake Eyre • Other lakes	• The lowest point of Australia (main catchment basin). • Lake Disappointment, Lake Austin and Lake Barlee.

> Major Canals of the World

Name of the Canal	Country / Region	Which Regions / Seas / Lakes does it Connect? (UPPCS Mapping Special)
Suez Canal	Egypt	• Connects the Mediterranean Sea with the Red Sea (the largest artificial canal in the world). • Northern end: Port Said; southern end: Port Suez. • Order of lakes from north to south: Timsah → Great Bitter → Little Bitter (<i>chronology fact</i>)
Panama Canal	Panama	• Connects the Pacific Ocean with the Atlantic Ocean.
Welland Canal	Canada	• Connects Lake Erie with Lake Ontario.
Erie Canal	United States of America	• Connects Lake Erie with the Hudson River.
Soo Canal	United States of America	• Connects Lake Superior with Lake Huron.
Kiel Canal	Germany	• Connects the Baltic Sea with the North Sea.
New Waterway Canal	Germany	• Connects the North Sea with Rotterdam.
North Sea Canal	Germany	• Located between Amsterdam and the North Sea.
Ludwig Canal	Germany	• Used as an inland waterway in southern Germany.
Göta Canal	Sweden	• Connects Gothenburg with Stockholm.
Manchester Canal	Great Britain (UK)	• Connects Liverpool with Manchester.
Volga-Don Canal	Russia	• Connects Stalingrad with Rostov.
White Sea Canal	Russia	• Connects the White Sea with Lake Onega.
Albert Canal	Belgium	• Connects Antwerp with Liège.
K.P. Canal	India	• Connects Tamil Nadu with Andhra Pradesh (<i>Indian geography special</i>).
Sault Ste. Marie Canal	Canada	• A principal canal located in Canada.

> Major Waterfalls of the World

Name of the Waterfall	Related River / Type	Location / Country (Direct Match)	Special UPPCS Fact
Salto Angel (Angel Falls)	Rio Caroní / Churún river	Venezuela	It is the highest waterfall in the world .
Victoria Falls	Zambezi river	Border of Zambia and Zimbabwe	The most famous waterfall of Africa.
Niagara Falls	St. Lawrence river system	United States of America (USA) and Canada border	Located near the Great Lakes region.
Tugela Falls	—	South Africa	The second highest waterfall of the African continent.
Kukenán	—	Venezuela	—
Mardalsfossen / Tyssestrengene	—	Norway	Principal waterfalls of Europe.
Yosemite / Ribbon	—	California (United States of America)	—
Gavarnie Falls	—	France	—
Sutherland	—	New Zealand	—

> Principal Dams and Rivers of the World

Name of the Dam	Related River	Location / Country (Direct Match Standard)
Rogun Dam	—	Tajikistan (one of the highest dams in the world)
Nurek Dam	—	Tajikistan
Aswan Dam	Nile river	Egypt
Owen Falls Dam	White Nile river	Uganda (<i>outflow region of the White Nile</i>)
Sennar Dam	Blue Nile river	Sudan
Cahora Bassa Dam	Zambezi river	Mozambique
Kariba Dam	Zambezi river	Zambia - Zimbabwe border (<i>data reference: Zambezi</i>)
Inga Dam	Congo / Zaire river	Democratic Republic of the Congo (DRC)
Kainji Dam	Niger river	Nigeria (<i>data reference: Niger</i>)
Akosombo Dam	Volta river	Ghana

> Principal Local Winds of the World

A. Cold Local Winds

Name of the Wind	Region / Country of Flow (UPPCS Matching Special)	Name of the Wind	Region / Country of Flow (UPPCS Matching Special)
Mistral	Spain and France	Bora	Adriatic Sea coast (Europe)
Norther / Norte	United States of America and Mexico	Tramontane	Central Europe
Williwaw	Alaska (USA)	Levanter	Spain
Pampero	Argentina and Uruguay	Gregale	Mediterranean region of southern Europe
Papagayo	Mexico	Joran	Jura Mountains (Switzerland)
Purga / Buran	Tundra region / Russia (Siberia)	Seistan	Eastern Iran
Bise	Eastern France, Switzerland	Southerly Buster	New South Wales (Australia)
Haboob	Sudan	Cape Doctor	Republic of South Africa

B. Warm Local Winds

Name of the Wind	Region / Mountain / Country of Flow	Name of the Wind	Region / Mountain / Country of Flow
Chinook	Rocky Mountains (USA) (<i>snow-eating wind</i>)	Santa Ana	Southern California (USA)
Foehn	Alps (Switzerland)	Shamal / Samoon	Iraq (Shamal), Iran (Samoon/Garmsil)
Sirocco	Italy (<i>wind that causes 'blood rain'</i>)	Khamsin / Ghibli	Egypt (Khamsin), Libya (Ghibli)
Simoom	Arabian desert	Berg / Leste	South Africa (Berg), Madeira and Canary (Leste)
Brickfielder	Australia	Sukhovey / Zonda	Russia-Kazakhstan (Sukhovey), Andes region (Zonda)
Black Roller	Plains of North America	Karaburan	Tarim Basin of Central Asia
Loo	Northern India	Yamo / Kumbang	Japan (Yamo), Java-Indonesia (Kumbang)
Aiala / Solano	France (Aiala), south-eastern Spain (Solano)	Temporal / Chili	Central America (Temporal), Tunisia (Chili)

> Principal Islands Located in the Oceans (Islands by Ocean)

Name of the Island	Ocean / Sea where Located	Name of the Island	Ocean / Sea where Located
Greenland	Arctic Ocean (<i>the largest island in the world</i>)	Baffin Island	Northern Arctic Ocean
New Guinea	Western Pacific Ocean	Victoria Island	North Polar (Arctic) Ocean
Honshu Island	North Pacific Ocean (<i>main island of Japan</i>)	Ellesmere Island	North Polar (Arctic) Ocean
North and South Islands	South-west Pacific Ocean (<i>New Zealand</i>)	Britain	North Atlantic Ocean
Tasmania	S.W. Pacific Ocean (<i>Australia</i>)	Iceland / Ireland	North Atlantic Ocean
Borneo / Madagascar	Indian Ocean	Quebec / Cuba	Caribbean Sea
Java / Sumatra	Indian Ocean (<i>Indonesia</i>)	Sri Lanka	Indian Ocean
Sulawesi Island	Indian Ocean		

> Principal Cold and Warm Ocean Currents

> Ocean Currents of the Pacific Ocean

Warm Currents of the Pacific Ocean	Cold Currents of the Pacific Ocean
• Kuroshio current (<i>the black current of Japan</i>) • Alaska current • El Niño current (<i>temporary warm current</i>) • Tsushima current • East Australian current • North Equatorial current / South Equatorial current • North Pacific current / Counter Equatorial current	• Humboldt or Peru current (<i>El Niño forms when it weakens</i>) • California current • Kurile / Oyashio current • Okhotsk current • Antarctic current

> Currents of the Atlantic and Indian Oceans

Ocean	Warm Currents	Cold Currents
Atlantic Ocean	• Gulf Stream (<i>the blanket of Europe</i>) • Brazil current • Florida current • Norwegian current / Irminger current • North Atlantic current / Guinea current • North and South Equatorial currents	• Canary current • Benguela current • Labrador current • Falkland current • East Greenland current • Antarctic current
Indian Ocean	• Mozambique current • Agulhas current • South Equatorial current	• West Australian current • West Wind Drift

> Principal Local Vegetation Types and their Natural Regions

Name of the Vegetation	Adapted Natural Region / Country (Direct Match Standard)
Maquis	Shrubby vegetation found in the coastal region of the Mediterranean Sea.
Chaparral	Vegetation of the Mediterranean climate of California (USA) .
Matorral	Vegetation of Chile (South America).
Fynbos	The distinctive vegetation of the Cape province of South Africa .
Baobab	Characteristic of the savanna grasslands (<i>the upside-down tree/bottle tree</i>).
Epiphytes	Vegetation of the equatorial (tropical rainforest) region.
Cedar (Deodar)	Vegetation of the Mediterranean mountain regions.

> Major Grasslands of the World

Type of Grassland	Name of the Grassland	Related Country / Geographical Region (UPPCS Matching Special)
1. Tropical Grasslands	• Savanna • Campos • Llanos	Africa Brazil Venezuela and Colombia
2. Temperate Grasslands	• Prairies • Pampas • Veld • Downs • Steppes	USA and Canada Argentina South Africa Australia Asia, Ukraine, Russia and the Manchuria region of China

> Major Valleys of the World

Name of the Valley	Geographical Location / Country	Special UPPCS Exam Pointer
Valley of the Kings	Egypt	A historical site located on the western bank of the Nile.
Silicon Valley	Northern California (USA)	The computer-chip production region located south of the San Francisco Bay.
Grand Valley	United States of America (USA)	Located between the Rocky Mountains and the Sierra Nevada.
Taylor Valley	Antarctica	A dry valley; it was discovered by Robert Falcon Scott .
Panjshir Valley	Afghanistan	A strategic valley located north of Kabul.
Chumbi Valley	India, Bhutan and Tibet border	A strategic triangle near the Doklam area of Sikkim (India).
Po Valley	Italy	Famous for rice production and industrial development.
Loire Valley	France	Famous for its historic castles (châteaux).
Sacred Valley	Peru	A sacred site of the Inca civilisation.
Chitwan / Kathmandu Valley	Nepal	Famous for Chitwan National Park.
Rabbit / Pleasant Valley	United States of America (USA)	—
Humber Valley	Canada	—
Ebro Valley	Spain	—
Blackwood / King / Wolgan	Australia	Principal Australian valleys.

> Other Principal Volcanoes and their Location

Name of the Volcano	Location (Country)	UPPCS Special Fact
Mount Kinabalu	Malaysia	The highest point on the island of Borneo.
Alborz (Elburz)	Iran	Mountain/volcanic region located south of the Caspian Sea.

Name of the Volcano	Location (Country)	UPPCS Special Fact
Colima	Mexico	A highly active volcano of Mexico.
Mount Rainier	United States of America (USA)	A large stratovolcano located in Washington state.
Taal Volcano	Philippines	A small but active volcano located within a lake on Luzon island.

> Major Oceanic Trenches of the World

Trench	Related Ocean	Location / Feature	Maximum Depth
Mariana Trench	Pacific Ocean	• Located east of the Philippines, it is the deepest point in the world.	11,034 metres
Tonga Trench	Pacific Ocean	• Located east of the Tonga islands in the South Pacific Ocean.	—
Sunda Trench <i>(former name: Java Trench)</i>	Indian Ocean	• Located south of the island of Java in Indonesia, it is the deepest point of the Indian Ocean.	7,725 metres
Puerto Rico Trench	Atlantic Ocean	• Located on the boundary of the Caribbean Sea and the Atlantic Ocean.	—

> Principal Rivers of the North American and African Continents

Name of the River	Continent / Location	UPPCS Special Main Features (Core Concept Pointers)
Mississippi-Missouri	North America	• They are two separate rivers which together form the fourth largest river system in the world. • Its delta is shaped like a 'bird's foot' (bird's-foot delta).
Rio Grande	North America	• This river forms the natural international boundary between the United States of America (USA) and Mexico.
St. Lawrence	North America	• Forms the busiest inland water transport route in the world. • The world-famous 'Niagara Falls' is located on this river (in the lake-connecting region).
Colorado	North America	• Flows in the western part of the continent. • The world-famous 'Grand Canyon' and the 'Hoover Dam' are on this river.

Name of the River	Continent / Location	UPPCS Special Main Features (Core Concept Pointers)
Niger	Africa	• Falls into the Gulf of Guinea and is famous in West Africa as the ' Oil River '.
Zambezi	Africa	• The famous Victoria Falls and the Kariba Dam are located on this river.
Congo / Zaire	Africa	• The only river in the world that crosses the Equator twice . • The famous Stanley and Livingstone Falls are located on it.
Limpopo	Africa	• Rises in the Highveld of South Africa and crosses the Tropic of Capricorn twice .

➤ Principal Rivers of Europe and their Features

Name of the River	Source	Where it Falls (Mouth)	Main Features (Quick Revision Notes)
Volga	Valdai Hills of Russia	Caspian Sea	• It is the longest river of Europe . • Connects the Baltic Sea with the Caspian Sea; the lifeline of Russian culture and trade.
Danube	Black Forest of Germany	Black Sea	• It is the second longest river of Europe (~2,850 km). • It passes through ten countries . Its delta is a UNESCO World Heritage Site.
Rhine	Swiss Alps	North Sea	• It is one of the busiest and most important waterways of Europe (1,233 km). • Separates Germany from the Netherlands.
Loire	French Cévennes range	Atlantic Ocean	• It is the longest river of France .
Po	Italian Cottian Alps	Adriatic Sea	• It is the longest river of Italy (more than 652 kilometres).

➤ Countries Located on the Principal Geographical Lines (Equator, Cancer, Capricorn, GMT)

Name of the Line	Countries Located (Continent-wise Classification - UPPCS Special)
Equator <i>(through 13 countries/territories)</i>	• Africa: São Tomé and Príncipe, Gabon, Republic of the Congo, Democratic Republic of the Congo (DRC), Uganda, Kenya, Somalia. • Asia: Maldives, Indonesia, Kiribati. • South America: Ecuador, Colombia, Brazil.

Name of the Line	Countries Located (Continent-wise Classification - UPPCS Special)
Tropic of Cancer	• North America: Hawaii islands (USA), Bahamas, Mexico. • Africa: Western Sahara, Mauritania, Algeria, Libya, Egypt, Mali, Niger, Chad. • Asia: India, Myanmar, Oman, Saudi Arabia, United Arab Emirates (UAE), Bangladesh, China, Taiwan.
Tropic of Capricorn	• South America: Chile, Argentina, Paraguay, Brazil. • Africa: Namibia, Botswana, South Africa, Mozambique, Madagascar. • Oceania: Australia (<i>your original data describes only 2 continents</i>).
Prime Meridian (GMT line)	• Europe: United Kingdom (UK), Spain, France. • Africa: Ghana, Algeria, Mali, Togo, Burkina Faso. • Other: Antarctica.

> List of Countries with the Most Time Zones

Rank	Name of the Country	Total Number of Time Zones
1	France	12 (<i>the most, owing to its overseas territories/islands</i>)
2	Russia	11
3	Antarctica	10
4	United States of America (USA)	9
5	Australia	8
6	Denmark	5

> Principal Geographical Triangles/Regions of the World

Name of the Region	Main Feature	Extent / Related Countries (UPPCS Core Map Point)
Bermuda Triangle	• A mysterious region of the north-west Atlantic Ocean.	Bermuda → Miami (southern Florida, USA) → San Juan (Puerto Rico).
Golden Triangle	• The principal opium-producing region of South-East Asia.	Myanmar → Laos → Thailand.
Golden Crescent	• The notorious opium-producing region of south-central Asia.	Iran → Afghanistan → Pakistan.
Coral Triangle	• A marine region with extremely rich biodiversity/corals .	Philippines → Indonesia → Malaysia → Papua New Guinea → Timor-Leste.

> Countries Sharing a Border with the Red Sea

The UPPCS special short trick to remember the countries touching the Red Sea is "**DESSEY**":

- **List of countries:** 1. Egypt 2. Sudan 3. Eritrea 4. Djibouti 5. Saudi Arabia 6. Yemen (*Note: your data mainly describes Egypt, Saudi Arabia, Yemen, Sudan and Djibouti*).

> Countries Included in the Arabian Peninsula

The Arabian Peninsula is the largest peninsula in the world. The countries included in it are as follows:

- **List of countries:** 1. Saudi Arabia 2. Yemen 3. Oman 4. United Arab Emirates (UAE) 5. Qatar 6. Kuwait 7. Jordan 8. Iraq (southern part).

> Nordic Countries

The Nordic countries are a geographical and cultural region located in northern Europe and the North Atlantic.

- **Countries included:** 1. Iceland 2. Norway 3. Denmark 4. Sweden 5. Finland

> The 6 Countries Sharing a Border with the Black Sea

The UPPCS special short trick to remember the countries touching the Black Sea is "**BURGER-T**":

Bulgaria → Ukraine → Russia → Georgia → Romania → Turkey

> Principal Tribes of the World and their Habitats

Name of the Tribe	Geographical Habitat / Country (Extremely Important for UPPCS Matching)
Pygmy	In the rainforests of the Congo Basin (live in huts built on trees).
Bushmen	Kalahari Desert (Botswana/South Africa) (<i>termites are called the 'rice of the Bushmen'</i>).
Masai	Plateau regions of Kenya, northern Tanzania and eastern Uganda in East Africa.
Eskimo	Arctic/icy regions of Alaska, Canada and Greenland (<i>igloo dwellings</i>).
Bora	Western Amazon Basin (Brazil, Peru and Colombia).
Kirghiz	Kyrgyzstan and adjoining areas in Central Asia (<i>yurt dwellings</i>).
Semang / Sakai	Forest regions of Malaysia .
Maori	The original inhabitants of New Zealand .
Afridi	North-western mountainous region of Pakistan .
Veddhas / Ainu	Sri Lanka (Veddhas), Japan (Ainu).
Berber / Nuba	Morocco (Berber), Sudan (Nuba).

Name of the Tribe	Geographical Habitat / Country (Extremely Important for UPPCS Matching)
Zulu / Bantu / Ikatha	South Africa (<i>the Bantu are also spread over the whole of central/southern Africa</i>).
Papuans / Lapps	Papua New Guinea (Papuans), the Sami region of Norway, Sweden and Finland (Lapps).
Yupik / Yukaghir / Tartar	Eastern Russia and Alaska (Yupik), north-eastern Siberia (Yukaghir), Siberia (Tartar/Kamasin).
Bedouin / Budh	Nomadic herders of the Arabian desert.
Maya / Red Indians	Central America (Maya), the North American continent (Red Indians).
Hadza	Tanzania.
Fulani / Hausa / Fula	West Africa and Nigeria (Fulani/Hausa), the Mediterranean region of Africa (Fula).
Kurds	The border mountain region of Iran, Iraq, Türkiye and Armenia.
Oromo	Ethiopia.
Luo and Baganda	The region around Lake Victoria .
Kikuyu and Kamba	The mountainous region of Kenya.
Aborigines / Black Fellows	The original inhabitants of Australia .
Fellah	Agricultural labourers of Egypt .
Lai / Uighur	Myanmar (Lai), the Tarim Basin/Xinjiang of China (Uighur).

> Classification of the Principal Vegetation of the World (Vegetation Types)

Name of the Vegetation	Adapted Geographical Region / Natural Environment (Direct Match Standard)
Xerophyte	• Thorny vegetation growing in tropical desert (arid) regions.
Hydrophyte	• Vegetation living in water or in waterlogged areas.
Hygrophyte	• Vegetation of marshy areas and of equatorial hot-humid (high-moisture) regions.
Mesophyte	• Vegetation of the temperate zone (normal rainfall and temperature).
Cryophyte	• Vegetation of the tundra and extremely cold (icy) regions.
Halophyte	• Vegetation found in salty / saline soils or coastal areas.

Name of the Vegetation	Adapted Geographical Region / Natural Environment (Direct Match Standard)
Lithophyte	• Vegetation growing on hard rocks or stones.
Tropophyte	• Grasses and vegetation of the tropical climate (changing seasons).

➤ Principal Industrial Cities of the World and their Related Industries

Industrial City	Country / Region	Principal Industry (UPPCS Matching Special)
Detroit	United States of America (USA)	Automobile (motor car) industry <i>(extremely important)</i>
Turin	Italy	Automobile / motor car industry
Nagoya	Japan	Automobile industry
Chicago	United States of America (USA)	Meat processing and the iron and steel industry
Buenos Aires	Argentina	Meat and dairy industry
Kinshasa	Congo (DRC)	Meat industry
Pittsburgh	United States of America (USA)	Iron and steel industry <i>(the steel capital of the world)</i>
Hamilton	Canada	Iron and steel industry
Essen / Tula	Germany (Essen), Russia (Tula)	Iron and steel industry
Kobe / Anshan	Japan (Kobe), China (Anshan)	Iron and steel industry
Magnitogorsk / Krivoy Rog	Russia (Magnitogorsk), Ukraine region	Iron and steel industry
Los Angeles	United States of America (USA)	Hollywood , oil refining and aircraft
New Orleans / Houston	United States of America (USA)	Oil refining (New Orleans), oil and natural gas (Houston)
San Francisco	United States of America (USA)	Oil refining, computers and ships
Maracaibo / Abadan	Venezuela (Maracaibo), Iran (Abadan)	Oil refining industry
Mosul	Iraq	Mineral oil industry

Industrial City	Country / Region	Principal Industry (UPPCS Matching Special)
Manchester / Osaka	Great Britain (Manchester), Japan (Osaka)	Cotton textile industry / textile industry
Shanghai / Cairo	China (Shanghai), Egypt (Cairo)	Textiles and machinery (Shanghai), textile industry (Cairo)
Milan / Lyons	Italy (Milan), France (Lyons)	Silk textile industry
Rio de Janeiro	Brazil	Textiles and the coffee industry
São Paulo	Brazil	Coffee industry (<i>the largest coffee market in the world</i>)
Havana	Cuba	Cigar industry
Antwerp	Belgium	Diamond industry (diamond cutting/trade)
Kimberley	South Africa	Diamond mining industry
Johannesburg	South Africa	Gold mining industry
Montreal / Belfast	Canada (Montreal), Great Britain (Belfast)	Shipbuilding industry
Rotterdam / Amsterdam	Netherlands	Shipbuilding
Glasgow / Hamburg	Great Britain (Glasgow), Germany (Hamburg)	Shipbuilding industry
Stockholm / Vladivostok	Sweden (Stockholm), Russia (Vladivostok)	Shipbuilding
Ottawa	Canada	Paper industry
Dundee	Scotland / Great Britain	Jute industry
Frankfurt / Zurich	Germany (Frankfurt), Switzerland (Zurich)	Engineering and transport (Frankfurt), engineering (Zurich)
Paris	France	Transport and aircraft
Vienna / Venice	Austria (Vienna), Italy (Venice)	Glass industry
Copenhagen	Denmark	Dairy industry
Santiago	Chile	Wine industry
Multan	Pakistan	Pottery

Industrial City	Country / Region	Principal Industry (UPPCS Matching Special)
Wuhan	China	Ships and iron and steel
Pelialia / Chelyabinsk	Russian region	Metal and military machinery industry

> Research Institutes

> National Research Institutes

Name of the Research Institute	Headquarters / Location (Place)
Institutes of Dehradun (Uttarakhand)	• Wildlife Institute of India (WII) • Forest Research Institute (FRI) • Forest Survey of India (FSI) • Wadia Institute of Himalayan Geology (WIHG) • Indian Council of Forestry Research and Education (ICFRE) • Indian Institute of Soil and Water Conservation (IISWC)
Institutes of Lucknow (Uttar Pradesh)	• National Botanical Research Institute (NBRI) • Birbal Sahni Institute of Palaeosciences (BSIP)
Institutes of Kolkata (West Bengal)	• Botanical Survey of India (BSI) • Zoological Survey of India (ZSI)
Institutes of New Delhi	• Indian Agricultural Research Institute (IARI) • National Institute of Plant Genome Research (NIPGR) • National Research Centre on Plant Biotechnology • National Environmental Science Academy (NESA) • National Physical Laboratory (NPL)
Institutes of Hyderabad (Telangana)	• National Geophysical Research Institute (NGRI) • Institute of Forest Biodiversity (IFB)
Institutes of Chennai (Tamil Nadu)	• National Institute of Ocean Technology (NIOT) • National Centre for Coastal Research (NCCR) • National Centre for Sustainable Coastal Management (NCSCM)
Institutes of Maharashtra	• National Environmental Engineering Research Institute (NEERI) — Nagpur (<i>a favourite UPPCS question</i>) • Indian Institute of Tropical Meteorology (IITM) — Pune • National Chemical Laboratory (NCL) — Pune
Institutes of Goa	• National Institute of Oceanography (NIO) — Panaji • National Centre for Polar and Ocean Research (NCPOR) — Vasco da Gama

Name of the Research Institute	Headquarters / Location (Place)
Institutes of Jharkhand	• Central Institute of Mining and Fuel Research (CIMFR) — Dhanbad • National Metallurgical Laboratory (NML) — Jamshedpur • Institute of Forest Productivity (IFP) — Ranchi
Jodhpur / Jaipur (Rajasthan)	• Central Arid Zone Research Institute (CAZRI) — Jodhpur • Arid Forest Research Institute (AFRI) — Jodhpur
Institutes of Haryana	• Central Soil Salinity Research Institute (CSSRI) — Karnal • National Dairy Research Institute (NDRI) — Karnal
Madhya Pradesh / Chhattisgarh	• Tropical Forest Research Institute (TFRI) — Jabalpur • National Institute of Biotic Stress Management (NIBSM) — Raipur
Other principal institutes	• G.B. Pant National Institute of Himalayan Environment — Almora (UK) • National Institute of Hydrology (NIH) — Roorkee (UK) • Himalayan Forest Research Institute (HFRI) — Shimla (HP) • Rain Forest Research Institute (RFRI) — Jorhat (Assam) • Institute of Wood Science and Technology (IWST) — Bengaluru (KN) • Institute of Forest Genetics and Tree Breeding (IFGTB) — Coimbatore (TN) • Institute of Minerals and Materials Technology (IMMT) — Bhubaneswar (OD) • North East Institute of Science and Technology (NEIST) — Jorhat (Assam) • National Institute for Interdisciplinary Science and Technology (NIIST) — Thiruvananthapuram (Kerala)

> Principal Global Environment and Water Resource Organisations

Name of the Organisation / Institution	Headquarters / Location (City, Country)	UPPCS Special Fact / Importance
United Nations Environment Programme (UNEP)	Nairobi, Kenya	<i>The headquarters asked most often in UPPCS.</i>
International Union for Conservation of Nature (IUCN)	Gland, Switzerland	Issues the Red Data Book.
World Wide Fund for Nature (WWF)	Gland, Switzerland	Its emblem is the 'Giant Panda'.
World Meteorological Organization (WMO)	Geneva, Switzerland	—
World Climate Research Programme (WCRP)	Geneva, Switzerland	—
World Resources Institute (WRI)	Washington D.C., USA	—

Name of the Organisation / Institution	Headquarters / Location (City, Country)	UPPCS Special Fact / Importance
World Water Council (WWC)	Marseille, France	—
International Water Management Institute (IWMI)	Colombo, Sri Lanka	—
International Institute for Environment and Development (IIED)	London, Britain	—
International Institute for Applied Systems Analysis (IIASA)	Laxenburg, Austria	—
International Geosphere-Biosphere Programme (IGBP)	Stockholm, Sweden	—
Stockholm Resilience Centre	Stockholm, Sweden	—
European Environment Agency (EEA)	Copenhagen, Denmark	—
Potsdam Institute for Climate Impact Research (PIK)	Potsdam, Germany	—
Smithsonian Environmental Research Center (SERC)	Edgewater, USA	—

> Global Energy, Maritime and Space Organisations

Name of the Organisation / Institution	Headquarters / Location (City, Country)
International Energy Agency (IEA)	Paris, France
Intergovernmental Oceanographic Commission (IOC-UNESCO)	Paris, France
European Space Agency (ESA)	Paris, France
International Renewable Energy Agency (IRENA)	Abu Dhabi, UAE
International Atomic Energy Agency (IAEA)	Vienna, Austria
International Maritime Organization (IMO)	London, Britain

> Principal Polar, Oceanographic and Meteorological Institutions (America and Europe)

Name of the Organisation / Institution	Headquarters / Location (City, Country)
National Oceanic and Atmospheric Administration (NOAA)	Silver Spring, USA

Name of the Organisation / Institution	Headquarters / Location (City, Country)
NASA Goddard Institute for Space Studies (GISS)	New York, USA
US Geological Survey (USGS)	Reston, USA
National Center for Atmospheric Research (NCAR)	Boulder, USA
Woods Hole Oceanographic Institution (WHOI)	Woods Hole, USA
International Polar Foundation (IPF)	Brussels, Belgium
Alfred Wegener Institute (AWI)	Bremerhaven, Germany
Scott Polar Research Institute (SPRI)	Cambridge, Britain
British Antarctic Survey (BAS)	Cambridge, Britain
Commonwealth Scientific and Industrial Research Organisation (CSIRO)	Canberra, Australia
Australian Institute of Marine Science (AIMS)	Townsville, Australia

> Principal International Agricultural and Forestry Research Institutes

Name of the Organisation / Institution	Headquarters / Location (City, Country)	UPPCS Special / Indian Context
International Crops Research Institute for the Semi-Arid Tropics (ICRISAT)	Patancheru (Hyderabad), India	Extremely important: the only international agricultural institute located in India that is included in your data.
International Rice Research Institute (IRRI)	Los Baños, Philippines	International rice research institute (a regional centre of it has also been set up at Varanasi, UP).
Center for International Forestry Research and World Agroforestry (CIFOR-ICRAF)	Nairobi, Kenya	—

Miscellaneous

> Principal Geographical Sobriquets

A. International Geographical Sobriquets

Geographical Sobriquet (Epithet)	Related City / Country / Place (UPPCS Direct One-liner)
Key to the Mediterranean / Pillars of Hercules	Strait of Gibraltar (<i>extremely important mapping fact</i>)
Sugar Bowl of the World / Pearl of the Antilles	Quebec / Cuba (<i>Cuba</i>)
Bread Basket of the World / Never-Never Land	Prairies / Prairies of the North (North America)
Sick Man of Europe / Gateway of the West	Turkey (Istanbul)
Playground of Europe	Switzerland
Cockpit of Europe	Belgium
Powder Keg of Europe	Balkan region
Garden of Lakes / Land of a Thousand Lakes	Finland
Empire City / City of Skyscrapers	New York (United States of America)
Smoky City / Windy City / Garden City	Chicago (United States of America)
Oil Capital of Europe / Granite City	Aberdeen (Scotland)
Queen of the Adriatic / City of Canals	Venice (Italy)
Babylon of the West	Rome (Italy)
Venice of the World	Stockholm (Sweden)
Venice of the North	St. Petersburg (Russia)
Pearl of Siberia	Lake Baikal (Russia)
Land of the Rising Sun / Britain of the East / Land of Delicate Flowers	Japan
Manchester of the East	Osaka (Japan)
Mistress of the Eastern Seas / Pearl of the East / Pearl of the Indian Ocean	Sri Lanka
Land of White Elephants / Paris of Asia	Thailand
Land of the Sunset / Land of the Midnight Sun	Norway
Land of the Morning Calm / Hermit Kingdom	Korea

Geographical Sobriquet (Epithet)	Related City / Country / Place (UPPCS Direct One-liner)
Land of Kangaroos / Land of the Golden Fleece (golden wool)	Australia
Britain of the South	New Zealand
Land of the Thunderbolt	Bhutan
Land of Golden Pagodas	Myanmar
Land of Canals / Land of Rivers	Pakistan (of canals), Bangladesh (of rivers)
Gate of Tears	Bab-el-Mandeb Strait
Island of Cloves	Zanzibar (Tanzania)
City of Eternal Springs	Quito (Ecuador)
Land of a Million Elephants	Laos
Loneliest Island in the World	Tristan da Cunha
Land of Promise	Mindanao (Philippines)
City of Dreaming Spires / Garden of England	Oxford (England), Kent (England)
Sorrow of China	Hwang Ho river (Yellow River)
Emerald Isle / Land of the Lily	Ireland (Emerald), Canada (Land of the Lily)
Pamir Plateau	Roof of the World
White Man's Grave	Guinea Coast (West Africa)
Land of Windmills	Netherlands
Pearl of the Orient / City of the Golden Gate	Singapore (Pearl), San Francisco (Golden Gate)
Quaker City / Forbidden City	Philadelphia (Quaker), Lhasa-Tibet (Forbidden)
Land of Cakes / Island of Pearls / City of Monuments	Scotland (Cakes), Bahrain (Pearls), Vienna (Monuments)
Blue Grass State / Paradise of the World	Kentucky-USA (Blue Grass), Paris-France (Paradise of the World)

B. Indian Geographical Sobriquets

Geographical Sobriquet	Related City / State / River (UP Special and India Special)
Backbone of the economy of North India	Ganga river
Sorrow of Bengal / Sorrow of Bihar	Damodar river (Bengal), Kosi river (Bihar)
Land of Five Rivers / City of the Golden Temple	Punjab (land), Amritsar (city)
Manchester of India	Ahmedabad (Gujarat)
Gateway of India / City of Seven Islands	Mumbai (Maharashtra)
Queen of the Arabian Sea / Venice of the East	Kochi / Cochin (Kerala)
Spice Garden of India	Kerala
Switzerland of India	Kashmir
Twin Cities	Hyderabad - Secunderabad
Queen of Chhota Nagpur	Netarhat (Jharkhand)
Queen of the Hills (India)	Mussoorie (Uttarakhand)

> Principal Straits of the World and their Connections

Name of the Strait	Which Seas/Gulfs does it Connect?	Which Land Masses does it Separate? (Data Reference)
Strait of Gibraltar	Mediterranean Sea and Atlantic Ocean	Separates Spain (Europe) and Morocco (Africa).
Bering Strait	Bering Sea and Chukchi Sea / Arctic	Separates Alaska (USA) and Russia (Asia).
Strait of Dover	English Channel and North Sea / Atlantic	Separates Britain and France.
Strait of Malacca	Andaman Sea / Bay of Bengal and Java Sea	Malaya (Malaysia) and Sumatra (Indonesia).
Palk Strait	Bay of Bengal and Arabian Sea (Gulf of Mannar)	Separates India and Sri Lanka.
Strait of Florida	Gulf of Mexico and Atlantic Ocean	Florida (USA) and the West Indies.
Strait of Hormuz	Persian Gulf and Gulf of Oman	Oman/UAE and the Iran region.
Bab-el-Mandeb	Red Sea and Arabian Sea (Gulf of Aden)	Yemen (Asia) and Djibouti (Africa).

Name of the Strait	Which Seas/Gulfs does it Connect?	Which Land Masses does it Separate? (Data Reference)
Bosphorus Strait	Black Sea and Sea of Marmara	The two parts of Istanbul/Turkey.
Dardanelles Strait	Sea of Marmara and Aegean Sea	The Turkish region.
Sunda Strait	Java Sea and Indian Ocean	The islands of Sumatra and Java.
Bass Strait	Tasman Sea and Southern Sea	The Australian mainland and Tasmania.
Davis Strait	Baffin Bay and Atlantic Ocean	Greenland and Canada.
Denmark Strait	North Atlantic and Arctic Ocean	Greenland and Iceland.
Torres Strait	Arafura Sea and Coral Sea	Papua New Guinea and Australia.
Makassar Strait	Java Sea and Celebes Sea	The islands of Borneo and Sulawesi.
Yucatán Strait	Gulf of Mexico and Caribbean Sea	Mexico and Cuba.
Strait of Otranto	Adriatic Sea and Ionian Sea	Italy and Albania.
North Channel	Irish Sea and Atlantic Ocean	Ireland and Scotland.
Luzon Strait	South China Sea and Philippine Sea	Taiwan and Luzon island (Philippines).
Hudson Strait	Hudson Bay and Atlantic Ocean	Baffin Island of Canada and the mainland.
Korea Strait	Sea of Japan and East China Sea	South Korea and Japan.
Strait of Magellan	Pacific Ocean and South Atlantic Ocean	The Chilean mainland and Tierra del Fuego.

➤ Important International Boundary Lines

Name of the Boundary Line	Between which Countries / Regions?
McMahon Line	India and China
Radcliffe Line	India and Pakistan
Durand Line	Pakistan and Afghanistan
38th Parallel	South Korea and North Korea
49th Parallel	United States of America (USA) and Canada
17th Parallel	Vietnam (North and South)

Name of the Boundary Line	Between which Countries / Regions?
31st Parallel	Iran and Iraq
20th / 22nd Parallel	Sudan-Libya / Sudan-Egypt
25th Parallel	Mauritania and Mali
Hindenburg / Oder-Neisse	Poland and Germany
Maginot / Siegfried Line	France and Germany
Mannerheim Line	Russia and Finland
Blue Line	Lebanon and Israel
Green Line / UN Buffer Zone	Republic of Cyprus and Turkish Cyprus

> Principal Global Cities Located on the Banks of Rivers

Name of the City	Name of the River on which Located	Name of the City	Name of the River on which Located
New York	Hudson	Hamburg	Elbe
Lahore	Ravi	Kyiv	Dnieper
Cairo	Nile	Rotterdam	Rhine
Khartoum	At the confluence of the White Nile and the Blue Nile	Quebec City	St. Lawrence
Perth	Swan	Warsaw	Vistula

> Classification of Soil Particles and their Diameter Sizes

- **Special note:** The Uttar Pradesh Public Service Commission (UPPSC) asks direct questions to arrange the sizes of these particles in **ascending** or **descending** order.

Name of the Soil Particle	Exact Diameter Size of the Particles
1. Gravel	Particles with a diameter of more than 5.000 mm (<i>the largest particles</i>)
2. Fine gravel	From 2.000 mm to 5.000 mm
3. Coarse sand	From 0.200 mm to 2.000 mm
4. Fine sand	From 0.020 mm to 0.200 mm
5. Silt	From 0.002 mm to 0.020 mm

Name of the Soil Particle	Exact Diameter Size of the Particles
6. Clay	Particles with a diameter of less than 0.002 mm (<i>the smallest particles - highest water-holding capacity</i>)

> Local Names of Shifting (Jhum) Cultivation in the World

Name of the Cultivation	Related Country / Geographical Region (UPPCS Direct Matching Special)
Ray	Vietnam
Masole	Congo (Zaire river valley)
Fang (Fagang)	Equatorial African countries
Logan	Western Africa
Milpa	Mexico and Central America
Conuco (Koko)	Venezuela
Roca	Brazil
Chitemene	Uganda, Zambia and Zimbabwe
Ladang	Java and Indonesia

> Various Regional Names of Cyclones

Regional Name of the Cyclone	Related Sea / Country / Geographical Region (UPPCS Direct Matching Special)
Cyclone	Indian Ocean
Hurricane	Caribbean Sea / Atlantic Ocean
Typhoon	China Sea
Taifu	Japan
Willy Willy	Northern Australia
Baguio	Philippines
Tornado	America (USA)

> Important Ports of the World (Major Global Ports)

Name of the Port	Location / Related Country	Name of the Port	Location / Related Country
Vancouver	Canada (Pacific coast)	Yokohama	Japan
New Orleans	United States of America (USA)	Manila	Philippines
Santos	Brazil (<i>the world-famous 'coffee port'</i>)	Jakarta	Indonesia
Cape Town / Nolloth	South Africa (<i>Nolloth: Port Diamond</i>)	Auckland	New Zealand
Cairo / Alexandria	Egypt	Fremantle	Australia (western coast)
Copenhagen	Denmark	Colombo	Sri Lanka
Lisbon	Portugal	Gwadar	Pakistan (Balochistan province)
La Skhira / Tripoli	Tunisia (La Skhira), Libya (Tripoli)	Murmansk / Igarka	Russia (<i>Murmansk remains open even in winter</i>)
St. Petersburg	Russia (Baltic coast)		

> Special Types of Culture (Special Culture Types)

Name of the Culture / Production	Subject Matter (Commercial production of what?)
Viticulture	Commercial-scale production of grapes .
Sericulture	Rearing of silkworms for silk production .
Pisciculture	The practice of fish farming on a commercial scale.
Apiculture	Beekeeping for the production of honey and wax.
Oliviculture	Commercial-scale cultivation of the olive .
Horticulture	Commercial-scale gardening (fruits, flowers, vegetables).
Arboriculture	Cultivation and propagation of special kinds of trees and shrubs .

> Transformation of Rocks (Metamorphic Rocks Classification)

Igneous Rock (Original)	Metamorphosed Form	Sedimentary Rock (Original)	Metamorphosed Form
Granite	Gneiss	Shale	Shale schist
Syenite	Syenite gneiss	Sandstone	Quartzite

Igneous Rock (Original)	Metamorphosed Form	Sedimentary Rock (Original)	Metamorphosed Form
Gabbro	Serpentine	Limestone	Marble
Basalt	Schist	Lignite coal	Anthracite coal

03

Indian Geography

GEOGRAPHY OF INDIA

> India's Neighbouring Countries and Border States/UTs

Neighbouring Country	Length of the Border	Associated Indian States / Union Territories (UT)
Bangladesh	4096.07 km	West Bengal, Assam, Meghalaya, Tripura, Mizoram. <i>(the longest border)</i>
China	3488.00 km	Ladakh (UT), Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh.
Pakistan	3323.00 km	Ladakh (UT), Jammu and Kashmir (UT), Punjab, Rajasthan, Gujarat.
Nepal	1751.00 km	Uttarakhand, Uttar Pradesh, Bihar, West Bengal, Sikkim.
Myanmar	1643.00 km	Arunachal Pradesh, Nagaland, Manipur, Mizoram.
Bhutan	699.00 km	Sikkim, West Bengal, Assam, Arunachal Pradesh.
Afghanistan	106.00 km	Ladakh (UT). <i>(the shortest border)</i>

> Important Channels and Straits of India (Water Channels)

Channel / Gulf / Passage	Geographical Location (between which?)
8 Degree Channel	Between the Maldives and Minicoy
9 Degree Channel	Between Lakshadweep and Minicoy
10 Degree Channel	Between Andaman and Nicobar <i>(repeated often in UPPCS)</i>
Duncan Passage	Between South Andaman and Little Andaman
Great Channel	Between Sumatra of Indonesia and Nicobar
Coco Strait	Between the Coco Islands of Myanmar and North Andaman
Gulf of Mannar	Between south-eastern Tamil Nadu and Sri Lanka

> Regional Classification of the Himalaya according to Sidney Burrard

Part of the Himalaya	Between which Rivers is it Located?	Total Length
Punjab Himalaya	Between the Indus and the Sutlej	560 kilometres
Kumaon Himalaya	Between the Sutlej and the Kali	320 kilometres
Nepal Himalaya	Between the Kali and the Teesta	800 kilometres (<i>the longest part</i>)
Assam Himalaya	Between the Teesta and the Dihang	750 kilometres

> Mountains and Highest Peaks of the Himalaya / Central and South India

> Classification of the Himalaya and Mountain Ranges (Chronology)

A. Order of Physiographic Division from North to South

Trans-Himalaya → Great Himalaya (Himadri) → Himachal (Lesser) Himalaya → Outer / Shiwalik Himalaya → Indo-Gangetic Plain

B. Correct Order of the Mountain Ranges from North to South

Region	Correct Order of the Ranges from North to South (UPPCS Chronology Special)
Trans-Himalayan ranges	Karakoram → Ladakh → Zaskar → Kailash
Lesser Himalayan ranges	Pir Panjal (J&K - the longest) → Dhauladhar (HP) → Mussoorie (UK) → Mahabharat (Nepal)

> Principal Mountain Peaks of the Great Himalaya

Mountain Peak	Official Height (metres)
Mount Everest	8,848.86 m
Kanchenjunga	8,598 m
Makalu	8,481 m
Dhaulagiri	8,172 m
Nanga Parbat	8,126 m
Annapurna	8,078 m
Nanda Devi	7,817 m

> The Five Highest Mountain Peaks of India

Name of the Mountain Peak	Location	Official Height (metres)
1. K2 (Godwin Austen)	Karakoram Range	8,611 m
2. Kanchenjunga	Sikkim	8,586 m
3. Nanda Devi	Uttarakhand	7,816 m
4. Kamet	Uttarakhand	7,756 m
5. Saltoro Kangri	Ladakh	7,742 m

> Mountains and Highest Peaks of Central and South India

Mountain Range / Hill	Name of the Highest Peak	Height (metres)	Special Details (UPPCS Facts)
Aravalli Range	Guru Shikhar	1,722 m	One of the oldest ranges in the world.
Satpura Range	Dhupgarh	1,350 m	Located on the Mahadeo hills.
Maikal Range	Amarkantak	1,065 m	It is the source of the Narmada and Son rivers .
Chhota Nagpur Plateau	Parasnath Hill	—	A famous pilgrimage centre of Jainism.
Nilgiri Hills	Doddabetta	2,637 m	The highest peak of the Nilgiris.
Anamalai Hills	Anamudi	2,695 m	The highest peak of South India.
Cardamom (Elaichi)	—	—	The southernmost mountain region of peninsular India.

Correct order of the South Indian hills from north to south: Nilgiri → Anamalai → Cardamom.

> Eastern Coastal Hills (Geographical Order from North to South)

Name of the Hill	Location (State)	Geographical Order (North to South)
Mahendragiri	Odisha	1 (northernmost)
Nallamalai / Velikonda / Palkonda / Nagari	Andhra Pradesh	2
Javadi / Shevaroy / Panchamalai / Sirumalai	Tamil Nadu	3 (southernmost)

> Principal Hills of North-East India

Name of the State	Principal Hills Located (UPPCS Matching Special)
Arunachal Pradesh	Dafla → Miri → Abor → Mishmi → Patkai Bum (<i>order from west towards east/north</i>)
Meghalaya	Garo → Khasi → Jaintia hills (<i>exact west-to-east order</i>)
Assam	Mikir hills, North Rengma, Barail hills
Nagaland / Manipur	Naga hills (Nagaland), Barail/Bal hills (Manipur)
Mizoram	Mizo hills, Lushai hills

> Short Quick Points on the Geographical Order of Mountain Ranges and Hills:

- **Correct order from north to south (N to S):** Karakoram → Ladakh → Zaskar → Pir Panjal.
- **Correct order from west to east (W to E):** Garo hills → Khasi hills → Jaintia hills → Mikir hills.

> Geographical Order of Mountain Ranges and Hills (Pointers)

A. Correct Order from North to South (Geographical Order - N to S)

Karakoram → Ladakh → Zaskar → Pir Panjal

B. Correct Order from West to East (Geographical Order - W to E)

Garo → Khasi → Jaintia → Mikir

> Exact North-to-South Order of the Principal Hills of the Eastern Ghats

State	Correct Geographical Order from North to South (N to S Chronology)
Andhra Pradesh	Nallamalai hills → Velikonda hills → Palkonda hills → Nagari hills
Tamil Nadu	Javadi hills → Shevaroy hills → Panchamalai hills → Sirumalai hills

> Coastal Plains and Delta Regions of India

Western Coastal Plain	Geographical Extent	Eastern Coastal Plain and Delta	Geographical Location
Konkan coast	Between Daman (Maharashtra) and Goa	Utkal coast / Mahanadi delta	Odisha
Kannada or Kanara coast	Between Goa and Mangaluru (Karnataka)	Northern Circars coast / Godavari and Krishna delta	Odisha and Andhra Pradesh
Malabar coast	From Mangaluru to Kanyakumari	Coromandel coast / Kaveri delta	Andhra Pradesh and Tamil Nadu

> Principal Valleys of India

State / UT	Names of the Principal Valleys (UPPCS Matching Special)
Ladakh	• Markha Valley • Suru Valley
Jammu and Kashmir	• Kashmir Valley (<i>between the Great Himalaya and the Pir Panjal</i>)
Himachal Pradesh	• Kinnaur Valley • Parvati Valley (<i>Kullu</i>) • Sangla Valley • Malana Valley • Kullu Valley • Pangi Valley • Spiti Valley • Lahaul Valley • Kangra Valley • Chamba Valley
Uttarakhand	• Doon Valley • Johar Valley • Valley of Flowers (Chamoli) • Dharma Valley • Sor Valley (<i>Pithoragarh</i>) • Nelang Valley (<i>Uttarkashi</i>)
West Bengal	• Neora Valley (<i>Kalimpong, Darjeeling</i>)
Sikkim	• Chumbi Valley • Yumthang Valley
Nagaland	• Dzukou Valley
Andhra Pradesh	• Araku Valley
Tamil Nadu	• Cumbum Valley
Kerala	• Silent Valley

> Principal Mountain Passes of India

State / UT	Names of the Principal Passes (UPPCS Matching Special)
Ladakh	Chang La pass, Khardung La pass
Uttarakhand	Lipulekh pass, Mana pass, Niti pass
Sikkim	Nathu La pass, Jelep La pass
Arunachal Pradesh	Bomdi La pass, Yang Yap pass, Diphu pass, Chankan pass
Manipur	Tuju pass

> Passes of the Western Ghats (correct chronological order from north to south):

Thal Ghat (Nashik-Mumbai) → Bhor Ghat (Mumbai-Pune) → Palghat (Coimbatore-Cochin) → Shencottah (Thiruvananthapuram-Madurai)

> Principal Rivers of India and their Sources

Name of the River	Source (Where does it rise?)	Location (State / Country)
Indus	Mansarovar lake (near the Bokhar Chu glacier)	Tibet (China)
Ganga	Gangotri glacier (as the Bhagirathi)	Uttarakhand
Yamuna	Yamunotri glacier (Bandarpunch peak)	Uttarakhand
Jhelum	Verinag spring (Sheshnag lake, Pir Panjal range)	Jammu and Kashmir
Ravi	Rohtang pass (Kullu hills)	Himachal Pradesh
Chambal	Janapav hill (near Mhow)	Madhya Pradesh
Kosi	Saptakoshi region	Nepal
Narmada	Amarkantak hills	Madhya Pradesh
Godavari	Trimbakeshwar hills (Nashik district)	Maharashtra
Krishna	From near Mahabaleshwar	Maharashtra
Kaveri	Brahmagiri hills (Coorg district)	Karnataka

> Ganga River System: Tributaries Joining from the Left and Right Banks

Rivers Joining the Ganga from the Left Bank	Rivers Joining the Ganga from the Right Bank
• Ramganga • Gomti • Ghaghara • Gandak • Burhi Gandak • Kosi • Mahananda • Brahmaputra (joins as the Padma in Bangladesh)	• Yamuna • Tons • Son

Correct order of the tributaries from west to east:

Yamuna → Tons → Gomti → Ghaghara → Son → Gandak → Burhi Gandak → Kosi → Mahananda → Hooghly → Brahmaputra

> Principal Rivers and Tributaries of South India (Peninsular India)

A. Decreasing Order of Length of the Rivers:

Godavari → Krishna → Narmada → Mahanadi → Kaveri

B. Tributary Systems of the Principal Rivers:

Name of the Main River	Its Principal Tributaries (Extremely Important for UPPCS Matching)
Godavari (<i>Vridha Ganga</i>)	Purna, Pranhita, Wainganga, Penganga, Wardha, Manjira, Indravati.

Name of the Main River	Its Principal Tributaries (Extremely Important for UPPCS Matching)
Krishna	Tungabhadra, Koyna, Ghatprabha, Malprabha, Musi, Bhima, Panchganga, Dudhganga.
Kaveri	Amaravati, Bhavani, Hemavati, Kabini.
Mahanadi	Shivnath, Hasdeo, Ib, Tel.

> Regional (River Valley) Division of the Himalaya

Part of the Himalaya	Between which Rivers is it Located?	Total Length
Punjab Himalaya	Between the Indus and the Sutlej	560 kilometres
Kumaon Himalaya	Between the Sutlej and the Kali	320 kilometres
Nepal Himalaya	Between the Kali and the Teesta	800 kilometres (<i>the longest part</i>)
Assam Himalaya	Between the Teesta and the Dihang (Brahmaputra)	750 kilometres

> Principal Waterfalls of India

Waterfall	Related River	Location / State	Special Exam Fact
Kunchikal	Varahi river	Karnataka	The highest waterfall in India.
Jog / Gersoppa / Mahatma Gandhi	Sharavati river	Karnataka	—
Shivasamudram	Kaveri river	Karnataka	—
Gokak	Ghataprabha river	Belagavi (Karnataka)	—
Dudhsagar	Mandovi river	Goa and Karnataka border	—
Dhuandhar	Narmada river	Jabalpur (Madhya Pradesh)	—
Dugdhdhara / Kapildhara	Narmada river	Madhya Pradesh	—
Chachai	Bihad river	Madhya Pradesh	—
Bhalkund / Rahatgarh	Bina river	Sagar (Madhya Pradesh)	—
Chitrakote	Indravati river	Bastar (Chhattisgarh)	Called " the Niagara of India ".

Waterfall	Related River	Location / State	Special Exam Fact
Tirathgarh	Kanger river	Chhattisgarh	—
Amritdhara / Devdhara	Hasdeo river	Chhattisgarh	—
Kendai	—	Chhattisgarh	—
Lodh / Burha Ghagh	Burha river	Latehar (Jharkhand)	—
Jonha / Gautamdharma	Rarhu river	Ranchi (Jharkhand)	—
Dassam / Dasong	Kanchi river	Ranchi (Jharkhand)	—
Sadni	Shankh river	Jharkhand	—
Chuliya	Chambal river	Rajasthan	—
Duduma	Machkund river	Koraput (Odisha)	—
Pykara	Pykara river	Tamil Nadu	In the Nilgiri region.

> Principal Lakes of India, State-wise

State / UT	Names of the Principal Lakes (Quick Pointers)
Jammu and Kashmir	Wular lake, Dal lake, Zanskar lake, Anantnag, Nagin, Verinag, Sheshnag, Tso Moriri, Manasbal.
Uttarakhand	Nainital, Bhimtal, Naukuchiatal, Khurpatal, Roopkund (<i>the mysterious lake</i>), Sattal, Naini, Malatal.
Rajasthan	Pichola, Sambhar, Pachpadra, Rajsamand, Jaisamand, Didwana, Pushkar, Fateh Sagar, Udai Sagar, Nakki, Dhebar, Lunkaransar.
Sikkim	Gurudongmar (<i>at the highest altitude</i>), Cholamu lake, Tsomgo lake.
Kerala	Sasthamkotta, Ashtamudi, Vembanad lake.
Tamil Nadu	Berijam, Ooty, Kodaikanal, Chembarambakkam, Veeranam lake.
Assam	Deepor Beel, Son Beel, Chapanal, Chandubi.
Maharashtra	Lonar (<i>crater lake</i>), Powai, Gorewada, Salim Ali Sarovar.
Telangana	Hussain Sagar, Osman Sagar, Himayat Sagar.
Gujarat	Nal Sarovar, Narayan Sarovar, Hamirsar.
Andhra Pradesh	Kolleru lake.
Manipur	Loktak lake (<i>the largest freshwater lake of the North-East</i>)

State / UT	Names of the Principal Lakes (Quick Pointers)
Meghalaya / Mizoram	Umiam (Meghalaya), Palak (Mizoram).
Haryana / Punjab	Surajkund (Haryana), Kanjli, Harike (Punjab).

› The Peninsular Plateau of India: At a Glance

Physiographic Division	Principal Features and Divisions (UPPCS Core Points)	Highest Peaks / Height
General introduction	- The oldest and most stable landmass of India (ancient igneous and metamorphic rocks). - Area: ~16 lakh sq km (~48% of the country - the largest physical region). - Shape: triangular; general slope: from west to east.	—
1. Central Highlands (north of the Narmada river)	- Main parts: the Malwa, Bundelkhand, Baghelkhand and Chhota Nagpur plateaus. Malwa Plateau: mainly in Madhya Pradesh, average height 300–600 metres. Chhota Nagpur Plateau: mainly in Jharkhand (average height 700 metres). Extremely rich in mineral wealth (iron, coal, mica, bauxite).	Principal high peak of Chhota Nagpur: Parasnath (1,365 metres) (a Jain pilgrimage centre).
2. Deccan Plateau (south of the Narmada river)	- Main extent: Maharashtra, Karnataka, Telangana and Andhra Pradesh. - Formation: mainly of basalt rocks, because of which black soil (regur) is widespread here.	—
3. Aravalli and Vindhya-Satpura	Aravalli: one of the oldest mountain systems in the world. Vindhya and Satpura: between them the Narmada flows westward in a rift valley.	Highest peak of the Aravalli: Guru Shikhar (1,722 metres).
Drainage system (river slope)	West-flowing rivers (rift valley): Narmada and Tapi (principal rivers). East-flowing rivers (according to the slope): Mahanadi, Godavari, Krishna and Kaveri.	—

> Western Ghats versus Eastern Ghats: Comparative Table

Criterion / Feature	Western Ghats	Eastern Ghats
Geographical extent	• Parallel to the western coast of India, from the Tapti valley to Kanyakumari (~1,600 km).	• Parallel to the eastern coast of India, from Odisha to Tamil Nadu (~1,500 km).
States covered	• Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu.	• Parts of Odisha, Andhra Pradesh, Telangana, Karnataka and Tamil Nadu.
Continuity and height	• Higher (average 1,200 metres) and continuous. Can be crossed only through passes.	• Lower and discontinuous/fragmented , because the east-flowing rivers have badly dissected them.
Highest peak	• Anamudi / Anaimudi (2,695 metres) - located in Kerala (<i>it is also the highest peak of South India</i>).	• Jindhagada peak (1,690 metres) - located in Andhra Pradesh. (<i>Note: your data also mentions Mahendragiri, 1,501 m, Odisha, which is the second principal peak</i>).
Monsoon and rainfall	• They block the south-west monsoon and cause heavy rain on the western slope, while the interior of the Deccan becomes a rain-shadow region .	• Being parallel to the monsoon, such orographic rainfall is low here.
Constitutional/international status	• Recognised as a UNESCO World Natural Heritage Site .	—
Meeting point	The Western Ghats and Eastern Ghats meet at the Nilgiri hills; the highest peak of the Nilgiris is Doddabetta (2,637 metres).	

> Coastal Plains of India

Western Coastal Plain	States Covered	Eastern Coastal Plain	States Covered
1. Gujarat coast	Gujarat	1. Utkal coast	Odisha
2. Konkan coast	Maharashtra and Goa	2. Andhra coast	Andhra Pradesh
3. Kannada / Kanara coast	Karnataka	3. Coromandel coast	Tamil Nadu

Western Coastal Plain	States Covered	Eastern Coastal Plain	States Covered
4. Malabar coast	Kerala		

> Major Soils of India

Type of Soil	Main Characteristics	Abundance and Deficiency of Minerals / Elements	Main Area of Extent
Red and yellow soil	- Developed on crystalline igneous rocks in areas of low rainfall. - In the eastern and southern parts of the Deccan Plateau (~10.6% of the geographical area).	Abundance: rich in potash. Deficiency: phosphorus, humus, nitrogen and magnesia.	Eastern and southern parts of the Deccan Plateau, Chhattisgarh, Odisha.
Black soil (<i>Regur / cotton soil / Chernozem</i>)	- Has the property of self-ploughing (high moisture-retention capacity). - Most suitable for cotton cultivation.	Abundance: lime, iron, magnesia, aluminium in abundance. Deficiency: phosphate, nitrogen and humus.	Maharashtra, Gujarat, Madhya Pradesh, Rajasthan, Chhattisgarh, Odisha, Tamil Nadu.
Laterite soil	Formed by the leaching of lime and silica in areas with rainfall of more than 200 cm.	Abundance: iron oxide and aluminium oxide in abundance. Deficiency: nitrogen, phosphoric acid, potash, lime and organic matter.	Eastern Ghats, Rajmahal hills, Satpura, the hills of Meghalaya and the Western Ghats.

> Principal Nuclear Power Plants of India

Nuclear Power Plant	Location (State)	Special Exam Fact
Narora	Uttar Pradesh	Located in Bulandshahr district of Uttar Pradesh (UP Special).
Tarapur	Maharashtra	India's very first nuclear power plant (1969).
Rawatbhata	Rajasthan	Established with the cooperation of Canada.
Kalpakkam	Tamil Nadu	Also called the Indira Gandhi Centre for Atomic Research.
Kakrapar	Gujarat	Located near the Tapi river.
Kaiga	Karnataka	In the Kali river region.

> Principal Hydroelectric Stations and Projects of India

State / UT	Name of the Hydroelectric Project / Plant	Related River	Special Facts for the UPPCS Exam (Exam Insights)
Andaman and Nicobar	Kalpong project	Kalpong river	• The first hydroelectric project of this Union Territory (UT).
Uttarakhand	• Tehri project • Koteswar • Tanakpur • Dhauliganga	Bhagirathi + Bhilangana Bhagirathi Sharda (Mahakali) Dhauliganga	• The Tehri dam is the highest dam of India (260.5 metres). • Tanakpur is located near the India-Nepal border.
Himachal Pradesh	• Bhakra Nangal • Nathpa Jhakri • Karcham-Wangtoo • Kol Dam • Dehar • Chamera (I, II, III) • Baira Siul • Parbati-III	Sutlej Sutlej Sutlej Sutlej Beas Ravi Baira (tributary of the Ravi) Parbati	• Bhakra Nangal is one of the largest multipurpose projects of India (Himachal-Punjab border). • Nathpa Jhakri is one of the largest underground projects of the country.
Jammu-Kashmir (UT)	• Salal • Uri (I, II) • Dulhasti • Kishanganga • Sewa-II	Chenab Jhelum Chenab Kishanganga (tributary of the Jhelum) Sewa river	• The Kishanganga project has been in dispute between India and Pakistan over the Indus Waters Treaty. • Salal and Dulhasti are 'run-of-the-river' projects.
Karnataka	• Shivasamudram • Sharavati • Kali Nadi • Varahi	Kaveri Sharavati Kali river Varahi	• Shivasamudram (1902) is one of the earliest hydroelectric plants of Asia. • The Sharavati project is located near the famous 'Jog Falls'.
Kerala	• Idukki project • Sabarigiri	Periyar Pamba	• Idukki is a distinctive arch dam (double curvature arch dam) on the Periyar river.
Gujarat	• Sardar Sarovar	Narmada	• It is a joint project of four states (Gujarat, Madhya Pradesh, Maharashtra and Rajasthan).
Andhra Pradesh and Telangana	• Srisaillam project • Nagarjuna Sagar • Pulichintala • Machkund	Krishna Krishna Krishna Machkund	• Nagarjuna Sagar is a giant masonry dam located on the border of Andhra Pradesh and Telangana. • Machkund is a joint project of Andhra Pradesh and Odisha.

State / UT	Name of the Hydroelectric Project / Plant	Related River	Special Facts for the UPPCS Exam (Exam Insights)
Madhya Pradesh	- Indira Sagar - Omkareshwar - Gandhi Sagar	Narmada Narmada Chambal	Gandhi Sagar is the first of the three principal dams built on the Chambal river.
Sikkim	- Teesta-III and Teesta-V - Rangit	Teesta Rangit (tributary of the Teesta)	The Teesta project is one of the principal hydroelectric projects of north-eastern India.
North-East India	- Loktak (Manipur) - Tuirial (Mizoram) - Kameng, Subansiri (Arunachal)	Leimatak/Loktak lake Tuirial river Kameng / Subansiri	- The Loktak project is linked with Loktak, the largest natural freshwater lake of the North-East. - The Subansiri Lower Project is the largest hydroelectric project under construction in India.
Maharashtra	- Koyna - Bhira	Koyna (tributary of the Krishna) Kundalika	The Koyna project is also called the 'lifeline' of Maharashtra.
Odisha and Jharkhand	- Hirakud, Balimela (Odisha) - Subarnarekha-I and II (Jharkhand)	Mahanadi / Sileru Subarnarekha	The Hirakud dam is the longest dam of India.
West Bengal	Teesta Low Dam-III and IV	Teesta	Located on the lower course of the Teesta river in the Darjeeling region.

> Principal Thermal Power Plants of India

Name of the State	Names of the Thermal Power Plants (UPPCS Matching Special)
Uttar Pradesh	Singrauli • Rihand • Anpara • Obra • NTPC Dadri (<i>extremely important for UP Special</i>)
Chhattisgarh / Odisha	Korba (Chhattisgarh) • Talcher (Odisha)
Madhya Pradesh	Vindhyachal • Satpura • Sanjay Gandhi
Maharashtra / Gujarat	Khaperkheda, Chandrapur Super, Koradi (MH) • Ukai, Mundra, Surat (GJ)
Jharkhand / West Bengal	Bokaro, Patratu (Jharkhand) • Farakka (West Bengal)
Delhi / Haryana / Rajasthan	Badarpur (Delhi) • Rajiv Gandhi (Haryana) • Kota, Suratgarh, Chhabra (Rajasthan)
Tamil Nadu	Neyveli (<i>lignite based</i>) • Ennore • Tuticorin • North Chennai

Name of the State	Names of the Thermal Power Plants (UPPCS Matching Special)
Telangana / Andhra Pradesh / UK	• Ramagundam (Telangana) • Simhadri (AP) • Kashipur (Uttarakhand)

> Minerals and their Principal Producing Areas in India (Mines - UPPCS Special)

Name of the Mineral	State	Principal Producing Areas / Famous Mines (Extremely Important for UPPCS Matching Questions)
Iron ore	Odisha Chhattisgarh Karnataka Jharkhand / Goa	• Keonjhar, Sundargarh, Mayurbhanj • Bailadila, Dantewada (<i>high-quality haematite</i>) • Ballari, Chitradurga, Tumakuru • Singhbhum (Jharkhand), North Goa (Goa)
Coal	Jharkhand West Bengal Odisha / Chhattisgarh MP / Telangana	• Jharia (<i>the largest</i>), Bokaro, Giridih, Karanpura • Raniganj (<i>the oldest coalfield of India</i>) • Talcher, Ib Valley (Odisha), Korba (Chhattisgarh) • Singrauli, Umaria (Madhya Pradesh), Singareni (Telangana)
Copper	Rajasthan Madhya Pradesh / Jharkhand	• Khetri, Jhunjhunu • Malanjkhand, Balaghat (MP), Singhbhum (Jharkhand)
Bauxite (aluminium ore)	Odisha Jharkhand / Gujarat / Chhattisgarh	• Koraput, Kalahandi, Rayagada • Lohardaga (Jharkhand), Kutch, Jamnagar (Gujarat), Surguja, Kabirdham (CG)
Manganese	Madhya Pradesh Maharashtra / Odisha / Karnataka	• Balaghat • Nagpur, Bhandara, Sindhudurg (MH), Sundargarh, Keonjhar (OD), Ballari (KN)
Uranium	Jharkhand Andhra Pradesh / Telangana	• Jaduguda, Bhatin, Narwapahar • Tummalapalle, Kadapa (AP), Lambapur-Peddagattu (Telangana)
Mica	Jharkhand / Andhra Pradesh / Raj.	• Koderma (<i>the mica capital</i>), Giridih, Hazaribagh (JH), Nellore (AP), Bhilwara, Ajmer (RJ)
Gold	Karnataka / Jharkhand	• Hutti, Kolar (KGF) (Karnataka), Singhbhum region (Jharkhand)
Lead and zinc	Rajasthan	• Zawar, Rampura-Agucha, Rajpura-Dariba (<i>mixed ore</i>)
Silver	Rajasthan / Karnataka	• Zawar, Rampura-Agucha (RJ), Hutti region (KN)
Chromite / nickel	Odisha	• Sukinda valley (Jajpur) (<i>India's top region for both chromite and nickel</i>)

Name of the Mineral	State	Principal Producing Areas / Famous Mines (Extremely Important for UPPCS Matching Questions)
Rock phosphate	Rajasthan / Madhya Pradesh	• Jhamarkotra, Udaipur (RJ), Jhabua (MP)
Diamond	Madhya Pradesh	• Panna (<i>the only active diamond mine in India</i>)
Tin	Chhattisgarh	• Bastar, Dantewada (<i>Chhattisgarh is the only tin-producing state in India</i>)
Gypsum	Rajasthan / Jammu-Kashmir	• Bikaner, Nagaur, Jaisalmer, Barmer (RJ), Kishtwar, Udhampur (J&K)
Limestone	Raj. / MP / CG / AP	• Chittorgarh (RJ), Satna, Katni (MP), Raipur (CG), Kadapa, Kurnool (AP)
Dolomite	Chhattisgarh / Odisha / MP	• Bastar, Bilaspur (CG), Sundargarh (OD), Katni, Jabalpur (MP)
Magnesite	Tamil Nadu / Uttarakhand	• Salem, Dharmapuri (TN), Bageshwar, Pithoragarh (UK)

> Types of Coal in India and their Features

Type of Coal	Carbon Content	Quality and Main Features (UPPCS Core Pointers)	Main Areas of Occurrence
1. Anthracite	90% to 95%	- It is a hard coal of the best / highest grade. - Gives no smoke while burning and provides the most heat.	From Jammu-Kashmir .
2. Bituminous	55% to 80%	- It is second-grade coal. Most of India's coal is of this grade. - The coal of the Gondwana period is of this type. It represents 80 of the country's total 113 coalfields.	Jharkhand, Odisha, Chhattisgarh, W. Bengal.
3. Lignite	40% to 55%	- It is also called "brown coal" and is of inferior quality. - Because of its high sulphur content (2 to 7%), it is unsuitable for some industries.	Tamil Nadu (Neyveli).
4. Peat	Less than 40%	- It is coal of the lowest grade (the poorest), with very high moisture. - On burning it gives off a great deal of ash and smoke.	Marshy areas.

> Principal Petroleum-Producing Areas of India (Oil Fields)

State / Offshore Area	Principal Oil-Producing Wells / Basins (Core Field Pointers)
Assam	• Digboi (<i>India's very first oil well</i>) • Naharkatiya • Moran • Rudrasagar
Gujarat	• Ankleshwar • Kalol • Mehsana • Navagam • Kosamba
Mumbai offshore	• Mumbai High • Bassein and satellite fields (<i>India's principal producer</i>)
Rajasthan	• Barmer • Mangala • Bhagyam • Aishwarya (<i>names have been asked directly in UPPCS</i>)
Andhra Pradesh / Tamil Nadu	• Krishna-Godavari basin (AP) • Kaveri basin, Nagapattinam area (TN)
Arunachal Pradesh / Tripura	• Kharsang, Namchik-Namphuk (AR), Baramura, Trishna, Rokhia (TR)

> Additional UPPCS Trending Facts Related to Minerals and Energy (Revision)

- **Largest producer of bauxite:** Odisha (Koraput-Kalahandi belt).
- **Monopoly of chromite:** The Sukinda valley of Odisha produces more than about 90% of the country's chromite.
- **Famous copper refinery:** Khetri (Rajasthan), operated by Hindustan Copper Limited.
- **Largest reserve of lignite coal:** Neyveli (Tamil Nadu).

> Miscellaneous

> Principal Tribes of India and their Geographical Distribution

Name of the Tribe	Main States / Union Territories of Distribution (UPPCS Matching Special)
Bhil	Rajasthan, Gujarat, Madhya Pradesh, Maharashtra (<i>one of the largest tribal groups of India</i>)
Gond	Madhya Pradesh, Chhattisgarh, Maharashtra, Telangana, Andhra Pradesh, Odisha, Uttar Pradesh
Santhal	Jharkhand, West Bengal, Bihar, Odisha
Munda / Oraon	Jharkhand, Odisha, West Bengal, Chhattisgarh
Ho / Kharia / Birhor / Paharia	Jharkhand, Odisha, Chhattisgarh
Garo, Khasi, Jaintia	Meghalaya (<i>the tribes are named after these very hills</i>)
Bodo	Assam

Name of the Tribe	Main States / Union Territories of Distribution (UPPCS Matching Special)
Naga	Nagaland, Manipur, Arunachal Pradesh
Lepcha / Bhutia	Sikkim, West Bengal
Toda / Kota	Tamil Nadu (<i>the Toda tribe lives on the Nilgiri hills and is famous for polyandry</i>)
Irula / Kurumba	Tamil Nadu, Kerala
Chenchu / Koya	Andhra Pradesh, Telangana, Chhattisgarh, Odisha
Warli / Katkari	Maharashtra, Gujarat
Sahariya / Baiga / Korku	Madhya Pradesh, Chhattisgarh, Rajasthan, Uttar Pradesh, Maharashtra
Jarawa / Sentinelese	Andaman and Nicobar Islands (<i>extremely vulnerable PVTG groups</i>)

> Principal International Airports of India

Name of the Airport	Location (City, State)	Name of the Airport	Location (City, State)
Chaudhary Charan Singh	Lucknow (Uttar Pradesh)	Indira Gandhi	New Delhi
Lal Bahadur Shastri	Varanasi (Uttar Pradesh)	Chhatrapati Shivaji Maharaj	Mumbai (Maharashtra)
Netaji Subhas Chandra Bose	Kolkata (West Bengal)	Kempegowda	Bengaluru (Karnataka)
Sardar Vallabhbhai Patel	Ahmedabad (Gujarat)	Rajiv Gandhi	Hyderabad (Telangana)
Jai Prakash Narayan	Patna (Bihar)	Chennai International	Chennai (Tamil Nadu)
Lokpriya Gopinath Bordoloi	Guwahati (Assam)	Cochin International	Kochi (Kerala)
Birsa Munda	Ranchi (Jharkhand)	Thiruvananthapuram	Thiruvananthapuram (Kerala)
Biju Patnaik	Bhubaneswar (Odisha)	Goa International	Mopa (Goa)
Jaipur International	Jaipur (Rajasthan)	Dr. Babasaheb Ambedkar	Nagpur (Maharashtra)
Srinagar International	Srinagar (Jammu-Kashmir)	Sri Guru Ram Dass Ji	Amritsar (Punjab)

> Railway Zones of India and their Headquarters

Railway Zone	Headquarters	Railway Zone	Headquarters
Central Railway	Mumbai (Chhatrapati Shivaji Maharaj Terminus)	South Western Railway	Hubballi (Karnataka)
Western Railway	Mumbai Central	West Central Railway	Jabalpur (Madhya Pradesh)
Northern Railway	New Delhi	North Western Railway	Jaipur (Rajasthan)
North Central Railway	Prayagraj (Uttar Pradesh)	North Eastern Railway	Gorakhpur (Uttar Pradesh)
Eastern Railway	Kolkata	East Central Railway	Hajipur (Bihar)
South Eastern Railway	Kolkata	East Coast Railway	Bhubaneswar (Odisha)
Metro Railway, Kolkata	Kolkata	South Coast Railway	Visakhapatnam (Andhra Pradesh)
Southern Railway	Chennai (Tamil Nadu)	Northeast Frontier Railway	Maligaon, Guwahati (Assam)
South Central Railway	Secunderabad (Telangana)	South East Central Railway	Bilaspur (Chhattisgarh)

> Water Transport in India: Principal National Waterways

- **Inland water transport:** Transport of goods and passengers within the country through rivers, canals, backwaters (kayals) and lakes.
- **Total number declared:** At present **111 National Waterways** have been declared in India.

National Waterway	River System / Area of Extent	Exact Extent (from where to where)	Total Length	Special Fact
NW - 1	Ganga-Bhagirathi-Hooghly	Prayagraj to Haldia	1,620 km	The longest National Waterway of India.
NW - 2	Brahmaputra river	Sadiya to Dhubri (Assam)	891 km	The main waterway of north-eastern India.
NW - 3	West Coast Canal	Kottapuram to Kollam (Kerala)	205 km	India's first waterway navigable for 24 hours.
NW - 4	Godavari-Krishna rivers	Kakinada to the Puducherry canal system	—	South-eastern coastal network.
NW - 5	Brahmani-Mahanadi river system	Along with the East Coast Canal	—	Odisha/West Bengal region.

National Waterway	River System / Area of Extent	Exact Extent (from where to where)	Total Length	Special Fact
NW - 16	Barak river	Bhanga to Lakhimpur (Assam)	—	A strategic waterway of the North-East.

> Principal Agricultural Commodity Boards of India

Name of the Board	Headquarters (Location)
Coffee Board	Bengaluru (Karnataka)
Rubber Board	Kottayam (Kerala)
Tea Board	Kolkata (West Bengal)
National Jute Board	Kolkata (West Bengal)
Tobacco Board	Guntur (Andhra Pradesh)
Grape Processing Board	Pune (Maharashtra)

> Principal Oil Refineries of India

Name of the Refinery	Year of Establishment	Location (State)
Digboi	1901	Assam (<i>the oldest refinery of India</i>)
Mumbai (Trombay)	1955	Maharashtra
Visakhapatnam	1957	Andhra Pradesh
Noonmati (Guwahati)	1962	Assam
Barauni	1964	Bihar
Koyali	1965	Gujarat
Kochi	1966	Kerala
Haldia	1975	West Bengal
Mumbai (BP)	1979	Maharashtra
Mathura	1982	Uttar Pradesh
Bangalore	1988	Karnataka
Nagapattinam	1993	Tamil Nadu
Panipat	1998	Haryana

Name of the Refinery	Year of Establishment	Location (State)
Numaligarh	1999	Assam
Tatipaka	2001	Andhra Pradesh
Paradip	2016	Odisha

> Principal Steel Plants and their Establishment

Steel Plant	Year of Establishment	Collaborating Country (with whose help?)	Location (State)
Bhilai	1955	Russia (former Soviet Union)	Chhattisgarh
Rourkela	1959	Germany	Odisha
Durgapur	1959	Britain	West Bengal
Bokaro	1964	Russia (former Soviet Union)	Jharkhand

> Principal Aluminium Companies of India

Name of the Company	Year of Establishment	Collaborating Country	Principal Industrial Centres
INDALCO	1938	Canada	J.K. Nagar (West Bengal), Muri (Jharkhand), Alwaye (Kerala)
HINDALCO	1958	United States of America (USA)	Renukoot (Uttar Pradesh)
BALCO	1965	Soviet Union (USSR)	Korba (Chhattisgarh), Koyna (Maharashtra)
MALCO	1965	Italy	Mettur, Salem (Tamil Nadu)
VEDANTA	1976	Germany	Jharsuguda (Odisha)
NALCO	1981	France	Damanjodi (Odisha), Angul (Odisha)

> Principal Mineral Institutions

Name of the Mineral Institution	Headquarters / Location (Place)
Hindustan Copper Limited (HCL)	Khetri (Rajasthan)
Hindustan Zinc Limited (HZL)	Udaipur (Rajasthan)
Indian Bureau of Mines (IBM)	Nagpur (Maharashtra)
Mineral Exploration Corporation Limited (MECL)	Nagpur (Maharashtra)

Name of the Mineral Institution	Headquarters / Location (Place)
National Metallurgical Laboratory	Jamshedpur (Jharkhand - Burmahmines area)
Central Institute of Mining and Fuel Research	Dhanbad (Jharkhand)

> Principal River Valley Projects of India

Project	Related River	Beneficiary States / UT (UPPCS Matching Special)
Sabarigiri	Kakki / Pamba	Kerala
Nagarjuna Sagar	Krishna	Andhra Pradesh and Telangana (<i>masonry dam</i>)
Pochampad	Godavari	Telangana
Idukki	Periyar	Kerala
Pavana	Pavana	Maharashtra
Nathpa Jhakri	Sutlej	Himachal Pradesh
Suil	Suil	Himachal Pradesh
Bansagar	Son	Bihar, Uttar Pradesh, Madhya Pradesh
Ramganga	Ramganga	Uttarakhand
Nimoo Bazgo	Indus	Ladakh (UT)
Uri / Tulbul	Jhelum	Jammu Kashmir (UT)
Chutak	Suru	Kargil (Ladakh)
Baglihar / Dulhasti	Chenab	Jammu Kashmir (UT)
Almatti	Krishna	Andhra Pradesh, Karnataka, Maharashtra
Chambal	Chambal	Madhya Pradesh, Rajasthan
Damodar Valley	Damodar and Barakar	Jharkhand and West Bengal
Bhakra Nangal	Sutlej	Delhi, Himachal Pradesh, Punjab, Haryana, Rajasthan
Hirakud	Mahanadi	Odisha
Tehri	Bhagirathi + Bhilangana	Uttarakhand
Narmada Valley	Narmada	Gujarat, Rajasthan, Madhya Pradesh, Maharashtra
Rihand	Rihand	Uttar Pradesh

Project	Related River	Beneficiary States / UT (UPPCS Matching Special)
Kishanganga	Kishanganga	Jammu Kashmir (UT)
Gandak	Gandak	Uttar Pradesh, Bihar
Srisaillam	Krishna	Andhra Pradesh and Telangana
Mettur	Kaveri	Tamil Nadu
Sharavati	Sharavati	Karnataka, Goa and Tamil Nadu
Shivasamudram	Kaveri	Karnataka
Hidkal	Ghataprabha	Karnataka
Purna	Purna	Maharashtra
Kakrapar	Tapti	Gujarat
Ranjit Sagar (Thein)	Ravi	Punjab
Pong Dam	Beas	Himachal Pradesh
Sanjay Sarovar	Wainganga	Madhya Pradesh
Koyna	Koyna	Maharashtra
Matatila / Rani Laxmibai	Betwa	Uttar Pradesh
Tungabhadra	Pennar and Tungabhadra	Andhra Pradesh and Karnataka

> States and their Principal Hill Stations

State / UT	Principal Hill Stations
Uttarakhand	Ranikhet, Lansdowne, Mussoorie, Mukteshwar, Nainital, Almora, Bhowali
Himachal Pradesh	Dalhousie, Kasauli, Manali, Mandi
Jammu Kashmir (UT)	Gulmarg, Pahalgam, Srinagar
Tamil Nadu	Coonoor, Ooty (Udhagamandalam), Kodaikanal, Yercaud
Karnataka	Nandi Hills, Kemmangundi
Kerala	Munnar
Madhya Pradesh	Pachmarhi

State / UT	Principal Hill Stations
Meghalaya / West Bengal	Shillong (Meghalaya), Kalimpong, Darjeeling (West Bengal)

> Regional Names of Jhumming / Shifting Cultivation in India

Local Name	Related Region / State (India)
Jhum	North-East India (Assam, Meghalaya, Mizoram, Nagaland)
Pamlou	Manipur
Deepa	Chhattisgarh (Bastar), Andaman and Nicobar Islands
Bewar or Dahiya (Dahiyar)	Madhya Pradesh (Bundelkhand region)
Podu or Penda	Andhra Pradesh
Pama Dabi / Koman / Bringa / Kaman / Vinga / Dhavi	Odisha
Kumari	Hilly areas of the Western Ghats of Kerala
Valre or Waltre / Batra	South-eastern Rajasthan
Khil	Himalayan region
Kuruwa	Jharkhand
Jara and Erka	Southern states

> Main Facts Related to Indian Standard Time (IST)

Criterion / Component	Exam-Useful Core Points (IST Facts)
Meridian line	India's standard time is measured from the 82°30' East longitude .
Exact location	This line passes through Mirzapur in Uttar Pradesh (near Allahabad/Prayagraj).
Difference from GMT	It is 5 hours 30 minutes ahead (+5:30) of Greenwich Mean Time (GMT).
Longitudinal rule	- A difference of 15 degrees of longitude causes a difference of 1 hour in local time. - By mutual global understanding, the standard meridian is chosen at a multiple of 7°30' of longitude.

Criterion / Component	Exam-Useful Core Points (IST Facts)
Other historical facts	Before independence Mumbai and Kolkata had their own local times; these were abolished after independence and only one standard time for the whole of India was adopted.

> Principal Mountain Peaks of Andaman and Nicobar

Name of the Mountain Peak	Geographical Location (exact island)
1. Saddle Peak	North Andaman (<i>the highest peak of the Andaman Islands</i>)
2. Mount Diavolo	Middle Andaman
3. Mount Koyob	South Andaman
4. Mount Harriet	South Andaman
5. Mount Thullier	Great Nicobar (<i>the highest peak of the Nicobar Islands</i>)

> Correct North-to-South Order of the Islands of Andaman and Nicobar

Order (North to South)	Name of the Island (Andaman and Nicobar chain)
1 (northernmost)	North Andaman
2	Middle Andaman
3	South Andaman
4	Little Andaman
5	Car Nicobar
6	Little Nicobar
7 (southernmost)	Great Nicobar (<i>Indira Point is located on it</i>)

> The Peninsular Plateau of India: Delimitation of Boundaries and Average Height

Boundary Area (Direction)	Natural / Geographical Boundary
North-western boundary	Aravalli range and Delhi
Eastern boundary	Rajmahal hills
Western boundary	Gir hills (Gujarat)
North-eastern boundary	Shillong (Meghalaya) and the Karbi-Anglong plateau

Boundary Area (Direction)	Natural / Geographical Boundary
Average height	The average height of this entire plateau region is 600-900 metres .

> Principal Nuclear Power Plants of India

Name of the Nuclear Plant	Location (State)	Present Status / Special Fact
Tarapur	Maharashtra	India's first nuclear power station (1969).
Narora	Uttar Pradesh	Located in Bulandshahr district (UPPCS special).
Rawatbhata	Rajasthan	Established with Canadian cooperation .
Kakrapar / Mithi Viridi	Gujarat	Mithi Viridi is proposed with the cooperation of the United States of America (USA) .
Kaiga	Karnataka	—
Kalpakkam / Kudankulam	Tamil Nadu	Kudankulam built with Russian cooperation .
Jaitapur	Maharashtra	—
Kovvada	Andhra Pradesh	Proposed plant.
Haripur	West Bengal	Proposed plant.
Fatehabad	Haryana	Proposed plant.
Chutka	Madhya Pradesh	Proposed plant.

04

Indian Polity

INDIAN POLITY AND CONSTITUTION

> Constitutional Development of India

Name of the Act	Principal Provisions and Main Points (UPPCS Prelims Special)
Regulating Act of 1773	- The first step to control the affairs of the East India Company in India. - The Governor of Bengal was designated 'Governor-General of Bengal' (Warren Hastings became the first Governor-General). A Supreme Court was established at Calcutta (1774) (1 Chief Justice + 3 other judges).
Pitt's India Act of 1784	- The political and commercial functions of the Company were separated. - A new body named the 'Board of Control' was created (beginning of dual government). - The British government was given complete control over the affairs and administration of the Company in India.
Charter Act of 1793	- The Company's trade monopoly was extended for another 20 years. - Members and staff of the Board of Control began to be paid out of Indian revenues.
Charter Act of 1813	- The Company's trade monopoly was ended, except for trade in tea and with China. Christian missionaries were permitted to come to India to preach/enlighten the people. - Local governments got the power to levy taxes on persons and to punish those who did not pay.
Charter Act of 1833	- The Governor-General of Bengal was made the 'Governor-General of India' (Lord William Bentinck became the first Governor-General of India). - The Governors of Madras and Bombay were deprived of legislative powers (complete centralisation of administration). - An (unsuccessful) attempt was made to introduce open competition for the selection of civil servants.
Charter Act of 1853	- For the first time the legislative and executive functions of the Governor-General's Council were separated. - The Macaulay Committee was appointed in 1854 for recruitment to the civil service.

Name of the Act	Principal Provisions and Main Points (UPPCS Prelims Special)
Government of India Act of 1858	- The governance of India passed directly under the British Crown; the designation of Governor-General was changed to 'Viceroy of India' (Lord Canning became the first Viceroy of India). - A new post of 'Secretary of State for India' was created. - A 15-member council (advisory committee) was formed to assist the Secretary of State.
Indian Councils Act of 1861	- A beginning was made of associating Indian representatives with the law-making process. - In 1862 Lord Canning nominated three Indians (the Raja of Banaras, the Maharaja of Patiala and Sir Dinkar Rao) to the legislative council. - The Viceroy was empowered to issue ordinances in an emergency. Legislative councils were set up in Bengal (1862), the North-Western Frontier Province (1866) and Punjab (1897).
Indian Councils Act of 1892	The functions of the legislative councils were increased; members got the right to discuss the budget and to ask questions of the executive.
Indian Councils Act of 1909 (<i>Morley-Minto Reforms</i>)	- The size and scope of functions of the central and provincial legislative councils were enlarged (asking supplementary questions, moving resolutions on the budget). - Provision was made for separate electorates / communal representation for Muslims.
Government of India Act 1919 (<i>Montagu-Chelmsford Reforms</i>)	- The administrative lists of the centre and the provinces were separated and the provinces were given the right to legislate on their own list. Dyarchy was introduced in the provinces; the subjects were divided into two parts—'Transferred' and 'Reserved'. Bicameralism and direct elections were introduced in the country for the first time. - In 1926 a Central Public Service Commission (on the recommendation of the Lee Commission) was set up for the recruitment of civil servants. The central budget was separated from the budgets of the provinces.
Government of India Act 1935	Dyarchy in the provinces was abolished and they were given autonomy, and a system of dyarchy at the centre was introduced. Residuary powers were given to the Viceroy. - The 'Council of India' established by the Act of 1858 was abolished. - Along with a Federal Public Service Commission (FPSC), Provincial Service Commissions and a Joint Service Commission (JPSC) for two or more provinces were established. - A Federal Court was established in 1937.

> The Constituent Assembly

> Constituent Assembly: Important Facts

Subject / Event	Principal Facts
Basis of formation	On the recommendation of the Cabinet Mission Plan, 1946 .
Allocation of seats	1 representative for a population of every 10 lakh (1 million) persons.
Time of election	Elections were completed in July–August 1946 .
Original strength	A total of 389 fixed (296 from British India + 93 from the princely states).
Strength after Partition	After the Muslim League separated, 299 members in all remained (229 from the Indian provinces + 70 from the princely states).
First meeting	On 9 December 1946 (only 207 members attended).
Temporary (provisional) President	Dr. Sachchidananda Sinha (on 9 December 1946; being the eldest member).
Permanent President	Dr. Rajendra Prasad (elected unopposed on 11 December 1946).
Vice-Presidents	H.C. Mukherjee and V.T. Krishnamachari (there were two Vice-Presidents).
Secretariat officers	• Secretary: H.V.R. Iyengar • Chief Draftsman: S.N. Mukherjee

> The Drafting Committee and its Members

Name of the Committee	Date of Formation	Total Number of Members	Members and their Special Details
Drafting Committee	29 August 1947	7 members (Chairman + 6 others)	1. Dr. Bhimrao Ambedkar (Chairman) 2. N. Gopalaswamy Ayyangar 3. Alladi Krishnaswamy Ayyar 4. Dr. K.M. Munshi (the only member who was originally a Congressman) 5. Syed Mohammad Saadulla 6. N. Madhava Rau (he took the place of B.L. Mitter, who withdrew for reasons of health) 7. T.T. Krishnamachari (became a member in 1948 after the death of D.P. Khaitan)

- **Publication of the draft (Quick Revision):** The first draft was published in **February 1948**, the second draft in **October 1948**, and the final draft was presented to the Assembly by Dr. Ambedkar on **4 November 1948**.

> Principal Committees of the Constituent Assembly and their Chairmen

Name of the Committee	Name of the Chairman
Union Powers Committee / Union Constitution Committee / Committee for (Negotiating with) the States	Jawaharlal Nehru
Provincial Constitution Committee	Sardar Vallabhbhai Patel
Drafting Committee	Dr. B.R. Ambedkar
Advisory Committee on Fundamental Rights and Minorities	Sardar Vallabhbhai Patel
• <i>Fundamental Rights Sub-Committee</i>	• J.B. Kripalani
• <i>Minorities Sub-Committee</i>	• H.C. Mukherjee
Rules of Procedure Committee / Steering Committee	Dr. Rajendra Prasad

> Parts of the Indian Constitution/ Schedules

> Parts of the Indian Constitution and the Related Articles

Part	Subject	Related Articles
I	The Union and its Territory	1 – 4
II	Citizenship	5 – 11
III	Fundamental Rights	12 – 35
IV	Directive Principles of State Policy (DPSP)	36 – 51
IV A	Fundamental Duties <i>(added by the 42nd Amendment)</i>	51A
V	The Union	52 – 151
VI	The States	152 – 237
VII	States in Part B of the First Schedule <i>(repealed by the 7th Amendment, 1956)</i>	238 (repealed)
VIII	Union Territories	239 – 242
IX	Panchayats <i>(by the 73rd Constitutional Amendment)</i>	243 – 243O
IX A	Municipalities <i>(by the 74th Constitutional Amendment)</i>	243P – 243ZG
IX B	Co-operative Societies <i>(by the 97th Amendment)</i>	243ZH – 243ZT

Part	Subject	Related Articles
X	Scheduled and Tribal Areas	244 – 244A
XI	Relations between the Union and the States (legislative and administrative)	245 – 263
XII	Finance, Property, Contracts and Suits (Art. 300A, the right to property, is in it)	264 – 300A
XIII	Trade, Commerce and Intercourse within the Territory of India	301 – 307
XIV	Services under the Union and the States (UPSC and UPPSC, etc.)	308 – 323
XIV A	Tribunals <i>(added by the 42nd Amendment)</i>	323A – 323B
XV	Elections	324 – 329A
XVI	Special Provisions Relating to Certain Classes (SC, ST, OBC, Anglo-Indians)	330 – 342
XVII	Official Language	343 – 351
XVIII	Emergency Provisions	352 – 360
XIX	Miscellaneous	361 – 367
XX	Amendment of the Constitution	368
XXI	Temporary, Transitional and Special Provisions (such as Art. 370 earlier)	369 – 392
XXII	Short Title, Commencement, Authoritative Text in Hindi and Repeals	393 – 395

> Schedules of the Constitution

Schedule	Main Subject and Provisions
First	• Names of the States and Union Territories (UTs) and their territorial jurisdiction/boundaries.
Second	• Salaries, allowances and privileges of various constitutional functionaries (such as the President, Governors, Speakers/Deputy Speakers of the Lok Sabha/Legislative Assemblies, judges of the Supreme Court/High Courts, and the CAG).
Third	• Forms of oaths and affirmations taken by ministers, members of Parliament/legislatures, candidates for election, judges of the Supreme Court/High Courts and the CAG .
Fourth	• Allocation of seats in the Rajya Sabha to the States and Union Territories.

Schedule	Main Subject and Provisions	
Fifth	Special provisions relating to the administration and control of Scheduled Areas and Scheduled Tribes (ST).	
Sixth	Provision of autonomous district councils for the administration of the tribal areas of four special states—Assam, Meghalaya, Tripura and Mizoram (AMTM).	
Seventh	Division of legislative powers (between the Centre and the States). Union List: laws by the Centre (present subjects: 100) State List: laws by the States (present subjects: 61) Concurrent List: laws by both the Centre and the States (present subjects: 52)	The original Constitution had 97 (Union), 66 (State) and 47 (Concurrent) subjects respectively.
Eighth	Languages recognised by the Constitution. - The original Constitution had 14 languages, which have now become 22.	Languages added later: Sindhi, Konkani, Manipuri, Nepali, Bodo, Dogri, Maithili and Santhali.
Ninth	Giving special constitutional protection to certain central and state laws (land reforms and abolition of zamindari).	Added by the First Constitutional Amendment, 1951 .
Tenth	Anti-defection provisions (relating to the disqualification of members of Parliament and legislatures).	Added by the 52nd Constitutional Amendment, 1985 .
Eleventh	Subjects of the Panchayats (functions of Panchayati Raj institutions). - It includes agriculture, drinking water, rural housing, roads, health, education, etc.	Added by the 73rd Constitutional Amendment, 1992 . There are 29 subjects in all.
Twelfth	Subjects of the Municipalities (functions of urban local bodies). - It includes urban planning, land use, water supply, sanitation, fire services, etc.	Added by the 74th Constitutional Amendment, 1992 . There are 18 subjects in all.

> Eighth Schedule: Important Constitutional Amendments Relating to Languages

Amendment Act	Year	Languages Added (trick to remember)
21st Amendment	1967	• Sindhi
71st Amendment	1992	• Nepali, Manipuri, Konkani (<i>trick: NaMaK - Na, Ma, Ka</i>)
92nd Amendment	2003	• Santhali, Bodo, Dogri, Maithili (<i>trick: BDSM or Bodo-Mas</i>)

> Seventh Schedule: Principal Subjects of the Three Lists

Name of the List	Present Subjects (Original Subjects)	Principal Law-maker	Principal Subjects Asked in the Exam (Core Examples)
Union List	100 (97)	Parliament only	• Defence • Foreign affairs • Posts and telegraphs • Telephones • Railways • Currency • Banking.
State List	61 (66)	State legislature	• Police • Prisons • Agriculture • Irrigation • Public health • Local government.
Concurrent List	52 (47)	Both Parliament + States	• Education • Forests • Marriage and divorce • Factories • Electricity • Economic planning • Waterways.

> Foreign Sources of the Indian Constitution

Source Country / Act	Features Incorporated in the Constitution
Government of India Act, 1935	• Federal scheme • Office of the Governor • Structure of the judiciary • Public Service Commissions • Emergency provisions • Administrative details.
Constitution of Britain	• Rule of Law • Parliamentary system of government • Legislative procedure • Single citizenship • Cabinet system • Prerogative writs • Parliamentary privileges • Bicameralism.
United States of America (USA)	• Fundamental Rights • Independence of the judiciary • Judicial review • Office of the President/impeachment • Removal of judges of the Supreme Court and High Courts • Federal supremacy.
Constitution of Ireland	• Directive Principles of State Policy (DPSP) • Method of election of the President • Nomination of members to the Rajya Sabha.
Constitution of Canada	• Federation with a strong Centre • Vesting of residuary powers in the Centre • Appointment of state Governors by the Centre • Advisory jurisdiction of the Supreme Court (Art. 143).
Constitution of Australia	• Concurrent List • Freedom of trade, commerce and intercourse • Joint sitting of the two Houses of Parliament.
Constitution of France	• Republican structure • The ideals of liberty, equality and fraternity in the Preamble.
Soviet Union (USSR / Russia)	• Fundamental Duties • The ideal of justice (social, economic and political) in the Preamble.
South Africa	• Procedure for amendment of the Constitution (Art. 368) • Election of members of the Rajya Sabha.

Source Country / Act	Features Incorporated in the Constitution
Weimar Constitution of Germany	• Suspension of Fundamental Rights during an Emergency.
Constitution of Japan	• Procedure established by law.

> The Preamble

> The Preamble: Important Components and Nature

Component / Dimension	Important Facts Related to the Preamble (Quick Revision)
Historical background	• The 'Objectives Resolution' was moved by Nehru on 13 December 1946. • It was adopted by the Constituent Assembly on 22 January 1947, and it later became the Preamble.
The only amendment	• Three new words were added by the 42nd Constitutional Amendment, 1976 : Socialist, Secular and Integrity .
Justice (3 kinds)	• Social, economic and political justice.
Liberty (5 kinds)	• Liberty of thought, expression, belief, faith and worship.
Equality (2 kinds)	• Equality of status and of opportunity.
Source (foreign)	• The idea/draft of the Preamble was taken from America and its distinctive language from Australia .
Nature / justiciability	• The Preamble is not justiciable (non-justiciable) (it cannot be enforced in a court).

> Facts Related to the Preamble and the 42nd Amendment

Subject	Words in Order / New Provisions Added
Correct order of the key words of the Preamble	Sovereign → Socialist → Secular → Democratic → Republic <i>(this order has been asked several times in UPPCS)</i>
Words added to the Preamble by the 42nd Amendment (1976)	1. Socialist 2. Secular 3. Integrity

Subject	Words in Order / New Provisions Added
New Articles added to the DPSP (Part IV) by the 42nd Amendment (1976)	• Article 39(A): equal justice and free legal aid to the poor. • Article 39(f): protection of children and youth against exploitation and their development in conditions of dignity. • Article 43(A): participation of workers in the management of industries. • Article 48(A): protection and improvement of the environment and safeguarding of wildlife.

> Historic Judgments of the Supreme Court (SC) on the Preamble

Case	Year	Main Verdict of the Supreme Court (Core Verdict)
Berubari Union case	1960	• The Preamble is not a part of the Constitution , hence Parliament cannot amend it.
Kesavananda Bharati case	1973	• The Preamble is an integral part of the Constitution and Parliament can amend it without damaging its 'basic structure'.
LIC of India case	1995	• The Preamble is an integral part of the Constitution.

Historic amendment: By the **42nd Constitutional Amendment (1976)** the Preamble was amended only once and three words, '**Socialist**', '**Secular**' and '**Integrity**', were added.

> The Preamble: Important Statements and Influences

Political Expert / Scholar	Important Statement Made about the Preamble
Thakurdas Bhargava / M. Hidayatullah	The Preamble is " the soul of the Indian Constitution ".
N.A. Palkhivala	The Preamble is the " identity card " of the Constitution.
K.M. Munshi	The Preamble is " the horoscope of the Constitution ".
Historical Event / Country	Main Influence on the Preamble / Element Borrowed
Objectives Resolution	• Moved by Nehru on 13 December 1946 ; passed on 22 January 1947 . It later became the Preamble.
Constitution of America	• The concept of " We the People of India " and the idea of a Preamble.
Constitution of Australia	• The language and distinctive style of the Preamble.
French Revolution	• The ideal words liberty, equality and fraternity included in the Preamble.
Russian Revolution	• The ideal of three kinds of justice in the Preamble: social, economic and political justice.

> Directive Principles of State Policy

> Directive Principles of State Policy (DPSP - Part IV)

Article	Main Provision and Directive Principle
39A	Equal justice and free legal aid to the poor <i>(by the 42nd Amendment)</i>
41	Right to work, to education and to public assistance in certain cases (unemployment, old age, sickness and disablement).
42	Provision for just and humane conditions of work and maternity relief .
43	Living wage and a decent standard of life for all workers.
43A	Ensuring participation of workers in the management of industries <i>(by the 42nd Amendment)</i> .
43B	Promotion of the voluntary formation and professional management of co-operative societies <i>(by the 97th Amendment)</i> .
48	Organising agriculture and animal husbandry on modern and scientific lines (prohibition of the slaughter of milch cattle).
48A	Protection and improvement of the environment and safeguarding of forests and wildlife <i>(by the 42nd Amendment)</i> .
49	Protection of monuments , places and objects of national importance .

> Famous Statements of Thinkers on the Directive Principles of State Policy (DPSP)

Name of the Scholar / Thinker	Famous Statement Made about the DPSP (UPPCS Special)
K.T. Shah	"The Directive Principles of State Policy are like a cheque on a bank, payable at the convenience of the bank. "
T.T. Krishnamachari	Called the DPSP " a veritable dustbin of sentiments ".
K.C. Wheare	Regarded them as " a manifesto of aims and aspirations ".
Sir Ivor Jennings	"These principles have no regular pattern of philosophy; they are Fabian socialism without the socialism. "
Granville Austin	"The Directive Principles and the Fundamental Rights together are the conscience of the Constitution. "
Dr. B.R. Ambedkar	"The Directive Principles are like instruments of instructions to the legislature and the executive."
B.N. Rau	"These principles have moral precept and educative value for the authorities of the State."

Name of the Scholar / Thinker	Famous Statement Made about the DPSP (UPPCS Special)
K. Santhanam	"These principles will cause constitutional conflict between the Centre and the States. "

> Fundamental Duties

> Fundamental Duties: Part IVA, Article 51A

Subject / Criterion	Important Facts and Features (UPPCS Special Points)
Historical background	- Introduced on the recommendations of the Sardar Swaran Singh Committee (1976). - Borrowed from the Constitution of the Soviet Union (USSR / Russia).
Constitutional amendment	- Initially 10 Fundamental Duties were added by the 42nd Constitutional Amendment, 1976. - The 11th Fundamental Duty (education of children aged 6-14) was added by the 86th Constitutional Amendment, 2002.
Principal features	Non-justiciable: the Fundamental Duties are non-justiciable (no punishment can be given directly by a court for their violation). - In nature some are moral duties and some are civic duties. - They are only for Indian citizens, not for foreigners. - Rights and duties are interrelated and inseparable.

> List of the 11 Fundamental Duties (in order)

1. To **abide by the Constitution** and respect its ideals and institutions, the National Flag and the National Anthem.
2. To cherish and follow the noble ideals **which inspired the national struggle** for freedom.
3. To uphold and protect the **sovereignty, unity and integrity** of India.
4. To **defend the country** and render national service when called upon to do so.
5. To promote **harmony and the spirit of common brotherhood** among all the people.
6. To value and preserve the rich heritage of our **composite culture**.
7. To protect and improve the **natural environment** (forests, lakes, rivers, wildlife).
8. To develop the **scientific temper**, humanism and the spirit of inquiry.
9. To safeguard **public property** and to abjure violence.
10. To strive continuously **towards excellence** in all spheres of individual and collective activity.
11. Provision of **primary education for children aged 6 to 14 years** by the parent or guardian (**86th Amendment, 2002**).

> Principal Articles

> The Union and its Territory (Articles 1-4) and the Integration of the Princely States

Article	Provision (Federal System)	Integration of the Princely States into India (Special Facts)
Article 1	India is a Union of States. (States cannot secede from the Union; single citizenship).	• At the time of independence 13 princely states were in Pakistan and 552 princely states were within the geographical boundary of India.
Article 2	Admission/establishment of new (foreign) states: Parliament may add foreign territory to India on the prior notice of the President.	• The responsibility of integrating the princely states into the Indian Union was entrusted to Sardar Patel, V.P. Menon and Lord Mountbatten .
Article 3	Formation of new states and alteration of names/boundaries: Parliament may alter the names, areas or boundaries of the existing states within India.	• Integration of the 3 rebellious princely states: 1. Hyderabad: by 'Operation Polo' (military action). 2. Junagadh: by a 'plebiscite'. 3. Jammu-Kashmir: by the signing of the 'Instrument of Accession'.
Article 4	Amendments made under Articles 2 and 3 will be by simple majority (they will not be deemed constitutional amendments under Art. 368).	• Fact about Sikkim: became an 'associate state' by the 35th Amendment, then a full state (the 22nd) by the 36th Amendment (1975) .

> Important Articles Related to Citizenship (Part II)

Article	Main Provision (Quick Revision Line)
Article 5	Citizenship at the commencement of the Constitution (26 January 1950).
Article 6	Rights of citizenship of persons who migrated to India from Pakistan .
Article 7	Rights of citizenship of persons who migrated from India to Pakistan .
Article 8	Citizenship of persons of Indian origin (PIO) residing outside India (abroad).
Article 9	Automatic termination of Indian citizenship on voluntarily acquiring the citizenship of a foreign state (rule of single citizenship).
Article 10	Continuance of the rights of citizenship (citizenship cannot be taken away except by a law made by Parliament).
Article 11	Full power of Parliament to make law/regulate citizenship.

> Vice-President: Constitutional Articles and Vacancy of Office (Part V)

Article	Main Provision	Causes of Vacancy of Office (Art. 67)
Article 63	• There shall be a Vice-President of India.	1. On expiry of the 5-year term .
Article 64	• The Vice-President shall be the ex-officio Chairman of the Rajya Sabha .	2. By resignation addressed to the President.
Article 65	• To act as President when the office of President falls vacant.	3. By removal through a resolution passed by the Rajya Sabha.
Article 66	• Procedure and qualifications for the election of the Vice-President.	4. On death during the term.
Article 67	• Term of office of the Vice-President (5 years from entering upon office).	5. When the Supreme Court declares the election void or him disqualified .
Article 69	• Oath by the Vice-President (before the President or a person appointed by him).	
Article 71	• Disputes connected with the election of the President or Vice-President shall be decided only by the Supreme Court .	

> Quick Pointers:

- The Vice-President is not a member of the Rajya Sabha, so he **has no ordinary right to vote**, but as Chairman he can give a **casting vote**.
- When the Vice-President acts as 'Acting President', he does not function as Chairman of the Rajya Sabha. During that period the Deputy Chairman conducts the House.
- **Vice-Presidents who later became President:** Dr. Sarvepalli Radhakrishnan, Dr. Zakir Husain, V.V. Giri, R. Venkataraman, Dr. Shankar Dayal Sharma.

> The Executive: Articles Relating to the President and the Prime Minister

Article	Subject / Name	Main Provision on the Relationship between the President and the Prime Minister
Article 74	Council of Ministers and the President	• There shall be a Council of Ministers, with the Prime Minister at its head, to aid and advise the President. The President is bound to act on its advice.
Article 75	Other provisions as to ministers	• The Prime Minister shall be appointed by the President , and the other ministers shall be appointed by the President on the advice of the Prime Minister.

Article	Subject / Name	Main Provision on the Relationship between the President and the Prime Minister
Article 77	Conduct of business of the Government of India	• All executive action of the Government of India shall be expressed to be taken in the name of the President .
Article 78	Duties of the Prime Minister	• It shall be the duty of the Prime Minister to communicate to the President all decisions of the Council of Ministers and matters relating to administration/legislation.

> Constitution and Internal Structure of Parliament (Part V: Articles 79 - 95)

Article	Subject Matter / Provision	Article	Subject Matter / Provision
79	Constitution of Parliament (President + Rajya Sabha + Lok Sabha)	88	Rights of ministers and the Attorney General as respects the Houses
80	Composition/organisation of the Rajya Sabha	89	Chairman (Vice-President) and Deputy Chairman of the Rajya Sabha
81	Composition/organisation of the Lok Sabha	90	Vacation, resignation and removal of the Deputy Chairman of the Rajya Sabha
82	Readjustment of seats after each census	92	Chairman/Deputy Chairman not to preside while a resolution for his removal is under consideration
83	Duration (term) of the two Houses of Parliament	93	Speaker and Deputy Speaker of the Lok Sabha
84	Essential qualifications for membership of Parliament	94	Vacation, resignation and removal of the Speaker and Deputy Speaker of the Lok Sabha
85	Sessions, prorogation and dissolution of Parliament	95	Power of the Deputy Speaker or another person to perform the duties of the Speaker
86	Right of the President to address and send messages to the Houses		
87	Special address by the President (first session after the Lok Sabha election and the first session of each year)		

> Principal Articles Relating to the Executive and the Legislature

Article	Office / Subject	Main Constitutional Provision
76	Attorney General of India	• The chief legal adviser of the Government of India. Has the right to speak in both Houses of Parliament (not to vote).

Article	Office / Subject	Main Constitutional Provision
110	Money Bill	• Definition of a Money Bill. Whether a bill is a Money Bill or not is decided by the Speaker of the Lok Sabha .
112	Annual Financial Statement	• The word 'budget' is not mentioned in the Constitution; in its place is the term 'Annual Financial Statement'.
123	Ordinance-making power of the President	• Power of the President to promulgate ordinances while Parliament is not in session (recess).
143	Consultation with the Supreme Court	• Power of the President to seek the advice of the Supreme Court on a question of law of public importance (the advice is not binding).
148	Comptroller and Auditor-General of India (CAG)	• Provision for the office of the Comptroller and Auditor-General of India (guardian of the public purse of India).
149	Duties and powers of the CAG	• Determination of the functions, duties and powers of the Comptroller and Auditor-General of India.

> Constitutional Provisions Relating to the Official Language (Part XVII)

Article	Main Provision and Direction
343	Official language of the Union: the official language of the Union shall be Hindi in the Devanagari script.
344	Provision for the constitution of a Commission (Official Language Commission) and a Committee of Parliament on the official language.
350A	Duty of the states to provide adequate facilities for instruction in the mother tongue at the primary stage.
350B	Provision for the appointment of a Special Officer for linguistic minorities .
351	Directive for the development of the Hindi language (the Union government shall work for the spread and development of Hindi).

Note

Note: The **Eighth Schedule of the Indian Constitution** at present contains **22 official languages**, including Hindi.

> Principal Articles on the Judiciary (Supreme Court and High Courts)

Supreme Court	Main Provision	High Court and Subordinate Courts	Main Provision
Article 124	Establishment and constitution of the Supreme Court.	Article 214	Provision of High Courts for the states.
Article 125	Salaries, allowances and facilities of judges.	Article 215	High Courts to be courts of record .
Article 126	Appointment of an Acting Chief Justice .	Article 226	Power of the High Courts to issue writs for Fundamental Rights and other purposes.
Article 127	Appointment of ad hoc judges (only in the SC).	Article 233	Appointment of District Judges by the Governor.
Article 128	Attendance of retired judges at sittings of the Supreme Court.	Article 235	Control over subordinate courts (administrative control of the High Court).
Article 129	The Supreme Court to be a court of record and its power to punish for contempt.		
Article 136	Special Leave to Appeal (Special Leave Petition - SLP) by the Supreme Court.		
Article 137	Review by the Supreme Court of its own judgments or orders (review/reconsideration).		
Article 141	Law declared by the Supreme Court shall be binding on all courts within India.		

> Principal Articles Relating to the Governor (Part VI)

Article	Main Provision Relating to the Governor (UPPCS Matching Special)
153	Governors of States: there shall be a Governor for each state (the same person may be Governor of two or more states).
154	The executive power of the state shall be vested in the Governor and exercised by him either directly or through officers subordinate to him.
155	Appointment of the Governor: the Governor is appointed by the President.
156	Term of office of the Governor: during the pleasure of the President (normally 5 years).
158	Conditions of the Governor's office.

Article	Main Provision Relating to the Governor (UPPCS Matching Special)
159	Oath or affirmation by the Governor (administered by the Chief Justice of the High Court of the state concerned).
160	Discharge of the functions of the Governor in certain contingencies.
161	Pardoning power of the Governor: power to grant pardons, reprieves, respites or remissions of punishment (except the death sentence and court-martial).
162	Extent of the executive power of the state.
163	There shall be a Council of Ministers , with the Chief Minister at its head, to aid and advise the Governor.
164	Appointment of ministers (by the Governor on the advice of the Chief Minister), their tenure, salaries and other provisions.

> Power of the President to Make Regulations for UTs (Article 240)

Article	Related Union Territory
240(1)(A)	Andaman and Nicobar Islands
240(1)(B)	Lakshadweep
240(1)(C)	Dadra and Nagar Haveli
240(1)(D)	Daman and Diu
240(1)(E)	Puducherry

> Centre-State Legislative and Administrative Relations

Article	Main Legislative / Administrative Provision (UPPCS Matching Special)
Article 245	Parliament may make laws for the whole or any part of the country, and a state legislature for its own state.
Article 246	Classification of the powers to make laws on the subjects of the three lists (Union, State and Concurrent) of the Seventh Schedule.
Article 247	Power of Parliament to establish certain additional courts.
Article 248	Residuary powers: the right to make laws on subjects not in any of the three lists (such as cyber crime, ISRO, DRDO) belongs only to the Union government (Parliament) .
Article 252	Power of Parliament to legislate on a State List subject with the consent/at the request of two or more states .

Article	Main Legislative / Administrative Provision (UPPCS Matching Special)
Article 253	Parliament has full power to legislate even on State List subjects for giving effect to international agreements/treaties .
Article 254	If a state law is repugnant (inconsistent) to a law of Parliament, the law of Parliament shall prevail and the state law shall be void.
Article 263	Constitution of an Inter-State Council (by the President); meetings with Chief Ministers for Centre-State coordination.

> Centre-State Financial Relations (Part XII: Articles 265 - 281)

Article	Main Financial Provision (UPPCS Fact Pointer)
265	• Taxes not to be imposed save by authority of law: no tax can be levied without a legal basis.
266	• Details of the Consolidated Funds and Public Accounts of India and of the states.
267	• Contingency Fund: a fund under the President/Governor for unforeseen expenditure.
268	• Duties levied by the Union but collected and appropriated by the states .
270	• Taxes levied and collected by the Union but distributed between the Union and the states .
274	• Prior recommendation of the President is required for bills affecting taxation in which states are interested.
275	• Grants-in-aid from the Union to certain states.
280	• Constitution of the Finance Commission: by the President at the expiration of every fifth year.
281	• The recommendations of the Finance Commission to be laid before both Houses of Parliament by the President.

> Public Service Commissions (UPSC and UPPSC: Part XIV, Articles 315 - 323)

Article	Main Constitutional Provision (Quick Pointers)
315	• Provision for the constitution of Public Service Commissions for the Union (UPSC) and for the states (SPSC/UPPSC).
316	• Appointment and term of office of the Chairman and members of the Commission.
317	• Removal and suspension of members of a Public Service Commission.
318	• Power to make regulations regarding the conditions of service of members and staff of the Commission.

Article	Main Constitutional Provision (Quick Pointers)
319	• Prohibition on members of the Commission holding any other government office on ceasing to be members.
320	• Functions of the Union and State Public Service Commissions (conducting examinations, etc.).
321	• Power of Parliament to extend the functions of the Public Service Commissions.
322	• Expenses: the expenses of the UPSC are charged on the Consolidated Fund of India, while those of an SPSC (UPPSC) are charged on the Consolidated Fund of the state concerned .
323	• Reports (annual reports) of the Public Service Commissions: the UPSC submits its report to the President and the UPPSC to the Governor.

› Election Commission of India (ECI): Constitutional Provisions (Part XV)

Article	Main Provision and Jurisdiction (Quick Pointers)
Establishment and nature	• An autonomous constitutional body ; established on 25 January 1950 .
Electoral responsibility	• Conducting the elections to the Lok Sabha, Rajya Sabha, state Legislative Assemblies, and of the President and Vice-President in the country. • Note: it has nothing to do with panchayat and municipal elections in the states (for these there is provision for a 'State Election Commission').
Article 324	• Superintendence, direction and control of elections shall be vested in the Election Commission.
Article 325	• Provision that no person shall be excluded from the electoral roll or declared ineligible on grounds of religion, race, caste or sex.
Article 326	• Elections to the Lok Sabha and state Legislative Assemblies shall be on the basis of universal adult suffrage .
Article 327	• Power of Parliament to make provision with respect to elections to legislatures.
Article 328	• Power of the state legislature to make laws for elections to the legislature of that state.
Article 329	• Bar to interference by courts in electoral matters/delimitation of constituencies.

› Special Provisions for States/Regions (Articles 370 and 371), Part 21

Article	Related State (Extremely Important for UPPCS Prelims Matching)
Article 370	Jammu Kashmir (rendered completely inoperative from 31 October 2019; Jammu-Kashmir is now an ordinary Union Territory of India)

Article	Related State (Extremely Important for UPPCS Prelims Matching)
Article 371	Maharashtra and Gujarat
Article 371A	Nagaland
Article 371B	Assam
Article 371C	Manipur
Article 371D and E	Andhra Pradesh (and Telangana)
Article 371F	Sikkim
Article 371G	Mizoram
Article 371H	Arunachal Pradesh
Article 371I	Goa
Article 371J	Karnataka (Hyderabad-Karnataka region)

> Important Constitutional Amendment Acts

Amendment	Year	Main Provisions and UPPCS Special Points
1st	1951	• The Ninth Schedule was added ; an attempt to keep land reform laws outside judicial review.
10th	1961	• Dadra and Nagar Haveli was incorporated into the Indian Union (UT).
11th	1961	• The election of the President/Vice-President cannot be challenged on the ground of any vacancy in the electoral college .
12th	1962	• Goa, Daman and Diu were incorporated into the Indian Union (UT).
31st	1973	• The number of Lok Sabha seats was increased from 525 to 545 .
36th	1975	• Sikkim was made a full (22nd) state of India .
80th	2000	• An alternative scheme of devolution of taxes between the Union and the states was made (on the recommendation of the 10th Finance Commission).
85th	2001	• Provision of consequential seniority in promotion for government employees of the SC/ST categories.
86th	2002	• By adding Article 21A , education was made a Fundamental Right for children aged 6 to 14 (the 11th Fundamental Duty was also added).

Amendment	Year	Main Provisions and UPPCS Special Points
97th	2012	• Constitutional status to co-operative societies . Part IX-B and Article 43B (DPSP) were added. Forming a society became a Fundamental Right under Article 19(1)(c).
99th	2014	• Creation of the National Judicial Appointments Commission (NJAC) in place of the collegium. <i>(The Supreme Court declared it unconstitutional and struck it down)</i> .
100th	2015	• India-Bangladesh Land Boundary Agreement (LBA) ; amendment of the First Schedule for the exchange of territories.
101st	2016	• The Goods and Services Tax (GST) was introduced.
102nd	2018	• Constitutional status was given to the National Commission for Backward Classes (NCBC) (Article 338B added).
103rd	2019	• Provision of 10% reservation for the economically weaker sections (EWS) (Articles 15(6) and 16(6) added).
104th	2020	• Reservation of SC/ST seats in the Lok Sabha/Assemblies extended by 10 years (till 2030). The Anglo-Indian quota was abolished .
105th	2021	• The power of the states to prepare the list of socially and educationally backward classes (SEBC) was restored .
106th	2023	• Women's Reservation Act (Nari Shakti Vandan Adhiniyam) ; provision for reserving 33% of seats for women in the Lok Sabha and state Legislative Assemblies.

> Reorganisation of States

> Commissions and Committees Related to the Reorganisation of States

Name of the Commission / Committee	Year	Principal Members	Main Recommendations / UPPCS Special Facts
Dhar Commission	1947-48	S.K. Dhar (Chairman)	• The basis of reorganisation of states should be administrative convenience, not language (opposed the linguistic basis).
JVP Committee	1948	Jawaharlal Nehru, Vallabhbhai Patel, Pattabhi Sitaramayya	• Regarded the security, unity and economic prosperity of the country as the main basis of reorganisation of states (rejected the immediate linguistic basis).

Name of the Commission / Committee	Year	Principal Members	Main Recommendations / UPPCS Special Facts
States Reorganisation Commission	1953	Syed Fazl Ali (Chairman), Hriday Nath Kunzru, K.M. Panikkar	• Accepted language as a basis for the reorganisation of states. The States Reorganisation Act, 1956 was enacted on its report.

Important fact: The first state formed on a linguistic basis in the country was **Andhra Pradesh (1 October 1953)**.

> Reorganisation of States (Fazl Ali Commission) and the Chronological Years of New States

Main Facts of the Fazl Ali Commission (1953-1956)	Name of the State	Year / Date of Formation (UPPCS Chronology Special)
• Constituted: 1953; report: 1956	Andhra Pradesh	1 October 1953
• Members: Syed Fazl Ali (Chairman), Hriday Nath Kunzru, K.M. Panikkar.	Gujarat and Maharashtra	1 May 1960
• Recommendation: accepted language as a basis but rejected the principle of 'one state-one language'.	Nagaland	1 December 1963
• Result: the 7th Constitutional Amendment, 1956 was made. The classification of states in the original Constitution was abolished and 14 states and 6 Union Territories were created.	Haryana	1 November 1966
	Himachal Pradesh	25 January 1971
	Meghalaya, Manipur, Tripura	21 January 1972 (formed together)
	Sikkim	16 May 1975 (full state)
	Mizoram and Arunachal	20 February 1987
	Goa	30 May 1987
	Chhattisgarh	1 November 2000
	Uttarakhand	9 November 2000
	Jharkhand	15 November 2000

Main Facts of the Fazl Ali Commission (1953-1956)	Name of the State	Year / Date of Formation (UPPCS Chronology Special)
	Telangana	2 June 2014

> Panchayati Raj

> Principal Committees and Development Related to Panchayati Raj

Name of the Committee	Year of Formation	Main Recommendations / UPPCS Special Facts
Balwant Rai Mehta Committee	1957	- Recommended a three-tier Panchayati Raj system (village, block, district) and 'democratic decentralisation'. Note: on this basis Pt. Nehru inaugurated Panchayati Raj for the first time in the country on 2 October 1959 in Nagaur district of Rajasthan.
Ashok Mehta Committee	1977	Suggested a two-tier panchayat system (Zila Parishad and Mandal Panchayat).
L.M. Singhvi Committee	1986	Strongly recommended giving constitutional status to Panchayati Raj institutions.
P.K. Thungon Committee	1988	Recommended constitutional recognition of Panchayati Raj and regular elections.

Constitutional status (Quick Fact): During the tenure of Prime Minister P.V. Narasimha Rao, the panchayats got constitutional status through the **73rd Constitutional Amendment Act, 1992**, which came into effect from **24 April 1993** (that is why 24 April is celebrated as National Panchayati Raj Day).

> Federal versus Unitary Features of the Constitution

Federal Features of the Constitution	Unitary Features of the Constitution
• Dual polity (governments at the central and state levels)	• A strong Centre (the Centre being more powerful)
• Written and rigid Constitution	• Single Constitution (states having no separate constitutions)
• Clear division of powers (Seventh Schedule)	• Single citizenship
• Supremacy of the Constitution	• Flexibility of the Constitution (in special circumstances)
• Independent judiciary	• Integrated judiciary

Federal Features of the Constitution	Unitary Features of the Constitution
• Bicameral legislature (Lok Sabha and Rajya Sabha)	• Appointment of Governors by the Centre
	• All-India Services (IAS, IPS, IFS)
	• Emergency provisions (Part XVIII)

> The Four Main Instruments of Direct Democracy

Type	Simple Definition / Meaning (Concept Pointer)
1. Referendum	Under it a proposed law or policy is sent directly to the voters for their vote (yes or no).
2. Initiative	Through it the people themselves can prepare the draft of a law and ask the legislature (Parliament) to make the law.
3. Recall	Through it the voters can remove any of their representatives or officials from office even before the end of his term.
4. Plebiscite	It is generally the process of ascertaining the opinion of the general public on an issue of great public importance or a territorial dispute.

1 Constitutional Writs: Types and their Jurisdiction

Name of the Writ	Literal Meaning	Against whom can it be Issued?	Against whom can it not be Issued?	Special UPPCS Prelims Fact
Habeas Corpus	"Let the body be produced"	• Against both public officials • and private individuals	—	• It is the greatest protector of personal liberty.
Mandamus	"We command"	• Public officials, bodies, corporations • Lower courts, tribunals and the government	• Private individuals or private bodies • President / Governor	• Issued only to enforce public duties.
Quo-Warranto	"By what authority or warrant"	• Against a person who has illegally usurped a public (government) office	• Private offices • Ministerial offices	• It can be filed by any interested person (need not be the aggrieved person).
Prohibition	"To forbid or prohibit"	• Judicial authorities (lower courts) • Quasi-judicial authorities	• Administrative authorities • Legislative bodies and private individuals	• It is issued only to stop proceedings while they are pending.

Name of the Writ	Literal Meaning	Against whom can it be Issued?	Against whom can it not be Issued?	Special UPPCS Prelims Fact
Certiorari	"To be informed / to be certified"	- Judicial and quasi-judicial authorities - Administrative authorities (<i>since 1991</i>)	- Legislative bodies - Private individuals or bodies	- Also called the "post-mortem" writ. - To call up a pending case or to quash a decision.

➤ Emergency Provisions in the Indian Constitution

Type of Emergency	Related Article	Source / Borrowed from Where?	Special Facts and Amendments (UPPCS Special)
National Emergency	Article 352	Government of India Act, 1935	By the 44th Amendment (1978) the words 'internal disturbance' were removed and replaced by 'armed rebellion'.
President's Rule (<i>failure of constitutional machinery</i>)	Article 356	Government of India Act, 1935	Imposed in the states on the report of the Governor or on failure to comply with the directions of the Centre (also under Art. 365).
Financial Emergency	Article 360	Government of India Act, 1935	A financial emergency has not been imposed even once in India so far.

➤ Emergency Provisions of the Indian Constitution (Comparative Table)

Criterion / Feature	National Emergency (Article 352)	President's Rule (Article 356)	Financial Emergency (Article 360)
Main ground of proclamation	War, external aggression or armed rebellion.	Failure of the constitutional machinery in the states (on the report of the Governor or without a report).	Threat to the financial stability or credit of India or of any part of it.
Role of the cabinet	Proclaimed by the President only on the written recommendation of the Union Cabinet.	On the ordinary advice of the Union Cabinet.	On the ordinary advice of the Union Cabinet.
Period for first approval by Parliament	Must be approved by Parliament within 30 days (1 month) of the date of proclamation.	Must be approved by Parliament within 2 months (60 days) of the date of proclamation.	Must be approved by Parliament within 2 months (60 days) of the date of proclamation.
Majority for parliamentary approval	Must be passed by both Houses of Parliament by a special majority.	Passed by both Houses of Parliament by a simple majority.	Passed by both Houses of Parliament by a simple majority.

Criterion / Feature	National Emergency (Article 352)	President's Rule (Article 356)	Financial Emergency (Article 360)
Duration of operation	• 6 months at first; with Parliament's approval it can continue indefinitely, being extended every 6 months .	• 6 months at first; it can remain in force only for a maximum total period of 3 years .	• Once approved, its duration is not fixed (it remains in force until Parliament revokes it).
Effect on Fundamental Rights	• Affected: Fundamental Rights are suspended under Articles 358 and 359 (but Articles 20 and 21 are never suspended).	• No effect on the Fundamental Rights of citizens.	• No effect on the Fundamental Rights of citizens.
Termination of the proclamation	• By the Lok Sabha passing a resolution by a simple majority , or by the President at any time.	• The President can himself withdraw the proclamation at any time without parliamentary approval.	• The President can revoke it at any time by a subsequent proclamation.
Financial effect	—	—	• The financial powers of the states pass to the Centre; reduction of salaries possible.

> Types of Bills and their Comparative Features

Feature / Criterion	Government Bill	Private Member's Bill	Ordinary Bill	Constitution Amendment Bill (Art. 368)
Introduced by	• A minister of the government.	• Any member of Parliament other than a minister.	• Any member of Parliament (minister or non-minister).	• Any member of Parliament.
House	• Introduced in either House.	• Introduced in either House.	• In either House (Lok Sabha or Rajya Sabha).	• Can be introduced in either House .
Prior approval of the President	—	—	• Prior permission of the President is not required .	• Prior permission of the President is not required .
Chances of passing	• High (because the government has a majority).	• Negligible / very low (only 14 have become law so far).	• Passed by a simple majority of Parliament.	• Passed separately by each House by a special majority .
Joint sitting in case of deadlock (Art. 108)	—	—	• There is provision for a joint sitting if the other House holds it up for 6 months.	• There is no provision for a joint sitting in case of disagreement.

Feature / Criterion	Government Bill	Private Member's Bill	Ordinary Bill	Constitution Amendment Bill (Art. 368)
Power of the President	—	—	The President may give assent, withhold it or return it for reconsideration.	The President must give assent (it is binding).

> Delimitation Commission

Criterion / Component	Important Facts (Core Pointers)
Appointment	Made by the President of India.
Main function	To determine the boundaries of constituencies throughout the country on the basis of the census.
Cooperation	This commission works together with the Election Commission of India (ECI) .
Composition (members)	1. Chairman: a retired judge of the Supreme Court 2. Chief Election Commissioner (CEC) 3. State Election Commissioners of the states concerned
Judicial review	The decisions of the Delimitation Commission cannot be challenged in any court.
Constituted so far	It has been constituted 4 times in India so far: 1952, 1963, 1972 and 2002.

> Composition of the Houses of Parliament

Name of the House	Maximum Strength	Elected / Representation of Union Territories	Number of Nominated Members
Lok Sabha (Lower House / House of the People)	550 (<i>after the 104th Amendment</i>)	530 (from the states) 20 (from the Union Territories - UTs)	0 (<i>Anglo-Indian nomination abolished</i>)
Rajya Sabha (Upper House / Council of States)	250	238 (representatives of the states and Union Territories)	12 (<i>nominated by the President from literature, science, art and social service</i>)

> Public Accounts Committee (PAC)

Component	Principal Feature / Rule
History / establishment	The oldest parliamentary committee; established in 1921 (under the Government of India Act 1919).

Component	Principal Feature / Rule
Total membership	22 members [15 from the Lok Sabha + 7 from the Rajya Sabha].
Term	The term of members is 1 year (elected by the system of proportional representation).
Membership of ministers	No minister can become a member of this committee.
Appointment of the Chairman	Made by the Speaker of the Lok Sabha from among its members (by convention a member of the opposition becomes Chairman).
Main function	To examine the appropriation accounts of the Government of India and the reports of the CAG (Comptroller and Auditor-General) on them.

> Attorney General of India

Criterion / Component	Constitutional Provision (Article 76)
Nature of the office	The highest law officer of the country (chief legal adviser to the Government of India).
Appointment	By the President on the advice of the Union Cabinet.
Term	Not fixed by the Constitution.
Removal	The procedure/grounds of removal are not mentioned in the Constitution. He holds office during the pleasure of the President.
Principal functions	• To advise the Government of India on legal matters. • To appear on behalf of the government in the Supreme Court or any High Court in cases concerning the government.

> Allocation of Lok Sabha Seats to the States and Union Territories (UTs)

Name of the State / UT	Lok Sabha Seats	Name of the State	Lok Sabha Seats	Name of the State	Lok Sabha Seats
Uttar Pradesh	80	Rajasthan	25	Tripura	2
Maharashtra	48	Odisha	21	Manipur	2
Bihar	40	Telangana	17	Goa	2
Tamil Nadu	39	Jharkhand	14	Arunachal Pradesh	2
Madhya Pradesh	29	Punjab	13	Meghalaya	2

Name of the State / UT	Lok Sabha Seats	Name of the State	Lok Sabha Seats	Name of the State	Lok Sabha Seats
Karnataka	28	Chhattisgarh	11	Dadra-Nagar Haveli and Daman-Diu	2
Gujarat	26	Haryana	10	Delhi (UT)	7
Andhra Pradesh	25	Jammu-Kashmir (UT)	5	Uttarakhand	5
Himachal Pradesh	4	States/UTs with 1 seat:	Ladakh, Sikkim, Chandigarh, Lakshadweep, Puducherry, Nagaland, Mizoram, Andaman-Nicobar.	1 (each)	

> Recognised National Political Parties of India (Current 2026 Updated)

Name of the Political Party	Year of Founding
1. Indian National Congress (INC)	1885
2. Communist Party of India - Marxist (CPI-M)	1964
3. Bharatiya Janata Party (BJP)	1980
4. Bahujan Samaj Party (BSP)	1984
5. Aam Aadmi Party (AAP)	2012
6. National People's Party (NPP)	2013

At present, according to the rules of the Election Commission of India, only **6 national political parties in all** are recognised in the country.

> Fundamental Rights Available Only to Indian Citizens (Not to Foreigners)

Article	Subject Matter of the Fundamental Right (Core Concept)
Article 15	Prohibition of discrimination on grounds only of religion, race, caste, sex or place of birth.
Article 16	Equality of opportunity in matters of public employment.
Article 19	6 democratic freedoms: speech and expression, peaceful assembly, forming associations/unions, movement throughout the country, residence, and practising any profession/occupation.

Article	Subject Matter of the Fundamental Right (Core Concept)
Article 29	Protection of the interests of minorities (language, script and culture).
Article 30	Right of minorities to establish and administer educational institutions.

> Right to Equality (Part III: Articles 14 - 18) and its Legal Concept

Article	Fundamental Right (Right to Equality)	'Equality before law' (Art. 14) <i>(inspired by Britain / negative concept)</i>	'Equal protection of laws' (Art. 14) <i>(inspired by America / positive concept)</i>
14	Equality before law and equal protection of laws.	• Absence of special privileges in favour of any person.	• Equal treatment in both the rights and the obligations conferred by laws.
15	Prohibition of discrimination on grounds only of religion, race, caste, sex or place of birth.	• The same uniform system for all persons under the ordinary courts.	• The same law applied without discrimination to persons similarly situated.
16	Equality of opportunity in matters of public employment (government jobs).	• No person (rich or poor, official or ordinary) is above the law.	• Rule of Law: the Supreme Court has regarded this principle of A.V. Dicey as part of the Basic Structure .
17	Abolition of untouchability , and its practice in any form is forbidden.		
18	Abolition of titles (except military and academic distinctions).		

> Principal Financial Committees of Parliament (Parliamentary Committees)

Name of the Committee	Membership and Composition	Term	Nomination / Selection of the Chairman	Main Functions and Other Exam Facts
Estimates Committee	30 members (all of them only from the Lok Sabha ; no representation of the Rajya Sabha)	1 year	Nominated by the Speaker of the Lok Sabha from among the members.	• To bring about economy in government expenditure, organisational efficiency and administrative reform. • It is the largest parliamentary committee.

Name of the Committee	Membership and Composition	Term	Nomination / Selection of the Chairman	Main Functions and Other Exam Facts
Public Accounts Committee (PAC)	22 members (15 from the Lok Sabha + 7 from the Rajya Sabha)	1 year	By convention the Speaker of the Lok Sabha appoints a leader of the opposition as Chairman.	- To examine the annual reports of the CAG, examine the appropriation accounts and give its report to the Lok Sabha. - Nicknames: 'mini Lok Sabha', 'twin sister of the Estimates Committee'.
Committee on Public Undertakings	22 members (15 from the Lok Sabha + 7 from the Rajya Sabha)	1 year	Chosen by the Speaker of the Lok Sabha only from among the Lok Sabha members.	- To examine the reports and accounts of government/public undertakings (PSUs) and the CAG's reports on them. Special rule: No minister can become a member of this committee.

05

General Science

GENERAL SCIENCE

> Important Discoveries Related to Science

> Important Discoveries and Scientists Related to Chemistry

Name of the Discovery / Theory	Related Scientist / Discoverer	UPPCS Exam Key Point
Periodic Table	Mendeleev	The first systematic table based on the atomic weights of elements.
Modern Periodic Table	Moseley	A table based on the atomic number of elements.
Discovery of the electron	J.J. Thomson	The negatively charged fundamental particle of the atom.
Charge of the electron	Millikan	Determination of the charge by the oil drop experiment .
Wave nature of the electron	De Broglie	Theory of the dual nature of matter.
Discovery of the proton	Rutherford	The positively charged fundamental particle of the atom (note: Goldstein made the naming/initial discovery).
Discovery of the neutron	Chadwick	The neutral (charge-less) fundamental particle of the atom.
Atomic model / detailed form of the periodic table	Niels Bohr	A stable model of the distribution of electrons in orbits (shells).
Wave theory of light / quantum theory	Max Planck	Light energy travels in the form of small packets (quanta/photons).
Discovery of X-rays	Roentgen	High-energy waves used in medicine and security.
Theory of the catalyst	Berzelius	Discovery of substances that change the rate of a chemical reaction.
Radioactivity	Henri Becquerel	Discovery of substances that disintegrate spontaneously (SI unit: becquerel).
Artificial radioactivity	Madame Curie and F. Joliot	Making elements artificially radioactive in the laboratory.

Name of the Discovery / Theory	Related Scientist / Discoverer	UPPCS Exam Key Point
Radioactive age determination (carbon dating)	W.F. Libby	Method of finding the age of fossils and old objects.
Principle of electrolysis	Michael Faraday	Laws of decomposition of chemical compounds by electric current.
Cyclotron	Lawrence	A device that accelerates charged particles.
Scattering of light by solid particles	John Tyndall	The path of light becoming visible in a colloidal solution (Tyndall effect).
Discovery of the meson	Hideki Yukawa	Sub-atomic particles that explain nuclear forces.
Discovery of argon (Ar)	Rayleigh and Ramsay	The inert gas present in the largest amount in the atmosphere.
Discovery of neon, krypton and xenon	Ramsay and Travers	Discovery of other important noble/inert gases.
pH value (pH scale)	Sorensen	A scale for measuring acidity and alkalinity (0 to 14).

› Order of Waves, their Discoverers and Principal Applications

Name of the Wave	Discoverer	Main Applications Useful for the Exam (Exam Notes)
Gamma rays	Becquerel	In nuclear reactions and for destroying cancer cells. (<i>highest energy/frequency</i>)
X-rays	Roentgen	In medicine (to see bone fractures, etc.) and in security checks.
Ultraviolet rays	Ritter	In disinfection, water purification and detecting forged documents.
Visible radiation	Newton	For seeing objects with human eyes and in photography.
Infrared radiation	Herschel	In remote controls and for heat treatment of patients' muscles.
Short radio waves / microwaves	Heinrich Hertz	In radar systems, satellite communication and microwave ovens.
Long radio waves	Marconi	In cellular phones and in the transmission (broadcasting) of radio and TV waves. (<i>longest wavelength</i>)

> Medical Equipment and Diagnostic Tools and their Applications

Medical Instrument	Main Function and Practical Use (Exam Facts)
Stethoscope	Used to hear the rate and sound of the heartbeat.
Pacemaker	To control the heart rate and bring it back to normal when it becomes too fast or too slow.
Sphygmomanometer	Used to measure the blood pressure of the human body.
CT Scan	To view bones, muscles, joints (knee, shoulder), internal organs (heart, stomach, intestines), fat tissue and blood vessels.
Electrocardiography (ECG)	Used to detect abnormalities and diseases of the heart.
Electroencephalogram (EEG)	To measure/examine disorders of the brain and its electrical activity.
Auto Analyser	Used to test biological constituents such as glucose, urea and cholesterol in the blood.
Magnetic Resonance Imaging (MRI)	To detect accurately any abnormality or internal disorder in the whole body or a particular organ.
Endoscope	An instrument for direct inspection of the internal organs of the body, working on the principle of optical fibres.

> Principal Measuring Instruments and their Uses (Scientific Instruments Master Matrix)

Measuring Instrument	Field of Measurement / Basic Use (Core Function)
Audiometer / Phonometer	Used to measure the intensity of sound.
Odometer	To measure the total distance covered by a wheeled vehicle.
Altimeter	To measure the altitude of aircraft or mountains.
Auxanometer / Crescograph	An instrument that measures the growth of plants.
Luxmeter / Photometer	An instrument to measure luminous intensity (intensity of light).
Lactometer	To measure the purity of milk and its relative density.
Hydrometer	To measure the relative density of liquids.
Hygrometer	To measure the humidity of the air or atmosphere.
Manometer	To measure the pressure of liquids or gases.
Galvanometer	To detect the presence and direction of an electric current.

Measuring Instrument	Field of Measurement / Basic Use (Core Function)
Ammeter	To measure the actual magnitude of an electric current.
Anemometer	To measure the speed and velocity of wind.
Voltmeter / Electrometer	Measures electric potential difference or electric charge.
Seismograph	To measure earthquake waves and their intensity.
Thermometer	To measure ordinary human or industrial temperature.
Pyrometer / Actinometer	To measure high temperature and solar radiation.
Pyrheliometer	A special instrument to measure very intense solar radiation.
Spectroheliograph	A special instrument used for photographing the sun.
Caratmeter	For exact determination of the purity of gold.
Fathometer	Used to measure the depth of the ocean.
Tachometer	An instrument to measure rotational speed or RPM (revolutions per minute).
Polygraph	An instrument for detecting lies in criminal investigation (lie detector).
Bolometer	Measures thermal or electromagnetic radiation.
Evaporimeter	Measures the rate of evaporation of water.
Calorimeter	Measures the quantity of heat absorbed or released in a chemical or physical process.
Cryometer	A thermometer that measures very low (cryogenic) temperatures.
Dynamometer	An instrument that measures the force, torque or power produced by an engine.
Electroscope	Detects the presence of electric charge in a body or object.
Gravimeter	An instrument for the precise measurement of the Earth's gravitational acceleration (g).
Gyroscope	An instrument measuring angular velocity and the direction of rotating bodies.
Hydrophone	An instrument that records sound waves under water.
Nephoscope	Measures the height of clouds, their velocity and direction of movement.
Microphone	Converts sound waves into electrical signals.

> Important Laws and Principles of Science (Scientific Laws)

Law / Principle	Basic Definition and Scientific Concept	Practical Applications and Instruments
Pascal's law	Pressure applied at any one point of a confined fluid in equilibrium is transmitted equally in all directions.	Hydraulic lift, hydraulic press, hydraulic brakes.
Hooke's law	Within the elastic limit, the stress on a body is always proportional to its strain (stress $\propto$ strain).	Young's modulus , which indicates the stiffness of skin or a material / its likelihood of deformation.
Archimedes' principle	The upthrust (buoyant force) on a solid immersed in a liquid is equal to the total weight of the liquid displaced by that solid.	In the hydrometer and in the construction of ships and submarines.
Boyle's law	At a fixed/constant temperature, the volume of a given quantity of a gas is inversely proportional to its pressure ($V \propto 1/P$).	Refilling system of gas cylinders and the human breathing process.

> Newton's Laws of Motion

Note: The physicist and astronomer Sir Isaac Newton presented these laws in **1687** in his fundamental work "**Philosophiae Naturalis Principia Mathematica**" (Mathematical Principles of Natural Philosophy).

Order of the Law	Name of the Law	Basic Scientific Concept	Practical Examples Useful for the Exam (Exam Hits)
First law	Law of inertia	A body continues in its state of motion or rest until an external unbalanced force is applied to it.	Passengers leaning forward when a moving bus stops suddenly, or falling backward when a stationary vehicle starts.
Second law	Law of momentum	The rate of change of momentum is proportional to the applied force, and the change takes place in the direction of the force. Formula: $F = ma$ (force = mass $\times$ acceleration).	A cricketer pulling his hands back while catching a ball so as to reduce injury.
Third law	Law of action and reaction	To every action there is an equal and opposite reaction.	* The backward jerk (recoil) when a bullet is fired from a gun. * A rocket moving forward by the pressure of fast-moving gases. * Falling backwards when the rope breaks while drawing water from a well.

> The Electromagnetic Spectrum in Increasing Order of Wavelength

Electromagnetic Wave	Wavelength Range
Gamma rays	Less than 10 picometres
X-rays	Between 0.01 nanometre (nm) and 10 nanometres (nm)
Ultraviolet (UV) rays	Between 10 nanometres (nm) and 400 nanometres (nm)
Visible light	Between 400 nanometres (nm) and 700 nanometres (nm)
Infrared rays	Between 700 nanometres (nm) and 1 millimetre (mm)
Microwaves	Between 1 millimetre (mm) and 1 metre (m)
Radio waves	Longer than 1 metre (m)

Decreasing order of wavelength: radio waves > microwaves > infrared waves > visible waves > ultraviolet waves > X-rays > gamma rays.

- **Decreasing order of energy / frequency:** gamma rays > X-rays > ultraviolet waves > visible waves > infrared waves > microwaves > radio waves.

> Spectrum Order

- **Order of increasing wavelength:** violet → blue → green → yellow → orange → red (*red has the longest wavelength*).
- **Order of increasing frequency / energy:** red → orange → yellow → green → blue → violet (*violet has the highest frequency/energy*).

> Electromagnetic Waves and their Specific Applications

Name of the Wave	Main Practical Uses Useful for the Exam (Exam Target)
Gamma rays	In medical science, to destroy harmful cancer cells.
X-rays	In diagnostic work, for seeing the structure of bone fractures and for examining the internal organs of the body.
Ultraviolet rays	In water purifiers, to destroy harmful germs and bacteria.
Infrared rays	In physiotherapy (treatment), TV remote controls and AC operating systems.
Microwaves	In radar systems, weather forecasting, microwave ovens, mobile phone networks and Wi-Fi.
Radio waves	In radio broadcasting, television (TV) telecommunication systems and for long-distance commands/signals.

> Classification of Sound Waves on the Basis of Frequency

Type of Wave / Motion	Frequency Range / Mach Number (Core Concept)
Infrasonic	The frequency of the wave is less than 20 hertz (Hz) , below the range of human hearing (e.g., earthquake waves).
Ultrasonic	The frequency of the wave is more than 20,000 hertz , beyond the range of human hearing (used by bats and in sonar).
Subsonic	When a body or wave travels slower than the speed of sound (Mach number < 1).
Supersonic	When a body moves faster than the speed of sound (more than Mach 1) (e.g., fighter jets).
Hypersonic	When a body travels at a speed far greater than the speed of sound (between Mach 5.0 and Mach 10.0) (e.g., missiles).

> Various Uses of Ultrasonic Waves (>20,000 Hz)

- **In SONAR equipment:** for locating hidden objects or submarines in the sea.
- **In cleaning work:** for removing dust from costly clothes, aircraft parts and the parts of expensive watches.
- **Industrial use:** for removing soot from factory chimneys and for heating liquids.
- **Medical field:** in the ultrasound machine, ECG, sonography and ultrasonography, etc.
- **Security and control:** for controlling automatic doors and for destroying harmful insects.

> Speed of Sound - Important Rules at 25°C

- **Rule of the medium:** the speed of sound is always in the order **solid > liquid > gas** (sound cannot travel in a vacuum).
- **Correct order in solids:** aluminium (highest: **6420 m/s**) > nickel > steel > iron > brass.
- **Descending order in various media (Exam Note):** glass > water > hydrogen > helium > air.

> Principal Fuel Gases and their Main Components

Name of the Fuel Gas	Principal Components / Constituents
CNG (Compressed Natural Gas)	Methane, ethane
LPG (Liquefied Petroleum Gas)	Butane, propane
Coal gas	Hydrogen, methane, carbon monoxide
Water gas	Carbon monoxide and hydrogen (CO + H ₂)
Biogas	Methane (CH ₄)

Name of the Fuel Gas	Principal Components / Constituents
Natural gas	Methane (with small amounts of ethane and propane)

> Types of Glass and their Principal Uses

Type of Glass	Main Practical Uses (Core Uses)
Flint glass	For making cameras, telescopes and lenses.
Pyrex glass	For making laboratory apparatus that withstands high temperatures.
Soda glass	In everyday tube lights, glass bottles and window panes.
Crown glass	For making electric bulbs.
Crookes glass	In the lenses of sunglasses (it blocks ultraviolet rays).
Potash glass	For making strong commercial glassware and beakers.
Soft glass	For making ordinary laboratory test tubes.

> Phenomena Caused by Surface Tension (Surface Tension Examples)

Serial No.	Practical Examples of Surface Tension Asked Repeatedly in the Exam
1	A thin steel needle or blade floating on the surface of water/mercury.
2	Adding soap or detergent lowers the surface tension of water, making washing of clothes easier.
3	Formation of large bubbles of soap solution.
4	Spraying kerosene on the surface of water lowers the surface tension, so that mosquito larvae sink and die.
5	Hot soup tasting delicious (because, owing to lower surface tension, it spreads fully over the tongue).
6	The bristles of a shaving brush cling together when taken out of water.
7	Oil being sprayed to calm the stormy waves of the sea.
8	A drop of oil spreading very quickly over the surface of water.
9	Falling raindrops being spherical.
10	Cooling produced by surface tension when water is broken into fine droplets by a spray.

> Important Epidemiological Terminology

Term	Simple Medical Definition	UPPCS Exam Point
Endemic	When a disease or infection is always present permanently (regularly) in a particular geographical area or population.	A disease confined to a definite area.
Epidemic	An infectious disease spreading very rapidly in a short time in a definite population or large area.	A sudden local outbreak.
Pandemic	When an epidemic takes a severe form and spreads over a very large part of the world, several continents or the whole world.	A global epidemic (e.g., COVID-19).

> Principal Isotopes and their Medical/Scientific Use

Isotope	Main Function / Practical Use Useful for the Exam
Uranium - 235	For finding the age of rocks, minerals and the ancient Earth (radiometric dating).
Sodium - 24	For accurately detecting blood clots inside the human body.
Cobalt - 60	For destroying cancer cells and controlling blood cancer (leukaemia).
Arsenic - 74	For detecting the position of tumours hidden inside the human body.
Phosphorus - 32	In the treatment of leukaemia (blood cancer) and some bone disorders.
Iodine - 131	For testing the activity of the thyroid gland (goitre) and its treatment.

> Important Natural Acids and their Sources

Name of the Acid	Natural Source / Place of Occurrence	Specific Practical Use (Exam Facts)
Acetic acid	Vinegar	In the preservation of pickles and foodstuffs.
Citric acid	Fruits of the citrus family such as orange, lemon, mosambi, etc.	For giving a sour taste to foodstuffs.
Lactic acid	Sour milk, curd	Produced by Lactobacillus in the process of milk setting into curd.
Oxalic acid	Spinach, tomato, sorrel plants	It is especially used for removing rust or ink (colour) stains from clothes .
Ascorbic acid (Vitamin C)	Amla, sour citrus fruits, guava	Helps increase the body's immunity.
Tartaric acid	Tamarind, grapes, raw mango	Helpful in the manufacture of baking powder.

Name of the Acid	Natural Source / Place of Occurrence	Specific Practical Use (Exam Facts)
Formic acid (methanoic acid)	Sting of red ants, bee sting, stinging nettle	Its sting causes burning and swelling on the skin.
Hydrochloric acid (HCl)	Gastric juice of animals/humans	Performs the main function of digesting the food taken by animals in the internal process of digestion.

> Principal Alkaline Compounds and their Natural Sources

Name of the Base	Commercial/Natural Source	Principal Practical Uses (Exam Points)
Calcium hydroxide	Lime water	In whitewashing, making bleaching powder and treating acidic soil.
Ammonium hydroxide [NH₄OH]	Detergents used for cleaning window glass (window cleaners)	In household cleaning and in chemicals that remove grease stains.
Sodium hydroxide / potassium hydroxide	Everyday soap	Caustic soda (NaOH) is used in hard soap (for washing clothes) and potassium (hydroxide) in soft soap (for bathing).
Magnesium hydroxide	Milk of Magnesia	As an excellent antacid to remove excess acidity of the stomach.

> International Standard Units of Measurement (SI Units)

Physical Quantity	Standard SI Unit (Symbol)
Length	Metre (m)
Mass	Kilogram (kg)
Time	Second (s)
Temperature	Kelvin (K)
Amount of substance	Mole (mol)
Luminous intensity	Candela (cd)
Power of a lens	Dioptre (D)
Heat / energy	Joule (J)
Pressure	Pascal (Pa)
Inductance	Henry (H)

Physical Quantity	Standard SI Unit (Symbol)
Resistivity or specific resistance	Ohm-metre ($\Omega \cdot m$)
Radioactivity	Becquerel (Bq)

> Various Phenomena of Light

Phenomenon of Light	Real Examples and Applications Useful for the Exam (Exam Hotspots)
Refraction of light	* An object or fish lying in water appearing less deep than its real depth. * The bottom of a vessel or pond filled with water appearing raised. * The sun being visible a few minutes before actual sunrise and after sunset. * The sun appearing flattened or oval near the horizon at sunrise and sunset. * Twinkling of stars in a clear sky (because light passes through atmospheric layers).
Reflection of light	* Seeing one's face/image in an ordinary mirror. * Images (reflections) of outside scenes appearing on the surface of calm or still water. * Light returning after striking any shiny, smooth surface.
Total internal reflection	* A diamond sparkling brilliantly because of its special cut. * Rapid communication without energy loss in optical fibres. * Formation of a 'mirage' in deserts or on roads in summer. * In the 'endoscope', an instrument used in medicine to examine the internal organs of the body.
Interference of light	* A thin film of soap foam or bubbles appearing coloured in sunlight. * A film of kerosene or diesel oil spread on the surface of water appearing coloured. * The surface of a CD (compact disc) appearing rainbow-coloured when light falls on it.
Scattering of light	* The clear sky appearing blue. * The water of the deep sea appearing blue. * The sun appearing red at sunrise and sunset. * Danger signals being made red on roads and railways (because red has the longest wavelength, it is scattered least and is visible from a distance).

> Scalar and Vector Quantities

Type of Quantity	Basic Definition and Feature	Exact Examples Useful for the Exam (Exam Examples)
Scalar quantities	They have only magnitude and no definite direction.	Distance, volume, mass, density, work, energy, time, temperature, power, charge, speed, electric potential, etc.

Type of Quantity	Basic Definition and Feature	Exact Examples Useful for the Exam (Exam Examples)
Vector quantities	They must have both magnitude and direction .	Displacement, velocity, acceleration, force, momentum, impulse, electric field, magnetic field, etc.

> Principal Characteristics of Sound Waves

Characteristic	Simple Physical Meaning and its Effect
Wavelength	The total distance travelled by a sound wave before it repeats itself (completes one cycle).
Amplitude	Indicates the maximum displacement of the particles of the medium. High amplitude = loud sound .
Frequency	The number of wave cycles produced per second. High frequency = thin/sharp sound (high pitch/shrillness) .
Time period	The total time required to complete one full wave cycle.
Velocity	The speed with which a sound wave travels through a medium.

> Important Facts about Convex and Concave Lenses (Lenses Matrix)

Feature / Point	Convex Lens (converging)	Concave Lens (diverging)
Action on light	Converges parallel rays of light to a point.	Diverges parallel rays of light outward.
Physical structure	This lens is thick in the middle and thin at the edges.	This lens is thin in the middle and thick at the edges.
Correction of vision defects	Spectacles of this lens are made to correct hypermetropia or long-sightedness .	Spectacles of this lens are made to correct myopia or short-sightedness .
Nature of the image	Depending on the position of the object, it can form both real and virtual images.	Always forms an image that is smaller than the object, erect and virtual .
Main practical applications	* Side mirrors (rear-view) of vehicles. * The natural lens inside the human eye. * In movie projectors, cameras and magnifying glasses. * In solar cookers for concentrating sunlight at one place. * In astronomical telescopes and microscopes.	* The mirror used by dentists to look inside the mouth (dentist mirror). * In spectacles for correcting short-sightedness. * In flashlights, peepholes (the small hole in a door) and laser instruments.

> Transformation of Energy

Device	Initial Form of Energy	Transformed Form of Energy (Final Output)
Electric motor	Electrical energy	Into mechanical energy
Engine (heat engine)	Heat energy	Into mechanical energy
Loudspeaker	Electrical energy	Into sound energy
Transmitter	Electrical energy	Into electromagnetic energy
Candle	Chemical energy	Into light and heat energy
Photoelectric cell	Light energy	Into electrical energy
Sitar	Mechanical energy	Into sound energy
Electric generator (dynamo/generator)	Mechanical energy	Into electrical energy
Electric bulb	Electrical energy	Into light and heat energy
Electric cell (battery)	Chemical energy	Into electrical energy
Tubelight	Electrical energy	Into light energy

> Practical Applications of Mirrors

Type of Mirror	Main Practical Uses and Examples (Exam Hotspot)
Concave mirror	* In torches, searchlights and vehicle headlights (to obtain a powerful parallel beam of light). * As a shaving mirror for seeing an enlarged image of the face. * Used by dentists to see enlarged images of patients' teeth. * As a reflector in solar furnaces for concentrating sunlight.
Convex mirror	* In the side-view (wing) mirrors of vehicles for a wide view of the rear (because it always forms an erect, diminished image and has a wide field of view). * In street lamps for spreading light over a large area. * In sunglasses.

> Principal Gases, Chemical Compounds and their Industrial Uses

Name of the Gas / Chemical	Main Industrial and Practical Uses Useful for the Exam (Exam Notes)
Hydrogen peroxide	* In medicine as a disinfectant and as a wash for cleaning wounds, gargling and cleaning ears. * In bleaching delicate materials such as hair, cotton, silk, leather and feathers. * As a tooth and mouth wash and as a rocket fuel (oxidiser). * In the food industry as a means of preserving milk, fish, meat and other foods from spoiling.
Helium (He)	* Being extremely light and non-inflammable, for filling the tyres of aircraft and balloons. * Used in place of pure air in delicate eye surgeries (medical surgeries).
Neon (Ne)	* Used in television screens, advertisement boards (neon signs) and plasma display tubes. * For bright light in high-voltage discharge tubes.
Xenon (Xe)	* It is also called the 'stranger gas'. It is used in high-speed photography. * In medical science as an anaesthetic, and in smartphone flashes and high-intensity discharge lamps.
Sulphur hexafluoride	* As an excellent gaseous electrical insulator (dielectric) in high-voltage electrical equipment. * In the construction of special modern windows and doors for sound-proofing and thermal insulation.
Sulphuric acid	* In the production of chemical fertilisers (such as superphosphate). It is called the 'king of acids'. * In the manufacture of industrial explosives and in the nylon/rayon synthetic textile industries. * In petroleum refining industries, in the manufacture of medicines and in the storage batteries (lead-acid batteries) of automobiles.
Nitrogen	* In the industrial refining of sulphur and the highly inflammable acetylene gas. * For creating an inert atmosphere to protect (preserve) canned grains and fruit juices from oxidation. * In the production of ammonia gas and of sodium cyanamide, sulphocyanide and calcium cyanamide fertilisers. * In military and industrial fertilisers, explosives and as high-security preservatives.
Ammonia	* In the large-scale production of nitrogenous chemical fertilisers (such as ammonium sulphate, ammonium nitrate, urea, ammonium phosphate and ammonium carbonate). * In the synthesis of nitric acid and other important industrial chemicals.

Name of the Gas / Chemical	Main Industrial and Practical Uses Useful for the Exam (Exam Notes)
Chlorine	* In the production of wood pulp in paper mills and in the manufacture of bleaching powder. * In the synthesis of various products of the chemical industry and of pesticides (such as PVC). * As a germicide, for making drinking water (water supply) completely pure and bacteria-free. * In the production of special industrial gases for photography, pharmaceuticals and the match industry.

➤ Principal Chemical Compounds: Names, Formulae and Specific Uses

Trade / Common Name	Chemical Name	Specific Uses and Key Points for the UPPCS Exam
Common salt	Sodium chloride	• For taste and preservation of food. • For making ORS solution in dehydration.
Baking soda	Sodium bicarbonate	• For leavening in the bakery industry (bread, cake). • In fire extinguishers and for removing acidity of the stomach (antacid).
Washing soda	Sodium carbonate	• In washing clothes and in the glass, soap and paper industries. • For removing the permanent hardness of water.
Bleaching powder	Calcium oxychloride	• For disinfecting drinking water. • As a bleaching agent in the textile and paper industries.
Plaster of Paris (POP)	Calcium sulphate hemihydrate	• For plastering broken bones (to set them). • For making statues, toys and ceiling designs (POP) in houses.
Gypsum	Calcium sulphate dihydrate	• For slowing down the setting rate of cement (retarder). • For the improvement/treatment of alkaline soil.
Quick lime	Calcium oxide	• In the manufacture of cement and glass. • For drying gases in the laboratory (drying agent).
Slaked lime	Calcium hydroxide	• For whitewashing houses. • For neutralising acidic soil.
Blue vitriol	Copper sulphate	• As a fungicide (e.g., Bordeaux mixture). • In electroplating and the dyeing industry.
Green vitriol	Ferrous sulphate	• In making medicine for anaemia and in making ink.

Trade / Common Name	Chemical Name	Specific Uses and Key Points for the UPPCS Exam
White vitriol	Zinc sulphate	• In eye medicines (eye drops) and the leather industry.
Caustic soda	Sodium hydroxide	• In the manufacture of soap, detergents and artificial silk (rayon).
Caustic potash	Potassium hydroxide	• In making soft (shaving) soap and liquid soap.
Laughing gas	Nitrous oxide	• As an anaesthetic in surgery and dentistry.
Chloroform	Trichloromethane	• As a solvent and formerly as an anaesthetic (for unconsciousness).
Heavy water	Deuterium oxide	• As a moderator and coolant in nuclear reactors.
Alum	Potassium aluminium sulphate	• For purification of water (by coagulation). • As an antiseptic (to stop bleeding).
Main constituent of glass	Silica (silicon dioxide)	• In the manufacture of glass and quartz.
Vinegar	Acetic acid (dilute)	• For preservation of foodstuffs and for giving sourness.
Shora / saltpetre	Potassium nitrate	• In making gunpowder , fireworks and fertilisers.
Lunar caustic	Silver nitrate	• In making voting ink (indelible ink) . • For silvering the back of mirrors.
Agent of artificial rain	Silver iodide	• For causing artificial rain through cloud seeding.

> Principal Organic Compounds and their Applications (Organic Chemistry Matrix)

Organic Compound	Main Practical Applications / Special Facts Useful for the Exam (Exam Notes)
Citric acid	For the long-term preservation of pickles, murabbas and fruit juices.
Naphthalene	Its white balls kept among clothes prevent insects (moths) from damaging the clothes.
Tear gas (chloroacetophenone)	Used by the police to disperse riots or uncontrolled crowds.
Carotene	The natural pigment that gives carrots their characteristic red/orange colour .

Organic Compound	Main Practical Applications / Special Facts Useful for the Exam (Exam Notes)
Gammexane (BHC) and DDT	As a powerful insecticide in agricultural and domestic use.
Mercaptan (ethyl mercaptan)	A strongly smelling substance added to detect leakage of LPG gas immediately .
Potassium permanganate (KMnO ₄)	For cleaning well water and in medicine as an antibacterial substance (red medicine).
Ethylene / calcium carbide	Chemical substances used for artificially (quickly) ripening raw green fruits.
Polycarbonates	Polymers used in the manufacture of very strong bullet-proof glass .
Oxalic acid	Used for removing ink stains from clothes and in the manufacturing process of photography.
Allethrin and pyrethrin	The main constituents used in making mosquito-repellent liquids and coils for homes.
Sodium benzoate / benzoic acid	The most widely used in the preservation of foodstuffs such as pickles, jams, sauces and fruit juices (food preservation).
Ethylene glycol	In cold countries as an anti-freeze in car radiators (prevents water from freezing).
Carbon tetrachloride (CCl ₄) / paraffin	In liquid fire extinguishers (Pyrene) and in industry as a solvent/coolant.
Bakelite	A thermosetting plastic; used in making handles, switches and plugs of electrical equipment .

> Principal Human Diseases and their Responsible Agents (Environmental Diseases)

Name of the Disease	Responsible Pollutant / Metallic Agent (Core Cause)
Blood cancer (leukaemia)	Due to industrial exposure to benzene .
Itai-Itai	Due to pollution caused by the metal cadmium (a disease of the bones).
Minamata	Due to consuming fish/water contaminated with mercury (Hg) (a disease of the nervous system).
Blue Baby Syndrome	Due to an excess of nitrate (–NO₃) in drinking water (methaemoglobinaemia).
Blackfoot disease	Due to drinking water contaminated with arsenic (As) for a long time (a disease of the skin and feet).

Name of the Disease	Responsible Pollutant / Metallic Agent (Core Cause)
Wilson's disease	Due to excessive accumulation/toxicity of the metal copper (Cu) in the body.
Siderosis	Due to the deposition of fine particles of iron (Fe) in the lungs (a respiratory disease).

> Gases and their Characteristics Useful for the Exam

Name of the Gas (Compound)	Main Characteristic Useful for the Exam (Exam Facts)
Carbon monoxide (CO)	It combines with haemoglobin in the blood to form 'carboxyhaemoglobin', which completely blocks the oxygen-carrying capacity of the blood and the person dies of suffocation. It is also called the 'suffocating gas'.
Nitrous oxide	It is called laughing gas ; used as an anaesthetic in minor surgery.
Carbon dioxide	The main gas responsible for the greenhouse effect . Its amount in the atmosphere is about 0.03% .
Methane	A gas emitted from marshy areas and paddy fields; the main constituent of biogas and CNG .
Radon	The only radioactive noble/inert gas that is not normally found in the atmosphere .
Chlorine	The gas used by municipal corporations to make drinking water pure and bacteria-free .
Butane	The main constituent of the domestic LPG cylinder (along with propane).

> Carbon and its Allotropes

- **The 3 main allotropes of carbon:** (1) diamond, (2) graphite, (3) buckminsterfullerene (C₆₀).
- **Graphene:** a layer of pure carbon **only one atom thick** (2D structure), which is extremely strong and a good conductor of electricity.
- **Diamond:** the **hardest allotrope** of carbon found naturally (it sparkles because of total internal reflection).
- **Graphite:** thermodynamically **the most stable form of carbon**; it is soft and a good conductor of electricity.

> Prelims Facts Related to Metals and Non-metals

- **Metal found in blood:** iron (at the centre of haemoglobin).
- **Lightest metal:** lithium (used in batteries).
- **Heaviest metal:** osmium.
- **Hardest metal:** platinum (also called white gold).
- **Filament of the electric bulb:** tungsten (because of its high melting point).

- **Best conductor of heat and electricity among metals:** silver.
- **Soft metals that can be easily cut with a knife:** sodium (Na) and potassium (K).
- **The only non-metal that remains liquid at room temperature (25°C):** bromine (a dark red liquid).

> Difference between Physical Change and Chemical Change

Point of Comparison	Physical Change	Chemical Change
New substance	No new substance is formed in it.	A new substance is necessarily formed, whose properties differ from those of the original substance.
Permanence and reversibility	It is a temporary change which is easily reversible .	It is a permanent change , which is usually irreversible .
Exchange of energy	Usually a very small amount of heat or light energy is absorbed or released.	A large amount of heat or light energy is absorbed or released.
Basic nature of the substance	The chemical composition/nature of the substance does not change.	The basic chemical nature of the substance changes completely.
Principal examples (Exam Hotspots)	* Water freezing into ice. * Water boiling into steam. * Melting of wax. * Sugar or salt dissolving in water.	* Digestion of food. * Burning of paper or wood. * Rusting of metals. * Silver utensils turning black/tarnishing. * Burning of fuel (LPG, petrol). * Curdling of milk or formation of curd from milk.

2 Energy-Transforming Devices (Energy Transducers)

Name of the Device	Initial Form of Energy	Transformed Form of Energy (Final Output)
Dynamo	Mechanical energy	Into electrical energy
Microphone	Sound energy	Into electrical energy
Loudspeaker	Electrical energy	Into sound energy
Solar cell	Solar energy	Into electrical energy
Tubelight	Electrical energy	Into light energy
Electric motor	Electrical energy	Into mechanical energy
Electric bulb	Electrical energy	Into light and heat energy

Name of the Device	Initial Form of Energy	Transformed Form of Energy (Final Output)
Electric cell (battery)	Chemical energy	Into electrical energy
Sitar	Mechanical energy	Into sound energy

> Principal Common Alloys: Chemical Composition and Uses (Master Alloys Matrix)

Alloy	Chemical Composition	Main Practical Uses (Core Uses)
Bronze	Copper (88%) + tin (12%)	In making statues, coins, medals and strong utensils.
Brass	Copper (70%) + zinc (30%)	In band instruments, radio parts, wires, mridangams and machine tools.
Gun metal	Copper (88%) + tin (8-10%) + zinc (2-4%)	In making statues, gears, bearings, shutters, gutters and weapons.
Solder	Tin (50-60%) + lead (40-50%)	For soldering (joining) metals in electronic circuits.
Bell metal	Copper (80%) + tin (20%)	In making temple bells, household utensils and statues.
Rolled gold	Copper (90%) + aluminium (10%)	In making cheap ornaments that look like gold .
Dutch metal	Copper (80-85%) + zinc (15-20%)	In making very thin foils or leaves resembling copper or gold.
Stainless steel	Chromium (15-20%) + nickel (8-10%) + iron	Rust-free household utensils, surgical instruments , blades, etc.
Chrome steel	Chromium (2-3%) + iron (95%) + carbon (0.5%) + manganese (0.5-1%)	In making parts of heavy machinery, cutting tools and strong safes.
Magnalium	Aluminium (95-96%) + magnesium (2%) + iron (2.3%)	In making light metallic mirrors and scientific instruments.
Duralumin	Aluminium (95%) + copper (4%) + magnesium (0.5%) + manganese (0.5%)	Being very light and strong, in making the frames of aircraft, buses and automobiles .
German silver	Copper (50%) + zinc (35%) + nickel (15%)	(Note: it contains 0% silver); in heating equipment, utensils, fancy belts, etc.
Chrome-vanadium steel	Iron (97-98%) + chromium (0.8-1.1%) + manganese (0.7-0.9%) + nickel (0.5%)	In making high-stress automobile springs, axles and strong tools.

Alloy	Chemical Composition	Main Practical Uses (Core Uses)
Tungsten steel	Iron + tungsten + carbon	In rocket-engine nozzles that withstand very high temperatures and in heavy machine tools.
Aluminium bronze	Copper (90%) + aluminium (10%)	In making corrosion-free (rust-free) utensils, jewellery and shiny paints.
Nichrome	Nickel (55-60%) + chromium (20%) + iron (20-25%)	Used in making the filaments of electric heaters and heating coils (because of its high resistivity).

> Branches of Biology

Branches of Biology	
Ornithology	■ Study of birds
Ecology	■ Study of the effect of the surrounding environment on living beings
Palaeobiology	■ Study of fossils
Phrenology	■ Study of the functioning and derangement of the various parts of the brain
Anthology	■ Study of flowers
Agrostology	■ Study related to grasses
Entomology	■ Scientific study of insects
Parasitology	■ Study of parasitic organisms
Malacology	■ Study of molluscs and their shells
Genetics	■ Study of the hereditary characters of organisms and their inheritance
Demography	■ Study of human population and of the vital statistics of mankind
Virology	■ Study of viruses
Ethnology	■ Comparative study of the scientific descriptions of various cultures
Ethnography	■ Study of a particular culture
Ethology	■ Study of the behaviour of all animals including humans

Branches of Biology	
Ethics	■ Psychological study of moral conduct and duty
Bionics	■ Study of the functions, properties and methods occurring in the biological world and their application to non-living systems
Bionomics	■ Study of the relationship of organisms with their environment
Bionomy	■ Study of the laws of life
Palaeobotany	■ Study of plant fossils
Agronomy	■ Cultivation of crops
Gerontology	■ A branch of biology related to old age and ageing in humans
Oncology	■ A branch of medical science for the study of tumours related to cancer
Teratology	■ Study of abnormalities in physical development
Pedology	■ Scientific study of soil formation and types
Osteology	■ Study of bones
Orology	■ Study of mountains
Serpentology	■ Study of snakes
Herpetology	■ Study of amphibians and reptiles
Ichthyology	■ Study of fishes
Dermatology	■ Study of the skin and its diseases
Trophology	■ Study of nutrition and methods of nutrition
Tetology	■ Study of acquired characters
Bacteriology	■ Study of bacteria
Entomology	■ Study of insects
Myrmecology	■ Study of ants
Pathology	■ Study of the nature, symptoms and causes of diseases

Branches of Biology

Lithotripsy	■ A medical procedure in which the patient is treated by breaking up stones in the kidney, gall bladder or urinary bladder with the help of rays.
Neurology	■ Study related to the nervous system
Haematology	■ Study of blood
Serology	■ Scientific study of blood serum
Chondrology	■ Study related to cartilage
Mammalogy	■ Study of mammals
Myology	■ Study of human muscles
Physiology	■ Study of the functions and working of the various parts of the body

Biofertilisers

Nitrogen-fixing	■ Rhizobium ■ Azotobacter ■ Azospirillum ■ Acetobacter ■ Blue-green algae ■ Azolla
Phosphorus-solubilising	■ Bacillus ■ Pseudomonas ■ Aspergillus
Plant-growth promoting	■ Rhizobacteria ■ Pseudomonas
Blue-green algae	■ Other name:- Cyanobacteria ■ Prokaryotic algae ■ Symbiont of Azolla ■ Helpful in nitrogen fixation ■ Used as a biofertiliser in the paddy crop
Azolla	■ Aquatic fern ■ Feeding it to milch cattle increases the quantity of milk

Fruits and their Edible Parts	
Edible part	■ Fruit
Thalamus	■ Apple and pear
Pericarp, placenta	■ Guava, grape and tomato
Mesocarp	■ Papaya and mango
Endosperm	■ Coconut
Mesocarp and endocarp	■ Banana
Endosperm and embryo	■ Wheat
Peduncle (pedicel), cotyledons	■ Cashew
Aril	■ Litchi
Cotyledons and embryo	■ Gram, groundnut
Succulent perianth	■ Mulberry
Perianth and seeds	■ Jackfruit
Perianth	■ Pineapple
Juicy hairs	■ Orange

> Human Diseases

Human Diseases	
Two classes	■ Congenital diseases ■ Acquired diseases:- infectious and non-infectious
Germ Theory of Disease	
Germ theory of disease	■ Propounded by Robert Koch ■ Diseases are caused by various micro-organisms and harmful substances ■ Robert Koch proved that the anthrax disease of animals is caused by a microbial bacterium
Infectious Diseases	

Human Diseases	
Introduction	Such diseases spread when organisms come into contact with one another
Agents	- Bacterial diseases - Viral diseases - Protozoan diseases - Fungal diseases
Viral Diseases	
Virus	- Origin of the word :- from the Latin word 'virus' - Chemical composition :- a segment of nucleic acid surrounded by a protein coat - Multiplies only in a living cell
Virology	Science of viruses (branch of study)
Smallpox	
Virus	Variola virus
Transmission	Through the patient's skin, sputum, air, clothes, etc.
Vaccine	Edward Jenner (father of immunology)
Mumps	
Other name	Galsua
Virus	Mumps virus
Affected organ	Parotid gland
Polio	
Agent	Poliomyelitis virus
Transmission	Through water and food
Affected organ	Due to destruction of nerve tissues, messages stop reaching the muscles (paralysis)
Programme	Polio eradication in India:- since 1995
Jonas Salk	Made the first polio vaccine in 1952

Human Diseases	
Albert Sabin	■ Made the first oral polio vaccine ■ Taken by mouth
India	■ Polio-free since 2014
Hepatitis – B	
Cause	■ Hepadnavirus family
Affected organ	■ Swelling and inflammation in the liver
Transmission	■ Infected body fluids
Dengue	
Other name	■ Break-bone fever
Virus	■ Dengue virus (Flaviviridae family)
Vector mosquito	■ Female Aedes aegypti ■ Aedes albopictus ■ Culex fatigans
Affected organ	■ Rapid fall in the number of platelets in the blood ■ Internal bleeding
Rabies	
Other names	■ Jalantak ■ Hydrophobia
Affected organ	■ Central nervous system
Agent	■ Rabies virus
Transmission	■ From dogs, cats, bats, foxes, raccoons, etc.
Vaccine	■ Discovery :- Louis Pasteur ■ Year :- 1885 CE
Main symptoms	■ Fear of water ■ High fever
Yellow Fever	

Human Diseases	
Cause	Yellow fever virus
Vector mosquito	Aedes aegypti
Aedes Mosquito	
Diseases carried	Yellow fever, dengue fever, chikungunya, Zika fever, Mayaro fever
Other name	Tiger mosquito
Identification	White stripes on the legs
Japanese Encephalitis	
Cause	A virus of the Flaviviridae family
Vector	Culex mosquito
Purvanchal (UP)	Endemic (epidemic) affected region
AIDS	
Cause	Human Immunodeficiency Virus
Transmission	- Unsafe sexual contact - Contaminated blood transfusion
Affected organ	- The body's immune system - Destruction of the immune white blood cells called T-lymphocytes
1 December	World AIDS Day
Diagnostic tests	- ELISA - Western blot
Influenza	
Agent	Influenza virus
Infection	Air-borne disease
Symptoms	Cough, phlegm, sneezing, fever, etc.
Measles	
Agent	Morbilli virus

Human Diseases	
Infection	Air-borne disease
Chickenpox	
Agent	Varicella zoster virus
Infection	Through breath or sneezes
Symptoms	Mild fever and pain in the joints
Herpes	
Agent	Herpes virus
Affected organ	Skin
Other Diseases	
Ebola	Ebola virus
Diarrhoea	Rotavirus
Rubella	A viral disease
Nipah virus disease	- Agent :- Nipah virus - Zoonotic disease
Bacterial Diseases	
Tuberculosis	
Other names	Yakshma, tuberculosis (TB),apedik
Discoverer	Robert Koch
Affected organ	Lungs
Agent	Mycobacterium tuberculosis
Treatment method	- DOTS :- Directly Observed Treatment - MDR TB:- Multi-Drug Resistant TB
On not completing the DOTS course	Antibiotic drugs stop having any effect on the tuberculosis bacteria
Vaccine	BCG :- Bacillus Calmette–Guérin

Human Diseases	
Plague	
Cause	Caused by the bacterium <i>Pasteurella pestis</i> or <i>Yersinia pestis</i>
Affected organ	Lungs and lymph glands
Vector	Fleas carrying the germs of infected rats
Salmonella Bacilli	
Structure	Rod-shaped
Disease	Food poisoning
Spread	- Infected water - Contaminated food items
Cholera	
Bacterium	<i>Vibrio cholerae</i> (discovery:- Robert Koch)
Spread	Contaminated water and food
Affected organ	Intestine
Symptoms	Continuous diarrhoea and vomiting
Diphtheria	
Bacterium	<i>Corynebacterium diphtheriae</i>
Other names	- Rohini - Galaghontu
Affected organ	Windpipe (respiratory tract)
Vaccine	- DPT (combined vaccine) - This vaccine protects against diphtheria, tetanus and whooping cough
Whooping Cough (Pertussis)	
Bacterium	<i>Haemophilus pertussis</i> or <i>Bordetella pertussis</i>
Other name	Kukur khansi

Human Diseases	
Affected organ	Respiratory system
Vaccine	DPT
Pneumonia	
Other names	Phupphus shoth or phupphus pradah (inflammation of the lungs)
Bacterium	Streptococcus pneumoniae
Symptoms	- High fever - Water in the lungs
Affected organ	Lungs
Vaccine	Pneumococcal vaccine
Typhoid	
Other names	- Antra jwar (enteric fever) - Miyadi bukhari
Bacterium	Salmonella typhi (discovery:- Dr. Daniel Salmon, 1885 CE)
Affected organ	Intestine
Vaccine	TAB vaccine
Transmission	Dirty drinking water
Symptoms	- High fever - Headache
Tetanus	
Bacterium	Clostridium tetani
Other names	Dhanushtankar, lockjaw
Vaccine	DPT vaccine
Affected organ	Nervous system
Sujak / Gonorrhoea	

Human Diseases	
Bacterium	Neisseria gonorrhoeae
Affected organ	Urinary tract
Syphilis	
Bacterium	Treponema pallidum
Affected organ	Penis
Leprosy	
Transmission	Through physical contact
Bacterium	Mycobacterium leprae or lepromatosis
Symptoms	- Patches on the body and loss of sensation at those places - Degeneration of tissues
Eradication	- National Leprosy Eradication Programme since 1983 - India declared leprosy-free in 2005
Treatment drugs	- Dapsone - Rifampicin - Clofazimine
Antibiotic Drugs	
Effective	Against various bacterial diseases
Examples	Chloromycetin, erythromycin
Penicillin	- Discoverer:- Alexander Fleming (1928) - The first antibiotic - Obtained:- from the fungus Penicillium notatum
Sulpha Drugs	
What it is	A group of synthetic drugs
Examples	Sulphonamide, sulphamerazine, sulphadiazine, etc.
Prontosil	- The first sulpha drug - Year 1932

Human Diseases	
Trachoma or Roha	
Bacterium	■ The bacterium Chlamydia trachomatis
Infection	■ By using articles infected with the above bacterium
Symptoms	■ The eyes become red and there is growth in the cornea
Fungal Diseases	
Athlete's Foot	
Fungus	■ The fungi Trichophyton and Tinea pedis
Transmission	■ Infected soil
Affected organ	■ Soft parts of the skin, especially the parts between the toes/fingers
Aspergillosis	
Fungus	■ Fungi of the Aspergillus species
Affected organ	■ Lungs
Symptoms	■ Suffocation, loss of smell
Ergotism	
Fungus	■ Claviceps purpurea
Transmission	■ Consumption of contaminated grain
Affected organ	■ Digestive system, nervous system
Other Fungal Diseases	
Asthma	■ The fungus Aspergillus fumigatus
Scabies (itch)	■ Acarus scabiei fungus
Baldness	■ Tinea capitis fungus
Ringworm	■ Trichophyton verrucosum fungus
Protozoan Diseases	

Human Diseases	
Kala-azar	
Other name	Leishmaniasis
Cause	The protozoan <i>Leishmania donovani</i>
Affected organ	Skin and the entire body
Vector	Sand fly
Sleeping Sickness	
Cause	The protozoan <i>Trypanosoma</i>
Vector	Tsetse fly
Affected organ	- Blood - Cerebrospinal fluid - Heart muscles - Gonads
Malaria	
Discovery	By Sir Ronald Ross
Affected organ	RBCs and spleen
Cause	The protozoan <i>Plasmodium</i>
Vector	Female <i>Anopheles</i> mosquito
25 April	World Malaria Day
Quinine	A natural drug obtained from the bark of the cinchona tree
Synthetic drugs	Camoquin, chloroquine, atebirin
Gambusia	A fish released into water bodies to eat mosquito larvae
Dysentery	
Cause	<i>Entamoeba histolytica</i>
Infection	Through impure water
Other Parasitic Diseases	

Human Diseases	
Diarrhoea	Giardia intestinalis - small intestine
Pyorrhoea	Entamoeba gingivalis - gums affected
Schistosomiasis	
Other name	Bilharzia
Agent	The fluke (flatworm) Schistosoma
Affected organ	- Liver - Intestine and other organs
Symptoms	Itching of the skin, fever, headache, stomach ache
Medium	Unsafe drinking water, lack of sanitation
Worm-borne Diseases	
Filaria	
Other names	Shleepad, elephantiasis (hathi paon), feel paon
Cause	From the roundworm Wuchereria bancrofti of the phylum Nematoda
Vector	Female Culex mosquito
Symptoms	Swelling and deformity of the hands, legs, genitals and other organs
Affected organ	Lymph glands
Ascariasis	
Agent	The roundworm Ascaris
Effect	- Improper nutrition of children - Adverse effect on physical and mental development
Affected organ	Intestine
Drugs	- Alcopar - Oil of chenopodium
Fascioliasis	

Human Diseases	
Cause	The flat liver fluke of the phylum Platyhelminthes
Location	In the large bile ducts of the liver of sheep, goats, pigs, buffaloes, etc.
Neurocysticercosis	
Cause	The tapeworm <i>Taenia solium</i>
Affected organ	Nervous system
Spread	Eating pork
Hereditary Diseases	
Colour Blindness	
Other names	- Red-green blindness - Daltonism
Effect	Inability to distinguish red and green colours
Carriers	Women
Sufferers	Men
Haemophilia	
Other names	- Bleeder's disease - Royal disease
Beginning	From Queen Victoria
Type	Sex-linked hereditary disease
Carriers	Women are carriers while men get the disease
Effect	Blood does not clot
Down Syndrome	
Other name	Mongolian idiocy
Cause	Three chromosomes instead of two in the 21st pair of chromosomes (trisomic)

Human Diseases	
Effect	■ Impotence in affected males
Symptoms	■ The person has a round head, thick neck, open mouth and slanting eyes
Turner Syndrome	
Cause	■ Only one chromosome present among the sex chromosomes (monosomic)
Effect	■ Female sterility
Klinefelter Syndrome	
Cause	■ Three sex chromosomes instead of two (trisomic)
Effect	■ Male sterility
Albinism	
Other name	■ Ranjak heenta (lack of pigment)
Effect	■ Absence of colour in the skin, hair and eyes
Other Diseases	
Cancer	
Oncogene	■ Responsible for cancer ■ First identified as SRC ■ Year of discovery:- 1970 ■ First discovered in chickens
Cause	■ Disorganised tissue growth due to uncontrolled division and growth of cells
Metastasis	■ Spread of cancer cells from one organ of the body to another nearby organ
Treatment	■ Radio-phosphorus (P-32) ■ Cobalt-60 ■ Arsenic ■ Gold-198

Human Diseases

Cancer stages

- First:- cancer cells confined to one part of the body
- Second:- cancer cells reach the lymph nodes of the infected organ
- Third:- lymph nodes carry the cancer cells to other organs of the body (metastasis)

Gout (Arthritis)

Cause

- Deposition of uric acid in the joints of the body

Effect

- Unbearable pain in the joints

Dropsy

Cause

- Adulteration of mustard oil with argemone oil

Effect

- Swelling in the body

Hysteria (Vatonmad)

Cause

- A nervous disorder

Affected

- Generally young and unmarried women

Alzheimer's Disease

Affected organ

- Brain

Discovery

- German psychiatrist Dr. Alzheimer

Effect

- Loss of memory

Parkinson's Disease

Cause

- Decrease in the level of dopamine due to damage to nerve cells in the brain

Affected organ

- Nervous system

Astigmatism

Symptoms

- Reduction in the overall vision of the person

Treatment

- Cylindrical lens

Cataract

Human Diseases	
Symptoms	■ The lens of the eye becomes opaque
Treatment	■ Operation and fitting of a new artificial lens
Myopia	
Other name	■ Short-sightedness (near-sightedness)
Symptoms	■ Distant objects are not seen
Treatment	■ Concave lens
Hypermetropia	
Other name	■ Long-sightedness (far-sightedness)
Symptoms	■ Nearby objects are not seen
Treatment	■ Convex lens
Diabetes	
Cause	■ Reduced secretion of insulin ■ Increase of sugar in the blood
Ayurvedic treatment	■ Fenugreek seeds ■ Curry leaves
Insulin	■ It was discovered by Banting and Macleod
Heart Attack	
Cause	■ Blood clot in the coronary arteries due to deposition of cholesterol and lipid substances in fibrous tissues
Affected organ	■ Heart
Treatment	■ Anticoagulant drug:- aspirin ■ Surgery:- bypass surgery

Other Miscellaneous Facts about Plants

Dichogamy	■ Maturation of the androecium and gynoecium at different times in a bisexual flower ■ Self-pollination is not possible in these flowers ■ Two types:- ■ 1. Protandry ■ 2. Protogyny
Stem cutting	■ Developing a new plant from a part of the stem of the parent plant ■ Examples:- cocoa and sugarcane
Layering technique	■ Pressing a stem branch of a plant so that it touches the ground, then cutting off the roots that grow from the stem and raising them as new plants ■ Example:- jasmine
Sunflower	■ Botanical name:- <i>Helianthus annuus</i> ■ Very low amount of fat:- beneficial for heart patients ■ An angiosperm plant of the family Asteraceae
Aromatic oil is obtained	■ From sandalwood, camphor, peppermint and clove
Neem	■ Azadirachtin chemical (insecticide and fungicide) ■ Production of mosquito repellents ■ Uses:- helpful in checking pest growth in crops, biodiesel production, used in coating urea, useful in many diseases
Biodiesel	■ Chemical composition:- abundant hydrocarbons and lipids ■ Feature:- produced from plants such as jatropha, karanj, nagchampa and neem
Karanj or Karanja	■ Example of a biodiesel plant ■ Botanical name:- <i>Pongamia pinnata</i>
Jatropha	■ Other name:- Ratanjot ■ Botanical name:- <i>Jatropha curcas</i> ■ Angiosperm plant ■ Requires less water compared to other cash crops
Sadabahar (periwinkle)	■ A flowering plant that blooms all 12 months ■ Helpful in the treatment of diabetes

Other Miscellaneous Facts about Plants

Tulsi	■ Dicotyledonous and herbaceous plant ■ Antibiotic properties, expectorant ■ Flavonoids, saponins, glycosides, phenols, etc. present in the extract of tulsi leaves
Clove	■ Place of origin:- Indonesia ■ Family:- Myrtaceae ■ Part used:- flower bud
Cricket bat	■ Made:- from willow wood
Hockey stick	■ Made:- from mulberry wood
Sugarcane	■ Biofuel (ethanol) is obtained from it
Chilgoza	■ Botanical name:- Pinus gerardiana ■ Abundant carbohydrates and proteins ■ Obtained from the seed of the pine plant
Brahmi	■ Grows in marshy land ■ Use:- nourishment of the brain, invigorating
Mint (pudina)	■ Botanical name:- Mentha arvensis ■ Use:- production of menthol oil
Caffeine	■ Chemical nature:- alkaloid substance ■ Main sources:- tea, coffee, cocoa
Seaweeds	■ Are sources of iodine
Asafoetida (hing)	■ Produced from the secretion of the stem and root
Camphor	■ From the wood of the evergreen tree Cinnamomum camphora
Vanilla	■ Obtained from plants of the orchid family

Various Plant Organs

Root

Function	■ Absorption of water and nutrients from the soil and providing support to plants
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Various Plant Organs

Types

Three types

- Tap root:- mustard plant
- Fibrous root:- wheat plant
- Adventitious root:- grass, banyan

Modification of Roots

Modification of tap roots

For storage of food materials

- Fusiform:- radish
- Napiform:- turnip
- Conical:- carrot
- Tuberous- sweet potato

Modification of adventitious roots

For storage of food materials

- Tuberous roots:- sweet potato
- Fasciculated roots:- dahlia, shatavar (asparagus)
- Nodulose roots:- amba haldi (mango ginger)
- Moniliform (beaded) roots:- grape, bitter gourd

Stem

Introduction

- The aerial part of the plant axis which is normally situated above the ground and bears branches, leaves and flowers.

Functions

- Spreading the branches
- Holding up the leaves, flowers and fruits
- Conduction of water, mineral salts and photosynthetic products

Modification of Stems

In three ways

- Underground
- Aerial
- Sub-aerial

Underground stem

- Rhizome:- turmeric, ginger
- Tubers:- potato
- Bulb :- onion, garlic
- Corm :- jimikand (elephant-foot yam), arbi (colocasia)

Leaf

Various Plant Organs

Functions

- Preparing food by photosynthesis
- Controlling transpiration
- Some leaves store food, such as onion and garlic

Flower

Introduction

- The reproductive organ of the plant

Whorls of a flower

- Calyx
- Corolla
- Androecium (male reproductive organ)
- Gynoecium (female reproductive organ)

Complete flower

- In which all four whorls are found

Incomplete flower

- In which all four whorls are not found

Fruit

Introduction

- A mature or fully formed ovary

Formation

- After fertilisation

False fruits

- When a floral part of the plant changes into a part of the fruit
- Examples:- apple and cashew

True fruits

- All other fruits formed from the ovary

Plant Hormones

Growth promoters

- Those which increase the rate of growth
- Such as:- auxin, gibberellin, cytokinin

Growth inhibitors

- Those which decrease the rate of growth
- Such as:- abscisic acid, ethylene

Auxin

Introduction

- The most important plant hormone
- Formed at the upper tips of plants

Plant Hormones

Uses

- A high amount promotes stem growth while a low amount promotes root growth
- Important role in vegetative propagation
- In destroying weeds by spraying in fields
- Prevents plants from falling
- Development of fruits even without pollination and fertilisation in orange, lemon, grape, banana, tomato, etc.

Gibberellin

Introduction

- Formed at the upper tips of plants and in the embryo
- Promotes growth in all parts of plants

Uses

- Increase the height of plants
- Help in the germination of seeds
- Flowering and parthenocarpy (development of fruits without fertilisation)

Cytokinin

Other name

- Phytokinin

Uses

- Controls organ formation in plants
- Increases the rate of cell division
- Reduces the dormancy period and prevents senescence
- Protects chlorophyll from being destroyed

Ethylene

Other name

- Fruit-ripening hormone

Introduction

- Exists in the gaseous state
- Formed in the apical part of the stem of the plant
- It is a growth-inhibiting hormone

Functions

- Ripening of fruits
- Accelerating the abscission of leaves, fruits and flowers

Absciscic Acid

Functions

- Inhibitor of plant growth and plant metabolism
- Growth-inhibiting hormone:- stops cell division

Florigen

Plant Hormones

Formation	■ Takes place in the leaves
Function	■ Induces flowering in plants

Essential Nutrients for Plants

Function	■ Necessary for the regular, complete and disorder-free life cycle and proper development of plants
Essential elements	■ 17
Macronutrients	■ Total number:- 09 ■ Elements:- carbon, hydrogen, oxygen, nitrogen, phosphorus, potassium, sulphur, calcium, magnesium
Micronutrients	■ Total number:- 08 ■ Requirement:- in small amounts ■ Elements:- boron, chlorine, copper, iron, manganese, molybdenum, nickel and zinc
Boron	■ Necessary for nodule formation in leguminous (pulse) crops

Plant Physiology

Photosynthesis

Definition	■ Production of carbohydrates and oxygen by plants using the energy of sunlight, carbon dioxide and water
Reaction	■ $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$
Energy	■ Radiant energy is converted into chemical energy and stored in food materials
From CO₂	■ Carbon is taken
From water	■ Hydrogen is taken
Oxygen	■ Released into the atmosphere as a by-product of the reaction
Glucose	■ The final product of photosynthesis ■ Soon changes into starch

Plant Physiology

Red colour

- Most effective in photosynthesis

Photosynthetic pigments

- Location:- in the grana of the chloroplast
- Colours absorbed:- orange, red, blue, violet

Chlorophyll

- Pigment essential in photosynthesis
- Main element:- magnesium

Carotenoids

- Pigment that protects plants from ultraviolet damage

Respiration in Plants

Process of respiration

- A continuous process
- In it the plant takes in oxygen and releases carbon dioxide

Respiring parts

- All parts of plants
- Such as:- root, stem, leaf

Reproduction in Plants

In two ways

- Sexual and asexual

Asexual Reproduction

Vegetative propagation

- By stem :- ginger, garlic, etc.
- By leaf:- Bryophyllum, Bignonia
- By root:- sweet potato, dahlia, etc.

Budding

- Reproduction in yeast

Fragmentation

- Reproduction in algae

Spore formation

- Reproduction in fungi

Sexual Reproduction

Two methods

- Pollination and fertilisation

Flower

- The reproductive part of plants

Parts of a flower

- Sepals
- Petals
- Stamens
- Carpels

Plant Physiology

Reproductive organs in a flower

- Stamens and carpels

Stamen

- Filament:- a stalk
- Anther:- a flattened top part

Anther

- Pollen grains are produced here

Carpel

Three main parts

- Ovary
- Style
- Stigma

Pollination

What is it?

- Transfer of pollen grains from the anther to the stigma

Two types

- Self-pollination
- Cross-pollination

Self-pollination

- When pollen grains are transferred from the anther of a flower to the stigma of the same flower or of another flower of the same plant

Cross-pollination

- When pollen grains are transferred from the anther of one flower to the stigma of a flower of another plant of the same species

Agents

- Abiotic agents (wind and water)
- Biotic agents (animals)

Honeybee

- The most important biotic pollinator

> Vitamins and Other Nutrients

Nutrients

Vitamins

Introduction

- They are complex organic compounds

Discoverer

- Casimir Funk (1911-12)

Other name

- Accessory food factors
- Propounded by Frederick Gowland Hopkins

Nutrients	
Functions	- Control and regulation of metabolic activities - Protection of the body from diseases
Water-soluble	- Vitamin B - Vitamin C
Fat-soluble	- Vitamin A - Vitamin D - Vitamin E - Vitamin K
Vitamin A	
Other name	Retinol
Deficiency diseases	Night blindness, xerophthalmia, changes in skin cells, keratomalacia, stunted growth
Sources	Carrot, green vegetables, milk, liver, fish oil
Vitamin B1	
Other name	Thiamine
Deficiency diseases	Beri-beri, stoppage of growth
Sources	Cereals, pulses, soyabean, milk, etc.
Vitamin B2	
Other name	Riboflavin
Deficiency diseases	Cheilosis
Sources	Cheese (paneer), eggs, meat, leafy vegetables, yeast
Vitamin B3	
Other name	Niacin
Deficiency diseases	Pellagra or skin dermatitis
Sources	Yeast, meat, fish, eggs, milk, pulses, etc.

Nutrients	
Vitamin B5	
Other name	■ Pantothenic acid
Deficiency diseases	■ Reduced growth, skin diseases, greying of hair, reduced fertility, asthma, stress
Sources	■ Eggs, milk, meat and groundnut
Vitamin B6	
Other name	■ Pyridoxine
Deficiency diseases	■ Anaemia, skin diseases, muscular cramps
Sources	■ Milk, yeast, cereals and meat
Vitamin B7	
Other name	■ Biotin
Deficiency diseases	■ Skin diseases, hair fall
Sources	■ Meat, wheat, egg, vegetables, fruits, etc.
Vitamin B9	
Other name	■ Folic acid
Deficiency diseases	■ Anaemia, stunted growth
Sources	■ Green vegetables, pulses, yeast, meat, eggs
Vitamin B12	
Other name	■ Cyanocobalamin
Deficiency diseases	■ Anaemia, disorders of the nervous system
Sources	■ Meat, fish, egg, milk, intestinal bacteria
Vitamin C	
Other name	■ Ascorbic acid

Nutrients	
Deficiency diseases	■ Scurvy
Sources	■ Amla, fruits of the lemon family (citrus fruits), tomato and vegetables
Vitamin D	
Other name	■ Calciferol
Deficiency diseases	■ Rickets (rickets, osteomalacia)
Sources	■ Butter, liver, eggs, sunlight, fish oil, milk
Vitamin E	
Other name	■ Tocopherol
Deficiency diseases	■ Reduced fertility, weakness of reproductive organs and muscles
Sources	■ Oils, wheat, egg yolk, soyabean
Vitamin K	
Other name	■ Naphthoquinone
Deficiency diseases	■ Blood does not clot on injury
Sources	■ Intestinal bacteria, green vegetables, egg and soyabean

Carbohydrates	
Introduction	■ Organic substances made of carbon, hydrogen and oxygen in the ratio 1:2:1
Functions	■ Providing energy to the body ■ Acting as an energy store ■ Formation of nucleic acids (RNA and DNA)
Monosaccharides	■ The simple form of carbohydrates ■ Glucose, galactose, fructose, triose, etc.
Disaccharides	■ Made of two molecules of monosaccharides ■ Sucrose, maltose, lactose, fructose

Carbohydrates	
Polysaccharides	- Made of many molecules of monosaccharides - Starch, chitin, glycogen
Proteins	
Naming	G. J. Mulder
Discoverer	J. J. Berzelius
Amino acids	- Made of 20 types of amino acids - Essential amino acids:- 9 (from food) - Non-essential amino acids:- 11 (in the body)
Essential amino acids	- Obtained only from food - Histidine, isoleucine, leucine, lysine, methionine, phenylalanine, tryptophan, valine, threonine
Non-essential amino acids	- Also synthesised in the cells of the body - Glycine, serine, tyrosine, alanine, glutamic acid, glutamine, proline, aspartic acid, asparagine, arginine, cysteine
Functions of protein	- Growth and repair of the body - Formation of cells, protoplasm and tissues - Also provides energy to the body
Elements contained in protein	- Carbon, hydrogen, oxygen, nitrogen - Sulphur, phosphorus, iodine, iron
Simple proteins	- Those made only of amino acids - Such as:- globulin, histone, etc.
Conjugated proteins	- Proteins whose molecules are joined with other groups - Such as:- glycoprotein, chromoprotein, etc.
Derived proteins	- Proteins formed by hydrolysis - Such as:- peptone, proteoses, peptides, etc.
Hair	Made of the protein keratin
Milk	White colour due to the protein casein
Beta carotene	The light yellow colour of cow's milk is due to the presence of this protein

Carbohydrates	
Soyabean	The only non-animal protein in which all the essential amino acids are found
Other facts	Proteins are denatured by heat, X-rays, heavy metal salts, etc., but not by infrared radiation
Fats	
Formation	- From a combination of carbon and hydrogen - Made of glycerol and fatty acids
Functions	- Providing energy to the body - Getting deposited under the skin and preventing body heat from escaping
Saturated fats	- Less reactive - Single bonds - Harmful to health - Examples:- palm and coconut oil - Called saturated fats because they are saturated with hydrogen
Unsaturated fats	- More reactive - Double bonds between carbon atoms - Rapid rate of decomposition - Beneficial to health - Example:- fish oil
Trans fats	- Increase in the level of bad LDL cholesterol - Damage to the cardiovascular system - Examples:- animal fats and hydrogenated oils
Mineral Salts	
Minor (Macro) Minerals	
Calcium	- Deficiency:- weak teeth and bones, stunted body growth - Importance:- formation of teeth and bones, blood clotting, functioning of nerves and muscles
Phosphorus	- Deficiency:- weak teeth and bones - Importance:- providing strength to teeth and bones and acid-base balance

Carbohydrates	
Sulphur	■ Deficiency:- protein deficiency ■ Importance:- constituent of many amino acids and some hormones and vitamins, and necessary for the synthesis of carotene
Chlorine	■ Deficiency:- loss of appetite, fatigue ■ Importance:- acid-base balance, water balance
Sodium	■ Deficiency:- low blood pressure, fatigue, muscular cramps ■ Importance:- acid-base balance, heartbeat
Magnesium	■ Deficiency:- the physiology of the nervous system is affected ■ Necessary for the functioning of the nervous and muscular systems
Trace Minerals	
Iodine	■ Deficiency:- goitre and slow development ■ Importance:- an important constituent of the hormone thyroxine
Iron	■ Deficiency:- haemoglobin deficiency > anaemia ■ Importance:- constituent of haemoglobin, myoglobin and the cytochrome enzymes
Zinc	■ Deficiency:- rough skin, loss of fertility, loss of appetite, stunted growth ■ Importance:- necessary for normal growth, development of reproductive organs and insulin physiology
Copper	■ Deficiency:- anaemia, dryness of hair ■ Importance:- iron metabolism, development of connective tissue and blood vessels
Manganese	■ Deficiency:- irregularity in the growth of cartilage, bone and connective tissues
Cobalt	■ Deficiency:- anaemia ■ Importance:- an important constituent of vitamin B12, formation of RBCs
Other Miscellaneous Facts Related to Nutrients	

Carbohydrates	
Water	- Calorie-free diet - Helps in absorbing the nutrients present in food
Milk	- Other name:- complete food - Rennin:- the enzyme that digests milk protein - The sourness of milk is due to Lactobacillus bacteria - Composition:- water (about 87%), fat, protein, carbohydrate, calcium, potassium
Starch	- Obtained from plants - Reacts with iodine solution to give a blue-black colour - Present in plant seeds in the form of amylose or amylopectin
Cellulose	- Obtained from plants - Does not react with iodine solution - Forms the cell wall
Chilli	Vitamin C is found in it
Liver	Vitamin A is stored here
Vit. B6	The vitamin that helps in remembering dreams
Vit. E	Important source:- wheat germ oil
Vit. D	It is also called a hormone
Essential macronutrients	Potassium, sodium, magnesium, calcium, chlorine, sulphur
Alpha-keratin	Protein present in horns, hair, nails, hooves and skin
Iron	Highest content in green leafy vegetables
Papaya	Mainly vitamin C is found in it
Water	- About 60 to 70% of the body weight - Controls body temperature - Helps in the excretion of waste materials
Tyrosine	Due to deficiency of this protein a person feels sad and sluggish.

Organic Evolution

Origin of Life

Materialism	- Hypothesis presented by a Russian biochemist - In the book 'The Origin of Life'
Lamarckism	- In the book 'Philosophie Zoologique' in 1809 CE - The first rational theory on the hypothesis of organic evolution - By the French biologist Jean-Baptiste Lamarck
Principles of Lamarckism	- Tendency to increase in size - Direct effect of the environment - Effect of use or disuse of organs - Inheritance of acquired characters
Darwinism	- By Charles Darwin of England - Other names:- evolutionism or the theory of natural selection - Book:- On the Origin of Species by Means of Natural Selection (1859)
Principles of Darwinism	- Capacity for prolific reproduction in organisms - Struggle for existence - Variation and its inheritance - Survival of the fittest or natural selection
Mutation theory	- By Hugo de Vries of the Netherlands - Experiment:- on the plant <i>Oenothera lamarckiana</i> - According to him, a new species originates in a single step as a result of clear, permanent and sudden mutations.

Other Facts

Archaeopteryx	- The most ancient bird of the Jurassic period - It was the connecting link between reptiles and birds.
Euglena	Considered the connecting link between plants and animals.
Neopilina	Considered the connecting link between Annelida and Mollusca.
Dinosaur	Gigantic reptiles of the Mesozoic era
Mammoth	It is considered the ancestor of the elephant.
Cro-Magnon	This man is considered the last direct ancestor of modern man, Homo sapiens.

Biology and Classification	
The word 'Biology'	First used:- by Lamarck (France) and Treviranus (Germany) in 1801
Aristotle	- Father of biology - Father of zoology - Book:- Historia Animalium
Theophrastus	- Father of botany - Book:- Historia Plantarum
Characteristics of Living Beings	
Cellular organisation	- The structural and functional unit of all living beings - A defining characteristic of organisms
Metabolism	- All the biochemical reactions taking place in a living being to sustain life - Two types - Anabolism :- formation of complex molecules from simple molecules (growth) - Catabolism:- formation of simple molecules from complex molecules (respiration)
Growth	Formation of new cells in a living being by using food
Reproduction	The ability to give birth to organisms similar to oneself
Binomial Nomenclature	
Beginning	- By Carolus Linnaeus (father of taxonomy) - From the two-kingdom theory (since 1753)
Linnaeus's classification	- Division into the animal kingdom and the plant kingdom - The first word is the genus name, the second word the species name - In Homo sapiens, Homo is the genus and sapiens the species
Five-Kingdom Classification	

Biology and Classification	
Propounder	■ R. H. Whittaker ■ The most widely accepted classification in modern biology
Five kingdoms	■ Monera, Protista, Fungi, Plantae and Animalia
Monera	
Organisms	■ Unicellular prokaryotic
Cell organelles	■ Absence of an organised nucleus ■ Undeveloped organelles ■ Absence of nuclear membrane
Mode of nutrition	■ Autotrophic
Examples	■ Bacteria ■ Blue-green algae / cyanobacteria ■ Mycoplasma (capable of reproduction)
Protista	
Organisms	■ Unicellular eukaryotic ■ Exception:- sea-weed (multicellular)
Link	■ Between the animal and fungal kingdoms
Mode of nutrition	■ Both autotrophic and heterotrophic
Cell organelles	■ Cell wall in some ■ Nuclear membrane is found
Examples	■ Unicellular algae, diatoms, protozoa, Euglena, Paramecium, Chlorella, Amoeba
Fungi	
Organisms	■ Multicellular eukaryotic ■ Without chlorophyll ■ Thallophytes
Nutrition	■ Dependent on decaying organic matter ■ Heterotrophic, saprophytic, parasitic

Biology and Classification	
Cell wall	■ Made of a complex sugar called chitin ■ Cellulose absent
Examples	■ Yeast, Penicillium, mushroom
Mycology	■ Study of fungi
Lichen	■ Fungus + alga
Micro-organisms	
Bacteria	■ Simple and unicellular prokaryotic micro-organisms (discovery:- Leeuwenhoek) ■ Two types:- heterotrophic and autotrophic
Virus	■ Subcellular structure:- DNA or RNA surrounded by a protein coat ■ Nutrition in a host, not on its own ■ The link between non-living and living ■ Discovery:- Dmitri Ivanovsky (1892 CE) ■ It was discovered through the study of the mosaic disease of tobacco
Plants and their Sub-groups	
Organisms	■ Cell wall made of cellulose ■ Multicellular eukaryotic organisms
Mode of nutrition	■ Autotrophic ■ Food is made by the process of photosynthesis
Thallophyta	
Structure	■ Not differentiated into root, stem and leaves
Examples	■ Mainly aquatic plants (algae) ■ Ulothrix, Spirogyra, Chara
Bryophyta	
Structure	■ Differentiated into stem- and leaf-like structures ■ Absence of vascular tissues
Nickname	■ Amphibians of the plant kingdom

Biology and Classification	
Examples	■ Moss (Funaria), Marchantia
Pteridophyta	
Structure	■ Body differentiated into root, stem and leaves ■ Vascular tissues also present
Examples	■ Fern, horsetail, Marsilea
Gymnosperms	
Structure	■ Naked-seeded ■ Perennial, evergreen and woody
Examples	■ Pinus and Cycas
Angiosperms	
Structure	■ The seeds of the plants develop inside fruits, in the ovary
Other name	■ Flowering plants
Two parts	■ Monocotyledons ■ Dicotyledons
Examples	■ Fruit-bearing plants ■ Mango, guava, papaya
Plants: Other Facts	
Cryptogams	■ Hidden reproductive organs ■ Unable to produce seeds ■ Examples:- Thallophyta, Bryophyta, Pteridophyta
Phanerogams	■ Fully developed reproductive organs ■ Able to produce seeds ■ Examples:- gymnosperms, angiosperms
Animal Kingdom	
Organisms	■ Eukaryotic, multicellular, heterotrophic

Biology and Classification	
Classification	■ On many bases ■ Mainly on the basis of the notochord:- Non-Chordata and Chordata
Porifera	
Sponges	■ Members of this phylum are known by this name
Structure	■ Body with innumerable pores ■ Water system or canal system in the body
Other facts	■ Intake of food, respiration, removal of wastes, etc. through the route of water transport ■ Water enters through fine pores (ostia) and goes out through large pores (osculum)
Examples	■ Sycon, Spongilla, Euspongia
Coelenterata or Cnidaria	
Organisms / Structure	■ Tissue-level body organisation (diploblastic) ■ Aquatic, marine ■ Thread-like structures (tentacles) around the mouth for catching food
Examples	■ Sea anemone, Hydra, coral, jellyfish
Hydra	■ Respiratory organs and blood absent ■ Respiration takes place through diffusion
Ctenophora	
Related facts	■ Tissue-grade body organisation ■ Emission of light by bioluminescence
Other names	■ Sea walnuts, comb jellies
Examples	■ Pleurobrachia, Ctenoplane
Platyhelminthes	
Other name	■ Flatworms

Biology and Classification	
Structure	■ Triploblastic, bilaterally symmetrical, hermaphrodite ■ Organ-level body organisation ■ True body cavity (coelom) absent ■ Undeveloped digestive system ■ Flame cells for excretion
Examples	■ Taenia solium (tapeworm), Fasciola
Aschelminthes or Nematoda	
Other name	■ Roundworms
Structure	■ Triploblastic, bilaterally symmetrical, true body cavity ■ Tissues are found but not true organs ■ Complete alimentary canal
Other facts	■ Male small, female large ■ Unisexual and internal fertilisation
Examples	■ Ascaris (roundworm), threadworm, Wuchereria (the filaria worm)
Annelida	
Structure	■ Bilaterally symmetrical, triploblastic and organ-level organisation (true body cavity) ■ Body divided into many segments ■ Both unisexual and hermaphrodite
Examples	■ Nereis, Pheretima (earthworm), Hirudinaria (blood-sucking leech)
Arthropoda	
Structure	■ Bilaterally symmetrical, triploblastic ■ Open circulatory system (blood vessels absent)
Blood	■ Colourless, absence of haemoglobin (haemolymph)
Body	■ Covered by an exoskeleton (chitin) ■ Divided into head, thorax and abdomen ■ Jointed legs ■ Sense organs are found

Biology and Classification	
Other facts	■ Mostly egg-laying ■ Male and female are separate
Crustacea	■ Mainly aquatic ■ 5 pairs of legs ■ Respiration by gills ■ Examples:- crab, prawn, crayfish
Myriapoda	■ Terrestrial ■ Many pairs of legs ■ Respiration by tracheae ■ Examples:- centipede (kankhajura), millipede
Arachnida	■ Terrestrial ■ 4 pairs of legs ■ Respiration by tracheae and book lungs ■ Examples:- spider, scorpion, mite, tick
Insecta	■ Generally terrestrial and rarely aquatic ■ 3 pairs of legs ■ Respiration by tracheae ■ Examples:- butterfly, housefly, grasshopper, lice, cockroach, termite, bed bug, moth
■ Arthropoda is the largest phylum of the animal kingdom. ■ About two-thirds of the species on the Earth are arthropods.	
Mollusca	
■ The second largest phylum of the animal kingdom ■ These animals are terrestrial or aquatic	
Related facts	■ Bilaterally symmetrical, triploblastic, coelomate ■ Body soft and covered by a hard calcium shell ■ Open circulatory system ■ Kidneys for excretion, gills for respiration ■ Blood is colourless

Biology and Classification	
Male / female	Male and female separate and egg-laying
Examples	- Snail (Pila), Sepia (cuttlefish), Loligo (squid), Octopus (devilfish) - Aplysia (sea hare), Dentalium (tusk shell), Chaetopleura (chiton)
Echinodermata	
Structure	- Endoskeleton made of calcareous ossicles - Triploblastic and coelomate animals - Complete digestive system - Distinct excretory system
Other facts	- Spiny skin - Marine animals - Capacity for regeneration
Examples	Asterias or starfish, Ophiura or brittle star, Echinus or sea urchin, sea cucumber, hagfish
Hemichordata	
Organisms	Worm-like marine organisms
Structure	- Organ-system level of organisation - Bilaterally symmetrical, triploblastic and coelomate - Closed circulatory system
Examples	Balanoglossus, Saccoglossus
Chordata	
Three sub-phyla	- Urochordata:- notochord present only in the tail - Cephalochordata:- notochord from head to tail - Vertebrata:- notochord present in the embryonic stage, changed into the vertebral column in the adult
Class Pisces (Fishes)	
Chondrichthyes	

Biology and Classification	
Structure	■ Powerful jaws ■ Two-chambered heart ■ Skeleton of cartilage
Other facts	■ Viviparous, i.e., give birth to young ones ■ Cold-blooded (poikilothermic)
Examples	■ Scoliodon (dogfish), Pristis (sawfish), blue shark, whale shark, Carcharodon
Osteichthyes	
Structure	■ The heart is two-chambered ■ Endoskeleton bony ■ Air bladder is present
Other facts	■ Cold-blooded ■ Mostly egg-laying
Examples	■ Marine :- Exocoetus (flying fish), Hippocampus (sea horse) ■ Freshwater :- rohu, catla, magur ■ Aquarium :- Pterophyllum (angel fish), Betta (fighting fish)
Class Tetrapoda	
Amphibia	
Related facts	■ Heart made of three chambers ■ Respiration through gills, lungs and skin ■ Cold-blooded animals and lay eggs ■ Live both in water and on land
Examples	■ Bufo (toad), frog, Hyla (tree frog), salamander, Ichthyophis
Lung fish	■ The link between fishes and amphibians
Frog	■ Being highly permeable, it breathes through its skin itself. ■ On land the adult frog breathes through its lungs

Biology and Classification	
Reptilia	
Related facts	■ Body covered with scales made of keratin ■ Heart generally three-chambered ■ Cold-blooded and egg-laying
Examples	■ Tortoise, tree lizard, house lizard, chameleon, gharial, cobra, krait, viper, alligator
Crocodile	■ It has a four-chambered heart
King cobra	■ A nest-building snake
Mesozoic	■ Is called the age of reptiles
Poisonous snakes	■ Their fangs are modified forms of the maxillary teeth
Birds (Aves)	
Related facts	■ The heart is made of 4 chambers ■ Are warm-blooded (homeothermic) ■ Breathe through lungs
Ostrich	■ The tallest and largest living bird
Emu	■ The second largest bird in the world ■ Found in Australia.
Penguin	■ A flightless aquatic bird
Hummingbird	■ The smallest bird in the world ■ The only bird that can fly both forwards and backwards
Flightless birds	■ Ostrich, emu, kiwi, penguin, rhea, cassowary
Mammals	
Related facts	■ Most important characteristic :- presence of mammary glands ■ Two pairs of limbs ■ The heart is four-chambered

Biology and Classification	
Prototheria	■ The link between mammals and reptiles ■ Egg-laying mammals ■ Examples:- platypus, echidna
Metatheria	■ Mammals giving birth to immature young ones ■ Examples:- kangaroo or Macropus, koala
Eutheria	■ Birth of fully developed young ones ■ Examples:- dog, man, elephant
Blue whale	■ The largest mammal in the world
African elephant	■ The largest wild land animal on the Earth
Primates	■ Man, great apes, gorilla, chimpanzee, orang-utan
Miscellaneous	
Different organisms resembling fishes in name	■ Jellyfish :- phylum Cnidaria ■ Starfish :- phylum Echinodermata ■ Silverfish :- phylum Arthropoda ■ Cuttlefish :- phylum Mollusca ■ Hagfish :- Cyclostomata ■ Devilfish:- phylum Mollusca ■ Crayfish:- phylum Arthropoda ■ Razorfish:- phylum Mollusca
True fishes	■ Flying fish, pipefish, paddlefish, goldfish, globefish, catfish
Aquatic mammals	■ Sea cow, dolphin, seal or Phoca, sea lion, walrus, sperm whale ■ Blue whale:- the largest mammal on the Earth
Nilgai	■ The largest antelope species
Bat	■ A flying mammal
Cockroach	■ An organism with white (colourless) blood

Biology and Classification

Kangaroo

- An animal that excretes solid urine
- Keeps its young in the marsupial pouch until they develop

Class Insecta

- Six-legged organisms

Scientific Names of Important Organisms

Man

- Homo sapiens

Cow

- Bos taurus

Cat

- Felis catus

Dog

- Canis lupus familiaris

Housefly

- Musca domestica

Lion

- Panthera leo

Mango

- Mangifera indica

Paddy (rice)

- Oryza sativa

Wheat

- Triticum aestivum

Gram

- Cicer arietinum

Pea

- Pisum sativum

> Cell and Tissue

Cell

What it is

- The basic structural and functional unit of every living being
- Has the capacity of self-reproduction

Discovery

- First by Robert Hooke in 1665 CE
- Living and free cells were discovered by Leeuwenhoek

Study

- Under cell biology or cytology

Smallest cell

- Mycoplasma gallisepticum

Cell	
Largest cell	That of the ostrich egg
Human body	- The largest cell is the ovum - The longest cell is the neuron of the nervous system - The smallest cell is the granule cell of the cerebellum
Types	Two types on the basis of the presence of a nucleus - Prokaryotic - Eukaryotic

Types of Cell		
	Prokaryotic	Eukaryotic
Nucleus	Absence of a distinct nucleus	Fully developed nucleus
Nucleolus and membrane	Nuclear membrane and nucleolus absent	Nuclear membrane and nucleolus present
Cytoplasm	Membrane-bound organelles absent in the cytoplasm	All membrane-bound organelles present
Ribosomes	Only of the 70S type	Both 70S and 80S types
Chromosomes	Only one	More than one
DNA	Separate from protein, histone protein absent	Attached to protein and histone protein present
Found in	Bacteria and blue-green algae	All plants and animals except viruses and bacteria

Principal Cell Organelles	
Mitochondria	
Discovery	By Altmann in 1886 CE
Naming	By Benda
Main function	Carrying out respiration
Other name	- Power house of the cell - Energy released by the breakdown of food in the presence of oxygen is stored here in the form of ATP

Principal Cell Organelles	
Other facts	■ The Krebs cycle takes place in it ■ Circular DNA and 70S ribosomes are found
Plastids	
Found in	■ Plant cells and Euglena
Chloroplasts	■ Green colour due to the presence of chlorophyll ■ Centres of photosynthesis ■ Location:- in leaves
Chromoplasts	■ Responsible for the colours other than green found in plants ■ Location:- in flowers, fruit walls, seeds, etc.
Leucoplasts	■ In plant parts deprived of sunlight ■ Colourless plastids, storage tissue ■ Location:- in roots and underground stems
Examples of chromoplast pigments	■ Lycopene:- red colour of tomato ■ Carotene:- colour of carrot ■ Betanin:- colour of beetroot ■ Carotene:- yellow colour of leaves ■ Caricaxanthin:- yellow colour of papaya ■ Anthocyanin:- red colour of apple ■ Phycocyanin:- light blue coloured; protects plants from ultraviolet rays
Endoplasmic Reticulum	
Location	■ Spread from the nucleus to the cell membrane
Types	Two types ■ Smooth endoplasmic reticulum ■ Rough endoplasmic reticulum
Other facts	■ Ribosomes are attached to it ■ The main site of lipid synthesis
Golgi Body	
Discoverer	■ Camillo Golgi in 1898 CE

Principal Cell Organelles	
Cisternae	Functional unit
Main function	Packaging, storage and secretion of macromolecules:- carbohydrates, lipids, proteins, nucleic acids
Other facts	The main site of formation of glycolipids and glycoproteins
Other name	Traffic manager of the molecules of the cell
Ribosomes	
Structure	A structure made of RNA and protein
Discovery	First in the animal cell by George Palade
Naming	By Roberts
Other name	Protein factory
Other facts	- Help in protein synthesis - Formation:- by the nucleolus
Lysosome	
Discoverer	de Duve (1958)
Formation	In the Golgi body
Other names	- Digestive bag or suicide bag - They destroy their own cell by digesting it
Other facts	Absent in the red blood corpuscles of mammals
Spherosomes	
	- Are called the lysosomes of the plant cell - Synthesise and store fat
Centrosome and Centrioles	

Principal Cell Organelles

- In the cells of all animals and lower plants
- The centrosome is made of two cylindrical structures — the centrioles

Vacuole

What is it?

- A structure in the cytoplasm that encloses water, sap and excretory substances

Other facts

- In the plant cell it occupies 90% of the space of the cell

Nucleus

Discovery

- By Robert Brown in 1831 CE

- It is the control centre of the cell.
- Chromosomes and genes are present in the nucleus.

Parts

Made of four parts

- Nuclear membrane
- Nucleoplasm
- Nucleolus
- Chromatin network

Nuclear membrane

- A double-layered membrane surrounding the nucleus
- It separates the nucleus from the cytoplasm

Nucleolus

- Of special importance in cell division
- Site of synthesis of ribosomal RNA
- Storehouse of RNA

Chromatin network

- Chromatin is a structure made of DNA and protein
- Fine and extensive thread-like structures
- At the time of division they become organised into chromosomes

Chromosome

Principal Cell Organelles

- In the form of fine, long and coiled threads
- Telomere:- the end part of the chromosome
- DNA about 40%
- Histone protein about 60%
- In humans 23 pairs = 46 chromosomes
- 22 pairs are the same in women and men
- The 23rd pair is XX in women and XY in men
- Sex chromosomes = the 23rd pair
- Carries hereditary information from one generation to the next
- Naming:- Waldeyer (1888 CE);
discovery:- Wilhelm Nägeli (1842)

Deoxyribonucleic Acid (DNA)

What is it?

- A nucleic acid which, together with protein, forms the structure of the chromosome
- Basically the hereditary material that transmits characters from parents to offspring

Location

- Besides the nucleus of the cell, also in mitochondria and chloroplasts

Constituent substances

- Nitrogenous bases (adenine, guanine, cytosine, thymine)
- Sugar
- Phosphoric acid

Watson and Crick

- Presented the double-helix structure of DNA in 1953
- Awarded the Nobel Prize in 1962

Friedrich Miescher

- Discoverer:- Swiss chemist
- In 1869

Dr. Har Gobind Khorana

- First synthesised it in the laboratory

Principal Cell Organelles	
DNA	Is present in the nucleus, mitochondria and chloroplasts.
Ribonucleic Acid (RNA)	
Structure	Single-stranded
Other facts	- Takes part in the process of protein formation - Is a non-genetic material
Location	In both the nucleus and the cytoplasm
t-RNA	- The smallest of all RNAs - Main function:- bringing amino acids to the site of protein synthesis
r-RNA	- Formation :- in the nucleolus - About 80% of the total RNA of the cell - Helps in protein synthesis
m-RNA	- Messenger RNA - Brings the information recorded in DNA to the site of protein synthesis.
RNA	Is present in the cytoplasm and the nucleoplasm.
Gene	
Introduction	- The unit of heredity - The part of DNA that performs a specific function
In viruses	Made of DNA and RNA
In eukaryotic organisms	Made only of DNA

Plant Cell and Animal Cell		
	Animal Cell	Plant Cell
Size	Usually small	Comparatively large
Shape	Round or irregular	Square or cuboidal
Energy storage	In the form of carbohydrate and glycogen	In the form of starch

Plant Cell and Animal Cell

Growth	By increase in the number of cells	By increase in the size of cells
Cell wall	Not found	Cell wall made of cellulose
Cilia	Provide movement to the cell	Not found in the plant cell
Plastids	Not found	Found
Vacuole	Many small vacuoles	One large central vacuole

Cell Division**Mitosis****Meiosis**

- | | |
|---|--|
| ■ Takes place in somatic cells | ■ Takes place in reproductive (germ) cells |
| ■ Completed in only one stage | ■ Completed in two stages |
| ■ Only one division of the nucleus | ■ The nucleus divides twice |
| ■ Two daughter cells are formed as a result of division | ■ Four daughter cells are formed as a result of division |
| ■ Mitosis is necessary for growth and repair | ■ Necessary for maintaining the same number of chromosomes in sexually reproducing organisms |

Miscellaneous Facts Related to the Cell

- | | |
|-----------------------------|---|
| Respiratory function | <ul style="list-style-type: none"> ■ Mesosome in the prokaryotic cell ■ Mitochondria in the eukaryotic cell |
| Chloroplast | ■ Kitchen of the cell |
| Golgi body | ■ Traffic controller of the molecules of the cell |
| Cell membrane | ■ Semi-permeable membrane |
| Ageing gene | ■ Is responsible for old age |
| Tissue | |
| The word | ■ First used by Bichat |

Miscellaneous Facts Related to the Cell

Introduction ■ A group of cells in which the origin, structure and function of the cells are the same

Plant Tissue

Two classes ■ Meristematic tissue
■ Permanent tissue

Meristematic Tissue

Function ■ Continuous formation of new cells by cell division

Apical ■ Increases length at the tip

Intercalary ■ Increases length between the nodes

Lateral meristem ■ Increases the width (girth) of the plant

Permanent Tissue

Introduction ■ Are living or dead cells
■ Shape is definite
■ Cytoplasm with vacuoles

Simple tissue ■ Tissues that perform a similar function

Parenchyma ■ Protects plants and prepares food in the leaves
■ Also stores excretory substances:- gum, resin and tannin

Collenchyma ■ Gives mechanical support to plants

Sclerenchyma ■ Found in abundance in fibrous plants
■ Gives mechanical support to plants

Complex tissue ■ Made of more than one type of cells
■ Two types:- xylem and phloem

Xylem ■ Carry water/mineral salts from the roots to the other parts of plants (one-way conduction)
■ Dead cells

Miscellaneous Facts Related to the Cell

Phloem

- Carry food from the leaves to the other parts of plants (two-way conduction)
- Living cells, in the inner layer of the bark

Animal Tissue

Types

- Epithelial tissue
- Connective tissue
- Muscular tissue
- Nervous tissue

Epithelial Tissue

Function

- Providing an outer covering or protection to the body, as well as carrying out secretion or excretion, sensation and absorption

Connective Tissue

True connective tissue

- Adipose connective tissue :- in the form of fat cells under the skin and around the kidneys
- Tendon :- joins muscles to bones or to other muscles
- Ligament:- does the work of joining bones at the joints

Skeletal tissue

- It forms the endoskeleton of the body
- Two types:- cartilage and bone tissue

Fluid tissue

- Blood is a fluid connective tissue
- Three types:- red blood cells, white blood cells, platelets

Muscular Tissue

Function

- Necessary for locomotion and various kinds of movements

Types

Are of three types:-

- Unstriated / involuntary
- Striated or voluntary
- Cardiac muscle

Special

- The strongest muscles are those of the jaw.

Nervous Tissue

Introduction

- Made of a special type of cells called nerve cells or neurons

Miscellaneous Facts Related to the Cell

Axon	■ A long process found in the neuron
Cell and Tissue: Miscellaneous	
Genetics	■ Father:- Gregor Johann Mendel of Austria ■ Through experiments on the garden pea
Mendel's laws	■ Law of dominance, law of independent assortment, law of segregation

> Systems of the Human Body

Digestive System

Alimentary Canal

Introduction	■ 30 feet long, extending from the mouth to the anus
Parts	■ Buccal cavity, pharynx, oesophagus, stomach, small intestine and large intestine
Oral Cavity / Buccal Cavity	
Organs included	■ Tongue and teeth
Digestion	■ Begins in the buccal cavity itself with the help of saliva
Saliva	■ Contains 99% water and 1% enzymes ■ Enzymes:- ptyalin and lysozyme
Ptyalin	■ Breaks starch into the disaccharide maltose
Lysozyme	■ Destroys micro-organisms and bacteria
Teeth	
Humans	■ Heterodont ■ Four types of teeth:- incisor, canine, premolar, molar
Incisors	■ The front-most teeth that cut food
Canines	■ Pointed teeth that tear food

Digestive System	
Premolars + molars	■ Cheek teeth ■ Teeth that grind food
Enamel	■ The outer layer of the tooth ■ The hardest part of the human body ■ 98% made of calcium phosphate and carbonate
Stomach	
Digestion	■ The main site of protein digestion
Gastric glands	■ Present on the inner wall of the stomach ■ Secrete highly acidic gastric juice
Digestive enzymes in gastric juice	■ Pepsin, rennin, hydrochloric acid and mucus
Pepsin	■ Protein into peptones and proteoses
Rennin	■ Milk protein caseinogen into casein
Tongue	
Taste buds	■ Taste Buds ■ Provide the sense of the taste of food
Sweet taste	■ Sensed by the front part of the tongue
Bitter (pungent) taste	■ Sensed by the back part of the tongue
Sour taste	■ Sensed by both side parts of the tongue
Small Intestine	
Introduction	■ Highly coiled ■ Longer than the large intestine ■ The digestion of food is completed here
Parts	Divided into three parts ■ Duodenum:- 30 cm ■ Jejunum:- 2.5 metres ■ Ileum:- 2.5 metres to 3.5 metres

Digestive System	
Digestion	As soon as food comes into the small intestine, three digestive juices (bile juice, pancreatic juice and intestinal juice) are mixed with it.
Function	Final digestion and absorption of food
Enzymes in intestinal juice	- Erepsin:- conversion of peptones into amino acids - Maltase:- conversion of maltose into glucose - Sucrase:- changes sucrose into glucose and fructose - Lactase:- changes lactose into glucose and galactose
Enzymes in pancreatic juice	- Trypsin:- protein into proteoses and peptones - Amylase:- digestion of starch - Lipase:- digests emulsified fat and converts it into fatty acids
Absorption	Digested food reaching the blood
Assimilation	The process by which absorbed food materials are put to use by the body
Liver	
Introduction	The largest gland of the human body
Weight	About 1.5 kg
Main functions	- Synthesis and production of bile juice - Stores glucose in the form of glycogen - Secretes bile continuously
Bile juice	- Is stored in the gall bladder - A thick greenish-yellow, mildly alkaline fluid - No digestive enzyme is found in it
Bile pigments in bile juice	Bilirubin and biliverdin
Emulsification of fat	By two salts present in bile juice:- sodium glycocholate and sodium taurocholate
Pancreas	

Digestive System	
Introduction	■ The second largest gland of the human body ■ A mixed gland (endocrine + exocrine)
Pancreatic juice	■ 98% water and 2% enzymes and salts ■ Is alkaline ■ A complete digestive juice
Islets of Langerhans	■ Discoverer:- Langerhans ■ Is a part of the pancreas ■ The hormones insulin, glucagon and somatostatin are secreted
Large Intestine	
Introduction	■ Wider than the small intestine but shorter in length
Three parts	■ Caecum ■ Colon ■ Rectum
Functions	■ Absorption of water ■ Storing undigested food for some time
Appendix	
Other name	■ Vermiform appendix
Example	■ Of a vestigial organ
Attached to	■ The caecum of the large intestine
Other Important Facts	
Zymase	■ Converts glucose into ethanol
Enzymes	■ Are basically biocatalysts
Respiratory System	
Respiratory organs	■ Nose, pharynx, larynx, trachea, lungs
Respiration	

Digestive System	
Process of respiration	- Energy is released by the combustion of glucose in the presence of oxygen - Exchange of oxygen and carbon dioxide
Respiratory rate	12 to 15 times per minute in an adult human - About 44 times per minute in infants
Other facts	- A biochemical process - An involuntary process - Main stages:- glycolysis, Krebs cycle and electron transport system
Lungs	- The right lung is bigger and wider - The left lung is smaller but longer - Pulmonary respiration:- exchange of gases - Protected or enclosed by the pleural membrane - The alveoli, the sites of gas exchange, are surrounded by blood capillaries - Keeps body temperature under control
Internal respiration	Gaseous exchange between tissue fluid and blood
External respiration	- Inspiration and expiration - Gaseous exchange in the lungs
Transport of gases	- Transport of oxygen and CO₂ from the lungs to the cells and back from the cells to the lungs - Oxygen is transported by haemoglobin - Transport of carbon dioxide:- by haemoglobin, dissolved in plasma, and in the form of bicarbonate
Control of respiration	By the medulla oblongata of the brain
Breakdown of Glucose by Different Pathways in Respiration	
In yeast	- Absence of oxygen - Ethanol + CO₂ + energy
In muscle cells	- Lack of oxygen - Lactic acid + energy

Digestive System	
In mitochondria	■ Presence of oxygen ■ CO₂ + water + energy
Circulatory System	
Discovery	■ By William Harvey
Blood circulation	■ Always in a fixed direction ■ Carried out by the heart
Heart	
Location	■ In the thoracic cavity between the two lungs ■ Enclosed by a double membrane called the pericardium
Chambers	■ Four chambers ■ Atria (right and left) ■ Ventricles (right and left)
Weight	■ About 300 grams
Heartbeat	■ 70 to 80 times per minute ■ The heart rests between two beats ■ Potassium is necessary for controlling the heartbeat
Blood pressure	■ 120/80 (normal) ■ More than atmospheric pressure
Tricuspid valve	■ Between the right atrium and the right ventricle
Bicuspid valve	■ Between the left atrium and the left ventricle
Systolic pressure	■ 120 mmHg
ECG	■ Electrocardiogram ■ A type of medical test that shows the activity of the heart
Blood Vessels	
Arteries	
Function	■ Carrying pure blood from the heart to the various organs

Digestive System	
Aorta	■ The largest artery of the body ■ Carries pure blood from the left ventricle into the body
Pulmonary artery	■ The only artery, going to the lungs, in which impure blood flows ■ It carries impure blood from the right ventricle to the lungs
Right atrium	■ It receives impure blood from the body
Left atrium	■ Pure blood from the lungs comes into the left atrium and goes into the left ventricle
Veins	
Function	■ Carry impure blood from the various organs to the heart
Pulmonary vein	■ The only vein in which pure blood flows ■ It carries pure blood from the lungs to the left atrium
Vena cava	■ Supplies oxygen-deficient blood to the right side of the heart ■ Two types:- anterior (superior) and posterior (inferior)
Right ventricle	■ From here impure blood goes into the pulmonary artery
Left ventricle	■ From here pure blood reaches all the tissues of the body
Blood	
What is it?	■ A fluid connective tissue ■ Made of plasma and blood cells ■ pH value:- 7.4 (alkaline)
Plasma	■ About 55% of the total blood volume ■ A colourless fluid (about 90% water) ■ Carries CO₂, hormones and wastes ■ Plasma proteins formed in the liver:- globulin, albumin, fibrinogen
Blood Cells	

Digestive System	
Blood cells	Of three types - Red blood cells (Erythrocytes) - White blood cells (Leukocytes) - Platelets
Red blood cells	- Round, and nucleus absent - Red colour due to haemoglobin - Carry oxygen - Lifespan:- about 120 days - Develop in the bone marrow; old blood corpuscles are destroyed by the spleen
White blood corpuscles	- Nucleated and of irregular shape - Other names:- soldiers or brave soldiers - Function:- protection of the body from infection - The largest of the blood cells - Lifespan:- 10 to 13 days - Monocyte:- the largest white blood corpuscle - Lymphocyte:- the smallest WBC - Formation:- in the lymph glands, spleen, thymus gland and bone marrow
Granular WBCs	- Eosinophils - Basophils - Neutrophils
Agranular WBCs	- Lymphocytes - Monocytes
Platelets	- Minute, colourless and without a nucleus - Functions:- blood coagulation, forming blood clots, repair of blood vessels
Blood Pressure	
Measurement	Blood pressure is measured by the sphygmomanometer
Normal blood pressure	- Systolic pressure:- 115 to 120 mmHg - Diastolic pressure:- 75 to 80 mmHg
With increasing age	Blood pressure also increases

Digestive System	
Other Facts Related to the Heart and Blood	
Red bone marrow	● Blood cells are formed in the red bone marrow
Spleen	● Is called the graveyard of RBCs
Adrenal gland	● Controls blood pressure
Ischaemia	● Insufficient supply of blood in the body
Quantity of blood	● Blood in a healthy human:- 5 to 6 litres
Heparin	● Because this substance is found in the blood, blood does not clot (inside the vessels)
Sugar in the blood	● Measurement:- glucometer ● Normal amount:- 100 milligrams per decilitre ● Insulin:- the hormone responsible for its regulation ● Excess causes diabetes ● Deficiency causes hypoglycaemia
SAN	● Situated in the right atrium of the heart, controls the heart rate (natural pacemaker)
Hormones controlling heart rate	● Thyroxine ● Adrenaline
Blood Groups	
Discovery	● By Karl Landsteiner in 1900
Rh factor	● In the RBCs of the rhesus monkey (1940) ● Discovery:- by Landsteiner and Wiener
Antigens	
Introduction	● Those that stimulate the formation of antibodies ● Glycoproteins transferred from generation to generation
Other name	● Agglutinogens
Location	● Attached to the plasma membrane of the blood corpuscles

Digestive System	
Types	- Antigen-A - Antigen-B
Antibodies	
Other name	Agglutinins
In what form	Exist in the form of pure proteins
Location	In the blood plasma or serum
Types	- Anti A or a - Anti B or b
Blood Groups in Humans	
A, B, O	- These three blood groups were discovered by Karl Landsteiner - In 1900 CE
AB	Discovery:- by Decastello and Sturli in 1901
Blood group A	- Antigen:- A; antibody:- b - Can donate blood to:- A and AB - Can receive from:- O and A
Blood group B	- Antigen:- B; antibody:- a - Can donate blood to:- B and AB - Can receive from:- O and B
Blood group AB	- Antigen:- AB; antibody:- x - Can donate blood to:- AB - Can receive from:- O, A, B and AB - Other name:- universal recipient
Blood group O	- Antigen:- x ; antibody:- ab - Can donate blood to:- A, B, AB, O - Can receive from:- only O - Other name:- universal donor
Other Facts	

Digestive System	
Lymphocytes	Responsible for the formation of antibodies in the blood plasma
Thrombin enzyme	Converts fibrinogen into fibrin in the formation of a blood clot
Lymph	
Introduction	A colourless, clear fluid in the empty spaces between the blood vessels and the tissues
Direction of flow	From the tissues towards the heart
Related facts	- RBCs are not found in it - WBCs are more in it - Centre of formation of lymphocytes
Functions	- Immunity (disease resistance) - Filtering out and separating dust particles, cancer cells, bacteria, viruses, etc. by the lymph nodes - Helps in healing wounds - Helps in the circulation of substances from the tissues into the veins
Excretory System	
Excretory organs	Kidney, lungs, skin, liver, large intestine
Kidney	
Nephron	- The structural and functional unit of the kidney - The filtration unit of the kidney - Bowman's capsule:- a small structure in the nephron that purifies the blood
Two parts	- Outer part:- cortex - Inner part:- medulla
Functions	Excretion and osmoregulation
Excretion	- Urea, uric acid, ammonia, etc. - Urochrome:- the yellow pigment found in urine
Dialysis	Purification of blood (in place of) the kidney

Digestive System	
Osmoregulation	Control of the amount of salts and water in the body
Human Urine	
Composition	- Water:- 95%, urea:- 2.6 per cent - Salts:- 2.0%, uric acid:- trace amount - Creatinine:- 0.3%
Quantity	About 1.5 litres per day
pH value	6 (mildly acidic)
Formation	Urea is formed in the liver, then urine is formed in the kidney by ultrafiltration
Liver	
Functions	- Synthesis of urea (ammonia > urea) - Detoxification of alcohol and other harmful substances from the body
Skin	
Function	Taking water, salts and urea through the sweat glands and expelling them from the body in the form of sweat
Lungs	
Function	Excretion of carbon dioxide dissolved in the blood and, besides it, water vapour
Excretion in Different Animals	
- Unicellular organisms:- by diffusion - Coelenterata:- directly by the cells - In flatworms:- by flame cells - Phylum Annelida:- by nephridia - Vertebrates:- by kidneys	
Central Nervous System	

Digestive System	
Function	Regulation and control of all the activities of the body
Unit	Neuron or nerve cell
Parts	- Central nervous system:- brain and spinal cord - Peripheral nervous system:- cranial and spinal nerves - Autonomic nervous system:- sympathetic nervous system, parasympathetic nervous system
Brain	
Weight	- About 1400 grams - About 2% of the total (body) weight
Functions	Thinking, control of heart rate, balance of the body, etc.
3 parts	Forebrain, midbrain, hindbrain
Forebrain	
Two parts	- Cerebrum - Diencephalon
Cerebrum	- About 80% of the brain (the largest part) - It has separate centres for seeing, touching and smelling - Voluntary actions are controlled from here - It is also the centre of human consciousness, reasoning, memory and intelligence
Diencephalon	- Has 2 parts - Thalamus and hypothalamus
Thalamus	- The upper part of the diencephalon - Recognises cold, heat and pain
Hypothalamus	- The lower part of the diencephalon - Controls emotional activities - Controls hormones - Regulation of hunger, thirst, love, hatred, etc. - Controls body temperature, blood pressure, sleep, reproductive capacity, heartbeat, etc.
Midbrain	

Digestive System	
Two parts	■ Cerebral peduncle ■ Corpora quadrigemina
Functions	■ Control over sight, hearing and reflex actions
Cerebral peduncle	■ The front part of the midbrain ■ Made of fibres
Corpora quadrigemina	■ The rear part of the midbrain ■ Related to visual perception and hearing
Hindbrain	
Parts	■ Cerebellum ■ Medulla oblongata
Cerebellum	■ The second largest part of the human brain ■ Controls movement and maintains the balance of the body ■ Control over the contraction of voluntary muscles
Medulla oblongata	■ The rear-most triangular part of the brain ■ Centre of swallowing food and vomiting ■ Centre of all involuntary actions ■ Respiration, metabolism, coughing, sneezing, etc.
Spinal Cord (Sushumna / Merurajju)	
Discoverer	■ Marshall Hall
Location	■ Emerges from the foramen magnum at the medulla oblongata and extends down through the vertebral column to the end
Functions	■ The main centre of reflex actions ■ Balances involuntary actions ■ Conduction of impulses coming to and going from the brain
Peripheral Nervous System	
Includes	■ 12 pairs of cranial nerves ■ 31 pairs of spinal nerves

Digestive System	
Cranial nerves	■ The first, second and eighth pairs are sensory ■ The rest are mixed, i.e., sensory and motor
Spinal nerves	■ Connect the muscles of the hands and legs, the skin, the sense organs, etc. with the spinal cord
Autonomic Nervous System	
Two parts	■ Sympathetic nervous system ■ Parasympathetic nervous system
Sympathetic nervous system	■ Speeds up the heartbeat ■ Dilates the pupils of the eyes ■ Secretion from the sweat glands ■ Increases sugar in the blood ■ Reduces the secretion of the salivary glands ■ Its collective effect gives the feeling of fear, pain and anger
Parasympathetic nervous system	■ Constricts the pupils of the eyes ■ Increases the secretion of saliva ■ Contraction in the muscles of the urinary bladder ■ Contraction in the intestinal wall ■ Increase in digestive juices ■ Slows down the heartbeat ■ Its collective effect gives the feeling of rest and pleasure
Skeletal System	
Introduction	■ A framework of bones and cartilages ■ Gives shape to the body
Two parts	■ Exoskeleton :- hair and nails ■ Endoskeleton :- the inner skeleton
Endoskeleton	■ Axial skeleton:- 80 bones ■ Appendicular skeleton:- 126 bones
Bone	
Total bones	■ 206 in an adult human ■ About 300 in a newborn infant

Digestive System	
Constituents	■ Important constituent:- calcium phosphate ■ Collagen protein:- about 33%
Marrow cavity	■ A hollow cavity in the middle of long and thick bones ■ The fluid bone marrow is in this cavity
Vitamin D	■ Necessary for the normal growth of bones
Femur	■ The thigh bone ■ The largest bone of the human body
Stapes	■ A bone of the ear ■ The smallest bone of the human body
Ligaments	■ The joint of bone with bone is called this
Tendon	■ The joint of muscle and bone is called this
Patella (janu-phalak)	■ The bone that protects the knee joint and helps in bending (knee cap) ■ The largest sesamoid bone
Axial Skeleton	
Human skull	29 bones in total ■ Cranium:- 8 bones ■ Face:- 14 bones ■ In the middle ear:- 6 bones
Vertebral column (backbone)	26 bones in total ■ Neck:- 7 ■ Thorax:- 12 ■ Waist (lumbar):- 5 ■ Sacrum:- 1 ■ Coccyx (tail):- 1
Sternum	■ 1 bone
Ribs	■ 24 bones (12 pairs)
Appendicular Skeleton	
Girdles	■ 6 bones

Digestive System	
Limbs	120 bones
Nails	
Formation	- From the matter of dead cells - From a protein called alpha keratin
Exoskeleton	- Absence of blood circulation - No pain on cutting
Other Facts Related to the Skeletal System	
Arthroplasty	Replacement of the hip joint is called this
Calcium phosphate	Is the salt found in the largest amount in human bones
Hyoid	The bone situated below the tongue between the jaw and the throat
Principal bones	- Clavicle:- jatrak (collar bone) - Scapula:- skandh phalak (shoulder blade) - Sternum:- urosthi (breast bone) - Patella:- knee
Endocrine System	
Glands	
Introduction	Organs of the human body that secrete various kinds of fluids
Function	Secretion of substances necessary for the body
Three types	- Endocrine - Exocrine - Mixed gland
Endocrine Glands	
Introduction	- Ductless glands - Secrete hormones

Digestive System	
Hormones	- Reach the target organ through the blood - Chemically they are of the nature of proteins, peptones, amino acids, steroids - They decompose after taking part in a chemical reaction once
Pituitary Gland and the Hormones Secreted by it	
Introduction	- The smallest endocrine gland of the human body - In the 'sella turcica' of the sphenoid bone of the skull
Master gland	It is so called because it controls a person's growth, health, sexual development, etc.
Somatotropic hormone	- Other name:- growth hormone - Function:- normal and balanced growth of the body - Hyposecretion causes dwarfism in children and Simmonds' disease in adults - Hypersecretion causes gigantism in adolescents
TSH hormone	- Thyroid stimulating hormone - Function:- stimulating the thyroid gland to secrete the hormone thyroxine
ACTH hormone	- Adrenocorticotrophic hormone - Stimulates the secretion of hormones from the adrenal gland - Regulation of the secretion of the hormone cortisol
LTH hormone	- Luteotropic hormone (Prolactin) - Other name:- lactogenic hormone - Stimulates the secretion of milk from the mammary glands
Vasopressin	- Anti-diuretic hormone - Synthesis:- in the hypothalamus - Storage:- in the pituitary gland - Function:- water balance in the body and concentration of urine

Digestive System	
FSH hormone	● Follicle stimulating hormone ● Stimulates the formation of sperms in the testes ● Regulation of the menstrual cycle in women
LH hormone	● Luteinizing hormone ● In women it is called luteotropin ● Develops sexual characteristics
Oxytocin	● Birth hormone:- stimulates the contraction of the uterine muscles at the time of delivery ● Milk ejection hormone:- stimulates the secretion of milk from the mammary glands after childbirth ● Love hormone:- affection between mother and infant
Thyroid Gland and the Hormones Secreted by it	
Location	● In the human neck, below the larynx
Thyroxine hormone	● Structure:- an iodine-containing amino acid ● Control of metabolic activities ● Stimulant for growth ● Regulation of the functions of the reproductive organs ● Control of water balance
Related diseases	● Excess:- exophthalmic goitre and Plummer's disease ● Deficiency :- cretinism in children, myxoedema and goitre in adults
Hashimoto's disease	● Attack on the thyroid gland by the body's immune system ● This disease is also called the suicide of the thyroid
Parathyroid Gland and the Hormones Secreted by it	
Location	● Just behind the thyroid gland situated in the neck
Derived	● From the endoderm of the embryo

Digestive System	
PTH hormone	■ Parathyroid hormone ■ Regulates the amount of calcium in the blood ■ Its hyposecretion causes cramps and stiffness in the skeletal muscles ■ Its hypersecretion causes osteoporosis
Hormones Secreted by the Adrenal Gland	
Location	■ On the upper end of both kidneys
Two parts	■ Adrenal cortex ■ Adrenal medulla
Adrenal cortex	■ Aldosterone hormone ■ Cortisol:- secreted at the time of stress; helps in the decision to fight or flee in adverse situations ■ Androgen:- responsible for male characteristics and reproduction ■ Its disorder causes Addison's disease
Adrenal medulla	■ Adrenaline:- called the 'fight and flight' hormone and the emergency hormone. ■ Noradrenaline hormone
Hormones Secreted by the Pineal Gland	
Melatonin hormone	■ Its main function is to maintain the sleep-wake cycle and the biological rhythm of the body
Melanin	■ It is a pigment ■ Found in the skin, eyes, hair and the central nervous system
Serotonin	■ Exchange of information between nerve cells
Other facts	■ By regulating the biological clock in the body, the pineal gland also determines the sleep-wake cycle, body temperature and the menstrual cycle.
Hormones Secreted by the Thymus Gland	

Digestive System	
Thymosin hormone	Helps the body's immune system by forming T-lymphocytes in the body
Other hormones	Thymin or thymopoietin
Exocrine Glands	
Introduction	- Glands with ducts - The fluid secreted by them reaches the concerned organ through ducts
Secretion	Of enzymes
Examples	Salivary glands, sweat glands, oil glands, etc.
Other Glands	
Hormones Secreted by the Pancreas Gland	
Introduction	- The second largest gland of the human body after the liver - A mixed gland (endocrine + exocrine)
Pancreatic juice	Is called the complete digestive juice
Cells of the islets of Langerhans	The endocrine part of the gland - Alpha cells:- glucagon - Beta cells:- insulin - Delta cells:- somatostatin - F cells:- pancreatic polypeptide
Alpha cells	- Secretion of the hormone glucagon - Glucagon:- keeps the amount of glucose in the blood normal and synthesises glucose from the glycogen stored in the liver
Beta cells	Secrete the hormone insulin.
Insulin	- The metal zinc is present - A protein chain or peptide hormone - Function:- converting excess sugar of the blood into glycogen - Its deficiency causes diabetes - Also obtained from the roots of dahlia

Digestive System	
Frederick Sanger	- Received the Nobel Prize for discovering the sequence of amino acids in insulin. - He was the first to give the complete chemical structure of bovine (cow) insulin.
Reproductive Hormones	
Female Hormones	
Estrogen	- Production:- from the ovary - Age:- from the onset of puberty up to the age of menopause - Estradiol:- the principal estrogen - Other name:- female development hormone - Secondary sexual characters of the female (proper development of breasts, uterus, mammary glands, etc.) - Controlled by the pituitary hormones FSH and LH
Relaxin	Secretion:- from the corpus luteum and the ovary during pregnancy
Progesterone	- Production:- by the corpus luteum - Function:- stimulating the development of the characteristics necessary for pregnancy.
Male Hormones	
Testosterone	- Other name:- male development hormone - Function:- development of masculine sexual characteristics
Human Body Systems: Miscellaneous	
Fertilisation	- Male gamete (sperm) > female gamete (ovum) > fertilisation > zygote > diploid > formation and development of the embryo - The process of fertilisation takes place in the oviduct or Fallopian tube
Development of the human embryo	- Heartbeat:- from the fourth-fifth week of development - Embryonic development:- from the third to the eighth week
Neuron	Is the unit of nervous tissue

Digestive System	
Bernstein	■ In 1924 he showed that the blood group is a hereditary character.
Heart	■ Main function:- supplying blood to the organs of the body
Anaerobic respiration	■ In the absence of oxygen ■ Glucose — bacteria / ethyl alcohol in yeast ■ In the end, pyruvic acid
Aerobic respiration	■ In the presence of oxygen ■ $\text{Glucose} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{energy}$ ■ Two parts :- glycolysis and the Krebs cycle
Krebs cycle	■ First described:- by Hans Krebs ■ Other names:- citric acid cycle / tricarboxylic acid cycle ■ Location:- mitochondria ■ Functions:- production of ATP, production of energy

06

Environment and Ecology

ENVIRONMENT AND ECOLOGY

> Ramsar Sites / Biosphere Reserves / National Parks / Leopard Status Report / UNESCO World Heritage Sites - UPDATED 2026

> India's 18 Biosphere Reserves (September 2025)

S.No.	Name of the Biosphere Reserve	Year of Establishment	Year of Inclusion in the UNESCO MAB List / Fact
1	Nilgiri Biosphere Reserve	1986	India's first biosphere reserve; also included in the UNESCO MAB list.
2	Nokrek Biosphere Reserve	1988	(Meghalaya); included in the UNESCO MAB list.
3	Nanda Devi Biosphere Reserve	1988	(Uttarakhand); included in the UNESCO MAB list.
4	Manas Biosphere Reserve	1989	(Assam); only in the national list (not included in MAB).
5	Gulf of Mannar Biosphere	1989	(Tamil Nadu); included in the UNESCO MAB list.
6	Great Nicobar Biosphere	1989	(Andaman and Nicobar); included in the UNESCO MAB list.
7	Sundarban Biosphere Reserve	1989	(West Bengal); included in the UNESCO MAB list.
8	Simlipal Biosphere Reserve	1994	(Odisha); included in the UNESCO MAB list.
9	Dibru-Saikhowa Biosphere	1997	India's smallest biosphere reserve by area (Assam).
10	Dihang-Dibang Biosphere	1998	(Arunachal Pradesh); not included in the MAB list.
11	Pachmarhi Biosphere Reserve	1999	(Madhya Pradesh); included in the UNESCO MAB list.
12	Kanchenjunga Biosphere Reserve	2000	(Sikkim); included in the UNESCO MAB list (mixed site).
13	Agasthyamalai Biosphere	2001	(Kerala-Tamil Nadu); included in the UNESCO MAB list.
14	Achanakmar-Amarkantak	2005	(Madhya Pradesh-Chhattisgarh); included in the UNESCO MAB list.

S.No.	Name of the Biosphere Reserve	Year of Establishment	Year of Inclusion in the UNESCO MAB List / Fact
15	Rann of Kutch Biosphere	2008	India's largest biosphere reserve by area (Gujarat).
16	Cold Desert	2009	(Himachal Pradesh); not included in the MAB list.
17	Seshachalam Hills Biosphere	2010	(Andhra Pradesh); not included in the MAB list.
18	Panna Biosphere Reserve	2011	(Madhya Pradesh); the latest (12th) site to be included in the UNESCO MAB list .

India's total of 13 biosphere reserves included in UNESCO's MAB (Man and Biosphere) Programme:

At present, out of India's total of **18 notified biosphere** reserves, **13 reserves** have become part of UNESCO's global network (MAB), in chronological order: (1) Nilgiri, (2) Gulf of Mannar, (3) Sundarban, (4) Nanda Devi, (5) Nokrek, (6) Pachmarhi, (7) Simlipal, (8) Achanakmar-Amarkantak, (9) Great Nicobar, (10) Agasthyamalai, (11) Kanchenjunga, and (12) Panna, (13) **Cold Desert Biosphere Reserve**

> NOTE:

- India's **13th** UNESCO (UNESCO MAB) site is the **Cold Desert Biosphere Reserve**.
- This new status was given in **September 2025** during UNESCO's 5th World Congress of Biosphere Reserves held in Hangzhou, China.
- It is situated in the Lahaul and Spiti district of **Himachal Pradesh**.

> A Total of 101 Ramsar Sites in India (as of August 2026)

NOTE - At present (as of August 2026) India has a total of **101 Ramsar Sites**, in which **Glaw Lake** of **Arunachal Pradesh** has become India's 101st Ramsar site.

> Year-wise and Serial List of India's Ramsar Sites (1981 - 2026)

S.No.	Name of the Ramsar Site	State / Union Territory	Year Declared
1	Chilika Lake	Odisha	1981
2	Keoladeo National Park	Rajasthan	1981
3	Wular Lake	Jammu and Kashmir	1990
4	Harike Wetland	Punjab	1990
5	Loktak Lake	Manipur	1990
6	Sambhar Lake	Rajasthan	1990
7	Kolleru Lake	Andhra Pradesh	2002
8	Deepor Beel	Assam	2002
9	Pong Dam Lake	Himachal Pradesh	2002

S.No.	Name of the Ramsar Site	State / Union Territory	Year Declared
10	Ashtamudi Wetland	Kerala	2002
11	Sasthamkotta Lake	Kerala	2002
12	Vembanad-Kol Wetland	Kerala	2002
13	Bhoj Wetland	Madhya Pradesh	2002
14	Bhitarkanika Mangroves	Odisha	2002
15	Ropar Wetland	Punjab	2002
16	Kanjli Wetland	Punjab	2002
17	Tsomoriri Lake	Ladakh	2002
18	Point Calimere Wildlife and Bird Sanctuary	Tamil Nadu	2002
19	East Kolkata Wetlands	West Bengal	2002
20	Chandra Taal	Himachal Pradesh	2005
21	Renuka Wetland	Himachal Pradesh	2005
22	Hokersar Wetland	Jammu and Kashmir	2005
23	Surinsar-Mansar Lakes	Jammu and Kashmir	2005
24	Rudrasagar Lake	Tripura	2005
25	Upper Ganga River (Brijghat to Narora stretch)	Uttar Pradesh	2005
26	Nalsarovar Bird Sanctuary	Gujarat	2012
27	Sundarban Wetland	West Bengal	2019
28	Nandur Madhameshwar	Maharashtra	2019
29	Nawabganj Bird Sanctuary	Uttar Pradesh	2019
30	Parvati Arga Bird Sanctuary	Uttar Pradesh	2019
31	Saman Bird Sanctuary	Uttar Pradesh	2019
32	Samaspur Bird Sanctuary	Uttar Pradesh	2019
33	Sandi Bird Sanctuary	Uttar Pradesh	2019
34	Sarsai Nawar Jheel	Uttar Pradesh	2019
35	Beas Conservation Reserve	Punjab	2019
36	Keshopur-Miani Community Reserve	Punjab	2019
37	Nangal Wildlife Sanctuary	Punjab	2019
38	Asan Conservation Reserve	Uttarakhand	2020

S.No.	Name of the Ramsar Site	State / Union Territory	Year Declared
39	Kanwar or Kabar Taal Wetland	Bihar	2020
40	Lonar Lake	Maharashtra	2020
41	Sur Sarovar (Keetham Lake)	Uttar Pradesh	2020
42	Tso Kar Wetland Complex	Ladakh	2020
43	Sultanpur National Park	Haryana	2021
44	Bhindawas Wildlife Sanctuary	Haryana	2021
45	Thol Lake Bird Sanctuary	Gujarat	2021
46	Wadhvana Wetland	Gujarat	2021
47	Haiderpur Wetland	Uttar Pradesh	2021
48	Khijadia Wildlife Sanctuary	Gujarat	2021
49	Bakhira Wildlife Sanctuary	Uttar Pradesh	2021
50	Karikili Bird Sanctuary	Tamil Nadu	2022
51	Pallikaranai Marsh Reserve Forest	Tamil Nadu	2022
52	Pichavaram Mangroves	Tamil Nadu	2022
53	Pala Wetland	Mizoram	2022
54	Sakhya Sagar	Madhya Pradesh	2022
55	Koonthankulam Bird Sanctuary	Tamil Nadu	2022
56	Gulf of Mannar Marine Biosphere Reserve	Tamil Nadu	2022
57	Vembannur Wetland Complex	Tamil Nadu	2022
58	Vellode Bird Sanctuary	Tamil Nadu	2022
59	Vedanthangal Bird Sanctuary	Tamil Nadu	2022
60	Udhayamarthandapuram Bird Sanctuary	Tamil Nadu	2022
61	Sirpur Wetland	Madhya Pradesh	2022
62	Satkosia Gorge	Odisha	2022
63	Nanda Lake	Goa	2022
64	Ranganathittu Bird Sanctuary	Karnataka	2022
65	Tampara Lake	Odisha	2022
66	Hirakud Reservoir	Odisha	2022
67	Ansupa Lake	Odisha	2022

S.No.	Name of the Ramsar Site	State / Union Territory	Year Declared
68	Yashwant Sagar	Madhya Pradesh	2022
69	Chitrangudi Bird Sanctuary	Tamil Nadu	2022
70	Suchindram Theroor Wetland Complex	Tamil Nadu	2022
71	Vaduvur Bird Sanctuary	Tamil Nadu	2022
72	Kanjirankulam Bird Sanctuary	Tamil Nadu	2022
73	Thane Creek	Maharashtra	2022
74	Hygam Wetland Conservation Reserve	Jammu and Kashmir	2022
75	Shallbugh Wetland Conservation Reserve	Jammu & Kashmir	2022
76	Nagi Bird Sanctuary	Bihar	2023
77	Nakti Bird Sanctuary	Bihar	2023
78	Aghanashini Estuary	Karnataka	2024
79	Ankasamudra Bird Conservation Reserve	Karnataka	2024
80	Magadi Kere Conservation Reserve	Karnataka	2024
81	Karaivetti Bird Sanctuary	Tamil Nadu	2024
82	Longwood Shola Reserve Forest	Tamil Nadu	2024
83	Tawa Reservoir	Madhya Pradesh	2024
84	Nanjarayan Bird Sanctuary	Tamil Nadu	2024
85	Kazhuveli Bird Sanctuary	Tamil Nadu	2024
86	Sakkarakottai Bird Sanctuary	Tamil Nadu	2024
87	Therthangal Bird Sanctuary	Tamil Nadu	2024
88	Khecheopalri Wetland	Sikkim	2024
89	Udhwa Lake Bird Sanctuary	Jharkhand	2024
90	Gogabeel Lake	Bihar	2025
91	Udaipur Jheel	Bihar	2025
92	Gokul Jalashay	Bihar	2025
93	Kopra Reservoir	Chhattisgarh	2025
94	Siliserh Lake	Rajasthan	2025
95	Menar Wetland Complex	Rajasthan	2025
96	Khichan Wetland	Rajasthan	2025

S.No.	Name of the Ramsar Site	State / Union Territory	Year Declared
97	Patna Bird Sanctuary	Uttar Pradesh	2026
98	Chhari-Dhand Wetland	Gujarat	2026
99	Shekha Jheel Bird Sanctuary	Uttar Pradesh	2026
100	Jayaprakash Narayan Bird Sanctuary (Surha Taal)	Uttar Pradesh	2026
101	Glaw Lake	Arunachal Pradesh	2026

> Key Statistics and Facts:

- **State with the most sites:** **Tamil Nadu** is in the lead with the highest number of **20 Ramsar sites**. In second place is **Uttar Pradesh** with **13 sites**.
- **Largest Ramsar site:** the **Sundarban Wetland** of West Bengal (about 4,230 sq km).
- **Smallest Ramsar site:** the **Renuka Wetland** of Himachal Pradesh (about 0.2 sq km / 20 hectares).

> Status of Leopards in India 2022 Report (Leopard Status Report)

Note: This updated report was released in **February 2024** by the Union Ministry of Environment, Forest and Climate Change. According to it, the total number of leopards in India has increased from 12,852 in 2018 to **13,874 in 2022**.

Indicator / Parameter	Main Findings of the Report and State-wise Data (Exam Trends)
State with the most leopards	Madhya Pradesh (total number: 3,907).
Descending order of states	Madhya Pradesh (3,907) > Maharashtra (1,985) > Karnataka (1,879).
Status of Uttar Pradesh	The total number of leopards in Uttar Pradesh is 371 (important for UP Special).
Tiger reserve with the highest population	Nagarjunsagar Srisailem Tiger Reserve (Andhra Pradesh).
Reserve with the highest density	Sariska Tiger Reserve (Rajasthan).

> Natural World Heritage Sites of India (UNESCO Natural Heritage Sites)

Name of the Heritage Site	Location (State)	Name of the Heritage Site	Location (State)
Kaziranga National Park	Assam	Nanda Devi and Valley of Flowers	Uttarakhand
Manas Wildlife Sanctuary	Assam	Great Himalayan National Park	Himachal Pradesh

Name of the Heritage Site	Location (State)	Name of the Heritage Site	Location (State)
Keoladeo National Park	Bharatpur, Rajasthan	Western Ghats	Kerala, Tamil Nadu, Karnataka, Goa, Maharashtra and Gujarat (jointly)
Sundarbans National Park	West Bengal		

› National Parks of India (State-wise National Parks)

State / UT	Names of National Parks (Quick Pointers)
Uttar Pradesh	• Dudhwa (<i>Lakhimpur Kheri - extremely important for UP Special</i>)
Uttarakhand	• Jim Corbett (<i>India's first national park</i>) • Rajaji • Nanda Devi • Valley of Flowers • Gangotri • Govind Pashu Vihar
Madhya Pradesh	• Kanha • Bandhavgarh • Madhav • Panna • Pench • Satpura • Sanjay • Kuno (<i>famous for the rehabilitation of cheetahs</i>) • Van Vihar • Ghughua Fossil • Dinosaur Fossil
Assam	• Kaziranga (<i>for the one-horned rhinoceros</i>) • Manas • Nameri • Orang (Rajiv Gandhi) • Dibru-Saikhowa • Dehing Patkai • Raimona
Bihar / Jharkhand	• Valmiki (Bihar) • Betla (Jharkhand)
Chhattisgarh	• Guru Ghasidas (Sanjay) • Indravati • Kanger Valley
Himachal / Ladakh	• Great Himalayan • Inderkilla • Khirganga • Pin Valley • Simbalbara (HP) • Hemis (Ladakh - <i>largest in area</i>)
Jammu-Kashmir	• Dachigam (<i>for the Hangul/Kashmir stag</i>) • Kishtwar • Kazinag • City Forest (Salim Ali)
Rajasthan	• Ranthambore • Sariska • Keoladeo (Bharatpur) (<i>for the Siberian crane</i>) • Desert National Park • Mukundra Hills
Gujarat / Haryana	• Gir (<i>for Asiatic lions</i>) • Velavadar Blackbuck • Marine (Gulf of Kutch) • Vansda (GJ) • Kalesar • Sultanpur (HR)
Karnataka	• Bandipur • Bannerghatta • Kudremukh • Nagarhole • Anshi (Kali)
Kerala	• Eravikulam • Periyar • Silent Valley • Anamudi Shola • Mathikettan Shola • Pambadum Shola
Tamil Nadu / Telangana	• Mudumalai • Mukurthi • Guindy • Indira Gandhi (Anamalai) • Gulf of Mannar Marine (TN) • Kasu Brahmananda Reddy • Mahavir Harina Vanasthali • Mrugavani (TL)
Odisha / W. Bengal	• Bhitarkanika • Simlipal (OD) • Sundarban • Buxa • Gorumara • Jaldapara • Neora Valley • Singalila (WB)

State / UT	Names of National Parks (Quick Pointers)
Andhra / Arunachal	• Papikonda • Rajiv Gandhi (Rameswaram) • Sri Venkateswara (AP) • Namdapha • Mouling (AR)
Other North-Eastern states	• Keibul Lamjao (<i>the world's only floating park</i>), Shirui (Manipur) • Nokrek, Balphakram (Meghalaya) • Murlen, Phawngpui/Blue Mountain (Mizoram) • Intanki/Ntangki (Nagaland) • Clouded Leopard, Bison/Rajbari (Tripura) • Khangchendzonga (Sikkim)
Goa / Andaman	• Bhagwan Mahavir/Mollem (Goa) • Campbell Bay • Galathea Bay • Mahatma Gandhi Marine • Mount Manipur (Mount Harriet) • North Button • Saddle Peak (Andaman & Nicobar)

› Tiger Reserves of India (State-wise Tiger Reserves)

State / UT	Names of Tiger Reserves (UPPCS Matching Special)
Uttar Pradesh	• Dudhwa • Pilibhit • Ranipur (<i>the first in the Bundelkhand region</i>)
Uttarakhand	• Corbett • Rajaji
Madhya Pradesh	• Kanha • Bandhavgarh • Panna • Pench • Satpura • Sanjay • Veerangana Durgavati • Ratapani • Madhav
Maharashtra	• Melghat • Tadoba-Andhari • Pench • Navegaon-Nagzira • Bor (<i>the smallest</i>) • Sahyadri
Rajasthan	• Ranthambore • Sariska • Mukundra Hills • Ramgarh Vishdhari • Dholpur-Karauli
Karnataka	• Bandipur • Bhadra • Kali (Anshi) • Nagarhole • Biligiriranganatha (BRT)
Tamil Nadu	• Kalakkad-Mundanthurai • Mudumalai • Anamalai • Sathyamangalam • Srivilliputhur-Megamalai
Chhattisgarh	• Indravati • Achanakmar • Udanti-Sitanadi • Guru Ghasidas-Tamor Pingla
Assam	• Kaziranga • Manas • Nameri • Orang
Arunachal Pradesh	• Namdapha • Pakke (Pakhui) • Kamlang
Andhra / Telangana	• Nagarjunsagar-Srisailem (<i>India's largest tiger reserve by area</i>) (AP) • Kawal • Amrabad (TL)
Kerala / Mizoram	• Periyar • Parambikulam (KL) • Dampa (MZ)
Odisha / W. Bengal	• Simlipal • Satkosia (OD) • Sundarban • Buxa (WB)
Bihar / Jharkhand	• Valmiki (BR) • Palamau (JH)

> Ecology

> Types of Ecological Species

Type of Species	Ecological Definition and Nature	Precise Exam-useful Examples (Exam Hits)
Keystone species	A species which, even if small in number or biomass in its ecosystem, is of very great importance in controlling the structure of the entire ecosystem.	Tiger , lion, elephant (top carnivores or large animals).
Indicator species	A species which is highly sensitive to any subtle chemical or physical change in the environment and gives immediate information.	Lichen - they automatically die out as soon as SO₂ pollution increases in the atmosphere.
Endemic species	A species which is naturally found only in one specific geographical region and is found nowhere else in the world.	The lion-tailed macaque of the Western Ghats.
Exotic / Alien species	Species which are not native to that local ecosystem but are brought from outside and established, unknowingly or deliberately.	Tea (brought from China), potato (brought by the Portuguese), water hyacinth.
Umbrella species	Species with an extensive range, efforts to conserve which automatically conserve the other smaller species of the entire ecosystem coming under them.	One-horned rhinoceros , giant panda, Bengal tiger.
Flagship species	An attractive or distinctive species used as a 'symbol' or identity for the conservation campaign of a particular ecosystem.	Sangai deer (for Keibul Lamjao of Manipur), Bengal tiger (for the Sundarbans).

8 Biological Interactions, Categories of Biodiversity and Conservation Systems

> Main Types of Biological Interactions

Type of Interaction	Effect on Both Species	Practical Ecological Example (Core Example)
Mutualism / Symbiosis	Both species benefit mutually (+ , +).	Honeybee and flower (the honeybee gets nectar and the plants are pollinated), lichen (fungus + alga).
Parasitism	One organism benefits and the other host organism is harmed (+ , -).	Amarbel (dodder, on plants), parasitic insects or worms in the human stomach.

Type of Interaction	Effect on Both Species	Practical Ecological Example (Core Example)
Commensalism	One organism benefits and the other organism is neither harmed nor helped (+, 0).	Epiphytic plants such as orchids , which grow on the branches of large trees only for shelter, without causing them any harm.

> Biodiversity

> Three Main Categories of Biodiversity

- 1. Species diversity:** focused on the total number of different species present within a particular area or ecosystem and their abundance.
- 2. Genetic diversity:** the internal diversity of genes and genetic information present within organisms of the same species (such as the thousands of varieties of paddy).
- 3. Ecosystem diversity:** the diversity of the different types of habitats, biological communities and ecological processes present within a large area.

> Types and Measurement of Biodiversity

Type of Biodiversity	Basic Scientific Definition and Concept (Core Logic)	Precise Exam-useful Example (Exam Example)
Alpha (α) diversity	The total number (richness) of species found within one particular place, local community or a single ecosystem .	The total bird species found in one particular forest area or bird sanctuary.
Beta (β) diversity	The amount of difference/change in species diversity found between two different places, habitats or communities .	Comparison of the bird communities of two different forests, or of a forest and a grassland.
Gamma (γ) diversity	The overall/total diversity of all the ecosystems present in one vast geographical region (such as a state, country or continent) .	The total plant species found in the whole of North India or the entire Western Ghats region.

> Two Main Systems of Biodiversity Conservation

Type of Conservation	Basic Definition and Scientific Philosophy	Included Examples (Exam Hotspots)
In-situ conservation	The system of conserving organisms, flora or species within their natural habitat itself .	National Parks, Biosphere Reserves, Wildlife Sanctuaries, Sacred Groves.

Type of Conservation	Basic Definition and Scientific Philosophy	Included Examples (Exam Hotspots)
Ex-situ conservation	The system of conserving genetic diversity by taking endangered organisms or plants away from their natural habitat to some artificial or man-made safe place.	Seed banks, gene banks, botanical gardens, zoological gardens (zoos), cryopreservation labs.

› The 3 Zones of a Biosphere Reserve

- 1. Core zone:** This is the innermost part, which includes legally strictly protected areas. Here **any kind of human activity or entry is completely prohibited**.
- 2. Buffer zone:** This is the area around the core zone. Only controlled scientific research, environmental education, eco-tourism and training activities are permitted here.
- 3. Transition zone:** This is the outermost part. In this area, with the cooperation of local communities, sustainable human settlements, socio-cultural activities, crop agriculture and other sustainable economic activities are fully permitted.

› India's Four Major Biodiversity Hotspots

- 1. The Himalayas / Eastern Himalayas**
- 2. Indo-Burma region**
- 3. Sundaland** (includes the Nicobar Islands)
- 4. Western Ghats and Sri Lanka**

› Projects for the Conservation of Fauna in India

Name of the Project	Year / Date of Launch	Updated and Specific Exam-useful Facts (Exam Notes)
Musk Deer Project	1970	Started from Kedarnath Sanctuary with the cooperation of the IUCN .
Hangul Project	1970	Started in Dachigam National Park for the conservation of the Kashmir stag (Hangul).
Project Tiger	1 April 1973	* Update (2026): At present India has a total of 58 tiger reserves . * Most tiger reserves: 9 in Madhya Pradesh (Tiger State).
Crocodile Conservation Project	1974	Started from Odisha with the cooperation of the United Nations Development Programme (UNDP).
Gharial Project	1975	For the breeding and conservation of gharials in the Chambal and wildlife sanctuaries.
Project Elephant	February 1992	* Update (2026): At present India has a total of 33 elephant reserves .

Name of the Project	Year / Date of Launch	Updated and Specific Exam-useful Facts (Exam Notes)
Project Rhino	1987	One-horned rhino action plan for the conservation of the one-horned rhinoceros.
Red Panda Project	1996	In the Padmaja Naidu Himalayan Zoological Park with the cooperation of WWF-India .
Sea Turtle Project	1999	By the Wildlife Institute of India (WII) , Dehradun, for turtles (Olive Ridley).
Project Vulture	2006	Action plan to stop the deaths of vultures caused by the drug Diclofenac .
Project Snow Leopard	2009	For conservation in the high Himalayan regions (such as Hemis National Park).
Haathi Mere Saathi Campaign	2011	By the Ministry of Environment and the Wildlife Trust; mascot: ' Gaju '.

> Sustainable Development

> 17 Sustainable Development Goals: 2030 (17 Sustainable Development Goals - SDGs)

Note: Adopted by the United Nations in 2015, with the aim of ending poverty in all its forms from the entire world and ensuring sustainable development by 2030.

SDG No.	Sustainable Development Goal (SDG Target)	SDG No.	Sustainable Development Goal (SDG Target)
Goal 1	No Poverty	Goal 10	Reduced Inequalities
Goal 2	Zero Hunger	Goal 11	Sustainable Cities and Communities
Goal 3	Good Health and Well-being	Goal 12	Responsible Consumption and Production
Goal 4	Quality Education	Goal 13	Climate Action
Goal 5	Gender Equality	Goal 14	Life Below Water
Goal 6	Clean Water and Sanitation	Goal 15	Life on Land
Goal 7	Affordable and Clean Energy	Goal 16	Peace, Justice and Strong Institutions
Goal 8	Decent Work and Economic Growth	Goal 17	Partnerships for the Goals
Goal 9	Industry, Innovation and Infrastructure		

> UNESCO World Heritage Sites

> Environmental Pollution and Pollutants

> Primary Pollutants versus Secondary Pollutants

Category of Pollutant	Chemical Nature of Formation	Precise Exam-useful Examples (Exam Hotspots)
Primary pollutants	Those which are emitted directly from the source (such as vehicle smoke or factories) and remain in the atmosphere in the same form.	Carbon monoxide (CO), carbon dioxide (CO₂), sulphur dioxide (SO₂), oxides of nitrogen (NO , NO₂), methane (CH₄), lead (Pb), chlorofluorocarbons (CFCs), and particulate matter (PM₁₀ , PM_{2.5}).
Secondary pollutants	Those which are not emitted directly but are formed in the atmosphere by chemical reactions among primary pollutants or by the effect of sunlight and moisture.	Ozone (O₃ - tropospheric/ground-level), peroxyacetyl nitrate (PAN), photochemical smog , sulphuric acid (H₂SO₄), nitric acid (HNO₃), and organic aerosols.

- **Biodegradable pollutants:** pollutants that are completely decomposed over time by biological agents (such as bacteria) (e.g., domestic sewage, paper, agricultural waste).

> Specific Pollutants and the Organs of the Human Body Affected by them

Pollutant Element / Particle	Human Organ / System Affected	Pathological Effect (Impact)
Asbestos dust	Lungs	Asbestosis (serious difficulty in breathing).
Lead (Pb)	Central nervous system (brain)	Damage to the nervous system, obstruction of children's mental development.
Carbon monoxide (CO)	Blood streams	Reacts with haemoglobin and stops the oxygen-carrying capacity.
Mercury (Hg)	Stomach / nervous system	Minamata disease, serious diseases of the stomach and kidneys.
Fluoride (fluoride ion)	Teeth and bones	Fluorosis (yellowing of teeth and bending of bones).

> Ozone-depleting Gases and their Industrial Sources

Ozone Depleting Substance (ODS)	Main Industrial / Practical Source	Ozone Depletion Potential (ODP) Effect
Chlorofluorocarbons (CFCs)	Domestic refrigerators (fridges), air conditioners (AC) and aerosol sprays.	Release chlorine in the stratosphere and destroy the ozone layer.
Halons	Fire extinguishers.	They contain bromine, which is even more destructive than chlorine.
Carbon tetrachloride (CCl₄)	Industrial cleaning solvents, dry-cleaning chemicals.	Highly toxic and causes heavy damage to ozone.
Hydrochlorofluorocarbons (HCFCs)	Transitional coolants (as temporary substitutes for CFCs).	Less harmful than CFCs, but in the category of complete ban.
Methyl bromide	Agricultural pesticide, soil fumigation process.	Used to kill soil-borne insects; acutely destructive for ozone.
Methyl chloroform	Industrial degreasing (cleaning) solvents and coatings.	Capable of evaporating and reaching the stratosphere.

> Principal Gases Emitted by Volcanic Eruptions

Gases Emitted from Volcanoes	Special Exam-useful Fact (Exam Hit)
Water vapour	The largest amount (60% to 90%) in a volcanic eruption is of water vapour.
Carbon dioxide (CO₂)	The second main gas, which increases the greenhouse effect.
Sulphur dioxide (SO₂)	Because of it, conditions of acid rain develop in the atmosphere.
Hydrogen sulphide (H₂S)	A gas with a strong smell like rotten eggs.
Other gases	Hydrogen (H), carbon monoxide (CO), hydrogen chloride (HCl), and helium (He).

> The Sequential Process of Soil Erosion

Type of Erosion	Process of Formation and Simple Definition	UPPCS Exam Key-point
1. Splash erosion	Scattering and detachment of soil particles due to heavy raindrops falling directly on the soil.	The initial stage of erosion.
2. Sheet erosion	When, due to excessive rainfall, the fertile top layer of soil begins to flow with water as a uniform sheet.	This erosion is not immediately visible to the farmer, but is the most harmful.
3. Rill erosion	When flowing water, due to the slope, makes small and shallow channels in the soil.	Fragmentation of agricultural land begins.
4. Gully erosion	When the small channels of rill erosion become very large, deep and wide due to the rapid flow of water.	Because of it the "ravines" (badlands) of the Chambal valley have been formed.
5. Stream channel erosion	When water, instead of flowing over the land, flows as a very thick, deep and large stream and cuts its banks.	The extreme form of erosion.

> India's Principal Environmental Protection Acts

Name of the Act / Policy	Year of Passing	Regulatory Area / Special Fact (Exam Context)
River Boards Act	1956	For the regulation and development of inter-state rivers and river valleys.
Wild Life (Protection) Act	1972	Complete ban on hunting and illegal trade of endangered wildlife (stringent amendment in 1991).
Water (Prevention and Control of Pollution) Act	1974	For maintaining water quality and establishing pollution control boards (amended in 1988).
Water (Prevention and Control of Pollution) Cess Rules	1977	For levying a cess on water consumed by local bodies and industries.
Forest (Conservation) Act	1980	Strict control on stopping the destruction of forests and on diverting forest land to non-forestry purposes.
Air (Prevention and Control of Pollution) Act	1981	For the prevention of air pollution; 'noise pollution' was also added to it by the 1987 amendment.

Name of the Act / Policy	Year of Passing	Regulatory Area / Special Fact (Exam Context)
Environment (Protection) Act	1986	(It is also called Umbrella Legislation) ; passed after the Bhopal gas tragedy (1984) for the comprehensive protection of the entire environment.
National Environment Tribunal Act	1995	For giving speedy compensation to the victims of accidents occurring while handling hazardous substances.
National Environment Appellate Authority Act	1997	For hearing appeals regarding areas where the setting up of industries is restricted under the Environment Protection Act.
Ozone Depleting Substances (Regulation) Rules	2000	
Biological Diversity Act	2002	For the conservation of the country's rich biological resources and the sustainable sharing of their benefits.
National Environment Policy	2004	A guiding policy for balancing sustainable development and environmental protection in the country.
Forest Rights Act (FRA)	2006	Legal recognition of the land rights of Scheduled Tribes (ST) and other traditional forest dwellers living in forests.

› India's Upcoming Environmental Target Years

Target Objective	Target Year Fixed by the Government of India
Goal of making India a fully 'Developed India' (Viksit Bharat)	2047 (on the completion of 100 years of independence).
Goal of making India energy independent / self-reliant	2047
Goal of reducing the use of hydrofluorocarbons (HFC) by 80%	2047 (in line with the Kigali Amendment).
Goal of increasing India's nuclear power capacity to 1 lakh megawatt (100 GW)	2047
Goal of achieving complete self-reliance in shipbuilding	2047
Goal of Mumbai city achieving 'zero carbon emission' (Net Zero)	2050
Goal of 'Net Zero' (Net Zero Carbon Emission) by the entire Indian nation	2070 (announced at Glasgow COP26).

> India's Upcoming Targets for the Year 2030 (India's Targets for 2030)

Target Objective	Specific Numerical Value / Description (Exam Target)
Coal gasification	Target of achieving 100 million tonnes of coal gasification.
Renewable energy capacity	Target of achieving 500 gigawatt (GW) of non-fossil/renewable energy capacity.
Solar power capacity	Target of obtaining 280 gigawatt (GW) of the total renewable energy from solar energy alone.
Reduction in carbon emissions	Reducing the projected total carbon emissions by 1 billion tonnes .
Electric vehicles (EVs)	Target of making 30% of total vehicles electric .
Malaria eradication	Complete eradication of malaria from India.
HIV AIDS	Target of completely ending HIV AIDS infection from the country.
Green Railway Mission	Target of making Indian Railways 'Net Zero Carbon Emission' or a 'Green Railway' .

> Environmental Terminology

> Important Terms, Concepts and the Experts who Propounded them

Specific Term / Environmental Concept	Name of the Propounder / Related Expert
Green Development	W. M. Adams
Inventor / father of natural farming	Masanobu Fukuoka (Japanese farmer and philosopher)
Ecosystem	A. G. Tansley
Ecological niche	Joseph Grinnell
Greenhouse gas (discovery of the greenhouse effect)	Joseph Fourier
Ecology	Ernst Haeckel
Biological Diversity	Raymond F. Dasmann
Biodiversity (popularising the term)	Walter G. Rosen
Biodiversity hotspot	Norman Myers

Specific Term / Environmental Concept	Name of the Propounder / Related Expert
Biosphere reserve (first use of the term 'biosphere')	Eduard Suess
Ecotone	F. E. Clements

> Important Environmental Terminology and Definitions

Term	Basic Scientific Definition and Concept (Core Logic)	Special Exam-useful Fact
Sustainable Development	Development that meets the needs of the present generation without compromising the needs of future generations.	This concept was given by the Brundtland Commission (1987) in its report ' Our Common Future '.
Ecomark	A mark certifying environment-friendly and sustainable household products.	Started by the Bureau of Indian Standards (BIS) in 1991 (symbol: earthen pot).
AGMARK	A mark certifying the quality and purity of agricultural and food products.	Issued by the Directorate of Marketing and Inspection (DMI).
Greenwash	A false or misleading claim made by a company or organisation to present its products as environment-friendly or green.	Mere marketing/show in the name of environmental protection.
Habitat	The specific physical place or environment where a particular organism or species naturally lives.	It indicates the 'address' of the organism.
Lentic habitat	Ecological habitats of still water .	Examples: ponds, lakes, wetlands , pools, etc.
Lotic habitat	Ecological habitats of moving or running water .	Examples: rivers, springs, streams , etc.
Niche	The specific functional role played by a species in its ecosystem and its position.	This concept was first given by " Joseph Grinnell ".
Ecosystem	The community of organisms in a definite area and the non-living environment around them, which interact with one another.	* This term was first used by A. G. Tansley (1935) . * The limit of environmental support is called " carrying capacity ".

Term	Basic Scientific Definition and Concept (Core Logic)	Special Exam-useful Fact
Biomagnification	The entry of non-biodegradable harmful chemicals / heavy metals into organisms through the food chain and the increase of their concentration at each higher trophic level.	E.g., accumulation of DDT or heavy metals in top consumers.

> Movements Related to Environmental Protection and their Leaders

Name of the Movement	Main Leader / Pioneer	Special Exam-useful Geographical Fact (Exam Hits)
Chipko Movement (1973)	Sunderlal Bahuguna, Chandi Prasad Bhatt	To stop the felling of trees in Chamoli in Uttarakhand (then UP).
Appiko Movement (1983)	Pandurang Hegde	A movement run on the lines of Chipko in Karnataka (South India).
Narmada Bachao Andolan	Medha Patkar	A broad movement against the Sardar Sarovar Dam being built on the Narmada river.
Opposition to the Narmada Valley Project	Baba Amte	For the human rights of dam-displaced people and environmental protection.
Silent Valley Movement	M. K. Prasad	To save the Silent Valley in Kerala from a hydroelectric project on the Kunthipuzha river.
Bishnoi Movement	Amrita Devi Bishnoi	Historic supreme sacrifice for the protection of Khejri trees in Rajasthan (1730).
Anti-Tehri Dam Movement	Sunderlal Bahuguna	Protective opposition to the huge dam built on the Bhagirathi river in Uttarakhand.
Dongri Adivasi Movement	Chandi Prasad Bhatt	For the protection of the forest rights of local tribal communities.
Nilgiri Movement	N. Gopalakrishnan	For the conservation of the Nilgiri ecosystem of the Western Ghats.
Chilika Lake Movement	S. K. Patnaik	Against prawn farming and commercial exploitation in the Chilika Lake of Odisha.
Mithun Movement	Nagarika Bahini and tribal groups	For local wildlife and tribal forest rights in the Jharkhand region.

> Principal Personalities Associated with the Environment

Name of the Personality	Famous Epithet / Related Field (Exam Pointer)
Ravindra Kumar Sinha	Dolphin Man of India.
Kailash Sankhala	Tiger Man of India.
Dr. Ramdeo Misra	Father of Ecology in India.
E. P. Odum	Global father of ecology and environmental science.
Richard St. Barbe Baker	He is called the " Man of the Trees ".
Dr. Norman Borlaug	Originator of the Green Revolution in the world (Nobel Prize winner).
Dr. M. S. Swaminathan	Father of the Green Revolution in India.
Dr. Verghese Kurien	Father of the White Revolution / Milk Revolution in India (Operation Flood).
Dr. Salim Ali	Famous Indian bird expert (ornithologist).
Sunderlal Bahuguna	Main leader associated with the famous Chipko Movement (1973) of Uttarakhand.
Pandurang Hegde	Leader of the Appiko Movement (1983) , run on the lines of Chipko in Karnataka.
Baba Amte and Medha Patkar	Main leaders of the Narmada Bachao Andolan.

> Important Environment Days

Date	Name of the Day
2 February	World Wetlands Day
3 March	World Wildlife Day
20 March	World Sparrow Day
21 March	World Forestry Day
22 March	World Water Day
23 March	World Meteorological Day
22 April	World Earth Day
22 May	International Day for Biological Diversity

Date	Name of the Day
31 May	World No Tobacco Day
5 June	World Environment Day
8 June	World Oceans Day
29 July	International Tiger Day
10 August	World Lion Day
12 August	World Elephant Day
29 August	National Sports Day
16 September	World Ozone Day (International Ozone Day)
1 to 7 October	National Wildlife Week
3 October	World Habitat Day
4 October	World Animal Welfare Day
13 October	International Day for Disaster Reduction
19 November	World Toilet Day
26 November	National Environment Protection Day
5 December	World Soil Day
11 December	International Mountain Day
14 December	National Energy Conservation Day

› Short Trick for UPPCS Prelims (Chronological Revision):

- **Sequence of March:** 20 (Sparrow) → 21 (Forest) → 22 (Water) → 23 (Meteorology).
- **The magic figure of 22:** 22 March (Water) → 22 April (Earth) → 22 May (Biodiversity).

› Energy Resources

› Classification of Energy Resources

› Non-renewable versus Renewable Energy Resources

Non-Renewable Resources (limited and polluting)	Renewable Energy Resources (unlimited and environment-friendly)
* Coal	* Wind power

Non-Renewable Resources (limited and polluting)	Renewable Energy Resources (unlimited and environment-friendly)
* Petroleum / mineral oil	* Solar power
* Natural gas	* Hydro-electric power
* Atomic energy / nuclear energy	* Biogas
* Shale gas	* Geothermal energy
	* Tidal power
	* Ocean thermal energy (OTEC)
	* Energy produced from waste (Waste-to-Energy)

> Conventional versus Non-conventional Energy Sources

Conventional Sources of Energy (historically used for a long time)	Non-Conventional Sources of Energy (modern / newly discovered technologies)
* Coal, lignite	* Solar energy
* Petroleum, natural gas	* Wind energy
* Wood / forest waste	* Biogas
* Animal waste (cow-dung cakes, etc.)	* Geothermal energy
* Mineral oil	* Tidal energy / ocean wave energy
	* Hydrogen energy / green hydrogen
	* Fuel cells
	* Atomic energy (nuclear energy)

- **Note (Exam Point):** As a source, **atomic energy** is 'non-renewable' because it is based on a limited mineral (uranium), but being a new technology it is counted among the '**non-conventional**' sources of energy.

> International Environmental Conferences and Treaties

Conference / Treaty (Place)	Year	Important Facts and Exam-useful Points
Stockholm Conference (Sweden)	1972	- It is called the Magna Carta of the Environment. - The United Nations Environment Programme (UNEP) was established. - It was declared that World Environment Day would be celebrated on 5 June.
Montreal Protocol (Canada)	1987	- The first international agreement for the protection of the ozone layer. - It was decided to celebrate Ozone Day on 16 September.
Rio Conference / Earth Summit (Brazil)	1992	Agenda-21 for sustainable development was adopted. - The UNFCCC on climate change was established.
Kyoto Protocol / Earth + 5 (Japan)	1997	- Reducing global warming by cutting the emission of greenhouse gases. - Decision of a 5.2% cut from the 1990 level.
Johannesburg Conference / Rio + 10 / Second Earth Summit (South Africa)	2002	- Agreement on establishing a "World Solidarity Fund" for poverty eradication. - Special emphasis on sustainable development.
Cancun Conference (Mexico)	2010	Decision to establish a "Green Climate Fund" (GCF) of 100 billion dollars.
Rio + 20 Conference / Earth + 20 (Brazil)	2012	- The draft "The Future We Want" was presented. - The Zero Hunger Challenge (necessary food security for all) was launched.
Climate Adaptation Summit (Netherlands)	2021	A new umbrella fund named the International Fund for Agricultural Development (IFAD) was established.

> Historical Chronology of the Climate Change Conferences (UNFCCC)

Historic Conference / Stage	Year / Date	Main Exam-useful Provisions and Milestones (Exam Facts)
Earth Summit	1992	Held in Rio de Janeiro, Brazil; it is called the United Nations Conference on Environment and Development. The UNFCCC, the CBD and Agenda-21 were born here.
Coming into force of the UNFCCC	21 March, 1994	It was ratified by 197 countries of the world. The COP meetings began after this.

Historic Conference / Stage	Year / Date	Main Exam-useful Provisions and Milestones (Exam Facts)
COP 1	1995	Held in Berlin, Germany; the first COP conference towards tackling climate change.
COP 3 (Kyoto Protocol)	1997	Kyoto, Japan; legal targets fixed for developed countries to reduce greenhouse gas emissions.
COP 8 (Delhi Declaration)	2002	New Delhi, India ; emphasis on the technological needs of developing countries and sustainable development.
COP 13	2007	Bali, Indonesia; formulation of the post-Kyoto Protocol roadmap (the 'Bali Action Plan').
COP 16	2010	Cancun, Mexico; decision to establish the ' Green Climate Fund ' (GCF).
COP 17	2011	Durban, South Africa; blueprint of a new legal agreement for all countries (to come into force from 2020).
COP 21 (Paris Agreement)	2015	Paris, France; the historic resolve to keep global warming below 2°C (preferably 1.5°C) above pre-industrial levels.
COP 22	2016	Marrakesh, Morocco; discussion on the implementation of the rules and priorities of the Paris Agreement.
COP 23	2017	Bonn, Germany (under the presidency of Fiji); the process of finalising the rulebook of the Paris Agreement.
COP 24	2018	Katowice, Poland; the detailed 'rulebook' for implementing the Paris Agreement was adopted.
COP 25	2019	Madrid, Spain (under the presidency of Chile); ended without any final decision on the rules of the global carbon market.

- **History of the Sustainable Development Goals:** These goals (SDGs) were first proposed at the **Rio+20 Conference in 2012**, and have been set globally to be achieved by 2030.

> Principal Environmental Conventions and Protocols

Name of the Convention / Protocol	Year of Adoption	Year of Entry into Force	Basic Subject / Regulatory Area (Core Subject)
Ramsar Convention	1971	1975	Related to the conservation of wetlands of international importance.
Bonn Convention (CMS)	1979	1983	Related to the conservation of migratory species of wild animals.
Vienna Convention	1985	1988	An inter-governmental framework agreement for the protection of the ozone layer .

Name of the Convention / Protocol	Year of Adoption	Year of Entry into Force	Basic Subject / Regulatory Area (Core Subject)
Montreal Protocol	1987	1989	For the phased elimination of substances that damage the ozone layer (ODS/CFCs).
Basel Convention	1989	1992	Related to the transboundary movement and control of hazardous wastes .
Convention on Biological Diversity (CBD)	1992	1993	Related to the sustainable conservation of biological diversity and the use of genetic resources.
Kyoto Protocol	1997	2005	Binding under the UNFCCC for reduction in the emission of greenhouse gases (GHGs) .
Rotterdam Convention	1998	2004	Prior Informed Consent (PIC) for certain hazardous chemicals and pesticides in international trade.
Cartagena Protocol	2000	2003	Supplementary rules related to biosafety under the Convention on Biological Diversity (CBD).
Stockholm Convention	2001	2004	Protecting human health from the dangers of persistent organic pollutants (POPs) .
Nagoya Protocol	2010	2014	Access to genetic resources and the fair and equitable sharing of benefits arising from their use (ABS).
Minamata Convention	2013	2017	Protection of human health from the adverse effects of mercury (Hg) and its compounds.
Kigali Agreement	2016	2019	Amendment to the Montreal Protocol; for the phase-out of HFC gases that increase global warming.

➤ Important COP Summits and their Venues

COP No.	Venue / Country	Special UPPCS Exam-useful Fact (Exam Hit)
COP - 26	Glasgow (United Kingdom / Scotland)	India announced the 'Panchamrit' targets and net-zero by 2070.
COP - 27	Sharm el-Sheikh (Egypt)	Agreement on establishing the 'Loss and Damage' fund.
COP - 28	Dubai (United Arab Emirates / UAE)	First review of the 'Global Stocktake' and a call for transitioning away from fossil fuels.
COP - 29	Baku (Azerbaijan)	Main discussion on new goals of climate finance.

COP No.	Venue / Country	Special UPPCS Exam-useful Fact (Exam Hit)
COP - 30	Brazil	Focused on the conservation of the Amazon forests and new pledges of the Paris Agreement.
COP - 33	India (proposed - 2028)	India has officially offered to host it in 2028.

> Important International Environmental Conferences and Conventions

Conference / Convention	Year	Basic Subject / Objective (Core Function)	Special Exam-useful Fact (Exam Points)
Stockholm Conference	1972	The first summit of the United Nations on the human environment.	In memory of this conference, 5 June was declared World Environment Day .
Kyoto Protocol	1997	Reducing the emission of greenhouse gases (control of global warming).	The concept of ' carbon credit ' developed under it.
Cartagena Protocol	2000	Regulation of biosafety and modern biotechnology.	A supplementary agreement under the Convention on Biological Diversity (CBD); effective from 2003 .
Nagoya Protocol	2010	Access to genetic resources and fair and equitable sharing of benefits (ABS).	Under it, 20 'Aichi Targets' were declared for biodiversity conservation.
Rio+20 Conference	2012	Held in Rio de Janeiro on the completion of 20 years of the Earth Summit (1992).	The main focus was on the ' Green Economy ' and sustainable development .

> Environmental Institutions

> Important National Environmental Institutions

Name of the Institution	Location / Headquarters
Indian Institute of Forest Management (IIFM)	Bhopal (Madhya Pradesh)
Wildlife Institute of India (WII)	Dehradun (Uttarakhand)
G. B. Pant Institute of Himalayan Environment and Development	Almora (Uttarakhand)
Centre for Ecological Sciences (CES)	Bengaluru (Karnataka)
Central Arid Zone Research Institute (CAZRI)	Jodhpur (Rajasthan)
Indian Institute of Farming Systems Research	Meerut

Name of the Institution	Location / Headquarters
National Bureau of Agriculturally Important Microorganisms (NBAIM)	Mau
Himalayan Forest Research Institute (HFRI)	Shimla
Rain Forest Research Institute (RFRI)	Jorhat
Tropical Forest Research Institute (TFRI)	Jabalpur

> Food Safety and Standards Authority of India (FSSAI)

Dimension / Point	Statutory Structure and Functioning of FSSAI (Core Features)
Nature of the institution	It is a statutory body .
Establishment and Act	Established in September 2008 under the Food Safety and Standards Act, 2006 .
Nodal ministry	It functions under the administrative control of the Ministry of Health and Family Welfare , Government of India.
Headquarters and regional offices	* Headquarters: New Delhi. * 6 regional offices: Delhi, Guwahati, Mumbai, Kolkata, Cochin and Chennai.
Organisational structure	It has a Chairperson and 22 members , of whom one-third (1/3) must be women .
Main responsibilities and functions	* Food certification in India and framing of science-based standards and guidelines. * Issuing licences to food businesses, conducting safety audits and increasing food safety literacy.

> Miscellaneous

> Principal Breeds of Cattle and Livestock

Name of the Animal	Indigenous / Indian Breeds	Exotic Breeds / Special Facts
Cow (Cattle)	Gir, Sahiwal, Red Sindhi, Tharparkar, Hariana, Deoni, Ongole, Krishna Valley, Gangatiri, Gaolao, Hallikar, Amritmahal, Khillari, Kangayam, Umblachery, Pulikulam, Alambadi.	Jersey, Holstein Friesian, Brown Swiss, Red Dane, Ayrshire, Guernsey, Ankole, Normande, Simmental.

Name of the Animal	Indigenous / Indian Breeds	Exotic Breeds / Special Facts
Buffalo	Murrah, Surti, Jaffarabadi, Bhadawari, Nili Ravi, Mehsana, Nagpuri, Toda, Banni, Bargur, Chhattisgarhi, Chilika, Gojri, Kalahandi, Luit (Swamp), Mesra, Marathwadi.	* Bhadawari breed: the fat content in its milk is the highest. * Murrah breed: it is also called 'Black Gold'.
Goat	Jamunapari, Barbari, Beetal, Black Bengal, Sirohi, Jakhrana, Surti, Mehsana, Chegu, Kutchi goat, Attappady Black, Kanni Adu, Berari.	Boer, Alpine, Saanen (it is called the queen of milk of the world), Toggenburg, Anglo-Nubian.

07

Indian Economy

INDIAN ECONOMY

> Banking and Money

> Reserve Bank of India (RBI)

Dimension / Point	Precise Exam-useful Facts and Administrative Structure (Exam Facts)
Establishment and Act	On 1 April 1935 under the provisions of the Reserve Bank of India Act, 1934 .
Basis of establishment	On the recommendation of the Hilton Young Commission (1926) .
Central office (history)	The initial central office was in Kolkata , which was permanently shifted to Mumbai in 1937 .
Nationalisation	It was fully nationalised on 1 January 1949 .
Headquarters	Mumbai, Maharashtra .
Four regional zones	Eastern Zone—Kolkata Western Zone—Mumbai Northern Zone—New Delhi Southern Zone—Chennai
Basic objectives and functions	Controlling credit/currency in the country, regulating the entire banking system and ensuring financial stability.
Determination of monetary instruments	To control inflationary pressure in the country, the repo rate and reverse repo rate are determined by the RBI , through which the flow of cash in the banking system is controlled.
Quantitative monetary policy instruments	Bank rate, repo rate, reverse repo rate, Cash Reserve Ratio (CRR), Statutory Liquidity Ratio (SLR), Open Market Operations (OMO), and Marginal Standing Facility (MSF).
Qualitative monetary policy instruments	Margin requirement, credit rationing, selective credit control, moral suasion, and direct action.
First Governor (British origin)	Sir Osborne Smith - the first Governor of the RBI.
First Indian Governor	C. D. Deshmukh (Chintaman Dwarkanath Deshmukh) - the first RBI Governor of independent India (he was also the Governor at the time of nationalisation).
Current Governor	Shaktikanta Das (the current Governor of the RBI).

> Principal Banking Terminology and Instruments

Banking Term / Instrument	Basic Banking Meaning and Exam-useful Definition (Core Meaning)
Liability	The money/deposits which the bank has to return to its customers or depositors in the future. E.g., deposits in the bank and various types of savings/small savings schemes.
Bank Rate	The interest rate at which the Reserve Bank of India (RBI) provides long-term loans to commercial banks.
Repo Rate	The interest rate at which the Reserve Bank of India (RBI) provides short-term loans to commercial banks for their short-term needs against approved securities.
Reverse Repo Rate	The interest rate at which the Reserve Bank of India (RBI) accepts the surplus liquidity/money available with commercial banks (i.e., borrows money from commercial banks).
Cash Reserve Ratio (CRR)	It is mandatory for commercial banks to keep a fixed part of their total deposits (NDTL) in the form of cash with the Reserve Bank of India (RBI) .
Statutory Liquidity Ratio (SLR)	It is mandatory for commercial banks to keep a prescribed percentage of their total deposits (in the form of cash, gold or approved securities) in liquid form with themselves .

> Four Measures of Money Supply in India: M1, M2, M3, M4 (Money Supply Measures)

Money Measure	Components	Liquidity Position and Nickname
M1	Currency in circulation (notes and coins) + demand deposits with banks (current and savings accounts) + other deposits with the RBI that can be withdrawn.	* The most liquid money (usable as cash immediately). * It is called " Narrow Money ".
M2	M1 + savings deposits of post office savings banks.	* Slightly less liquid than M1, but still easily convertible into cash.
M3	M1 + time/long-term deposits with the banking system (fixed deposits - FD).	* Its liquidity is less than that of M1 and M2. * It is called " Broad Money ". (<i>most used in monetary policy analysis</i>)
M4	M3 + total deposits of post office savings organisations (National Savings Certificates, i.e., NSC, are not included in it).	* The least liquid measure of money supply (because it includes time deposits that cannot be converted into cash immediately).

- **Decreasing order of liquidity: M1 > M2 > M3 > M4** (M1 is the most liquid and M4 the least liquid).

> Historical Chronology of the Establishment of Banks (Chronological Order)

> Principal Banks Established before Independence

Name of the Bank	Year of Establishment	Historical and Exam-useful Significance (Exam Facts)
Bank of Hindustan	1770	The first bank established in India on the European pattern (by Alexander and Company).
General Bank of Bengal and Bihar	1773	An early attempt of the initial colonial period.
General Bank of Bengal	1786	A banking unit of the early British rule.
Bank of Bengal	1806	The first presidency bank (initially formed under the name Bank of Calcutta).
Bank of Bombay	1840	The second important presidency bank.
Bank of Madras	1843	The third presidency bank. <i>(Note: these three presidency banks were merged to form the Imperial Bank of India in 1921).</i>
Allahabad Bank	1865	India's first functioning joint stock bank.
Oudh Commercial Bank	1881	Was the first commercial bank managed by Indians on the basis of limited liability .
Punjab National Bank (PNB)	1894	The first bank formed entirely with Indian capital and management (Lala Lajpat Rai was one of its founders; it started from Lahore).
People's Bank of India	1901	An early banking unit of the period of the Swadeshi Movement.
Bank of India	1906	Established by leading businessmen of Bombay, inspired by the Swadeshi Movement.
Canara Bank	1906	A swadeshi bank established in Mangalore by Ammembal Subba Rao Pai.
Indian Bank	1907	Established in Madras as part of the Swadeshi Movement.
Bank of Baroda (BoB)	1908	Established by Sayajirao Gaekwad III (Maharaja of Baroda).
Central Bank of India	1911	The first swadeshi commercial bank, wholly owned by Indians.

Name of the Bank	Year of Establishment	Historical and Exam-useful Significance (Exam Facts)
Bank of Mysore	1913	Established through the efforts of the Maharaja of Mysore and Sir M. Visvesvaraya.

> Principal Banks Established after Independence

Name of the Bank	Year of Establishment	Regulatory Reform / Current Status
State Bank of Hyderabad	1941	The bank of the Nizam of Hyderabad; after independence it became an associate of SBI under the SBI (Subsidiary Banks) Act, 1959 and was finally merged into SBI .
UCO Bank	1943	Established by Ghanshyam Das Birla; it was nationalised in 1969.
United Bank of India	1950	Formed by the merger of some banks of Bengal; (it has now been merged into Punjab National Bank).
State Bank of India (SBI)	1955	On the recommendation of the Gorwala Committee, the Imperial Bank of India was nationalised and made the largest bank of the country.
State Bank of Bikaner and Jaipur	1963	An associate bank formed by the merger of the banks of Bikaner and Jaipur; (now merged into SBI).
Bank of Maharashtra	1935	(Established 1935); after independence it came under government ownership in the first phase of nationalisation in 1969.
Union Bank of India	1919	(Established in 1919, inaugurated by Mahatma Gandhi); nationalised in 1969.
Punjab and Sind Bank	1908	Came under government control in the second phase of nationalisation in 1980 (the group of 6 banks).
Syndicate Bank	1925	(Original name Canara Industrial and Banking Syndicate); nationalised in 1969; (now merged into Canara Bank).

> New Private Sector and Commercial Banks (Post-1994 Reforms)

Name of the Bank	Year of Establishment	Banking Category / Specific Nature
ICICI Bank	1994	Established under the new-generation private sector banks on the recommendation of the Narasimham Committee.

Name of the Bank	Year of Establishment	Banking Category / Specific Nature
HDFC Bank	1994	A leading new-generation private sector commercial bank (at present the country's largest private bank).
Axis Bank	1994	Its initial name was UTI Bank , which was later renamed Axis Bank.
IndusInd Bank	1994	One of the earliest private banks to obtain a licence after the new economic reforms.
Kotak Mahindra Bank	2003	The first institution to convert from a Non-Banking Financial Company (NBFC) into a commercial bank.
Yes Bank	2004	A private sector scheduled commercial bank.
Bandhan Bank	2015	The first bank to convert from a microfinance institution (MFI) into a full commercial bank.
AU Small Finance Bank	2017	A Rajasthan-based finance company that received the status of a Small Finance Bank (SFB).
IDFC First Bank	2018	A commercial bank formed by the merger of the infrastructure finance company IDFC and Capital First.

> Financial Institutions

> Principal Financial Institutions and their Establishment (Financial Institutions Matrix)

Name of the Financial Institution	Date / Year of Establishment	Main Exam-useful Facts (Exam Notes)
Reserve Bank of India (RBI)	1 April 1935	Established on the recommendation of the Hilton Young Commission; nationalised on 1 January 1949.
Industrial Credit and Investment Corporation of India (ICICI)	January 1955	An early major financial institution formed with the cooperation of the private sector.
State Bank of India (SBI)	1 July 1955	On the recommendation of the Gorwala Committee, the Imperial Bank of India (1921) was nationalised and renamed.
Life Insurance Corporation of India (LIC)	September 1956	Formed by nationalising about 245 Indian and foreign companies (headquarters: Mumbai).

Name of the Financial Institution	Date / Year of Establishment	Main Exam-useful Facts (Exam Notes)
Regional Rural Banks (RRB)	2 October 1975	Established for rural credit on the recommendation of the M. Narasimham Committee.
National Bank for Agriculture and Rural Development (NABARD)	12 July 1982	The apex bank for agriculture and rural development, formed on the recommendation of the Sivaraman Committee .
Small Industries Development Bank of India (SIDBI)	April 1990	Headquarters: Lucknow (Uttar Pradesh); extremely important for the UPPCS exam.

> Principal Financial and Regulatory Institutions

Name of the Financial Institution	Year of Establishment	Headquarters (HQ)	Main Exam-useful Facts / Nature (Exam Notes)
Industrial Finance Corporation of India (IFCI)	1948	New Delhi	It is India's first Development Financial Institution (DFI) .
Industrial Development Bank of India (IDBI)	1964	Mumbai	Established to give long-term financial assistance to industries.
Unit Trust of India (UTI)	1964	Mumbai	The country's first mutual fund institution.
General Insurance Corporation of India (GIC)	1972	Mumbai	The apex reinsurance provider of non-life (general) insurance.
Deposit Insurance and Credit Guarantee Corporation (DICGC)	1978	Mumbai	The institution that insures depositors' money in banks up to ₹5 lakh (a wholly owned subsidiary of the RBI).
Export-Import Bank of India (EXIM Bank)	1982	Mumbai	The apex institution for financing foreign trade (import-export).
Securities and Exchange Board of India (SEBI)	1988	Mumbai	The regulatory body of the stock market (statutory status: 1992).
National Housing Bank of India (NHB)	1988	New Delhi	The supreme regulatory body of the housing finance sector.
National Securities Depository Limited (NSDL)	1996	Mumbai	India's first electronic securities depository.
Insurance Regulatory and Development Authority of India (IRDAI)	1999	Hyderabad	Regulator of the entire insurance sector (life and non-life) (recommendation of the Malhotra Committee).
Central Depository Services Limited (CDSL)	1999	Mumbai	A leading securities depository of the country.

Name of the Financial Institution	Year of Establishment	Headquarters (HQ)	Main Exam-useful Facts / Nature (Exam Notes)
Credit Guarantee Fund Trust (CGTMSE)	2000	Mumbai	A trust for collateral-free loan facility to micro and small enterprises.
Clearing Corporation of India (CCIL)	2001	Mumbai	Provider of clearing and settlement facilities for financial market transactions.
Agriculture Insurance Company of India (AIC)	2002	New Delhi	A specialised company of the agriculture and crop insurance sector.
Pension Fund Regulatory and Development Authority (PFRDA)	2003	New Delhi	The supreme regulator of the pension fund sector in the country (statutory status: 2013).
India Infrastructure Finance Company Limited (IIFCL)	2006	New Delhi	For giving long-term loans to infrastructure projects.
National Payments Corporation of India (NPCI)	2008	Mumbai	The apex body operating retail payments in the country (such as UPI, IMPS, RuPay).
Micro Units Development and Refinance Agency (MUDRA)	2015	Mumbai	An NBFC providing refinance support to micro enterprises.
Insolvency and Bankruptcy Board of India (IBBI)	2016	New Delhi	Regulator of the Insolvency and Bankruptcy Code.
National e-Governance Services Limited (NeSL)	2016	Mumbai	India's first Information Utility company.
International Financial Services Centres Authority (IFSCA)	2020	Gandhinagar	The unified regulator of international financial centres located in GIFT City (Gujarat) .
National Bank for Financing Infrastructure and Development (NaBFID)	2021	Mumbai	A new Development Financial Institution (DFI) for infrastructure development.

> Poverty

> Important Committees Related to the Measurement of Poverty

Name of the Committee	Year of Formation	Main Exam-useful Recommendations and Criteria (Exam Points)
Y. K. Alagh Committee	1979	* For the first time in India, poverty was determined on the basis of nutrients (calorie consumption). * For rural areas: 2400 kilocalories (per person per day) * For urban areas: 2100 kilocalories were fixed.

Name of the Committee	Year of Formation	Main Exam-useful Recommendations and Criteria (Exam Points)
D. T. Lakdawala Committee	1993	* Recommended making state-specific poverty lines for each state. * Asked for these lines to be updated using the consumer price indices CPI-IW (industrial workers) in urban areas and CPI-AL (agricultural labourers) in rural areas.
Suresh Tendulkar Committee	2009	* For the first time, monthly per capita consumption expenditure (MPCE) was made the basis instead of calorie consumption. * For rural households: ₹816 (per person per month) * For urban households: ₹1000 was fixed.
C. Rangarajan Committee	2014	* Gave a new criterion by revising consumption expenditure at the all-India level: * For rural areas: ₹972 per month (₹32 per day) * For urban areas: ₹1407 per month (₹47 per day) was fixed.

> Unemployment

> Types and Characteristics of Unemployment

Note: In India the **National Sample Survey Organisation (NSSO)** / **National Statistical Office (NSO)** compiles and publishes data related to employment and unemployment.

Type of Unemployment	Basic Meaning and Characteristic (Core Concept)	Special Exam-useful Facts (Exam Facts)
Frictional unemployment	* The transitional period between leaving one job and getting another job. * Unemployment generated due to technological change, etc. (such as retrenchment in companies).	* It is considered "search" or "voluntary" unemployment. * It is mainly found in developed countries.
Structural unemployment	* Unemployment generated due to a change in the structure of the economy or the lack of skills/qualifications among workers in line with industrial demand.	* This is the unemployment found in the largest amount in India.
Disguised / hidden unemployment	* When more people than required are engaged in a work or occupation (even if some people are removed, there is no difference in total production).	* In it the "marginal productivity of the extra worker is zero." * It is found especially in agriculture and the unorganised sector.
Seasonal unemployment	* Unemployment generated because work is available according to the weather or season and not available in the remaining months/time.	* This too is seen especially in the agriculture sector (except at the time of sowing/harvesting).

Type of Unemployment	Basic Meaning and Characteristic (Core Concept)	Special Exam-useful Facts (Exam Facts)
Cyclical unemployment	* Unemployment generated by a fall in demand due to an economic recession in the market. It ends automatically when demand in the market rises.	* Its figures in India are negligible (zero). * It is mainly found in the industrial sectors of developed countries .

> Inflation

> Instruments for Controlling Inflation (Monetary Policy Tools)

Quantitative Tools	Qualitative Tools
* Repo rate	* Rationing of credit
* Reverse repo rate	* Margin requirement
* Bank rate	* Moral suasion
* Cash Reserve Ratio (CRR)	
* Statutory Liquidity Ratio (SLR)	
* Open Market Operations (OMO)	
* Marginal Standing Facility (MSF rate)	

> Agriculture

> Important Revolutions Related to Agriculture and Production (Agricultural Revolutions)

Name of the Revolution	Related Production / Sector
Green Revolution	Foodgrain production (especially wheat and rice)
White Revolution	Milk production and dairy development
Blue Revolution	Fish production
Yellow Revolution	Oilseed production
Silver Revolution	Egg and poultry production
Brown Revolution	Production of chemical fertilisers
Black Revolution	Production of petroleum / crude oil
Golden Revolution	Fruit, horticulture and honey production

> Five-Year Plans

> Main Priority Sectors in the Five-Year Plans (Five-Year Plans Core Matrix)

Plan	Period	Model / Architect	Main Priority and Historic Work (Exam Target)
First Five-Year Plan	1951 - 56	Harrod-Domar model	* The highest priority was given to the agriculture sector. * Large irrigation projects such as Bhakra-Nangal and Damodar Valley were started.
Second Five-Year Plan	1956 - 61	P. C. Mahalanobis model	* Priority was given to heavy industries and minerals. * Large steel plants were established in the country at Bhilai (Russia), Durgapur (Britain) and Rourkela (Germany), along with hydroelectric projects.
Third Five-Year Plan	1961 - 66	Gadgil Yojana / Sukhamoy Chakravarty	* Making both agriculture and industry self-reliant was its main objective. * In 1964 the Bokaro Iron and Steel Industry was established with the cooperation of the former Soviet Union (USSR).
Fourth Five-Year Plan	1969 - 74	Ashok Rudra and A. S. Manne	* Emphasis was laid on growth with stability and progress towards self-reliance. * The goal of ending inequalities of income and wealth in society was set.
Fifth Five-Year Plan	1974 - 79	D. P. Dhar model	* Poverty eradication (Garibi Hatao) and self-reliance / population control were its main objectives. * (Note: due to political change this plan was ended in 1978 itself, 1 year before time).

> Principal Schemes

> Principal Schemes from 2000 to 2013

Name of the Scheme	Exact Date / Year of Launch	Main Exam-useful Objective (Exam Context)
Annapurna Yojana	1 April 2000	Providing free foodgrains to extremely poor senior citizens.
Sarva Shiksha Abhiyan	November 2000	Universalisation of elementary education for children of 6-14 years.
Pradhan Mantri Gram Sadak Yojana	25 December 2000	Connecting rural habitations with all-weather roads (100% centrally funded).
Sampoorna Grameen Rozgar Yojana	25 September 2001	Ensuring employment and food security in rural areas.

Name of the Scheme	Exact Date / Year of Launch	Main Exam-useful Objective (Exam Context)
Hariyali Yojana	27 January 2003	Rural development scheme for water harvesting and watershed management.
Varishtha Pension Bima Yojana	14 July 2003	Assured pension / social security scheme for senior citizens.
National Horticulture Mission	2005-06	Integrated development of the horticulture sector (now part of MIDH).
National Rural Health Mission	12 April 2005	Providing accessible and quality health services to the rural population.
Jawaharlal Nehru National Urban Renewal Mission (JNNURM)	3 December 2005	Integrated development of urban infrastructure and basic services.
National Rural Employment Guarantee Scheme (NREGS)	2 February 2006	Legal guarantee of 100 days of employment (later renamed MGNREGA).
National Food Security Mission	2007-08	Increasing self-sufficiency in the production of wheat, rice and pulses.
Rashtriya Krishi Vikas Yojana (RKVY)	2007-08	Achieving an annual growth rate of 4% in agriculture and allied sectors.
Rashtriya Swasthya Bima Yojana	1 October 2007	Providing health insurance to BPL families of the unorganised sector.
Aam Aadmi Bima Yojana	2 October 2007	Social security insurance for rural landless households.
Indira Gandhi National Old Age Pension Scheme	19 November 2007	Financial assistance / pension to the elderly of the BPL category.
Jawaharlal Nehru National Solar Mission	11 January 2010	Increasing solar power generation capacity in the country (initial target of 20 gigawatt).
National Rural Livelihood Mission (NRLM)	June 2011	Removing rural poverty through Self-Help Groups (SHGs).
Nirmal Bharat Abhiyan	1 April 2012	For increasing sanitation coverage in rural areas (old name Total Sanitation Campaign).
Direct Benefit Transfer Scheme (DBT)	1 January 2013	Sending the subsidy of government schemes directly into the bank accounts of beneficiaries.
Deendayal Antyodaya Yojana–National Urban Livelihoods Mission	24 September 2013	Giving urban poor households opportunities for skill development and self-employment.

> Principal Schemes of 2014 and 2015

Name of the Scheme	Exact Date / Year of Launch	Main Exam-useful Objective (Exam Context)
Pradhan Mantri Jan-Dhan Yojana	28 August 2014	The world's largest campaign of financial inclusion.
Make in India	25 September 2014	Making India a global manufacturing hub.
Swachh Bharat Mission	2 October 2014	Making the country open-defecation free (ODF) and complete cleanliness.
Saansad Adarsh Gram Yojana	11 October 2014	Each MP adopting villages and developing them as model villages.
Mission Indradhanush (immunisation)	25 December 2014	Ensuring full immunisation coverage of children and pregnant women.
UJALA Yojana	5 January 2015	Energy conservation by distributing LED bulbs at low prices.
Beti Bachao, Beti Padhao Abhiyan	22 January 2015	Improving the sex ratio and ensuring the education of girls (launched from Panipat).
Sukanya Samriddhi Yojana	22 January 2015	A small savings account scheme for the secure future of girls.
Pradhan Mantri Mudra Yojana	8 April 2015	Providing loans to micro enterprises in three categories (Shishu, Kishore, Tarun).
Pradhan Mantri Jeevan Jyoti Bima Yojana	9 May 2015	Voluntary life insurance of ₹2 lakh (for the age group 18-50 years).
Pradhan Mantri Suraksha Bima Yojana	9 May 2015	Accidental insurance of ₹2 lakh (for the age group 18-70 years).
Atal Pension Yojana	9 May 2015	Minimum assured pension scheme for workers of the unorganised sector.
Pradhan Mantri Awas Yojana–Urban	25 June 2015	Providing pucca houses to the poor and middle class of urban areas.
AMRUT Mission	25 June 2015	Ensuring water supply, sewerage and urban reform in cities.
Smart City Mission	25 June 2015	Making 100 cities of the country smart and sustainable.
Pradhan Mantri Krishi Sinchayee Yojana	1 July 2015	Goal of 'Har Khet Ko Pani' and 'Per Drop More Crop' (water-use efficiency).

Name of the Scheme	Exact Date / Year of Launch	Main Exam-useful Objective (Exam Context)
Digital India Programme	1 July 2015	Transforming India into a knowledge-based, digitally empowered society.
Pradhan Mantri Kaushal Vikas Yojana	15 July 2015	Providing youth with skill training in line with industries.
National Skill Development Mission	15 July 2015	Speeding up skill training activities and building an institutional framework.

> Principal Schemes from 2016 to 2020

Name of the Scheme	Exact Date / Year of Launch	Main Exam-useful Objective (Exam Context)
Startup India	16 January 2016	Promoting innovation and new enterprises in the country.
Pradhan Mantri Fasal Bima Yojana	18 February 2016	Comprehensive insurance cover to farmers for crop losses from natural disasters.
Stand-Up India	5 April 2016	Giving loans to SC, ST and women entrepreneurs for greenfield enterprises.
Pradhan Mantri Ujjwala Yojana	1 May 2016	Giving free LPG connections to women of BPL households.
Pradhan Mantri Awas Yojana–Gramin	20 November 2016	Giving pucca houses to all homeless households and those with kutcha houses in rural areas.
Pradhan Mantri Kisan Sampada Yojana	2017	Development of food processing and mega food park infrastructure.
Saubhagya Yojana (Sahaj Bijli Har Ghar)	11 October 2017	Ensuring complete electrification of all rural and urban houses of the country.
Deendayal SPARSH Yojana	3 November 2017	Scholarship to promote philately (stamp collecting) among school children.
Ayushman Bharat–PM Jan Arogya Yojana	23 September 2018	Cashless secondary and tertiary health insurance cover of up to ₹5 lakh per family.
Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)	24 February 2019	Direct income support of ₹6,000 per year to eligible farmer families (3 instalments).
Pradhan Mantri Shram Yogi Maandhan Yojana	5 March 2019	Monthly pension scheme of ₹3,000 for elderly workers of the unorganised sector.

Name of the Scheme	Exact Date / Year of Launch	Main Exam-useful Objective (Exam Context)
Pradhan Mantri Kisan Urja Suraksha evam Utthan Mahabhiyan (PM-KUSUM)	19 February 2019	Helping farmers set up solar pumps and grid-connected solar power plants.
Jal Jeevan Mission (Rural)	15 August 2019	Giving a functional household tap connection (FHTC) to every rural household by 2024.
Pradhan Mantri Kisan Maandhan Yojana	12 September 2019	Old-age pension scheme for small and marginal farmers (after the age of 60 years).
National Pension Scheme for Traders and Self-Employed Persons	12 September 2019	Voluntary social security scheme for small shopkeepers and the self-employed.
Atal Bhujal Yojana	25 December 2019	Improving groundwater management in the stressed blocks of identified states.
Pradhan Mantri Garib Kalyan Anna Yojana	26 March 2020	Giving 5 kg of free foodgrains per person to the poor during the Covid-19 pandemic.
PM SVANidhi Yojana	1 June 2020	Giving working capital loans to street vendors.
Pradhan Mantri Matsya Sampada Yojana	10 September 2020	Sustainable development of the fisheries sector in the country through the Blue Revolution.

> Principal Schemes from 2021 to 2024

Name of the Scheme	Exact Date / Year of Launch	Main Exam-useful Objective (Exam Context)
Ayushman Bharat Digital Mission	27 September 2021	Digital Health ID for citizens and an integrated health framework.
Pradhan Mantri Poshan Shakti Nirman Yojana (PM-POSHAN)	29 September 2021	The new form of the Mid-Day Meal in government primary schools.
PM Gati Shakti National Master Plan	13 October 2021	An integrated plan for multi-modal connectivity and infrastructure development.
PM Ayushman Bharat Health Infrastructure Mission	25 October 2021	Strengthening primary, secondary and tertiary health systems across the country.
Pradhan Mantri Schools for Rising India (PM-SHRI)	7 September 2022	Upgrading more than 14,500 schools of the country into modern model schools.

Name of the Scheme	Exact Date / Year of Launch	Main Exam-useful Objective (Exam Context)
National Logistics Policy	17 September 2022	Reducing the cost of freight transport and improving last-mile connectivity.
Pradhan Mantri Vishwakarma Yojana	17 September 2023	Giving skill training and financial assistance to traditional artisans and craftsmen.
PM-JANMAN (Janjatiya Adivasi Nyaya Maha Abhiyan)	15 November 2023	Socio-economic welfare of Particularly Vulnerable Tribal Groups (PVTGs).
Pradhan Mantri Kisan Drone Yojana / Namo Drone Didi	2023	Training women SHGs in drone operation for agricultural work (fertiliser spraying).
Pradhan Mantri Surya Ghar: Muft Bijli Yojana	13 February 2024	300 units of free electricity to 1 crore houses through rooftop solar.
Dharti Aaba Janjatiya Gram Utkarsh Abhiyan	2 October 2024	Development of infrastructure on a saturation basis in tribal-majority villages.

➤ Recent New Schemes and Expansions (Current Context: 2025–2026)

Name of the Scheme	Exact Date / Year of Launch	Main Exam-useful Objective (Exam Context)
Pradhan Mantri Viksit Bharat Rozgar Yojana (PM-VBRY)	1 August 2025	Incentive for providing new employment opportunities to youth in the formal workforce.
Restructured PM SVANidhi Scheme	1 August 2025	New expansion for street vendors with a higher loan limit and incentives for digital transactions.
Pradhan Mantri Dhan-Dhaanya Krishi Yojana	11 October 2025	Promoting the spread of high-yielding, climate-resilient crop varieties and sustainable agriculture.
Pradhan Mantri Surya Ghar: Model Solar Village Initiative	2025	Development of model villages in every district of the country to demonstrate rural solar power generation.
PM Awas Yojana–Urban 2.0	2025 (implementation)	The second phase of construction of pucca houses for the urban poor and middle class.
PM-e-Bus Sewa	2025 (expansion)	Large-scale operation of electric buses to promote public transport in cities.
Viksit Bharat–Guarantee for Rozgar and Aajeevika (VB-G RAM JI)	1 July 2026	Hundred-per-cent guarantee of employment to rural youth and a skill-development livelihood mission.

Name of the Scheme	Exact Date / Year of Launch	Main Exam-useful Objective (Exam Context)
BHAVYA Rasayan Scheme	24 July 2026	Making India self-reliant at the global level in the manufacturing of petrochemicals and chemicals.
Pradhan Mantri Surya Sarovar Yojana	31 July 2026	Setting up floating solar power plants on the banks of Amrit Sarovars and on reservoirs.

> Important Committees

> Important Tax and Banking Reform Committees (Tax & Financial Reform Committees)

Name of the Committee	Year of Formation	Main Exam-useful Recommendations and Objectives (Exam Points)
L. K. Jha Committee	1977	* Suggestions for reform in indirect taxation. * Recommended strict laws and administrative measures to curb black money and tax evasion.
Raja Chelliah Committee	1991	* Comprehensive reform in the direct and indirect tax system. * Making the direct tax system simple and broad-based and rationalising the tax rates.
Narasimham Committee (I and II)	1991 and 1998	* Historic recommendations for structural reforms in the financial and banking sector. * Emphasis on bringing transparency into the financial/tax system.
Kelkar Committee	2002	* Restructuring direct and indirect taxes and simplifying tax concessions.
Adhia Committee	2016	* Administrative measures for implementing a uniform tax system (GST - Goods and Services Tax) throughout the country. * Simplifying indirect taxes and implementing a single tax structure.

> Important Economic and Financial Committees (Committees Master Matrix)

Name of the Committee	Year of Formation	Related Sector / Main Subject (Core Subject)
Y. K. Alagh Committee	1979	Determination of the poverty line on the basis of nutrients (calories).
Sukhamoy Chakravarty Committee	1985	Review of the Indian monetary system and monetary policy.

Name of the Committee	Year of Formation	Related Sector / Main Subject (Core Subject)
Vaghul Committee	1987	Development and institutional framework of the Indian money market.
M. Narasimham Committee-I	1991	Structural reforms in the financial and banking sector (first phase).
Raja J. Chelliah Committee	1991–93	Comprehensive reforms in direct and indirect taxes (tax reforms).
C. Rangarajan Committee	1993	Suggestions on the disinvestment of public sector undertakings (PSUs).
D. T. Lakdawala Committee	1993	Methodology for determining state-specific poverty lines.
R. N. Malhotra Committee	1993	Reforms in the insurance sector and approval of private investment.
Tarapore Committee-I	1997	Full convertibility of the rupee on the capital account (Capital Account Convertibility).
M. Narasimham Committee-II	1998	Second-phase reforms for financial strength in the banking sector.
J. R. Varma Committee	1998	Risk management in the derivatives financial market.
L. C. Gupta Committee	1996	Regulatory framework of the derivatives market and futures trading.
Prof. Vijay Kelkar Committee	2002	Restructuring of direct and indirect taxes (Direct & Indirect Tax Reforms).
Tarapore Committee-II	2006	Roadmap for moving towards full capital account convertibility.
Raghuram Rajan Committee	2008	Comprehensive structural reforms in the financial sector (Financial Sector Reforms).
Suresh Tendulkar Committee	2009	New estimate of poverty on the basis of monthly per capita consumption expenditure.
Malegam Committee	2010	Regulation of micro-finance institutions.
Damodaran Committee	2011	Improvement of customer service in the banking sector (Customer Service in Banks).
Nachiket Mor Committee	2013	Financial inclusion and universal financial access.

Name of the Committee	Year of Formation	Related Sector / Main Subject (Core Subject)
Parthasarathi Shome Committee	2013	Reform in tax administration and GAAR (General Anti-Avoidance Rules).
Dr. Urjit Patel Committee	2014	Strengthening the monetary policy framework and inflation targeting based on the Consumer Price Index (CPI).
P. J. Nayak Committee	2014	Governance of public sector banks and bank boards.
Shanta Kumar Committee	2014	Restructuring of the Food Corporation of India (FCI) and reform in the grain procurement system.
Deepak Mohanty Committee	2015	Medium-term practical action plan for financial inclusion.
N. K. Singh Committee	2016	Review of the FRBM Act and a new framework of fiscal responsibility.
Bimal Jalan Committee	2018	Economic Capital Framework of the RBI and transfer of surplus.

> Important Curves and Coefficients of Economics

> Important Curves and Coefficients of Economics (Economic Curves & Indices)

Name of the Curve / Coefficient	Shape (if applicable)	Field of Measurement / Basic Relationship (Core Logic)
Lorenz Curve	Graphical line	A curve that studies the disparity (inequality) of wealth and income in an economy.
Gini Coefficient	Value between 0 and 1	It is the mathematical form of the Lorenz curve, which shows the precise measure of income disparity (0 = complete equality, 1 = complete inequality).
Kuznets Curve	Inverted 'U' shape	It shows that as a country's GDP (per capita income) increases, economic inequality rises at first , but after a certain point inequality begins to decline.
Environmental Kuznets Curve	Inverted 'U' shape	It shows that in the initial stage of economic growth (GDP) environmental damage increases , but after a peak point the damage begins to decline.
Phillips Curve	Negative slope	Shows the inverse (opposite) relationship between the rates of unemployment and inflation . When inflation rises, unemployment falls.

Name of the Curve / Coefficient	Shape (if applicable)	Field of Measurement / Basic Relationship (Core Logic)
Laffer Curve	Inverted 'U' shape	Shows the relationship between tax rates and the total tax revenue obtained from them. At first, raising taxes increases revenue , but after a point raising taxes causes revenue to fall.

> Various Deficits of the Economy

> Various Deficits of the Economy (Budgetary Deficits Formulas)

Type of Deficit	Simple Exam-useful Definition and Explanation
Budget Deficit	The excess of total expenditure (revenue + capital expenditure) over total receipts (revenue + capital receipts).
Revenue Deficit	The excess of the government's day-to-day administrative revenue expenditure over its ordinary revenue receipts.
Fiscal Deficit	The excess of total expenditure over total receipts excluding the government's borrowings. It shows the government's entire borrowing liability in the year.
Primary Deficit	The deficit obtained by subtracting the interest payments made by the government on past loans from the fiscal deficit.

> Indices

> Global Multidimensional Poverty Index (MPI)

Note: This index was started in **2010** by the **United Nations Development Programme (UNDP)** and the **Oxford Poverty and Human Development Initiative (OPHI)**. This index measures deprivation in 10 indicators across 3 dimensions:

Main Dimensions	The 10 Indicators Included
1. Health	(1) Nutrition, (2) Child mortality
2. Education	(3) Years of schooling, (4) School attendance
3. Standard of Living	(5) Cooking fuel, (6) Sanitation (toilet), (7) Drinking water, (8) Electricity, (9) Housing, (10) Assets

> Economics: Important Terminology

> Important Terminology Related to Economics (Economic Core Concepts)

Terminology	Simple Exam-useful Definition and Explanation (Core Meaning)
Gross Domestic Product (GDP)	The total monetary value of all final goods and services produced within the geographical/domestic boundary of a country in a given financial year.
Inflation	A continuous rise in the general prices of goods and services in the market, which indicates a fall in the purchasing power of money.
Deflation	A continuous fall in the general prices of goods and services in the market, which usually occurs during a decline in demand and an economic recession.
Monetary Policy	The policy of the country's central bank (such as the RBI in India) to influence liquidity through control of the money supply and interest rates in the economy.
Fiscal Policy	The policy of the country's government to influence and run the economy through taxation, public expenditure (government spending) and public debt.
Balance of Payments (BoP)	A systematic record of all kinds of international economic/monetary transactions (imports, exports, financial flows) of the residents of a country with the rest of the world over a given period (usually 1 year).

> Census 2011

> Exact Figures and Administrative Facts of Census 2011

Historical sequence	It is the 15th census of India and the 7th census of independent India.
Census Commissioner	Dr. C. Chandramouli.
Statutory basis	Conducted under the provisions of the Census Act, 1948.
Reference Date	1 March 2011.
Total population of India	121.09 crore (about 17.5% of the world's total population).
Gender division	* Male population: 62.32 crore * Female population: 58.76 crore
Decadal growth rate (2001–11)	A growth of 17.64% was recorded.
Population density	382 persons per sq km.
Overall sex ratio	943 females per 1000 males.
Child sex ratio (0–6 years)	919 females per 1000 males.

Historical sequence	It is the 15th census of India and the 7th census of independent India.
Literacy rate (total)	74.04%
Literacy rate (gender-wise)	* Male literacy: 82.14% * Female literacy: 65.46%
Rural versus urban ratio	* Rural population: 83.37 crore (68.84%) * Urban population: 37.71 crore (31.16%)
Urban growth (fact)	Compared to 2001, an increase of about 9.1 crore was recorded in the urban population in 2011.

› State-wise Demographic Categories (Top 3 versus Bottom 3)

Demographic Indicator	Top 3 States / UTs (Descending Order)	Bottom 3 States / UTs (Ascending Order)
Total population	1. Uttar Pradesh (19.98 crore) 2. Maharashtra (11.24 crore) 3. Bihar (10.41 crore)	1. Sikkim (6.11 lakh) 2. Mizoram (10.97 lakh) 3. Arunachal Pradesh (13.84 lakh)
Population growth rate	1. Meghalaya (27.82%) 2. Arunachal Pradesh (25.92%) 3. Bihar (25.42%)	1. Nagaland (–0.58% - negative growth) 2. Kerala (4.91%) 3. Goa (8.23%)
Population density (persons/sq km)	1. Bihar (1,102) 2. West Bengal (1,028) 3. Kerala (859)	1. Arunachal Pradesh (17) 2. Mizoram (52) 3. Sikkim (86)
Sex ratio	1. Kerala (1,084) 2. Tamil Nadu (996) 3. Andhra Pradesh (993)	1. Haryana (879) 2. Jammu-Kashmir (889) 3. Sikkim (890)
Total literacy rate	1. Kerala (93.91%) 2. Lakshadweep (91.85%) 3. Mizoram (91.33%)	1. Bihar (61.80%) 2. Arunachal Pradesh (65.38%) 3. Rajasthan (66.11%)
Female literacy rate	1. Kerala (91.98%) 2. Mizoram (89.40%) 3. Lakshadweep (87.95%)	1. Rajasthan (52.12%) 2. Bihar (51.50%) 3. Jharkhand (55.42%)

› Child Sex Ratio (0-6 Age Group)

Geographical Unit	Child Sex Ratio Value	Special Exam-useful Fact
All-India average	919	Considerably lower than the overall sex ratio (943).

Geographical Unit	Child Sex Ratio Value	Special Exam-useful Fact
Highest (first)	Arunachal Pradesh (972)	The situation is better in tribal areas.
Highest (second)	Mizoram (970)	The second top state of the North-East.
Lowest (bottom 1)	Haryana (834)	The most backward state of the country in child sex ratio.
Lowest (bottom 2)	Punjab (846)	The second lowest after Haryana.
Lowest (bottom 3)	Jammu-Kashmir (862)	A worrying decline in the border hill states.

> Urbanisation Infrastructure

Urbanisation Parameter	Actual Figures / State (Exam Points)
Urban population (total)	37.71 crore (31.16%)
Rural population (total)	83.37 crore (68.84%)
State with the largest urban population	Maharashtra (first in terms of number/quantity).
State with the highest urbanisation percentage	Goa (62.17%) (first in terms of ratio/percentage).
State with the lowest urbanisation percentage	Himachal Pradesh (10.04%) (i.e., 89.96% rural population).
Total towns / urban units (2011)	7,935 units.
Statutory Towns	4,041
Census Towns	3,894

> Religious Census 2011

Religious Community / Faith	Total Population (in crore / lakh)	Share in Total Population (per cent)
Hindu	96.63 crore	79.8%
Muslim	17.22 crore	14.2%
Christian	2.78 crore	2.3%
Sikh	2.08 crore	1.7%

Religious Community / Faith	Total Population (in crore / lakh)	Share in Total Population (per cent)
Buddhist	84.43 lakh	0.7%
Jain	44.52 lakh	0.4%
Other religions and faiths	79.38 lakh	0.7%
Religion not stated	28.67 lakh	0.2%

> Social Categories: Scheduled Castes (SC) and Scheduled Tribes (ST)

Social Category	Total Population (in the country)	Per cent of Total Population	First State in terms of Population
Scheduled Castes (SC)	20.14 crore	16.63%	Uttar Pradesh (highest in number)
Scheduled Tribes (ST)	10.45 crore	8.63%	Madhya Pradesh (highest in number)

08

Uttar Pradesh Special

UTTAR PRADESH SPECIAL

> Uttar Pradesh: Polity and Administrative Facts

> Uttar Pradesh: Historical Capitals, Naming, Formation and Reorganisation

Subject / Event	Period / Date
Sequence of the state's capitals	• Agra (from 1836) • Prayagraj (from 1858) • Lucknow (partially) (from 1921) • Lucknow (fully) (from 1935)
Change of the state's name	• North-Western Provinces (1836) • United Provinces of Agra and Oudh (1877) • United Provinces (1937) • Present Uttar Pradesh (24 January 1950) - <i>(that is why UP Day is celebrated on 24 January).</i>
Division of the state & formation of 'Uttarakhand'	9 November, 2000 (Uttaranchal (present Uttarakhand) was formed by separating 13 hill districts of Uttar Pradesh.)
National Capital Region (NCR)	A total of 8 districts (Meerut, Ghaziabad, Gautam Buddh Nagar, Bulandshahr, Hapur, Baghpat, Muzaffarnagar, Shamli.)

> State Symbols of Uttar Pradesh

State Category	Name of the Symbol	Special Description
State emblem	Ganga-Yamuna confluence, a pair of fish, bow and arrow	Arranged within a circle
State tree	Ashoka	-
State animal	Barasingha	Swamp deer
State bird	Sarus crane	-
State flower	Palash / Tesu	It is also called the ' Flame of the Forest '
State sport	Hockey	-

> Uttar Pradesh: Polity and Administrative Facts

Administrative Unit / Indicator	Number of Seats / Description	Special Exam-useful Fact
Form of the state legislature	Bicameral	Both the Legislative Assembly and the Legislative Council exist.
Lok Sabha seats	80	The state with the most Lok Sabha seats in the country.
Rajya Sabha seats	31	The state with the most Rajya Sabha seats in the country.
Total members of the Legislative Assembly (LA)	403	-
Total members of the Legislative Council (LC)	100	-
Formation of the first Legislative Assembly	July 1937	At the time of the pre-independence United Provinces.
First President's Rule	25 February 1968 to 26 February 1969	The state's first period of President's Rule.
Assembly Speaker the most times	Shri Atmaram Govind Kher and Shri Keshari Nath Tripathi	Both leaders were elected Speaker 3 times each .
District with the most Assembly seats	Prayagraj (12 seats)	The number of seats is the highest because of the larger population.
Districts with the fewest Assembly seats	Shravasti, Mahoba, Chitrakoot (2 seats each)	-

> Uttar Pradesh: Geography

> Geographical Location of Uttar Pradesh

Geographical Indicator	Description / Figures
Total area	2,40,928 square kilometres (about 7.33% of India's total area)
Longitudinal extent	From 77°3' East longitude to 84°39' East longitude
East-west length	650 km
North-south width	240 km
Total bordering states/UTs	9 (8 states and 1 Union Territory - Delhi)

Geographical Indicator	Description / Figures
States adjoining Sonbhadra	4 states (Madhya Pradesh, Chhattisgarh, Jharkhand and Bihar)

> Districts with Distinctive Borders

District	Feature of the Geographical Border
Sonbhadra	The only district in the country bordering four states (Madhya Pradesh, Bihar, Jharkhand, Chhattisgarh).
Saharanpur	Borders three states (Haryana, Himachal Pradesh, Uttarakhand).
Lalitpur	Surrounded on three sides by the state of Madhya Pradesh .

> Extreme Points of Area and Population

Indicator / Category	District	Additional Description
Largest in area	Lakhimpur Kheri	7,680 sq km
Smallest in area	Hapur	660 sq km
Easternmost district	Ballia	On the Bihar border
Westernmost district	Shamli	On the Haryana border
Northernmost district	Saharanpur	On the Himachal/Uttarakhand border
Southernmost district	Sonbhadra	On the Chhattisgarh/MP border
Top 5 most populous districts	Prayagraj > Moradabad > Ghaziabad > Azamgarh > Lucknow	<i>(in decreasing order)</i>

> Districts Adjoining Neighbouring States

Neighbouring State	Number of Adjoining Districts	Names of the Districts
Madhya Pradesh	11 districts (<i>the most</i>)	Agra, Etawah, Jalaun, Jhansi, Lalitpur, Mahoba, Banda, Chitrakoot, Prayagraj, Mirzapur, Sonbhadra
Bihar	7 districts	Maharajganj, Kushinagar, Deoria, Ballia, Ghazipur, Chandauli, Sonbhadra
Haryana	6 districts	Saharanpur, Shamli, Baghpat, Gautam Buddh Nagar, Aligarh, Mathura

Neighbouring State	Number of Adjoining Districts	Names of the Districts
Rajasthan	2 districts	Agra, Mathura

> Principal Rivers of Uttar Pradesh and the Cities Situated on their Banks

Name of the River	Principal Towns / Cities Situated on its Banks
Ganga river	Bithoor, Varanasi, Prayagraj, Kannauj, Farrukhabad, Ghazipur, Kanpur, Mirzapur, etc.
Yamuna river	Kaushambi, Mathura-Vrindavan, Agra, Etawah, Hamirpur.
Gomti river	Lucknow, Jaunpur, Sultanpur (<i>asked many times</i>)
Rapti river	Gorakhpur , Balrampur, Bahraich, Shravasti.

> History, Art and Culture of Uttar Pradesh

> Folk Dances of Uttar Pradesh

Name of the Folk Dance	Related Region / District	Main Feature and Occasion (Key Facts)
Karma dance	Mirzapur and Sonbhadra	A dance performed by the Kharwar tribe group.
Charkula dance	Braj region	Performed on the festival of Holi with burning lamps arranged in several tiers on the head.
Raas dance	Braj region	Extremely famous in Braj culture.
Mayur dance	Braj region	A special dance performed during the Raslila.
Jhoola dance	Braj region	Performed in temples/areas in the month of Shravan (Sawan).
Mohaniya dance	Bundelkhand	A dance performed during the festival of Deepawali .
Rai dance	Bundelkhand	By women as thanksgiving after harvesting the crop.
Khayal dance	Bundelkhand	On the celebration/auspicious occasion of the birth of a son .
Shaura / Shaira dance	Bundelkhand	By the farmers of Bundelkhand while harvesting crops .
Ghoda (horse) dance	Bundelkhand	Performed by decorated horses on auspicious occasions.
Dhuriya dance	Bundelkhand	By men of the potter (Prajapati) caste dressed as women .

Name of the Folk Dance	Related Region / District	Main Feature and Occasion (Key Facts)
Devi dance	Bundelkhand	Associated with the traditional folk rituals of Bundelkhand.
Pai Danda dance	Bundelkhand	A popular dandiya-like dance performed by Ahirs (Yadavs).
Kartik Geet dance	Bundelkhand	With devotion in the month of Kartik.
Deepawali dance	Bundelkhand	At the time of the Deepawali festival.
Dhobiya dance	Purvanchal	An artistic portrayal of the livelihood relationship between the washerman and the donkey .
Kathghodwa dance	Purvanchal	On auspicious occasions the dancer, seated on an artificial (bamboo) mare, dances inside a circle.
Natwari dance	Purvanchal	A dance performed by people of the Ahir (Yadav) community .
Chaular dance	Mirzapur and Sonbhadra	In tribal and local society for good rain/crops.
Dharkahari dance	Sonbhadra	By local tribes on special occasions.
Dhedhiya dance	Fatehpur	Performed with a lamp in a perforated earthen pot on the head.
Chaurasiya dance	Jaunpur	Mainly by the Kahar community of Jaunpur.
Dheevar dance	General (state level)	By people of the Kahar caste on auspicious occasions.
Nautanki	Kanpur, Lucknow, Hathras	It is a folk drama-dance form, famous in the styles of these cities.

> Folk Songs and Classical Dance of Uttar Pradesh

A. Principal Folk Songs

Folk Song	Main Region	Occasion / Special Context
Kajri	Purvanchal (especially Mirzapur, Varanasi, Jaunpur)	Related to the rainy season and Sawan
Birha	Eastern Uttar Pradesh	Based on separation, love and social life
Alha	Bundelkhand	Heroic ballad; description of the valour of Alha-Udal

Folk Song	Main Region	Occasion / Special Context
Sohar	Whole of Uttar Pradesh	On the auspicious occasion of birth and childbirth
Banna-Banni	Various regions	Songs sung on the occasion of marriage
Chaiti	Eastern Uttar Pradesh	A seasonal song sung in the month of Chaitra
Phag / Phagua	Braj and Bundelkhand	Related to Holi and the spring season
Rasiya	Braj region (Mathura, Vrindavan, Agra)	Related to Radha-Krishna and the romantic sentiment (shringar)
Dhola-Maru	Western Uttar Pradesh (Rajasthan border)	A folk ballad of love and separation
Jogira	Purvanchal	Especially on the occasion of Holi
Lachari	Eastern Uttar Pradesh	By women on marriage and social occasions
Devi Geet	Purvanchal, Bundelkhand and the whole of UP	Related to the worship and devotion of gods and goddesses
Kaharwa	Eastern Uttar Pradesh	A song style used in folk dance and folk song
Nautanki songs	Western and Central Uttar Pradesh	Folk theatre and traditional entertainment

B. The Only Classical Dance of Uttar Pradesh: Kathak

- **Original gharana:** Lucknow gharana.
- **Royal patronage:** It received special encouragement and patronage in the court of **Wajid Ali Shah**, the last Nawab of Awadh.
- **Technical feature: Ghungroos** have a special place in it. Many kinds of subtle sounds are produced on the stage by the tapping of the feet (footwork).
- **Principal artistes (dancers):** Pandit Birju Maharaj, Lachhu Maharaj, Sitara Devi, Rama Devi, Shambhu Maharaj, etc.

C. Principal Music Gharanas and Related Artistes

Music Gharana	Principal Artistes / Musicians
Lucknow gharana	Ustad Wajid Hussain Khan, Achhan Maharaj, Shambhu Maharaj, Lachhu Maharaj, Birju Maharaj
Banaras gharana	Sitara Devi, Gopi Krishna, Kishan Maharaj, Anokhelal Mishra, Pandit Rajan Mishra, Pandit Sajan Mishra

Music Gharana	Principal Artistes / Musicians
Rampur-Sahaswan gharana	Inayat Hussain Khan, Mushtaq Hussain Khan, Nisar Hussain Khan, Rashid Khan
Kairana gharana	Abdul Karim Khan, Abdul Wahid Khan, Sawai Gandharva, Bhimsen Joshi (<i>important</i>)
Agra gharana	Faiyaz Khan, Vilayat Hussain Khan, Latafat Hussain Khan, Pandit Jagannathbuwa Purohit
Etawah / Imdadkhani gharana	Imdad Khan, Inayat Khan, Vilayat Khan, Shahid Parvez
Atrauli-Jaipur gharana	Alladiya Khan, Mogubai Kurdikar, Kishori Amonkar
Saharanpur gharana (sarangi)	Ustad Bundu Khan, Ustad Ashiq Ali Khan

> Ancient Gupta-period Temples, Stupas and Inscriptions of Uttar Pradesh

Historical Heritage / Site	Location (Place)	Period / Discoverer	Historical Significance and Facts
Dashavatara Temple	Deogarh village, Lalitpur	Gupta period (discovered by Captain Charles Strahan)	It was named by Alexander Cunningham. It is an excellent example of the Gupta-period Panchayatana style.
Bhitargaon Temple	Kanpur district	Sixth century (Gupta period)	It is the oldest surviving Hindu temple built of bricks.
Dhamek Stupa	Sarnath, Varanasi (13 km away)	Mauryan/Gupta period	It marks the 'Deer Park' or Rishipattana , where Mahatma Buddha gave his first sermon (Dharmachakrapravartan).
Bhitari Inscription Pillar	Village Bhitari, Saidpur (Ghazipur)	Reign of Skandagupta (about 455-467 CE)	This pillar inscription gives important information about Skandagupta's victory over the Hunas and the history of the Gupta empire.

> Ancient / Historical Names of Principal Cities

Present Name	Ancient / Historical Name
Varanasi	Kashi
Kannauj	Kanyakubja Nagar, Mahodayashri (<i>Mahodaya Nagar</i>)
Gorakhpur	Ramgram, Gorakshapur
Aligarh	Harigarh, Koil or Kol

Present Name	Ancient / Historical Name
Kanpur	Kanhpur
Kapilavastu (Siddharth Nagar)	Piprahwa
Lucknow	Lakshmanpur, Lakhanpur
Kushinagar	Kushinara
Mathura	Madhupuri

> Uttar Pradesh Environment

> Wildlife Sanctuaries and Bird Sanctuaries of Uttar Pradesh

A. Wildlife Sanctuaries

Name of the Wildlife Sanctuary	Related District	Special Fact (from the UPPCS exam point of view)
Chandraprabha Wildlife Sanctuary	Chandauli	The first wildlife sanctuary (1957)
Hastinapur Wildlife Sanctuary	Muzaffarnagar, Meerut, Hapur, Amroha, Bijnor	The largest wildlife sanctuary
Mahavir Swami Wildlife Sanctuary	Lalitpur	The smallest wildlife sanctuary (1977)
Kishanpur Wildlife Sanctuary	Lakhimpur Kheri	1972
Chambal Wildlife Sanctuary	Etawah and Agra	1989
Kaimur Wildlife Sanctuary	Mirzapur and Sonbhadra	-
Katarniaghat Wildlife Sanctuary	Bahraich	-
Ranipur Wildlife Sanctuary	Banda	<i>(it has now also received the status of a tiger reserve)</i>
Sohagibarwa Wildlife Sanctuary	Maharajganj	-
Sohelwa Wildlife Sanctuary	Gonda, Balrampur, Shravasti	-
Turtle Wildlife Sanctuary	Varanasi	-

> Principal Bird Sanctuaries of Uttar Pradesh (Updated 2026)

S.No.	Name of the Bird Sanctuary	Related District	Year of Establishment	Main Exam-useful Features and Special Facts
1	Shaheed Chandrashekhar Azad Bird Sanctuary	Unnao	1984	- Former name: Nawabganj Bird Sanctuary. - It is the first (oldest) bird sanctuary of Uttar Pradesh.
2	Samaspur Bird Sanctuary	Raebareli	1987	- A system of 6 interconnected lakes situated in the middle Ganga plain. - Also included as a Ramsar site.
3	Lakh Bahoshi Bird Sanctuary	Kannauj	1988	It is the largest bird sanctuary of Uttar Pradesh by area.
4	Sandi Bird Sanctuary	Hardoi	1990	- It is a natural wetland near the Garra river. - Included as a Ramsar site.
5	Bakhira Bird Sanctuary	Sant Kabir Nagar	1990	- It is also called the 'Bakhira Wildlife Sanctuary'. - It is the largest natural freshwater lake in Purvanchal.
6	Okhla Bird Sanctuary	Gautam Buddh Nagar	1990	An urban wetland (urban oasis) situated on the Delhi-Noida border.
7	Saman Bird Sanctuary	Mainpuri	1990	One of the largest habitats and breeding areas of the state bird, the Sarus Crane.
8	Parvati Arga Bird Sanctuary	Gonda	1990	Current affairs update: It has been declared an Eco-Sensitive Zone (ESZ) by the Ministry of Environment, Forest and Climate Change. - Formed of two interconnected oxbow lakes.
9	Vijay Sagar Bird Sanctuary	Mahoba	1990	Situated on the bank of a picturesque lake in Mahoba district of the Bundelkhand region.
10	Patna Bird Sanctuary	Etah	1991	Ramsar site update (2026): It has recently been included in the Ramsar list.
11	Jai Prakash Narayan Bird Sanctuary	Ballia	1991	- Popular/local name: Surha Taal Bird Sanctuary. Current affairs update (June 2026): It has been declared India's 100th Ramsar site.

S.No.	Name of the Bird Sanctuary	Related District	Year of Establishment	Main Exam-useful Features and Special Facts
12	Sur Sarovar Bird Sanctuary	Agra	1991	- It is also known as Keetham Lake. - India's famous 'Sloth Bear Rescue Facility' is also here.
13	Dr. Bhimrao Ambedkar Bird Sanctuary	Pratapgarh	2003	It was formerly called the Bainti Taal Bird Sanctuary.
14	Shekha Jheel Bird Sanctuary	Aligarh	2016 (notified)	UP's smallest bird sanctuary by area. Current affairs update (April 2026): It has been designated India's 99th Ramsar site.

➤ Principal Lakes, Taals and Reservoirs of Uttar Pradesh

District	Lake / Taal / Kund / Reservoir Located
Pilibhit	Phulhar Lake (<i>source of the Gomti river</i>)
Sonbhadra	Govind Ballabh Pant Sagar (<i>the largest artificial lake</i>)
Agra	Keetham Lake (Sur Sarovar)
Mathura	Noh Lake, Mansi Ganga Kund-taal, Radha and Shyam Lake
Unnao	Nawabganj Lake, Kundra Samandar
Kanpur Nagar	Moti Jheel
Jhansi	Barua Sagar and Lakshmi Taal
Budaun	Sagar Taal
Meerut	Ramtaal
Muzaffarnagar	Shukra Taal
Mahoba	Bela Sagar Lake, Madan Sagar
Fatehpur	Morai Taal
Raebareli	Samaspur Lake, Visaidha Taal
Pratapgarh	Bainti or Naiya Lake
Mirzapur	Tanda Dari Taal and Sirsi Reservoir
Varanasi	Aundhi Taal
Ballia	Surha Taal

District	Lake / Taal / Kund / Reservoir Located
Sultanpur	Raja ka Bandh, Bhojpur Taal
Gorakhpur	Ramgarh Taal
Sant Kabir Nagar	Bakhira Lake
Ambedkar Nagar	Darwan Lake
Gonda	Parvati and Arga Lake
Bahraich	Chitaura Lake
Lucknow	Karela Lake
Kannauj	Lakh Lake, Bahoshi Lake, Deoria Taal
Shahjahanpur	Bada Taal (oxbow lake)
Bareilly	Leelaur Lake
Rampur	Gaur Lake
Aligarh	Shekha Lake

➤ Updated List of the 13 Ramsar Sites of Uttar Pradesh (2026)

At present Uttar Pradesh has a total of 13 Ramsar Sites. In **2026, 3 new sites were added in UP (Patna Bird Sanctuary, Shekha Jheel Bird Sanctuary and Jai Prakash Narayan Bird Sanctuary / Surha Taal)**, due to which this list has now **increased from 10 to 13**.

S.No.	Name of the Ramsar Site	Related District	Year of Declaration	Main Exam-useful Features / Special Facts
1	Upper Ganga River (Brijghat to Narora stretch)	Bulandshahr, Hapur, Amroha	2005	- UP's first and largest Ramsar site. - A major habitat of the Ganges dolphin and the gharial.
2	Nawabganj Bird Sanctuary	Unnao	2019	- New name: Shaheed Chandrashekhar Azad Bird Sanctuary. - A monsoon-fed natural oxbow lake.
3	Sandi Bird Sanctuary	Hardoi	2019	A famous resting place for migratory birds (especially the common teal).
4	Samaspur Bird Sanctuary	Raebareli	2019	Situated in Salon tehsil, part of the middle Ganga plain.
5	Saman Bird Sanctuary	Mainpuri	2019	- A seasonal oxbow lake. - A major breeding site of the Sarus Crane.

S.No.	Name of the Ramsar Site	Related District	Year of Declaration	Main Exam-useful Features / Special Facts
6	Parvati Arga Bird Sanctuary	Gonda	2019	• A unique system of two interconnected oxbow lakes.
7	Sarsai Nawar Jheel	Etawah	2019	• The densest population and breeding of the Sarus Crane in India take place here.
8	Sur Sarovar (Keetham Lake)	Agra	2020	• A man-made reservoir situated on the bank of the Yamuna river.
9	Haiderpur Wetland	Muzaffarnagar - Bijnor	2021	• A man-made wetland formed by the construction of the Madhya Ganga Barrage. • Situated within the Hastinapur Wildlife Sanctuary.
10	Bakhira Wildlife Sanctuary	Sant Kabir Nagar	2021	• Uttar Pradesh's largest natural freshwater lake .
11	Patna Bird Sanctuary	Etah	January 2026	• Latest addition . In the original wildlife sanctuary category it is UP's smallest bird sanctuary.
12	Shekha Jheel Bird Sanctuary	Aligarh	April 2026	• Latest addition . UP's smallest Ramsar site by area (only 40 hectares).
13	Jai Prakash Narayan Bird Sanctuary	Ballia	June 2026	• Popular name: Surha Taal . • It has the distinction of becoming India's 100th Ramsar site .

> Uttar Pradesh: Forest Area

> Top and Bottom Districts in Forest Area

(Note: According to the Forest Status Report, figures are recorded in area in sq km)

Category (Forest Area)	Name of the District	Forest Area (sq km)
Minimum forest area <i>(in increasing order)</i>	(1) Sant Ravidas Nagar (Bhadohi)	3.12
	(2) Mau	11.00
	(3) Mainpuri	13.64
	(4) Sant Kabir Nagar	14.00
	(5) Deoria	15.21
Maximum forest area <i>(in decreasing order)</i>	(1) Sonbhadra	2540.29 (highest)
	(2) Lakhimpur Kheri	1273.06
	(3) Mirzapur	803.73
	(4) Pilibhit	687.11

> Uttar Pradesh: Economy

> Uttar Pradesh: Agriculture, Economy and Energy Resources

> Economic and Agricultural Indicators

Indicator	UP's Figure / Rank	Special Remark (for the Exam)
Contribution to the country's GDP	About 8%	Among the leading states at the national level
Contribution of agriculture to the state's GDP	26%	Strong position of the primary sector
Growth rate of agriculture and allied sector	10%	-
Rank in agricultural exports	Third place	-
First place in production	Milk production and sugarcane cultivation	Uttar Pradesh is number 1 in the country in both

> Energy Resources

Project / Agency / Centre	Location / Year	Special Fact
Formation of the State Electricity Board	1959	-
Establishment of the Electricity Regulatory Commission	1998	-
New and Renewable Energy Development Agency	1983	-
Order of total electricity generation	Thermal power > Hydro power > Nuclear power > Solar power	<i>Remember the chronology (asked repeatedly)</i>
Gas-based projects of NTPC	3 projects	Auraiya, Aonla, Dadri
Oldest thermal project	Harduaganj Thermal Power Station	Established in 1942
The only nuclear power station	Narora (Bulandshahr)	Established: 1991-92

> Principal Thermal Power Plants of Uttar Pradesh

Name of the Thermal Power Station / Project	Location (Related District)	Fuel / Specific Type
Dadri Thermal Power Station	Gautam Buddh Nagar	Gas-based project
Aonla Thermal Power Project	Bareilly	Gas-based project
Harduaganj Thermal Power Project	Aligarh	One of the oldest thermal projects of the state
Anpara Thermal Power Station	Sonbhadra	-
Singrauli Thermal Power Project	Sonbhadra	-
Meja Thermal Power Station	Prayagraj	-
Tanda Thermal Power Project	Ambedkar Nagar	-

> Expressways of Uttar Pradesh (Length & Map Data)

Name of the Expressway	Length	Special Fact / Current Status
Ganga Expressway	594 km	At present the longest under-construction/proposed expressway
Gorakhpur-Siliguri Expressway	519 km	Inter-state connectivity
Ghaziabad - Kanpur Expressway	380 km	-
Purvanchal Expressway	341 km	-
Agra-Lucknow Expressway	302 km	-
Bundelkhand Expressway	296 km	-
Delhi-Saharanpur-Dehradun Expressway	210 km	-
Yamuna Expressway	165 km	The first expressway of Uttar Pradesh
Ghazipur-Ballia-Manjhihat Expressway	117 km	-
Delhi-Meerut Expressway	96 km	-
Gorakhpur Link Expressway	91 km	-
Lucknow - Kanpur Expressway	63 km	-
Noida-Greater Noida Expressway	25 km	-

> Principal Industrial Parks and Cities of Uttar Pradesh (Parks & Cities)

Name of the Park / City	Location (Place)
Defence Park	Kanpur, Agra, Jhansi and Lucknow
Aerospace Park	Lucknow, Agra, Barabanki
Leather Park	Agra
Leather Technology Park	Banthar (Unnao)
Textile and Hosiery Park	Kanpur
Mega Food Park	Baheri (Bareilly)
Agro Park	Varanasi
Biotechnology Park	Lucknow
Regional Science City	Lucknow
Knowledge Park	Greater Noida
Software Technology Park of India	Noida
Night Safari Park	Greater Noida
Plastic City	Dibiyapur (Auraiya)
Saraswati Hi-Tech City	Prayagraj
Tronica City	Ghaziabad
Trans Delhi Signature City	Ghaziabad
Wave City	Ghaziabad
NIKIDA City	Between Kanpur and Unnao
Trans Ganga City	Unnao
Tourism Theme Park	Agra

> Principal Institutions in Uttar Pradesh

> Principal National Research Institutes

Name of the Institute	Location (City)
Indian Institute of Vegetable Research (IIVR)	Varanasi

Name of the Institute	Location (City)
Indian Institute of Handloom Technology (IIHT)	Chaukaghat, Varanasi
Indian Institute of Pulses Research (IIPR)	Kanpur
National Sugar Institute (NSI)	Kanpur
Indian Institute of Sugarcane Research (IISR)	Lucknow
Indian Veterinary Research Institute (IVRI)	Izzatnagar, Bareilly
Indian Grassland and Fodder Research Institute	Jhansi
Indian Grain Storage Management and Research Institute	Hapur

> Central Universities of Uttar Pradesh

Name of the University	Location	Year of Establishment / Special Note
University of Allahabad	Prayagraj	1887 <i>(the oldest university of the state)</i>
Banaras Hindu University (BHU)	Varanasi	1916
Aligarh Muslim University (AMU)	Aligarh	1920
Babasaheb Bhimrao Ambedkar University	Lucknow	1996
Rajiv Gandhi National Aviation University	Raebareli	Established under the 2013 Act
Rani Lakshmi Bai Central Agricultural University	Jhansi	2014

D. Principal Cultural / Historical Institutions of Uttar Pradesh

Name of the Institution	Location (City)	Year of Establishment / Special Fact
State Museum	Lucknow	1863 <i>(the oldest museum)</i>
College of Arts and Crafts	Lucknow	1911
Bharat Kala Bhavan	Varanasi	1920
Bhatkhande Music Institute	Lucknow	1926 <i>(former name: Marris College of Music; name changed in 1966; received the status of a state university in 2021)</i>
Hindustani Academy	Prayagraj	1927
Uttar Pradesh State Archives	Prayagraj	1949

Name of the Institution	Location (City)	Year of Establishment / Special Fact
Motilal Nehru Children's Museum	Lucknow	1957
State Lalit Kala Akademi	Lucknow	1962
Uttar Pradesh Sangeet Natak Akademi	Lucknow	1963
Uttar Pradesh Urdu Academy	Lucknow	1972
Bhartendu Natya Akademi	Lucknow	1975
Uttar Pradesh Film Corporation	-	1975
Uttar Pradesh Hindi Sansthan	Lucknow	1976
Uttar Pradesh Sanskrit Sansthanam	Lucknow	1976
Ayodhya Shodh Sansthan	Ayodhya	1988
International Ramkatha Museum	Ayodhya	1988
National Kathak Institute	Lucknow	1988-89
Folk Art Museum	Lucknow	1989
Uttar Pradesh Jain Vidya Shodh Sansthan	Lucknow	1990
District Museum	Sultanpur	1990
Uttar Pradesh Bhasha Sansthan	Lucknow	1994
Tribal and Folk Art Culture Museum	Lucknow	1996
State Freedom Struggle Museum	Meerut	1997
Uttar Pradesh Sindhi Academy	Lucknow	1997

Name of the Institution	Location (City)	Year of Establishment / Special Fact
Uttar Pradesh Punjabi Academy	Lucknow	1999
Museum of Kushan and Gupta Period Culture	Mathura	The only specialised museum
Aryabhatta Planetarium	Rampur	-
Indira Gandhi Planetarium	Lucknow	-

> Scheduled Castes (SC) and Scheduled Tribes (ST) in Uttar Pradesh

> Principal Tribes of Uttar Pradesh

Tribe / Group	Related Main Area of Habitation (District)
Gond group (<i>Gond, Dhuria, Nayak, Ojha, Pathari, Rajgond</i>)	Maharajganj, Siddharth Nagar, Basti, Gorakhpur, Deoria, Mau, Azamgarh, Jaunpur, Ballia, Ghazipur, Varanasi, Mirzapur, Sonbhadra, Sant Kabir Nagar, Kushinagar, Chandauli, Bhadohi.
Kharwar group	Deoria, Ballia, Ghazipur, Varanasi, Sonbhadra.
Tharu	Gorakhpur and Balrampur (mainly the Terai region).
Buksa / Bhoksa	Bijnor.
Mahigir	Bijnor.
Saharia	Lalitpur (<i>asked repeatedly</i>).
Parahiya, Baiga, Agariya, Patari, Bhuiya / Bhuiyan	Sonbhadra (<i>all these are concentrated in Sonbhadra</i>).
Pankha / Panika	Sonbhadra and Mirzapur.
Chero	Sonbhadra and Varanasi.

> Scheduled Castes (SC) and Scheduled Tribes (ST) in Uttar Pradesh

Indicator / Category	UP's Figure / District	Rank at the National Level / Special Note
SC % of total population	20.69%	Share of the Dalit population in the total population of the state.
Rank in the country in SC population	First place	The highest share, 16.6% , of India's total number of Scheduled Castes lives in UP.

Indicator / Category	UP's Figure / District	Rank at the National Level / Special Note
District with the highest SC population	Sitapur	-
District with the lowest SC population	Baghpat	-
ST % of total population	0.6%	Share of the tribal population in the total population of the state.
District with the highest ST population	Sonbhadra	-
District with the lowest ST population	Baghpat	Baghpat has the lowest population of both SC and ST.
Districts without Scheduled Tribes	Ayodhya and Jalaun	No Scheduled Tribe (ST) lives in these two districts.

> 'Firsts' in Uttar Pradesh (First in UP)

A. 'Firsts' in Uttar Pradesh

Category / Post / Place	Name of the First Person / Place
First Governor	Sarojini Naidu (<i>also India's first woman Governor</i>)
First Chief Minister	Govind Ballabh Pant
First woman Chief Minister	Sucheta Kripalani
First Chief Justice of the Allahabad High Court	Sir Walter Morgan
First Speaker of the Legislative Assembly	Purushottam Das Tandon
First Information Commissioner	Mohammad Asghar
First Lokayukta	Justice Vishambhar Dayal
First state emergency (President's Rule)	From February 1968 to February 1969
Chief Minister the most times	Mayawati (4 times in all)
First Ramsar site	Upper Ganga River (2005)
First National Park	Dudhwa National Park
First wildlife sanctuary	Chandraprabha Wildlife Sanctuary (Chandauli)
First tiger reserve	Dudhwa Tiger Reserve

Category / Post / Place	Name of the First Person / Place
First zoo	Nawab Wajid Ali Shah Zoological Garden, Lucknow
First UNESCO World Heritage Site	Agra Fort and the Taj Mahal (included in 1983)

B. Firsts in the Field of Education

Institution / College	Place / Name	Year of Establishment and Special Fact
First university	University of Allahabad	1887
First engineering college	Civil Engineers' (college) at Roorkee	1847 (<i>present Uttarakhand</i>) BHU (1916)
First medical college	Sarojini Naidu Medical College	1854 (Agra - former name: Thomason College)
First agricultural college	G. B. Pant University of Agriculture	Pantnagar (<i>then UP, present Uttarakhand</i>)
First Sanskrit college	Banaras	1791 (through the efforts of Resident Jonathan Duncan)
First university for the disabled	Chitrakoot	Jagadguru Rambhadracharya Handicapped University (India's first)
First Urdu Academy	Lucknow	India's first Urdu Academy

C. Firsts in History and Politics

Post / Event	Name / Place	Period / Special Description
First Governor of the United Provinces	Sir Harcourt Butler	During 1921-1950
First Governor of Agra	Sir C. T. Metcalfe	-
First Lieutenant Governor of the United Provinces	Sir James John Digges La Touche	Agra and Oudh region
First Nawab of Awadh	Saadat Ali Khan	Founder of the autonomous state of Awadh
First session of the Congress in UP	Allahabad	1888 (under the presidency of George Yule)
First Hindi newspaper	Banaras Akhbar	-
Buddha's first sermon	Sarnath (Varanasi)	In Buddhist history it is called ' Dharmachakrapravartan '

D. 'Firsts' in Uttar Pradesh: Awards and Honours

Category of Award	First Recipient (Male)	First Recipient (Female)
Bharat Ratna	Jawaharlal Nehru and Govind Ballabh Pant (1955)	Indira Gandhi (1971)
Padma Vibhushan	Vijaya Lakshmi Pandit (1962)	-
Padma Bhushan	Amarnath Jha, Shanti Swarup, Maithili Sharan Gupt, S. Narayan Shastri (1954)	Rameshwari Nehru (1955)
Padma Shri	Akhil Chandra Mitra (1954)	Mona Chandravati Gupta (1965)

E. Firsts in Environment and Ecology

Site / Sanctuary	Name	Special Fact
First Ramsar site	Upper Ganga River	Included in 2005
First National Park	Dudhwa National Park	Lakhimpur Kheri
First wildlife sanctuary	Chandraprabha Wildlife Sanctuary	Chandauli
First tiger reserve	Dudhwa Tiger Reserve	-
First zoo	Nawab Wajid Ali Shah Zoological Garden	Lucknow

F. Firsts in Infrastructure

Infrastructure Field	First Place / Name in Uttar Pradesh	Historical Year / Fact
First metro service	Lucknow Metro	-
First expressway	Yamuna Expressway	-
First National Highway (NH)	NH-34	Starts near Meerut and connects to Loharu in Haryana
First railway station	Kanpur railway station	1859
First train operation	Between Prayagraj and Kanpur	Run in 1859
First biotechnology park	Bakshi Ka Talab, Lucknow	-
Asia's first DNA bank	Biotech Park, Lucknow	Important at the global level

Infrastructure Field	First Place / Name in Uttar Pradesh	Historical Year / Fact
First Wi-Fi monument	Taj Mahal	Wi-Fi facility since 2015
First film centre	Noida (Gautam Buddh Nagar)	-

> Uttar Pradesh: Census (Census 2011)

A. Principal Statistical Figures

> Population

Category / Indicator	Total Figures	Percentage / Figure of Males	Percentage / Figure of Females
Total population	19,98,12,341 (16.51% of the country)	52.29%	47.71%
Area-wise population %	-	Rural: 77.73%	Urban: 22.27%

Indicator	UP's Figure / Rate	Special Note (for UPPCS)
Population growth rate	20.23%	Decadal growth rate (2001-2011)
Population density	829 persons/km ²	Considerably higher than the national average (382)
Urbanisation	22.3%	Rural population is 77.7%
Sex ratio	912	Females per 1000 males
Total literacy rate	67.7%	Male: 77.3% Female: 57.2% (the gap is exactly 20%)
Area-wise literacy	-	Rural: 65.5% Urban: 75.1%
Scheduled Castes (SC)	20.7%	Share in the total population of the state
Scheduled Tribes (ST)	0.6%	Share in the total population of the state

B. Decreasing/Increasing Order of Districts (Top 5 District Order)

(Note: > means decreasing order and < means increasing order)

Category / Indicator	Top 5 Districts (Highest to Lowest)	Bottom 5 Districts (Lowest to Highest)
Total population	Prayagraj > Moradabad > Ghaziabad > Azamgarh > Lucknow	Mahoba < Chitrakoot < Hamirpur < Shravasti < Lalitpur

Category / Indicator	Top 5 Districts (Highest to Lowest)	Bottom 5 Districts (Lowest to Highest)
Decadal growth rate	Gautam Buddh Nagar > Ghaziabad > Shravasti > Bahraich > Balrampur	Kanpur Nagar < Hamirpur < Baghpat < Fatehpur < Deoria
Population density	Ghaziabad > Varanasi > Lucknow > Sant Ravidas Nagar > Kanpur Nagar	Lalitpur < Sonbhadra < Hamirpur < Mahoba < Chitrakoot
Literacy rate	Gautam Buddh Nagar > Kanpur Nagar > Auraiya > Etawah > Ghaziabad	Shravasti < Bahraich < Balrampur < Budaun < Rampur
Sex ratio	Jaunpur > Azamgarh > Deoria > Pratapgarh > Sultanpur	Gautam Buddh Nagar < Hamirpur/Baghpat < Kanpur Nagar/Banda/Mathura
Rural population	Prayagraj > Azamgarh > Jaunpur > Sitapur > Gorakhpur	<i>Minimum rural %:</i> Ghaziabad < Lucknow < Kanpur Nagar < Gautam Buddh Nagar < Meerut
Urban population %	Ghaziabad > Lucknow > Kanpur Nagar > Gautam Buddh Nagar > Meerut	<i>Minimum urban %:</i> Shravasti < Kushinagar < Maharajganj < Sultanpur < Pratapgarh
Scheduled Castes (SC)	<i>Population:</i> Sitapur > Prayagraj > Hardoi > Azamgarh > Lakhimpur Kheri <i>Percentage (%):</i> Kaushambi > Sitapur > Hardoi > Unnao > Raebareli	<i>Population:</i> Baghpat < Shravasti < Gautam Buddh Nagar < Mahoba < Lalitpur <i>Percentage (%):</i> Baghpat < Bareilly < Balrampur < Gautam Buddh Nagar < Rampur/Varanasi
Scheduled Tribes (ST)	Sonbhadra > Ballia > Deoria > Kushinagar > Lalitpur	<i>(Sonbhadra has the highest ST population)</i>

B. Top Districts with the Highest and Lowest Values (Extreme Points)

Indicator	Highest (Top Districts / Figures)	Lowest (Bottom Districts / Figures)
Total population	(1) Prayagraj (59,54,351) (2) Moradabad (3) Ghaziabad	(1) Mahoba (8,75,958) (2) Chitrakoot (3) Hamirpur
Population density	(1) Ghaziabad (3971) (2) Varanasi (3) Lucknow	(1) Lalitpur (242) (2) Sonbhadra (3) Hamirpur
Total literacy rate	(1) Gautam Buddh Nagar (80.12%) (2) Kanpur Nagar (3) Auraiya	(1) Shravasti (46.74%) (2) Bahraich (3) Balrampur

Indicator	Highest (Top Districts / Figures)	Lowest (Bottom Districts / Figures)
Male literacy	Gautam Buddh Nagar (88.06%)	Shravasti (57.16%) <i>(the same order as total literacy)</i>
Female literacy	Kanpur Nagar (75.05%)	Shravasti (34.78%)
Decadal growth rate	Gautam Buddh Nagar (49%)	Kanpur Nagar (9.9%)
Sex ratio	Jaunpur (1024)	Gautam Buddh Nagar (851)

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